Informatica® Data Ingestion and Replication
October 2026

Connectors and Connections

Informatica Data Ingestion and Replication Connectors and Connections
October 2026

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Preface

Use *Data Ingestion and Replication Connectors and Connections* to determine the types of connectors that you need to download to be able to access sources and targets for each type of ingestion and replication task. It also describes the properties that you configure when you define a connection to be used by a task. You can download connectors and define connections in Administrator.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Connectors and Connections

Connections provide access to data in cloud and on-premises applications, platforms, databases, and flat files. Before you can define a connection, ensure that the connector for the source or target type is installed in IDMC.

If multiple connectors are available for a source or target type, get the one that your ingestion type supports. Some connectors are pre-installed. If you need a connector that is not pre-installed, you can download it from the **Add-On Connectors** page in Administrator.

CHAPTER 2

Data Ingestion and Replication connectors

You must have the correct connectors to create connections for the sources and targets you use in your Data Ingestion and Replication tasks.

Before you can define connections, your organization administrator must ensure that the source and target connectors that the organization uses are installed. Also, you must enable connectors for your runtime environment.

Some connectors are shared with other services, and some connectors are specific to Data Ingestion and Replication only. The connectors that are specific to Data Ingestion and Replication typically have names that include "Mass Ingestion" or "Database Ingestion."

For more information about connectors and connections, see "Licenses," "Runtime Environment," and "Connections" in the Administrator help.

Application Ingestion and Replication connectors

Before you define a connection for application ingestion and replication tasks, verify that the connectors for your source and target types are available in IDMC.

The following table lists the connectors that Application Ingestion and Replication requires to connect to a source or target:

Source or target type	Connector	Use for
Amazon Aurora PostgreSQL	PostgreSQL	Targets, all load types[1] Use only with Salesforce sources and with SAP ECC and SAP S4/HANA sources that use the SAP Mass Ingestion connector.
Amazon Redshift	Amazon Redshift V2	Targets, all load types
Amazon S3	Amazon S3 V2	Targets, all load types
Apache Kafka	Kafka	Targets, incremental load type
Databricks	Databricks	Targets, all load types

Source or target type	Connector	Use for
Google BigQuery	Google BigQuery V2	Targets, all load types
Google Cloud Storage	Google Cloud Storage V2	Targets, all load types
Marketo	Marketo V3	Sources, all load types
Microsoft Azure Data Lake Storage Gen2	Microsoft Azure Data Lake Storage Gen2	Targets, all load types
Microsoft Azure Synapse Analytics[3]	Microsoft Azure Synapse Analytics Database Ingestion	Targets, all load types
Microsoft Azure SQL Database, Microsoft SQL Server	SQL Server	Targets, all load types
Microsoft Fabric Data Warehouse	Microsoft Fabric Data Warehouse	Targets, all load types
Microsoft Fabric Lakehouse	Microsoft Fabric Lakehouse	Targets, all load types
Microsoft Fabric OneLake	Microsoft Fabric OneLake	Targets, all load types
Microsoft Dynamics 365	Microsoft Dynamics 365 Mass Ingestion	Sources, all load types
NetSuite	NetSuite Mass Ingestion	Sources, all load types
Oracle	Oracle Database Ingestion	Targets, all load types
Oracle Fusion Cloud Applications	Oracle Fusion Cloud Mass Ingestion	Sources - **REST**: All load types. - **BICC**: All load types.
Salesforce	Salesforce Mass Ingestion	Sources, all load types
Salesforce Data 360	Salesforce Data 360	Targets, all load types
Salesforce Marketing Cloud	Salesforce Marketing Cloud	Sources, initial load type
SAP ECC	- SAP Mass Ingestion - SAP ODP Extractor	Sources: - **SAP Mass Ingestion**: All load types. - **SAP ODP Extractor**: All load types.
SAP S/4HANA	SAP OData V2	Sources, all load types.
SAP S/4HANA	SAP ODP Extractor	Sources, all load types
ServiceNow	ServiceNow Mass Ingestion	Sources, all load types
Snowflake	Snowflake Data Cloud	Targets, all load types

Source or target type	Connector	Use for
Workday	Workday Mass Ingestion	Sources: - **SOAP**: All load types. - **RaaS**: Initial load type.

1. Load types are initial load, incremental load, and combined initial and incremental load.
2. For Microsoft Azure Synapse Analytics targets, Application Ingestion and Replication uses Microsoft Azure SQL Data Lake Storage Gen2 to store staging files. Before you configure a connection for a Microsoft Azure Synapse Analytics target, ensure that you have Microsoft Azure SQL Data Lake Storage Gen2 installed.

Database Ingestion and Replication connectors

Before you begin defining connections for database ingestion and replication tasks, verify that the connectors for your source and target types are available in IDMC.

The following table lists the connectors that Database Ingestion and Replication requires to connect to a source or target that can be configured in a database ingestion and replication task:

Source or target type	Connector	Use for
Amazon Redshift	Amazon Redshift V2	Targets, all load types[1]
Amazon S3	Amazon S3 V2	Targets, all load types
Databricks	Databricks	Targets, all load types
Db2 for i	Db2 for i Database Ingestion	Sources, all load types
Db2 for Linux, UNIX, and Windows	Db2 for LUW Database Ingestion	Sources, initial loads, and incremental loads and combined loads that use Log-based or Query-based CDC
Db2 for z/OS	Db2 for zOS Database Ingestion	Sources, all load types
Flat file	No connector required	Targets, initial load jobs
Google BigQuery	Google BigQuery V2	Targets, all load types
Google Cloud Storage	Google Cloud Storage V2	Targets, all load types
Kafka, including Apache Kafka, Confluent Kafka, Amazon Managed Streaming for Apache Kafka, and Kafka-enabled Azure Event Hubs	Kafka	Targets, incremental load type
Microsoft Azure Data Lake Storage Gen2	Microsoft Azure Data Lake Storage Gen2	Targets, all load types

Source or target type	Connector	Use for
Microsoft SQL Server, including on-premises SQL Server, RDS for SQL Server, Azure SQL Database, and Azure SQL Managed Instance	SQL Server	Sources, all load types. For Azure SQL Database sources, you must use the **Query-based** or **CDC Tables** capture method for incremental load and combined load jobs. Targets, all load types
Microsoft Azure Synapse Analytics[2]	Microsoft Azure Synapse Analytics Database Ingestion	Targets, all load types
Microsoft Fabric Data Warehouse	Microsoft Fabric Data Warehouse	Targets, all load types
Microsoft Fabric Lakehouse	Microsoft Fabric Lakehouse	Targets, all load types
Microsoft Fabric OneLake	Microsoft Fabric OneLake	Targets, all load types
MongoDB	MongoDB Mass Ingestion	Sources, all load types
MySQL, including RDS for MySQL	MySQL	Sources, initial load and incremental load types. RDS for MySQL in initial load jobs only.
Netezza	Netezza	Sources, initial load type
Open Table	Open Table	Targets, all load types
Oracle, including RDS for Oracle	Oracle Database Ingestion	Sources, all load types Targets, all load types
Oracle Autonomous Database	Oracle Autonomous Database	Targets, initial load type
Oracle Cloud Infrastructure (OCI) Object Storage	Oracle Cloud Object Storage	Targets, all load types
PostgreSQL, including on-premises PostgreSQL, Amazon Aurora PostgreSQL, Azure Database for PostgreSQL - Flexible Server, RDS for PostgreSQL, and Cloud SQL for PostgreSQL	PostgreSQL	Sources, all load types Targets (Amazon Aurora PostgreSQL targets only), all load types
Salesforce Data 360	Salesforce Data 360	Targets, all load types
SAP HANA, including on-premises SAP HANA and SAP HANA Cloud	SAP HANA Database Ingestion	Sources, all load types
Snowflake	Snowflake Data Cloud	Targets, all load types

Source or target type	Connector	Use for
Teradata Data Warehouse Appliance	Teradata	Sources, initial load type

1. Load types are initial load, incremental load, and combined initial and incremental load.
2. For the Microsoft Azure Synapse Analytics target type, Database Ingestion and Replication uses Microsoft Azure SQL Data Lake Storage Gen2 to store staging files. Ensure that you have Microsoft Azure SQL Data Lake Storage Gen2 installed.

Mock connectors

Database Ingestion and Replication supports mock, or sample, connections for some of the sources and targets. Use mock connections to learn how to create database ingestion and replication initial load tasks without creating real connections to the database.

A mock connector does not connect to a real database. Instead, a source mock connector uses flat files with sample data. A target mock connector reports the information about processed source data to Database Ingestion and Replication user interface, but it does not write any data to the target.

The sample connections appear in the source and target connection lists in Database Ingestion and Replication if you have the MockConnector license.

The following table lists mock connections that you can use for Database Ingestion and Replication sources and targets:

Connection name	Source or Target
Sample Oracle Connection	Source
Sample SQL Server Connection	Source
Sample S3 Connection	Target
Sample ADLS Gen2 Connection	Target

Note: You must use sample connections for both the source and target databases. You cannot use a sample connection for only one of them, for example, for the source but not for the target.

Source data

The source data for sample connections is stored in CVS files in the following directory:

Secure_Agent_installation/downloads/package-MockConnector.*version*/package/sampleData/source/*database_type*/

Each file represents a single table. A mock table name matches the file name. The first line in a file determines column headers, and the subsequent lines contain row data.

File Ingestion and Replication connectors

Before you define connections for file ingestion and replication tasks, ensure that you have a license for the connectors that File Ingestion and Replication requires for the source and target types.

The following table lists the connectors that a file ingestion and replication task supports based on the source and target types:

Source or target name	Connector	Source or target type
Local folder	No connector required	Source and Target
Advanced FTP	Advanced FTP V2	Source and Target
Advanced FTPS	Advanced FTPS V2	Source and Target
Advanced SFTP	Advanced SFTP V2	Source and Target
Amazon S3	Amazon S3 V2	Source and Target
Amazon Redshift	Amazon Redshift V2	target
Cloud Integration Hub	Cloud Integration Hub	Source and Target **Note:** You can't configure a file ingestion and replication task using Cloud Integration Hub as both source and target.
Databricks	Databricks	Target
Google BigQuery	Google BigQuery V2	Target
Google Cloud Storage	Google Cloud Storage V2	Source and Target
Hadoop Files	Hadoop Files V2	Source and Target
Microsoft Azure Blob Storage	Microsoft Azure Blob Storage V3	Source and Target
Microsoft Azure Data Lake Store	Microsoft Azure Data Lake Store Gen2	Source and Target
Microsoft Azure Data Lake Store	Microsoft Azure Data Lake Store Gen1 V3	Source and Target
Microsoft Azure Synapse SQL	Microsoft Azure Synapse SQL	Target
Microsoft Fabric OneLake	Microsoft Fabric OneLake	Source and Target
Oracle Cloud Object Storage	Oracle Cloud Object Storage	Source and Target
Snowflake	Snowflake Data Cloud	Target

Streaming Ingestion and Replication connectors

Before you define connections for the streaming ingestion and replication tasks, ensure that you have a license for the required connectors for your source and target types.

The following table lists the connectors that a streaming ingestion and replication task supports based on the source and target type:

Source or target name	Connector	Source or target type
Amazon Kinesis Data Firehose	Amazon Kinesis	Target
Amazon Kinesis Data Streams	Amazon Kinesis	Source and Target
Amazon Managed Streaming for Apache Kafka (Amazon MSK)	Kafka	Source and Target
Amazon S3	Amazon S3 V2	Target
AMQP	AMQP	Source
Apache Kafka	Kafka	Source and Target
Azure Event Hubs Kafka	Kafka	Source
Business 360 Events	Business 360 Events	Source
Confluent Kafka	Kafka	Source and Target
Databricks	Databricks	Target
Flat file	No connector required	Source and Target
Google BigQuery V2	Google BigQuery V2	Target
Google Cloud Storage	Google Cloud Storage V2	Target
Google PubSub	Google PubSub	Source and Target
JDBC V2	JDBC V2	Target
JMS	JMS	Source
Microsoft Azure Data Lake Storage	Azure Data Lake Store Gen2	Target
Microsoft Azure Event Hub	Azure Event Hubs	Target
MQTT	MQTT	Source
OPC UA	OPCUA	Source
REST V2	REST V2	Source

Note: While importing a streaming ingestion and replication task, both read and write connection types appear in the drop-down list on the **Import Review** page. You can also see connections to connectors that are not supported by Streaming Ingestion and Replication.

CHAPTER 3

Data Ingestion and Replication connection properties

Connections provide access to data in cloud and on-premises applications, platforms, databases, and flat files. Connection definitions include the location of the source or target, the runtime environment, and the other properties specific to the connection type.

Before you can create a connection, ensure that the correct connectors for your sources and targets are available in IDMC. The supported connectors vary by type of ingestion task.

To create a connection or search for an existing connection, use the Administrator service.

After you configure connection properties, the connection becomes available for use within the organization.

Configuring a connection

Configure a source or target connection on the **Connections** page in Administrator.

1. In Administrator, select **Connections**.
2. On the **Connections** page, click **New Connection**.
3. Configure the following connection details:

Property	Description
Connection Name	The name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Maximum length is 100 characters. Connection names are not case sensitive.
Description	A description of the connection.

4. Configure the connection-specific properties.

   For example, if you are configuring an Amazon S3 connection, enter the Amazon S3 connection properties. Click the Help icon for a description of each connection property.
5. To test the connection, click **Test Connection**.

   The results of the test are displayed at the top of the page.

If a connection fails, contact the database administrator, or recheck your settings and verify that the selected runtime environment has the status of Up and Running.

6. Click **Save** to save the connection.

Advanced FTP V2 connection properties

When you set up an Advanced FTP V2 connection, you must configure the connection properties.

The following table describes the Advanced FTP V2 connection properties:

Connection property	Description
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.
Runtime Environment	The name of the runtime environment where you want to run the tasks. Specify a Secure Agent.
Host	The host name or IP address of the FTP server.
Port	The port number to use for connecting to the FTP server. If left blank, the default port number 21 is used.
Username	User name to connect to the FTP server.
Password	Password to connect to the FTP server.
Folder Path	The directory to use after connecting to the FTP server.
Use passive mode	Indicates whether the connection uses **Passive** or **Active** mode. Specify **Yes** to use **Passive** mode. Specify **No** to use **Active** mode. The default value is **Yes**. In Passive mode, the server does not need to connect back to a port on the connection client, which is a firewall-friendly mode. If you have problems with connecting to the server, you might want to change the mode to Passive by selecting **Yes** for this option. In Passive mode, depending on the FTP server, the connection may require high port range based on the port availability to transfer data. In Active mode, the server attempts to connect back to a port on the connection client to perform the data transfer.
Data Connection Start Port	The starting port number to use for the data connection.
Data Connection End Port	The ending port number to use for the data connection.

Connection property	Description
Timeout	The number of seconds to wait when attempting to connect to the server. A timeout occurs if the connection cannot be established in the specified amount of time. If left blank, the default value of 120 seconds is used.
Connection Retry Attempts	The number of times to connect to retry the FTP connection if a connection cannot be established. This setting is used for both the initial connection and any reconnect attempts due to lost connections. If left blank, no retries will be attempted.
Connection Retry Interval	The number of seconds to wait between each connection retry attempt. **Note:** For instance, if you want to retry to connect up to 10 times with a five second delay between retries, then specify **10** for the **Connection Retry Attempts** and **5** for the **Connection Retry** Interval.
Control Encoding	If left blank, the connection uses the ISO standard ISO-8859-1. If supported by the server, other encodings such as UTF-8 can be specified to support international characters.
List Parser	The list parser to use for the server connection. If the field is left blank, the Advanced FTP V2 Connector attempts to use the MLSD parser. If the MLSD parser is not supported by the server, the UNIX parser is used. If you experience problems listing directories, select a different list parser.
Date Format	This date format is applied if the server returns a date that is different from the selected list parser default. If your location requires a different date format (for example, d MMM yyyy), specify the date format in this field. Not all list parsers support the date format setting. List parsers that do not support the date format setting ignores any user specified value.
Recent Date Format	Specify the date format to use when parsing the recent last modified date for each file. The recent date format applies in UNIX-based systems and appears on entries less than a year old. If your location requires a specific date format (for example, d MMM HH:mm), specify that pattern in this field. Not all list parsers support the recent date format setting. List parsers that do not support the recent date format setting ignores any user-specified value.
Bandwidth	Controls the maximum amount of network resources used for file transfers. The value is applicable for file uploads and downloads. Default is 0. 0 indicates that the bandwidth is not restricted.
Bandwidth Unit	The unit of the network bandwidth used for file transfer. You can choose one of the following units: - Kilobytes per second (KBps) - Megabytes per second (MBps)

Note: Advanced FTP V2 connector doesn't support NTLM proxy authentication.

Advanced FTPS V2 connection properties

When you set up an Advanced FTPS V2 connection, you must configure the connection properties.

The following table describes the Advanced FTPS V2 connection properties:

Connection property	Description
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.
Runtime Environment	The name of the runtime environment that contains the Secure Agent that you want to run the tasks.
Host	The host name or IP address of the server.
Port	The port number to use for connecting to the server. If left blank, the default port number is 21.
Username	User name to connect to the FTPS server.
Password	Password to connect to the FTPS server.
Folder Path	The directory to use after connecting to the server.
Use passive mode	Indicates whether the connection uses **Passive** or **Active** mode. Specify **Yes** to use **Passive** mode. Specify **No** to use **Active** mode. The default value is **Yes**. In Passive mode, the server does not need to connect back to a port on the connection client, which is a firewall-friendly mode. If you have problems with connecting to the server, you might want to change the mode to Passive by selecting **Yes** for this option. In Passive mode, depending on the FTPS server, the connection may require high port range based on the port availability to transfer data. In Active mode, the server attempts to connect back to a port on the connection client to perform the data transfer.
Data Connection Start Port	The starting port number to use for the data connection.
Data Connection End Port	The ending port number to use for the data connection.
Timeout	The number of seconds to wait when attempting to connect to the server. A timeout occurswill if the connection cannot be established in the specified amount of time. If left blank, the default value of 120 seconds is used.
Connection Retry Attempts	The number of times to connect to retry the Advanced FTP V2 connection if a connection cannot be established. This setting is used for both the initial connection and any reconnect attempts due to lost connections. If left blank, no retries will be attempted.

Connection property	Description
Connection Retry Interval	The number of seconds to wait between each connection retry attempt. **Note:** For instance, if you want to retry to connect up to 10 times with a five second delay between retries, then specify **10** for the **Connection Retry Attempts** and **5** for the **Connection Retry** Interval.
Control Encoding	If left blank, the connection uses the ISO standard ISO-8859-1. If supported by the server, other encodings like UTF-8 can be specified to support international characters.
Trusted Server	Specify whether the FTPS server is a trusted server. The Advanced FTP V2 Connector only supports a trusted server.
List Parser	The list parser to use for the server connection. If the field is empty, the Advanced FTP V2 Connector tries to use the MLSD parser. If the server does not support the MLSD parser, the connector uses the UNIX parser. If you experience problems listing directories, select a different list parser.
Date Format	This date format is applied if the server returns a date that is different from the selected list parser default. If your location requires a different date format (for example, d MMM yyyy), specify the date format in this field. Not all list parsers support the date format setting. List parsers that do not support the date format setting ignores any user specified values.
Recent Date Format	Specify the date format to use when parsing the recent last modified date for each file. The recent date format applies in UNIX-based systems and appears on entries less than a year old. If your location requires a specific date format (for example, d MMM HH:mm), specify that pattern in this field. Not all list parsers support the recent date format setting. List parsers that do not support the recent date format setting ignores any user-specified values.
Connection Type	Indicates if the connection type is IMPLICIT_SSL or EXPLICIT_SSL. - IMPLICIT_SSL. The connection automatically starts as an SSL connection. - EXPLICIT_SSL. After initial authentication with the FTPS server, the connection is encrypted with SSL or TLS depending on the security protocol you select. Default is IMPLICIT_SSL.
SecurityProtocol	Indicates whether SSL or TLS is used for EXPLICIT_SSL connections. Default is SSL.
Key Store File	The path and file name of the keystore file. The keystore file contains the certificates to authenticate the FTPS server.
Key Store Password	The password for the keystore file required to access the Trusted Server Certificate Store.
Key Alias	The alias of the individual key.
Key Store Type	Indicates if the type of the keystore is Java KeyStore (JKS) or Public Key Cryptology Standard (PKCS12). Default is JKS.
Bandwidth	Controls the maximum amount of network resources used for file transfers. The value is applicable for file uploads and downloads. Default is 0. 0 indicates that the bandwidth is not restricted.
Bandwidth Unit	The unit of the network bandwidth used for file transfer. You can choose one of the following units: - Kilobytes per second (KBps) - Megabytes per second (MBps)

Note: Advanced FTPS V2 connector doesn't support NTLM proxy authentication.

Advanced SFTP V2 connection properties

When you set up an Advanced SFTP V2 connection, you must configure the connection properties.

The following table describes the Advanced SFTP V2 connection properties:

Connection property	Description
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.
Runtime Environment	The name of the runtime environment that contains the Secure Agent that you want to run the tasks.
Host	The host name or IP address of the server. The host name is case insensitive and must be unique within the domain. The name cannot exceed 24 characters. It can contain letters (A to Z), digits (0 to 9), period (.) special character, and minus (-) sign.
Port	The port number to use to connect to the server. Default is 22.
Username	User name to connect to the SFTP server.
Password	Password to connect to the SFTP server.
Folder Path	The directory to use after connecting to the server.
Timeout	The number of seconds to wait when attempting to connect to the server. A timeout occurs if the connection cannot be established in the specified amount of time. If left blank, the default value of 120 seconds is used.
Connection Retry Attempts	The number of times to connect to retry the SFTP connection if a connection cannot be established. This setting is used for both the initial connection and any reconnect attempts due to lost connections. If left blank, no retries will be attempted.
Connection Retry Interval	The number of seconds to wait between each connection retry attempt. For example, if you want to retry to connect up to 10 times with a five second delay between retries, then specify 10 for the Connection Retry Attempts and 5 for the Connection Retry Interval.
Private Key File	The name of the SSH private key file along with the path where the file is stored. Ensure that the file path is on the machine that hosts the Secure Agent. For example, `C:/SSH/my_keys/key.ppk`
Private Key Passphrase	Specify the passphrase to encrypt the SSH private key.

Connection property	Description
Use Curve Kex Algorithm	Controls which key exchange algorithms are used. If you enable this option, the connection supports the following algorithms: - curve25519-sha256 - hmac-sha2-512 - hmac-sha2-256 If you do not enable this option, the connector supports the following algorithms: - ecdh-sha2-nistp256 - ecdh-sha2-nistp384 - ecdh-sha2-nistp521 - diffie-hellman-group14-sha1 - diffie-hellman-group-exchange-sha256 - diffie-hellman-group-exchange-sha1 - diffie-hellman-group1-sha **Note:** You can also use this connection to access SFTP servers that require the aes128-gcm@openssh.com or aes256-gcm@openssh.com GCM cipher. You can authenticate to servers that require the rsa-sha2-512 or rsa-sha2-256 RSA SHA-2 host key algorithm.
Bandwidth	Controls the maximum amount of network resources used for file transfers. The value is applicable for file uploads and downloads. Default is 0. 0 indicates that the bandwidth is not restricted.
Bandwidth Unit	The unit of the network bandwidth used for file transfer. You can choose one of the following units: - Kilobytes per second (KBps) - Megabytes per second (MBps)
Use File Integration Proxy Server	The connector connects to the SFTP server through the file integration proxy server. Verify that the following prerequisites are met: - You must have the File Integration Service license to use this option. - You must define a proxy server in File Servers. - If you don't have the File Integration Service proxy, you need to use the agent proxy through the proxy.ini file.
Proxy Server Host Name	Host name or IP address of the outgoing File Integration Service proxy server.
Proxy Server Port	Port number of the outgoing File Integration Service proxy server.

Note: Advanced SFTP V2 connector doesn't support NTLM proxy authentication.

Amazon Kinesis connection properties

The Amazon Kinesis connection is a messaging connection. Use the Amazon Kinesis connection to access Amazon Kinesis Data Streams or Amazon Kinesis Data Firehose as targets.

Amazon Kinesis Firehose connection properties

When you set up an Amazon Kinesis Firehose connection, you must configure the connection properties.

Important: Amazon Kinesis Firehose Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release.

The following table describes the Amazon Kinesis Firehose connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. Description that you can use to identity the connection. The description cannot exceed 4,000 characters.
Type	The Amazon Kinesis connection type. If you do not see the Amazon Kinesis connection type, go to the **Add-On Connectors** page to enable the connector.
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Service	The type of Kinesis Service that you want to use. Select **Kinesis Firehose**.
AWS Access Key ID	The access key ID of the Amazon AWS user account.
AWS Secret Access Key	The secret access key for the Amazon AWS user account.
Region	Region where the endpoint for your service is available. You can select one of the following values: - us-east-2. Indicates the US East (Ohio) region. - us-east-1. Indicates the US East (N. Virginia) region. - us-west-1. Indicates the US West (N. California) region. - us-west-2. Indicates the US West (Oregon) region. - ap-northeast-1. Indicates the Asia Pacific (Tokyo) region. - ap-northeast-2. Indicates the Asia Pacific (Seoul) region. - ap-northeast-3. Indicates the Asia Pacific (Osaka-Local) region. - ap-south-1. Indicates the Asia Pacific (Mumbai) region. - ap-southeast-1. Indicates the Asia Pacific (Singapore) region. - ap-southeast-2. Indicates the Asia Pacific (Sydney) region. - ca-central-1. Indicates the Canada (Central) region. - cn-north-1. Indicates the China (Beijing) region. - cn-northwest-1. Indicates the China (Ningxia) region. - eu-central-1. Indicates the EU (Frankfurt) region. - eu-west-1. Indicates the EU (Ireland) region. - eu-west-2. Indicates the EU (London) region. - eu-west-3. Indicates the EU (Paris) region. - sa-east-1. Indicates the South America (São Paulo) region. - us-gov-west-1. Indicates AWS GovCloud (US-West) region. - us-gov-east-1. Indicates AWS GovCloud (US-East) region. A streaming ingestion and replication task does not support ap-northeast-3 region.

Property	Description
Connection TimeOut (ms)	Optional. Number of milliseconds that the Data Ingestion and Replication service waits to establish a connection to the Kinesis Firehose after which it times out. Default is 10,000 milliseconds.
AWS Credential Profile Name	An AWS credential profile defined in the credentials file. A mapping accesses the AWS credentials through the profile name at run time. If you do not provide an AWS credential profile name, the mapping uses the access key ID and secret access key that you specify when you create the connection.
ARN of IAM Role	The Amazon Resource Name specifying the role of an IAM user. Applies to Cross-Account IAM Roles authentication.
External ID	The external ID for an IAM role is an additional restriction that you can use in an IAM role trust policy to designate who can assume the IAM role. Applies to Cross-Account IAM Roles authentication.

Amazon Kinesis Streams connection properties

When you set up an Amazon Kinesis Streams connection, you must configure the connection properties.

Important: Amazon Kinesis Streams Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release.

The following table describes the Amazon Kinesis Streams connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. Description that you can use to identity the connection. The description cannot exceed 4,000 characters.
Type	The Amazon Kinesis connection type. If you do not see the Amazon Kinesis connection type, go to the **Add-On Connectors** page to install the connector.
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Service	The type of Kinesis Service that you want to use. Select **Kinesis Streams**.
AWS Access Key ID	The access key ID of the Amazon AWS user account.
AWS Secret Access Key	The secret access key for your Amazon AWS user account.

Property	Description
Region	Region where the endpoint for your service is available. You can select one of the following values: - us-east-2. Indicates the US East (Ohio) region. - us-east-1. Indicates the US East (N. Virginia) region. - us-west-1. Indicates the US West (N. California) region. - us-west-2. Indicates the US West (Oregon) region. - ap-northeast-1. Indicates the Asia Pacific (Tokyo) region. - ap-northeast-2. Indicates the Asia Pacific (Seoul) region. - ap-northeast-3. Indicates the Asia Pacific (Osaka-Local) region. - ap-south-1. Indicates the Asia Pacific (Mumbai) region. - ap-southeast-1. Indicates the Asia Pacific (Singapore) region. - ap-southeast-2. Indicates the Asia Pacific (Sydney) region. - ca-central-1. Indicates the Canada (Central) region. - cn-north-1. Indicates the China (Beijing) region. - cn-northwest-1. Indicates the China (Ningxia) region. - eu-central-1. Indicates the EU (Frankfurt) region. - eu-west-1. Indicates the EU (Ireland) region. - eu-west-2. Indicates the EU (London) region. - eu-west-3. Indicates the EU (Paris) region. - sa-east-1. Indicates the South America (São Paulo) region. - us-gov-west-1. Indicates AWS GovCloud (US-West) region. - us-gov-east-1. Indicates AWS GovCloud (US-East) region. A streaming ingestion and replication task does not support ap-northeast-3 region.
Connection TimeOut (ms)	Optional. Number of milliseconds that the Data Ingestion and Replication service waits to establish a connection to the Kinesis Streams after which it times out. Default is 10,000 milliseconds.
AWS Credential Profile Name	An AWS credential profile defined in the credentials file. A mapping accesses the AWS credentials through the profile name at run time. If you do not provide an AWS credential profile name, the mapping uses the access key ID and secret access key that you specify when you create the connection.
ARN of IAM Role	The Amazon Resource Name specifying the role of an IAM user. Applies to Cross-Account IAM Roles authentication.
External ID	The external ID for an IAM role is an additional restriction that you can use in an IAM role trust policy to designate who can assume the IAM role. Applies to Cross-Account IAM Roles authentication.

AWS Credential Profile

You can define AWS credential profiles in the credentials file. Each credential profile consists of secret access key and access key ID.

Users can use the AWS credential profile names to use different AWS credentials at run time than the AWS credentials that they specify when they create an Amazon Kinesis connection with an Amazon Kinesis Streams as a source and target and Amazon Kinesis Firehose as a target.

Create AWS credentials for the users, such as access key ID and secret access key. Users can select an authentication type while creating an Amazon Kinesis connection, such as AWS credential profile. The default authentication type is AWS credential profile.

Generate an Access Key ID and Secret Access Key for the users in AWS.

Amazon Redshift V2 connection properties

Create an Amazon Redshift V2 connection to read from or write data to Amazon Redshift.

Prepare for authentication

You can configure **Default** and **Redshift IAM Authentication via AssumeRole** authentication types in an Amazon Redshift V2 connection to connect to Amazon Redshift. Additionally, you need to complete the S3 staging prerequisites to access S3 resources. You can also configure encryption, if required, to connect to Amazon Redshift.

Note: Application ingestion and replication, database ingestion and replication, and file ingestion and replication tasks do not support Redshift IAM authentication via AssumeRole unless you use an EC2 instance to assume the role.

See the following sections for a summary of the authentication, staging, and encryption prerequisites.

Authentication prerequisites

Before you begin, you need to have a registered user account with Amazon Redshift.

Get the minimum required details from your Amazon Redshift account from the AWS Console for the authentication type that you want to configure, as listed in the following table:

Default authentication	Redshift IAM Authentication via Assume Role
- JDBC URL - User name - Password	- JDBC URL - User name - Database name - Cluster identifier - Redshift IAM role ARN*
*To use the Redshift IAM role ARN, configure the Redshift IAM role ARN with the required trust policies to generate temporary security credentials to access Amazon Redshift. For instructions, see "Configure an assume role for Amazon Redshift" on page 31.	

Staging prerequisites

To enable staging on Amazon S3 and to gain access to S3 resources when you read or write data, you need to configure the staging properties in the Amazon Redshift V2 connection.

The following table summarizes the staging options that you can configure in the connection for both default and Redshift IAM Authentication via AssumeRole authentication and the tasks that you need to perform to get the required details for S3 staging:

S3 staging options	Tasks
Generate temporary credentials for the IAM user who assumes the S3 IAM role to access S3 staging.	**AWS configurations** Enable IAM users to assume an S3 IAM role and generate temporary credentials. For instructions, see the following references: - "Generate temporary security credentials using AssumeRole for Amazon S3 staging" on page 35. - Using an assume role for Amazon S3 resources How-To Library article. **Redshift V2 connection configurations** - Enter the value of the **S3 IAM Role ARN**. - Enter the **S3 Access Key ID** and **S3 Secret Access Key** values.
Generate temporary security credentials for an EC2 instance that assumes an S3 IAM role to access S3 staging.	**AWS configurations** Define an EC2 instance to assume an S3 IAM role and generate the temporary credentials for S3 staging. For instructions, see "Generate temporary security credentials using AssumeRole for EC2" on page 37. **Redshift V2 connection configurations** Configure the following minimum required properties: - Enable **Use EC2 Role to Assume Role**. - Enter the value of the **S3 IAM Role ARN**.
Generate the S3 access and secret access keys for the IAM user with access to the S3 bucket.	**AWS configurations** To generate the credentials, perform the following tasks: 1. "Create a minimal Amazon IAM policy" on page 30. 2. Create an IAM user, assign the policy to that user, and then generate the S3 access key ID and S3 secret access key in the AWS console. For more information about how to create an IAM user and generate keys, see the AWS documentation. **Redshift V2 connection configurations** Enter the **S3 Access Key ID** and **S3 Secret Access Key** values.
Configure IAM authentication	**AWS configurations** If you have an EC2 instance, and do not want to specify the keys or use the IAM role ARN, then assign the minimum policy to the EC2 with access to the S3 bucket. For instructions, see "Configure IAM authentication" on page 31. **Redshift V2 connection configurations** In this case, you do not need to enable or specify any of the staging properties in the connection.

Encryption prerequisites

To configure client-side and server-side encryption for the Default authentication and Redshift IAM authentication via AssumeRole during staging, see "Enable encryption" on page 38.

Create a minimal Amazon IAM policy

To stage the data in Amazon S3, you need to create an IAM policy with the minimum required permissions to access the S3 resources.

You can either attach the policy to the IAM user and generate the S3 access key ID and S3 secret access keys to access S3 resources. Or, if you have an EC2 instance, you can assign the minimum policy to the EC2 instance to access the S3 bucket for staging.

You need the following minimum required permissions in the policy:

- PutObject
- GetObject
- DeleteObject
- ListBucket

You can use the following sample Amazon IAM policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:DeleteObject",
                "s3:ListBucket",
                "s3:ListBucketMultipartUploads"
                    ],
            "Resource": [
                "arn:aws:s3:::<bucket_name>/*",
                "arn:aws:s3:::<bucket_name>"
            ]
        }
    ]
}
```

Note: The **Test Connection** does not validate the IAM policy assigned to users. Hence, ensure that the policy assigned to the user is valid.

Configure IAM authentication

Configure AWS Identity and Access Management (IAM) authentication and create a minimal Amazon IAM policy for both the EC2 role and Redshift role.

For instructions, see the following How-to-Library article: Configuring AWS IAM Authentication

Configure an assume role for Amazon Redshift

To use the Redshift IAM role ARN, configure the Redshift IAM role ARN with the required trust policies to generate temporary security credentials to access Amazon Redshift.

You can use one of the following options to generate the temporary security credentials:

AWS configurations	Connection details
Option 1. Configure an AssumeRole to enable an IAM user.	To use the AssumeRole for the IAM user, specify the following IAM user details: - Redshift Access Key ID - Redshift Secret Access Key - Redshift IAM Role ARN
Option 2. Define an EC2 instance to assume a Redshift IAM role.	To use the AssumeRole for Amazon EC2: - Specify the **Redshift IAM Role ARN** value. - Enable the **Use EC2 Role to Assume Role** check box.

For application ingestion and replication tasks and database ingestion and replication tasks, use Option 2 to have an EC2 role assume the Redshift IAM role.

For more information about configuring an AssumeRole, see the following How-to-Library article: Configure AssumeRole authentication for Amazon Redshift V2 Connector

Generate temporary security credential policies for Amazon Redshift

To use the temporary security credentials to connect to Amazon Redshift, both the IAM user and IAM role require policies.

The following section lists the policies required for the IAM user and IAM role:

IAM user

An IAM user must have the `sts:AssumeRole` policy to use the temporary security credentials in the same or different AWS account. The IAM user credentials are used to key-in the Redshift access key and Redshift secret key in the connection properties.

The following sample policy allows an IAM user to use the temporary security credentials in an AWS account:

```
{
"Version":"2012-10-17",
"Statement":
   { "Effect":"Allow",
"Action":"sts:AssumeRole",
"Resource":"arn:aws:iam::<ACCOUNT-HYPHENS>:role/<REDSHIFT-IAM-ROLE-NAME>"
   }
}
```

Redshift IAM role trust policy

The Redshift IAM role policy pertains to the role that is specified in the Redshift IAM Role ARN. An IAM role must have a trust policy attached with it to allow the IAM user to access Redshift using the temporary security credentials.

The following policy is a sample trust policy:

```
{
"Version":"2012-10-17",
"Statement":[
         {
         "Effect":"Allow",
         "Principal":{ "AWS":"arn:aws:iam::AWS-account-ID:<IAM-USER>" },
         "Action":"sts:AssumeRole"
          }
        ]
       }
}
```

For example, you can specify the role or user in the following format:

```
{
"Version": "2012-10-17",
"Statement": [
    {
"Effect": "Allow",
"Principal": {
"AWS": "arn:aws:iam::<AWS-account>:role/<name-of-the-role>"
    },
       "Action": "sts:AssumeRole"
    }
 ]
}


{
"Version": "2012-10-17",
"Statement": [
    {
"Effect": "Allow",
"Principal": {
"AWS": " arn:aws:iam::<AWS-account>:user/<name-of-the-user>"
    },
       "Action": "sts:AssumeRole"
    }
 ]
}
```

Minimum permission policies of the Redshift IAM role

The following policy shows the permissions required to the Redshift IAM Role, which will be assumed by an IAM user to connect to the Redshift database using an existing Amazon Redshift user:

```
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": [
          "redshift:GetClusterCredentials",

          "redshift:DescribeClusters"
            ],

"Resource": [
          "arn:aws:redshift:<REGION>:<ACCOUNT-ID>:dbuser:<Cluster_Identifier>/
<USER_NAME>",

          "arn:aws:redshift:<REGION>:<ACCOUNT-ID>:dbname:<Cluster_Identifier>/
<DATABASE_NAME>"

            ]

}
}
]
}
}
```

The following policy shows the permissions needed to be attached to the Redshift IAM Role, which will be assumed by an IAM user to connect to the Redshift database with a newly created user by the `Auto create DBUser` check box:

```
{

"Version": "2012-10-17",

"Statement": [
```

```
{

"Effect": "Allow",

"Action": [
          "redshift:GetClusterCredentials",

          "redshift:DescribeClusters",

          "redshift:CreateClusterUser",

          "redshift:JoinGroup"
            ],

"Resource": [
          "arn:aws:redshift:<REGION>:<ACCOUNT-ID>:dbuser:<Cluster_Identifier>/
<USER_NAME>",

          "arn:aws:redshift:<REGION>:<ACCOUNT-ID>:dbname:<Cluster_Identifier>/
<DATABASE_NAME>",

          "arn:aws:redshift:<REGION>:<ACCOUNT-ID>:dbgroup:<Cluster_Identifier>/
<GROUP_NAME>"
            ]
}

}

]

}
```

Generate temporary security credentials using AssumeRole for EC2

You can use temporary security credentials using AssumeRole for an Amazon EC2 role to connect to Amazon Redshift from the same or different AWS accounts.

The Amazon EC2 role can assume another IAM role from the same or different AWS account without requiring a Redshift access key and Redshift secret key.

Consider the following prerequisites when you use temporary security credentials using AssumeRole for EC2:

- To use temporary security credentials using AssumeRole for EC2, install the Secure Agent on an AWS service such as Amazon EC2.
- The EC2 role attached to the AWS EC2 service must not have access to Amazon Redshift but needs to have permission to assume another IAM role.
- The IAM role that needs to be assumed by the EC2 role must have a permission policy and a trust policy attached to it.

To configure an EC2 role to assume the IAM Role provided in the Redshift IAM Role ARN connection property, select the **Use EC2 Role to Assume Role** check box in the connection properties.

EC2 service role trust policy

The following is a sample trust policy that is defined in a trust relationship of the EC2 role attached to the EC2 instance:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "ec2.amazonaws.com"
```

```
                },
                "Action": "sts:AssumeRole"
            }
        ]
    }
```

The following is a sample trust policy of the Redshift IAM role when you enable EC2 assume role:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "Service": "redshift.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        },
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<Account-ID:role>/ec2_role_attached_to_ec2_instance"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

The permission policy that is required to be attached to the EC2 instance is same as the policy defined for the IAM user.

Configure an assume role for Amazon S3 staging

To configure AssumeRole authentication for S3 staging, you need to attach the minimum permission policies and trust policies for the IAM user and IAM role in the AWS console.

An IAM user can use the AssumeRole to temporarily gain access to the Amazon S3 resources. For more information about using an assume role for Amazon S3 resources, you can also refer to the How-to-Library article: Using an assume role for Amazon S3 resources

You can generate temporary security credentials using AssumeRole for Amazon S3 staging to access the Amazon S3 staging bucket. If you want EC2 instances to assume an IAM role to gain access to the S3 staging bucket securely, use the temporary security credentials generated using AssumeRole for EC2 instances.

Note: Do not use the root user credentials of the AWS account to generate the temporary security credentials. You need to use the credentials of an IAM user to generate the temporary security credentials.

Generate the temporary security credentials based on your requirement.

Generate temporary security credentials using AssumeRole for Amazon S3 staging

You can use the temporary security credentials using AssumeRole to access the Amazon S3 staging bucket from the same or different AWS accounts.

Ensure that you have the **sts:AssumeRole** permission and a trust relationship established within the AWS accounts to use the temporary security credentials. The trust relationship is defined in the trust policy of the IAM role when you create the role. The IAM role adds the IAM user as a trusted entity allowing the IAM users to use the temporary security credentials and access the AWS accounts. For more information about how to establish the trust relationship, see the AWS documentation.

When the trusted IAM user requests for the temporary security credentials, the AWS Security Token Service (AWS STS) dynamically generates the temporary security credentials that are valid for a specified period and provides the credentials to the trusted IAM users. The temporary security credentials consist of access key ID, secret access key, and secret token.

To use the dynamically generated temporary security credentials, provide the value of the **S3 IAM Role ARN** connection property when you create an Amazon Redshift V2 connection. The IAM Role ARN uniquely identifies the AWS resources. Then, specify the time duration in seconds during which you can use the temporarily security credentials in the **Temporary Credential Duration** advanced source and target properties.

External ID

You can specify the external ID for a more secure access to the Amazon S3 bucket when the Amazon S3 bucket is in a different AWS account than the IAM user or EC2 instance.

Note: Application ingestion and replication and database ingestion and replication tasks do not support use of External ID.

You can optionally specify the external ID in the AssumeRole request to the AWS Security Token Service (STS).

The external ID must be a string. The following sample shows an external ID condition in the assumed IAM role's trust policy:

```
"Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::AWS_Account_ID : user/user_name"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId": "dummy_external_id"
        }
      }
    }
  ]
```

Temporary security credentials policy

To use the temporary security credentials to access the Amazon S3 staging bucket, both the IAM user and IAM role require policies.

The following section lists the policies required for the IAM user and IAM role:

IAM user

An IAM user must have the `sts:AssumeRole` policy to use the temporary security credentials in the same or different AWS account.

The following sample policy allows an IAM user to use the temporary security credentials in an AWS account:

```
{
"Version":"2012-10-17",
"Statement":
        { "Effect":"Allow", "Action":"sts:AssumeRole",
          "Resource":"arn:aws:iam::<ACCOUNT-HYPHENS>:role/<ROLE-NAME>" }
}
```

IAM role

An IAM role must have a `sts:AssumeRole` policy and a trust policy attached with the IAM role to allow the IAM user to access the Amazon S3 bucket using the temporary security credentials. The policy specifies the Amazon S3 bucket that the IAM user can access and the actions that the IAM user can

perform. The trust policy specifies the IAM user from the AWS account that can access the Amazon S3 bucket.

The following policy is a sample trust policy:

```
{
"Version":"2012-10-17",
"Statement":[
         {
         "Effect":"Allow",
         "Principal":{ "AWS":"arn:aws:iam::AWS-account-ID:<ROLE-NAME" },
         "Action":"sts:AssumeRole"
          }
        ]
       }
}
```

Temporary security credentials for KMS

To use the temporary security credentials with AWS Key Management Service (AWS KMS)-managed customer master key and enable encryption with KMS, you must create a KMS policy.

You can perform the following operations to use the temporary security credentials and enable encryption with KMS:

- GenerateDataKey
- DescribeKey
- Encrypt
- Decrypt
- ReEncrypt

You can use the following sample policy:

```
{
"Version":"2012-10-17",
"Statement":[
{ "Effect":"Allow",
"Action":
[ "kms:GenerateDataKey", "kms:DescribeKey", "kms:Encrypt", "kms:Decrypt",
"kms:ReEncrypt*" ],
"Resource": ["arn:aws:kms:region:account:key/<KMS_key>"]
    }
  ]
   }
```

Generate temporary security credentials using AssumeRole for EC2

You can use temporary security credentials using AssumeRole for an Amazon EC2 role to access the Amazon S3 staging bucket from the same or different AWS accounts.

The Amazon EC2 role can assume another IAM role from the same or different AWS account without requiring a permanent access key and secret key. The Amazon EC2 role can also assume another IAM role from a different region.

Consider the following prerequisites when you use temporary security credentials using AssumeRole for EC2:

- To use temporary security credentials using AssumeRole for EC2, install the Secure Agent on an AWS service such as Amazon EC2.
- The EC2 role attached to the AWS EC2 service must not have access to Amazon S3 but needs to have permission to assume another IAM role.
- The IAM role that needs to be assumed by the EC2 role must have a permission policy and a trust policy attached to it.

To configure an EC2 role to assume the IAM Role provided in the **IAM Role ARN** connection property, select the **Use EC2 Role to Assume Role** check box in the connection properties.

Enable encryption

You can enable server-side encryption in the Amazon Redshift V2 connection for staging data in Amazon S3. To enable server-side encryption, create an AWS Key Management Service (AWS KMS)-managed customer master key.

Generate the customer master key ID for the same region where your Amazon S3 staging bucket resides.

For more information about generating a customer master key, see the AWS documentation.

To enable encryption with the customer master key, you need to create a minimal KMS policy. You can specify the customer master key ID when you create an Amazon Redshift V2 connection.

Create a minimal policy for using AWS KMS

To use the AWS Key Management Service (AWS KMS)-managed customer master key and enable the encryption with KMS, you must create a KMS policy.

You can perform the following operations to enable encryption with KMS:

- GenerateDataKey
- DescribeKey
- Encrypt
- Decrypt
- ReEncrypt

Sample policy:

```
{
"Version":"2012-10-17",
"Statement":[
{ "Effect":"Allow",
"Action":
[ "kms:GenerateDataKey", "kms:DescribeKey", "kms:Encrypt", "kms:Decrypt",
"kms:ReEncrypt*" ],
"Resource": ["arn:aws:kms:region:account:key/<KMS_key>"]
    }
  ]
   }
```

Connect to Amazon Redshift

Let's configure the Amazon Redshift V2 connection properties to connect to Amazon Redshift.

Before you begin

Before you get started, you'll need to get information from your Amazon Redshift account based on the authentication type you want to configure.

Check out "Prepare for authentication" on page 29 to learn about the authentication requirements before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _. + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	Name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use a Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. For file ingestion and replication tasks, use a Secure Agent. You can't run an application ingestion and replication task, database ingestion and replication task, or file ingestion and replication task on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can configure default and Redshift IAM AssumeRole authentication types to access Amazon Redshift.
Note: Application ingestion and replication tasks and database ingestion and replication tasks do not support Redshift IAM AssumeRole authentication without an EC2 instance.

Select the required authentication method and then configure the authentication-specific parameters.

Default authentication

The following table describes the basic connection properties for default authentication:

Properties	Description
JDBC URL	The JDBC URL to connect to the Amazon Redshift cluster. You can get the JDBC URL from your Amazon AWS Redshift cluster configuration page. Enter the JDBC URL in the following format: `jdbc:redshift://<cluster_endpoint>:<port_number>/<database_name>`, where the endpoint includes the Redshift cluster name and region. For example, `jdbc:redshift://infa-rs-cluster.abc.us-west-2.redshift.amazonaws.com:5439/rsdb` In the example, - infa-rs-qa-cluster is the name of the Redshift cluster. - us-west-2.redshift.amazonaws.com is the Redshift cluster endpoint, which is the US West (Oregon) region. - 5439 is the port number for the Redshift cluster. - rsdb is the specific database instance in the Redshift cluster to which you want to connect.
Username	User name of your database instance in the Amazon Redshift cluster.
Password	Password of the Amazon Redshift database user.
Use EC2 Role to Assume Role	Enables the EC2 instance that assumes an S3 IAM role to access the S3 resources to stage data using the temporary security credentials. The EC2 role must have a policy attached with permissions to assume an S3 IAM role. The S3 IAM role and the EC2 instance can be in the same or different AWS account. Select the check box to enable the EC2 role to assume an S3 IAM role specified in the S3 IAM Role ARN option to access the S3 resources for staging data. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks. By default, this check box is not selected. For instructions, see "Generate temporary security credentials using AssumeRole for EC2" on page 37.
S3 IAM Role ARN	The Amazon Resource Number (ARN) of the IAM role assumed by the IAM user or EC2 to use the dynamically generated temporary security credentials to stage data in Amazon S3. This property applies when you want to generate temporary security credentials to access the S3 staging buckets by using either the EC2 instance or the IAM user who assumes the S3 IAM role. Specify the S3 IAM role name to use the temporary security credentials to access the Amazon S3 staging bucket. For more information about how to get the ARN of the S3 IAM role, see the AWS documentation. **Note:** If you use the connection for application ingestion and replication or database ingestion and replication tasks that use role-based authentication, but not the default role for the AWS cluster, specify an IAM role ARN. If you use the default role, leave this field blank.

Advanced settings

The following table describes the advanced connection properties for default authentication:

Properties	Description
S3 Access Key ID	Access key ID of the IAM user to access the Amazon S3 staging bucket. Enter the access key ID when you use the following methods for S3 staging: - When the IAM user has access to S3 staging. - When the IAM user who assumes the S3 IAM role uses the temporary security credentials to access S3. The S3 access key ID is only validated at runtime, so verify its accuracy before saving the connection to prevent runtime errors. You do not need to enter the S3 access key ID if you use IAM authentication or the assume role for EC2 to access S3. **Note:** If you use the connection for application ingestion and replication or database ingestion and replication tasks that use key-based authentication, provide the access key value.
S3 Secret Access Key	Secret access key to access the Amazon S3 staging bucket. The secret access key is associated with the access key ID and uniquely identifies the account. Enter the secret access key value when you use following methods for S3 staging: - When the IAM user has access to S3 staging. - When the IAM user who assumes the S3 IAM role uses the temporary security credentials to access S3. The S3 secret access key is only validated at runtime, so verify its accuracy before saving the connection to prevent runtime errors. You do not need to enter the S3 secret access key if you use IAM authentication or the assume role for EC2 to access S3. **Note:** If you use the connection for application ingestion and replication or database ingestion and replication tasks that use key-based authentication, provide the access key value.
S3 VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for Amazon S3. You can use a VPC endpoint to enable private communication with Amazon S3. Select one of the following options: - Default. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon S3 through an interface endpoint which uses a private IP address from the IP address range of your subnet. It serves as an entry point for traffic destined to an AWS service.
Endpoint DNS Name for Amazon S3	The DNS name for the Amazon S3 interface endpoint. Replace the asterisk symbol with the bucket keyword in the DNS name. Enter the DNS name in the following format: `bucket.<DNS name of the interface endpoint>` For example, `bucket.vpce-s3.us-west-2.vpce.amazonaws.com`
STS VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for AWS Security Token Service. You can use a VPC endpoint to enable private communication with Amazon Security Token Service. Select one of the following options: - Default. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon Security Token Service through an interface endpoint which uses a private IP address from the IP address range of your subnet.

Properties	Description
Endpoint DNS Name for AWS STS	The DNS name for the AWS STS interface endpoint. For example, `vpce-01f22cc14558c241f-s8039x4c.sts.us-west-2.vpce.amazonaws.com`
KMS VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for AWS Key Management Service. You can use a VPC endpoint to enable private communication with Amazon Key Management Service. Select one of the following options: - Default. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon Key Management Service through an interface endpoint which uses a private IP address from the IP address range of your subnet.
Endpoint DNS Name for AWS KMS	The DNS name for the AWS KMS interface endpoint. For example, `vpce-0e722f5c721e19232-g2pkm2r7.kms.us-west-2.vpce.amazonaws.com`
External ID	The external ID associated with the IAM role. You can specify the external ID if you want to provide a more secure access to the Amazon S3 bucket. The Amazon S3 staging bucket and the IAM role can be in the same or different AWS accounts. If required, you also have the option to specify the external ID in the AssumeRole request to the AWS Security Token Service (STS) using an external ID condition in the assumed IAM role's trust policy. For more information about using an external ID, see External ID when granting access to your AWS resources. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks.

Properties	Description
Cluster Region	The AWS cluster region in which the Redshift cluster resides. Select the cluster region from the list if you choose to provide a custom JDBC URL with a different cluster region from that specified in the **JDBC URL** field property. To continue to use the cluster region name specified in the **JDBC URL** field property, select **None** as the cluster region in this property. You can only read data from or write data to the cluster regions supported by the AWS SDK. Select one of the following cluster regions: None Asia Pacific(Mumbai) Asia Pacific(Seoul) Asia Pacific(Singapore) Asia Pacific(Sydney) Asia Pacific(Tokyo) Asia Pacific(Hong Kong) AWS GovCloud (US) AWS GovCloud (US-East) Canada(Central) EU(Ireland) EU(Frankfurt) EU(Paris) EU(Stockholm) South America(Sao Paulo) Middle East(Bahrain) US East(N. Virginia) US East(Ohio) US West(N. California) US West(Oregon) Default is **None**. **Note:** A region value is required for application ingestion and replication tasks and database ingestion and replication tasks.
Connection Environment SQL	The SQL statement to set up the database environment that applies for the entire session. Separate multiple values with a semicolon (;). Specify only the configurations for the database environment in the SQL statement. Do not specify any DDL or DML commands in the SQL statement.
Master Symmetric Key	You cannot use client-side encryption type for Amazon Redshift V2 Connector. Hence, if you specify the master symmetric key, it is ignored. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks.
Customer Master Key ID	The customer master key ID generated by AWS Key Management Service (AWS KMS) or the ARN of your custom key for cross-account access when you stage data in Amazon S3. The customer master key serves to encrypt your data at the destination before they are saved in Amazon S3. You can either enter the customer-generated customer master key ID or the default customer master key ID. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks.

Redshift IAM Authentication via AssumeRole

The Redshift AssumeRole authentication enables the user to assume an IAM role or define an EC2 role configured with required trust policies to generate temporary security credentials to access Amazon Redshift.

Note: For application ingestion and replicationtasks and database ingestion and replication tasks, you must use an EC2 role.

The following table describes the basic connection properties for Redshift IAM AssumeRole authentication:

Properties	Description
JDBC URL	The JDBC URL to connect to the Amazon Redshift cluster. You can get the JDBC URL from your Amazon AWS Redshift cluster configuration page. Enter the JDBC URL in the following format: `jdbc:redshift://<cluster_endpoint>:<port_number>/<database_name>`, where the endpoint includes the Redshift cluster name and region. For example, `jdbc:redshift://infa-rs-cluster.abc.us-west-2.redshift.amazonaws.com:5439/rsdb` In the example, - infa-rs-qa-cluster is the name of the Redshift cluster. - us-west-2.redshift.amazonaws.com is the Redshift cluster endpoint, which is the US West (Oregon) region. - 5439 is the port number for the Redshift cluster. - rsdb is the specific database instance in the Redshift cluster to which you want to connect.
Username	User name of your database instance in the Amazon Redshift cluster.
Cluster Identifier	The unique identifier of the cluster that hosts Amazon Redshift. Specify the Amazon Redshift cluster name.
Database Name	Name of the Amazon Redshift database where the tables that you want to access are stored.
Redshift IAM Role ARN	The Amazon Resource Number (ARN) of the IAM role assumed by EC2 to use the dynamically generated temporary security credentials to access Amazon Redshift. Enter the Redshift IAM role ARN to access the Amazon Redshift cluster.

Properties	Description
Use EC2 Role to Assume Role	Enables the EC2 role to assume an IAM role, either to connect to Redshift or to stage data using the temporary security credentials: **Connect to Redshift with IAM authentication using the EC2 role** Select the check box to enable the EC2 role that assumes a Redshift IAM role specified in the **Redshift IAM Role ARN** field to access Amazon Redshift. The EC2 role must have a policy attached with permissions to assume a Redshift IAM role from the same or different account. **Access S3 resources to stage data** Select the check box to enable the EC2 role to assume an S3 IAM role specified in the **S3 IAM Role ARN** field and dynamically generate the temporary security credentials to access the S3 staging buckets. The EC2 role must have a policy attached with permissions to assume an S3 IAM role from the same or different AWS account.
S3 IAM Role ARN	The Amazon Resource Number (ARN) of the S3 IAM role assumed by the IAM user or EC2 to use the dynamically generated temporary security credentials to stage data in Amazon S3. This property applies when you want to generate the temporary security credentials to access the S3 staging buckets by using either the EC2 instance or the IAM user who assumes the S3 IAM role. Specify the S3 IAM role name to use the temporary security credentials to access the Amazon S3 staging bucket. For more information about how to get the ARN of the IAM role, see the AWS documentation. **Note:** If you use the connection for application ingestion and replication or database ingestion and replication tasks that uses role-based authentication, but not the default role for the AWS cluster, specify an IAM role ARN. If you use the default role, leave this field blank.

Advanced settings

The following table describes the advanced connection properties for Redshift IAM AssumeRole authentication:

Properties	Description
Redshift Access Key ID	The access key of the IAM user that has permissions to assume the Redshift IAM AssumeRole ARN. This property doesn't apply to Amazon Redshift AssumeRole authentication with EC2 role.
Redshift Secret Access Key	The secret access key of the IAM user that has permissions to assume the Redshift IAM Assume Role ARN. This property doesn't apply to Amazon Redshift AssumeRole authentication with EC2 role.
Database Group	The name of the database group to which you want to add the database user when you select the **Auto Create DBUser** option in this connection property. The user that you add to this database group inherits the specified group privileges. If you do not specify a database group name, the user is added to the public group and inherits its associated privileges. You can also enter multiple database groups, separated by a comma, to add the user to each of the specified database groups.

Properties	Description
Expiration Time	The time duration that the password for the Amazon Redshift database user expires. Specify a value between 900 seconds and 3600 seconds. Default is 900.
Auto Create DBUser	Select to create a new Amazon Redshift database user at run time. The agent adds the user you specified in the **Username** field to the database group. The added user assumes the privileges assigned to the database group. Default is disabled.
S3 Access Key ID	Access key of the IAM user to access the Amazon S3 staging bucket. Enter the access key ID when you use the following methods for S3 staging: - When the IAM user has access to S3 staging. - When the IAM user who assumes the S3 IAM role uses the temporary security credentials to access S3. The S3 access key ID is only validated at runtime, so verify its accuracy before saving the connection to prevent runtime errors. You do not need to enter the S3 access key ID if you use IAM authentication or the assume role for EC2 to access S3. **Note:** If you use the connection for application ingestion and replication or database ingestion and replication tasks that use key-based authentication, provide the access key value.
S3 Secret Access Key	Secret access key to access the Amazon S3 staging bucket. The secret key is associated with the access key and uniquely identifies the account. Enter the secret access key value when you use following methods for S3 staging: - When the IAM user has access to S3 staging. - When the IAM user who assumes the S3 IAM role uses the temporary security credentials to access S3. The S3 secret access key is only validated at runtime, so verify its accuracy before saving the connection to prevent runtime errors. You do not need to enter the S3 secret access key if you use IAM authentication or the assume role for EC2 to access S3. **Note:** If you use the connection for application ingestion and replication or database ingestion and replication tasks that use key-based authentication, provide the access key value.
S3 VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for Amazon S3. You can use a VPC endpoint to enable private communication with Amazon S3. Select one of the following options: - Default. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon S3 through an interface endpoint which uses a private IP address from the IP address range of your subnet. It serves as an entry point for traffic destined to an AWS service.
Endpoint DNS Name for Amazon S3	The DNS name for the Amazon S3 interface endpoint. Replace the asterisk symbol with the bucket keyword in the DNS name. Enter the DNS name in the following format: `bucket.<DNS name of the interface endpoint>` For example, `bucket.vpce-s3.us-west-2.vpce.amazonaws.com`

Properties	Description
STS VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for AWS Security Token Service. You can use a VPC endpoint to enable private communication with Amazon Security Token Service. Select one of the following options: - Default. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon Security Token Service through an interface endpoint which uses a private IP address from the IP address range of your subnet.
Endpoint DNS Name for AWS STS	The DNS name for the AWS STS interface endpoint. For example, `vpce-01f22cc14558c241f-s8039x4c.sts.us-west-2.vpce.amazonaws.com`
KMS VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for AWS Key Management Service. You can use a VPC endpoint to enable private communication with Amazon Key Management Service. Select one of the following options: - Default. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon Key Management Service through an interface endpoint which uses a private IP address from the IP address range of your subnet.
Endpoint DNS Name for AWS KMS	The DNS name for the AWS KMS interface endpoint. For example, `vpce-0e722f5c721e19232-g2pkm2r7.kms.us-west-2.vpce.amazonaws.com`
External ID	The external ID associated with the IAM role. You can specify the external ID if you want to provide a more secure access to the Amazon S3 bucket when the Amazon S3 staging bucket is in same or different AWS accounts. If required, you also have the option to specify the external ID in the AssumeRole request to the AWS Security Token Service (STS) using an external ID condition in the assumed IAM role's trust policy. For more information about using an external ID, see External ID when granting access to your AWS resources. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks.

Properties	Description
Cluster Region	The AWS geographical region in which the Redshift cluster resides. Select the cluster region from the list if you choose to provide a custom JDBC URL with a different cluster region from that specified in the **JDBC URL** field property. To continue to use the cluster region name specified in the **JDBC URL** field property, select **None** as the cluster region in this property. You can only read data from or write data to the cluster regions supported by the AWS SDK. Select one of the following cluster regions: None Asia Pacific(Mumbai) Asia Pacific(Seoul) Asia Pacific(Singapore) Asia Pacific(Sydney) Asia Pacific(Tokyo) Asia Pacific(Hong Kong) AWS GovCloud (US) AWS GovCloud (US-East) Canada(Central) EU(Ireland) EU(Frankfurt) EU(Paris) EU(Stockholm) South America(Sao Paulo) Middle East(Bahrain) US East(N. Virginia) US East(Ohio) US West(N. California) US West(Oregon) Default is **None**. **Note:** A region value is required for application ingestion and replication tasks and database ingestion and replication tasks.
Connection Environment SQL	The SQL statement to set up the database environment that applies for the entire session. Separate multiple values with a semicolon (;). Specify only the configurations for the database environment in the SQL statement. Do not specify any DDL or DML commands in the SQL statement.
Master Symmetric Key	You cannot use client-side encryption type for Amazon Redshift V2 Connector. Hence, if you specify the master symmetric key, it is ignored. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks.
Customer Master Key ID	The customer master key ID generated by AWS Key Management Service (AWS KMS) or the ARN of your custom key for cross-account access when you stage data in Amazon S3. The customer master key serves to encrypt your data at the destination before they are saved in Amazon S3. You can either enter the customer-generated customer master key ID or the default customer master key ID. For more information about how to configure server-side encryption, see "Enable encryption" on page 38. This property doesn't apply to application ingestion and replication tasks and database ingestion and replication tasks.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.
You can configure the Secure Agent to use the proxy server on Windows and Linux. You can use only an unauthenticated proxy server.

Note: The Secure Agent uses the proxy server only when it connects to Amazon S3 to stage the data, not when it connects to Amazon Redshift.

To configure the proxy settings for the Secure Agent, perform the following tasks:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

The proxy settings configured through the Secure Agent Manager are applicable when you create a connection or mapping, and the proxy settings configured through the JVM options are applicable when you run the mapping.

Private communication with Amazon Redshift

If you do not want to expose your traffic to the public internet, you can enable private communication with Amazon Redshift by configuring a gateway endpoint on the AWS console.

To establish a private connection with Amazon Redshift, ensure that the Secure Agent is a part of the subnet in the AWS Virtual Private Cloud (VPC). You can create a gateway endpoint and stage the Amazon S3 data to Amazon Redshift.

To configure private communication to connect to Amazon Redshift, you need to perform the following tasks:

- Create a cluster subnet group.
- Create a Redshift-managed VPC endpoint.
- Configure the gateway endpoint.

You can then specify the gateway endpoint in the Amazon Redshift V2 connection properties.

For more information, see
Configuring private communication with Amazon Redshift using the Amazon Redshift V2 Connector.

Private communication with Amazon S3

You can configure an Amazon Redshift V2 connection to establish private communication with Amazon S3 for staging.

You need to configure an interface endpoint on the AWS console to enable private communication to stage data in Amazon S3. The AWS S3 VPC endpoint enables an S3 request to be routed to the Amazon S3 service, without connecting to the internet.

Consider the following guidelines to establish private communication with Amazon S3:

- When an Amazon Redshift Cluster is spawned, an Elastic Network Interface (ENI) is generated for the cluster within the subnet. Ensure that the route table of the S3 gateway endpoint corresponds to the same subnet where your Redshift cluster ENI is created.

- To connect to an Amazon S3 bucket using a VPC endpoint, the Amazon Redshift cluster and the Amazon S3 bucket that it connects to must be in the same AWS region.

Amazon S3 V2 connection properties

Create an Amazon S3 V2 connection to read from and write to Amazon S3.

Prepare for authentication

You can configure multiple authentication types to access Amazon S3.

Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

- Basic authentication requires access key and secret key values from your AWS account.
- IAM authentication requires attaching policies to the EC2 role to grant access to specific folder paths and access Amazon S3 objects .
- AssumeRole with EC2 role authentication requires you to enable the EC2 role to assume another IAM role specified by the IAM Role ARN.
- AssumeRole with IAM user authentication requires the access key and secret key values of the IAM user and the ARN of the IAM role.
- Credential profile file authentication requires the credential profile file path and profile name. The credential profile file contains the access keys.
- IAM Roles Anywhere authentication requires the credential profile file path and profile name. The credential profile file contains the ARNs and paths to CA certificate and key files.

  To configure the credential profile file path and profile name, you must select Credential Profile File Authentication as the authentication type in the Amazon S3 V2 connection properties.
- Federated user single sign-on authentication requires the user name and password of the federated user, IdP SSO URL, ARN of the SAML identity provider, and ARN of the IAM role assumed by the federated user. You can only use ADFS 3.0 (IDP) for SSO.

Create a minimal Amazon IAM policy

You can configure an IAM policy through the AWS console. Use AWS IAM authentication to securely control access to Amazon S3 resources.

Use the following minimum required policies for users to read data from an Amazon S3 bucket:

- GetObject
- ListBucket

Use the following minimum required policies for users to write data to an Amazon S3 bucket:

- PutObject
- GetObject
- DeleteObject
- ListBucket
- ListBucketMultipartUploads. Applicable only for mappings in advanced mode.

The following sample policy shows the minimal Amazon IAM policy to write data to an Amazon S3 bucket:

```
{
    "Version": "2012-10-17",
    "Statement":
    [
        {
            "Effect": "Allow",
            "Action":[
                "s3:PutObject",
                "s3:GetObject",
                "s3:DeleteObject",
                "s3:ListBucket",
                "s3:ListBucketMultipartUploads"
                    ],
            "Resource": [
                "arn:aws:s3:::<bucket_name>/*",
                "arn:aws:s3:::<bucket_name>"
                        ]
        }
    ]
}
```

For bucket level access in advanced mode, you need to provide the `AllowListBucketMultipartUploads` permission at the bucket level in addition to the `ListBucketMultipartUploads` permission.

The following sample policy shows the minimal Amazon IAM policy to access the S3 bucket at the bucket level in advanced mode:

```
    {
            "Sid": "AllowListBucketMultipartUploads",
            "Action": [
                "s3:ListBucketMultipartUploads"
            ],
            "Effect": "Allow",
            "Resource": [
                "arn:aws:s3:::infa.qa.minimum.access.bucket"
            ]
        },
```

For mappings in advanced mode, you can use different AWS accounts within the same AWS region. Make sure that the Amazon IAM policy confirms access to the AWS accounts used in the mapping.

IAM authentication

To configure IAM authentication, the Secure Agent needs to run on an Amazon Elastic Compute Cloud (EC2) system. If you prefer not to specify the keys or use the IAM role ARN, then assign the minimum policy to the EC2 with access to the S3 bucket.

If you do not provide the access key and the secret key in the connection, Amazon S3 V2 Connector uses AWS credentials provider chain that looks for credentials in the following order:

1. The **AWS_ACCESS_KEY_ID** and **AWS_SECRET_ACCESS_KEY** or **AWS_ACCESS_KEY** and **AWS_SECRET_KEY** environment variables.
2. The **aws.accessKeyId** and **aws.secretKey** java system properties.
3. The credential profiles file at the default location, `~/.aws/credentials`.
4. The instance profile credentials delivered through the Amazon EC2 metadata service.

Perform the following steps to configure IAM authentication on EC2:

1. Create a minimal Amazon IAM policy.
2. Create the Amazon EC2 role. The Amazon EC2 role is used when you create an EC2 system. For more information about creating the Amazon EC2 role, see the AWS documentation.

3. Link the minimal Amazon IAM policy with the Amazon EC2 role.
4. Create an EC2 instance. Assign the Amazon EC2 role that you created in step 2 to the EC2 instance.
5. Install the Secure Agent on the EC2 system.

AssumeRole using EC2 role and IAM user

You can configure AssumeRole using EC2 role or IAM user to connect to Amazon S3.

You can use the temporary security credentials using AssumeRole to access AWS resources from the same or different AWS accounts.

When you configure AssumeRole using EC2 role or IAM user, ensure that you have the **sts:AssumeRole** permission and a trust relationship established within the AWS accounts to use the temporary security credentials. The trust relationship is defined in the trust policy of the IAM role when you create the role. The IAM role adds the EC2 role or IAM user as a trusted entity allowing the EC2 role or IAM user to use the temporary security credentials and access the AWS accounts.

For more information about how to establish the trust relationship, see the AWS documentation.

When the trusted EC2 role or IAM user requests for the temporary security credentials, the AWS Security Token Service (AWS STS) dynamically generates the temporary security credentials that are valid for a specified period and provides the credentials to the trusted EC2 role or IAM user.

AssumeRole using EC2 role

To configure an EC2 role to assume the IAM role provided in the **IAM Role ARN** connection property, select the **Use EC2 Role to Assume Role** check box in the Amazon S3 V2 connection properties.

The Amazon EC2 role can assume another IAM role from the same or different AWS account without requiring a permanent access key and secret key. The Amazon EC2 role can also assume another IAM role from a different region.

Consider the following prerequisites before you configure AssumeRole using EC2 role:

- Install the Secure Agent on an AWS service such as Amazon EC2.
- The EC2 role attached to the AWS EC2 service must not have access to Amazon S3 but needs to have permission to assume another IAM role.
- The IAM role that needs to be assumed by the EC2 role must have a permission policy and a trust policy attached to it.

AssumeRole using IAM user

To configure AssumeRole using IAM user, provide the value of the **IAM Role ARN** connection property when you create an Amazon S3 V2 connection. The IAM Role ARN uniquely identifies the AWS resources. Then, specify the time duration in seconds during which you can use the temporarily security credentials in the **Temporary Credential Duration** advanced source and target properties.

You need to follow some guidelines when you configure AssumeRole using IAM user. For more information, see "Rules and guidelines for AssumeRole via IAM user authentication" on page 69.

External ID

You can specify the external ID of your AWS account for a more secure access to the Amazon S3 bucket when the Amazon S3 bucket is in same or different AWS account.

You can optionally specify the external ID in the AssumeRole request to the AWS Security Token Service (STS).

The external ID must be a string.

The following sample shows an external ID condition in the assumed IAM role's trust policy:

```
"Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::AWS_Account_ID : user/user_name"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId": "dummy_external_id"
        }
      }
    }
  ]
```

AssumeRole policy

To use the temporary security credentials to access the AWS resources, both the IAM user and IAM role require policies.

The following section lists the policies required for the IAM user and IAM role:

IAM user

An IAM user must have the `sts:AssumeRole` policy to use the temporary security credentials in the same or different AWS account.

The following sample policy allows an IAM user to use the temporary security credentials in an AWS account:

```
{
"Version":"2012-10-17", "Statement":{ "Effect":"Allow", "Action":"sts:AssumeRole",
"Resource":"arn:aws:iam::<ACCOUNT-HYPHENS>:role/<ROLE-NAME>" }
}
```

IAM role

An IAM role must have a `sts:AssumeRole` policy and a trust policy attached with the IAM role to allow the IAM user to access the AWS resource using the temporary security credentials. The policy specifies the AWS resource that the IAM user can access and the actions that the IAM user can perform. The trust policy specifies the IAM user from the AWS account that can access the AWS resource.

The following policy is a sample trust policy:

```
{
"Version":"2012-10-17", "Statement":[{ "Effect":"Allow", "Principal":
{ "AWS":"arn:aws:iam::AWS-account-ID:root" },
"Action":"sts:AssumeRole" }
 ]
  }
   }
```

Here, in the `Principal` attribute, you can also provide the ARN of IAM user, which allows the designated user to dynamically generate temporary security credentials and helps to restrict further access.

For example,

```
"Principal" : { "AWS" : "arn:aws:iam:: AWS-account-ID :user/ user-name " }
```

Credential profile file authentication

You can provide the credentials required to establish the connection with Amazon S3 through the credential profile file.

Consider the following rules for a credential profile file:

- The credential file must be on the same machine where you installed the Secure Agent.
- The credential profile file name must end with `.credentials`.
- If you do not specify the credential profile path, the Secure Agent uses the credential profile file present in the following default location in your home directory:
  `~/.aws/credentials`

  Note: On Windows, you can refer to your home directory by using the environment variable `%UserProfile%`. On Unix-like systems, you can use the environment variable `$HOME`.
- If you do not specify the profile name, the credentials are used from the default profile in the credential file.

The following sample shows a credential profile file:

```
[default]

aws_access_key_id = 1233333

aws_secret_access_key = abcabcabc


[test-profile]

aws_access_key_id = 1233333

aws_secret_access_key = abcabcabc

aws_session_token = jahaheieomdrtflmlioerp
```

The `aws_access_key_id` and `aws_secret_access_key` are the AWS access key and secret key used as part of credentials to authenticate the user.

The `aws_session_token` is the AWS session token used as part of the credentials to authenticate the user. A session token is required only if you specify temporary security credentials.

IAM Roles Anywhere authentication

You can configure AWS IAM Roles Anywhere authentication to generate temporary security credentials to securely access Amazon S3 resources from your servers, applications, or containers outside of AWS using X.509 certificates instead of long-term credentials.

You need to complete certain prerequisites such as creating a CA certificate, generating temporary credentials, and creating a credential profile file to configure IAM Roles Anywhere authentication.

The credential profile file must include the ARNs for the trust anchor, profile, and IAM role, as well as the file paths for the CA certificate and private key.

Complete the following prerequisites to use IAM Roles Anywhere authentication:

1. Set up a Certificate Authority.
   Use AWS Private Certificate Authority to create a certificate authority and install a CA certificate to activate the certificate authority.
2. Configure IAM Roles Anywhere.
   a. In IAM Roles Anywhere, create a Trust Anchor using the certificate authority that you created.
   b. Create an IAM role with the minimal S3 permissions and a trust policy that allows IAM Roles Anywhere to assume it.
   c. Create a profile in the IAM Roles Anywhere console, linking the IAM role to the trust anchor.

3. Generate and download the certificate.
   a. In AWS Certificate Manager, request and obtain a private certificate associated with your certificate authority.
   b. Download and export the certificate body, certificate chain, and certificate private key.
   c. Decrypt the private key using OpenSSL.
   d. Place the private key, decrypted private key, and the certificate in a local directory.
4. Install IAM credentials helper and use the `aws_signing_helper` tool to request temporary credentials through the CreateSession API.
5. Create an AWS credentials file with the profile details including ARNs and paths to certificate and key files.
   Specify the following parameters in the credential profile file:
   - **<agent_loc>\aws_signing_helper.exe**. The path to the AWS Credentials Helper tool.
   - **--certificate <agent_loc>\certificate.pem**. The path to the X.509 certificate file. This certificate is used to authenticate the identity requesting temporary credentials from IAM Roles Anywhere.
   - **--private-key <agent_loc>\decrypted_private_key.pem**. The path to the decrypted private key corresponding to the certificate.
   - **--trust-anchor-arn <arn of trust anchor>**. The Amazon Resource Name (ARN) of the trust anchor that establishes trust between IAM Roles Anywhere and the certificate authority.
   - **--profile-arn <arn of profile>**. The ARN of the profile configured in IAM Roles Anywhere. Profiles specify which IAM roles can be assumed and what operations can be performed with the temporary security credentials issued.
   - **--role-arn <arn of role>**. The ARN of the IAM role that the credentials helper assumes.
   - **--session-duration <time in seconds>**. The duration during which the temporary session credentials is valid. This parameter is optional.

   The following sample shows a credential profile file:

```
[profile_name]
credential_process = <Secure Agent installation drirectory>\aws_signing_helper.exe
credential-process --certificate <agent_loc>\certificate.pem --private-key
<agent_loc>\decrypted_private_key.pem --trust-anchor-arn <arn of trust anchor> --
profile-arn <arn of profile> --role-arn <arn of role> --session-duration <time in
seconds>
```

When you create an Amazon S3 V2 connection, in the **Other Authentication Type** field, select **Credentials Profile File authentication**, and then specify the credential profile file path and profile name.

Rules for credential profile file

Consider the following rules for a credential profile file:

- The credential profile file must be on the Secure Agent machine.
- The credential profile file must have a `.credentials` extension.
- If you do not specify the credential profile path, the Secure Agent uses the credential profile file present in the following default location in the home directory:
  `~/.aws/credentials`

  Note: On Windows, you can refer to your home directory by using the environment variable `%UserProfile%`. On Unix-like systems, you can use the environment variable `$HOME`.

For more information on how to configure IAM Roles Anywhere authentication, see the AWS documentation.

Connect to Amazon S3

Let's configure the Amazon S3 connection properties to connect to Amazon S3.

Before you begin

Before you get started, you'll need to get information from your Amazon S3 account based on the authentication type that you want to configure.

Check out "Prepare for authentication" on page 50 to learn more about the authentication prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use a Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. For file ingestion and replication tasks and streaming ingestion and replication tasks, use a Secure Agent. You cannot run an application ingestion and replication task, database ingestion and replication task, file ingestion and replication task, or streaming ingestion and replication task on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can configure basic, AWS Identity and Access Management (IAM), temporary security credentials, assume role for EC2, credential profile file, and federated user single sign-on authentication types to access Amazon S3.

Select the required authentication method and then configure the authentication-specific parameters.

Basic authentication

Basic authentication requires access key and secret key values from your AWS account.

The following table describes the basic connection properties for basic authentication:

Property	Description
Access Key	Access key to access the Amazon S3 bucket.
Secret Key	Secret key to access the Amazon S3 bucket. The secret key is associated with the access key and uniquely identifies the account.
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>` For application ingestion and replication and database ingestion and replication tasks, add a trailing slash. For example: `<bucket name>/<my folder name>/`.
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).

IAM authentication

IAM authentication requires only the folder path to the Amazon S3 objects. The EC2 role must have access to the folder.

The following table describes the basic connection properties for AWS IAM authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>` For application ingestion and replication and database ingestion and replication tasks, add a trailing slash. For example: `<bucket name>/<my folder name>/`.
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).

AssumeRole via EC2 role authentication

AssumeRole via EC2 role authentication requires you to enable the EC2 role to assume another IAM role specified in the IAM Role ARN option.

The following table describes the basic connection properties for AssumeRole via EC2 role authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>` For application ingestion and replication and database ingestion and replication tasks, add a trailing slash. For example: `<bucket name>/<my folder name>/`.
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).
IAM Role ARN	The Amazon Resource Name (ARN) of the AWS Identity and Access Management (IAM) role assumed by the user to use the dynamically generated temporary security credentials. Enter the ARN value if you want to use the temporary security credentials to access AWS resources. **Note:** Even if you remove the IAM role that grants the agent access to the Amazon S3 bucket, the test connection is successful. For more information about how to get the ARN of the IAM role, see the AWS documentation.

Property	Description
External ID	The external ID of your AWS account. External ID provides a more secure access to the Amazon S3 bucket when the Amazon S3 bucket is in a different AWS account.
Use EC2 Role to Assume Role	Enables the EC2 role to assume another IAM role specified in the IAM Role ARN option. By default, this property is not selected. **Note:** The EC2 role must have a policy attached with permissions to assume an IAM role from the same or different account. **Note:** Enter a value for the IAM Role ARN property when you enable this property for a streaming ingestion and replication task.

AssumeRole via IAM user authentication

AssumeRole via IAM user authentication requires the access key and secret key values of the IAM user and the ARN of the IAM role.

The following table describes the basic connection properties for AssumeRole via IAM user authentication:

Property	Description
Access Key	Access key to access the Amazon S3 bucket.
Secret Key	Secret key to access the Amazon S3 bucket. The secret key is associated with the access key and uniquely identifies the account.
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>` For application ingestion and replication and database ingestion and replication tasks, add a trailing slash. For example: `<bucket name>/<my folder name>/`.

Property	Description
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).
IAM Role ARN	The Amazon Resource Name (ARN) of the AWS Identity and Access Management (IAM) role assumed by the user to use the dynamically generated temporary security credentials. Enter the value of this property if you want to use the temporary security credentials to access the AWS resources. **Note:** Even if you remove the IAM role that enables the agent to access the Amazon S3 bucket and create a connection, the test connection is successful. For more information about how to get the ARN of the IAM role, see the AWS documentation.
External ID	The external ID of your AWS account. External ID provides a more secure access to the Amazon S3 bucket when the Amazon S3 bucket is in a different AWS account.

Credential profile file authentication

Credential profile file authentication requires the credential profile file path and profile name.

The following table describes the basic connection properties for credential profile file authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>` For application ingestion and replication and database ingestion and replication tasks, add a trailing slash. For example: `<bucket name>/<my folder name>/`.
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).

Property	Description
Other Authentication Type	Determines whether you want to use the credential profile file authentication to connect to Amazon S3. Select one the following authentication types: - NONE. Select if you do not want to use credential profile file authentication. - Credential Profile File Authentication. Select to use credential profile file authentication to access the Amazon S3 credentials from a credential file. **Note:** To use IAM Roles Anywhere authentication also, you must select Credential Profile File Authentication as the authentication type. Enter the credential profile file path and profile name to connect to Amazon S3. You can use permanent IAM credentials or temporary session tokens when you configure the credential profile file authentication. Default is NONE.
Credential Profile File Path	The path to the credential profile file. If you don't enter the credential profile path, the Secure Agent uses the credential profile file available in the following default location in your home directory: `~/.aws/credentials` **Note:** Database Ingestion and Replication has not been certified with the **Credential Profile File Path** and **Profile Name** connection properties. Database Ingestion and Replication finds AWS credentials by using the default credential provider chain that is implemented by the DefaultAWSCredentialsProviderChain class, which includes the credential profile file.
Profile Name	Name of the profile in the credential profile file used to get credentials to access Amazon S3 resources. If you don't enter the profile name, the credentials from the default profile in the credential profile file are used.

Federated single sign-on authentication

Federated user single sign-on authentication requires the user name and password of the federated user, IdP SSO URL, ARN of the SAML identity provider, and ARN of the IAM role assumed by the federated user. You can only use ADFS 3.0 (IDP) for SSO.

The following table describes the basic connection properties for federated single sign-on authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>` For application ingestion and replication and database ingestion and replication tasks, add a trailing slash. For example: `<bucket name>/<my folder name>/`.
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).
Federated SSO IdP	SAML 2.0-enabled identity provider for the federated user single sign-on to use with the AWS account. You can only use ADFS 3.0 (IDP) for SSO. Select **None** if you don't want to use federated user single sign-on. **Note:** Federated user single sign-on is not applicable to application ingestion and replication tasks,database ingestion and replication tasks, and streaming ingestion and replication tasks.
Federated User Name	User name of the federated user to access the AWS account through the identity provider.

Property	Description
Federated User Password	Password for the federated user to access the AWS account through the identity provider.
IdP SSO URL	Single sign-on URL of the identity provider for AWS. Doesn't apply to astreaming ingestion and replication task.
SAML Identity Provider ARN	ARN of the SAML identity provider that the AWS administrator created to register the identity provider as a trusted provider.
Role ARN	ARN of the IAM role assumed by the federated user.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
S3 Account Type	The type of the Amazon S3 account. Select from the following options: - Amazon S3 Storage. Enables you to use the Amazon S3 services. - S3 Compatible Storage. Enables you to use the endpoint for a third-party storage provider. You can use only Scality RING and MinIO. Default is Amazon S3 storage.
REST Endpoint	The S3 storage endpoint required for S3 compatible storage. Enter the S3 storage endpoint in HTTP or HTTPs format. For example, http://s3.isv.scality.com.
S3 VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for Amazon S3. You can use a VPC endpoint to enable private communication with Amazon S3. Select one of the following options: - None. Select if you do not want to use a VPC endpoint. - Gateway Endpoint. Select to establish private communication with Amazon S3 through an interface endpoint. A gateway endpoint is a target for a route in your route table that is used to forward S3 traffic to the S3 gateway endpoint. - Interface Endpoint. Select to establish private communication with Amazon S3 through an interface endpoint which uses a private IP address from the IP address range of your subnet. It serves as an entry point for traffic destined to an AWS service. Default is None. Doesn't apply to anapplication ingestion and replication task or database ingestion and replication task.
Endpoint DNS Name for Amazon S3	The DNS name for the Amazon S3 interface endpoint. Enter the DNS name in the following format: `bucket.<DNS name of the interface endpoint>` Doesn't apply to anapplication ingestion and replication task or database ingestion and replication task.

Property	Description
STS VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for AWS Security Token Service. This option applies when you select the S3 VPC interface endpoint and when use AssumeRole via IAM user or EC2 role authentication or Federated SSO IdP authentication. Doesn't apply to anapplication ingestion and replication task, streaming ingestion and replication task, ordatabase ingestion and replication task.
Endpoint DNS Name for AWS STS	The DNS name for the AWS STS interface endpoint. Doesn't apply to an application ingestion and replication task or database ingestion and replication task.
KMS VPC Endpoint Type	The type of Amazon Virtual Private Cloud endpoint for AWS Key Management Service. This option applies when you select the S3 VPC interface endpoint and required when you specify the customer master key ID. Doesn't apply to an application ingestion and replication task or database ingestion and replication task.
Endpoint DNS Name for AWS KMS	The DNS name for the AWS KMS interface endpoint. Doesn't apply to an application ingestion and replication task or database ingestion and replication task.
Master Symmetric Key	A 256-bit AES encryption key in the Base64 format when you use client-side encryption. You can generate a key using a third-party tool. Doesn't apply to streaming ingestion and replication tasks.
Customer Master Key ID	The customer master key ID or alias name generated by AWS Key Management Service (AWS KMS) or the Amazon Resource Name (ARN) of your custom key for cross-account access. You must generate the customer master key for the same region where the Amazon S3 bucket resides. You can specify the following master keys: - Customer generated customer master key. Enables client-side or server-side encryption. - Default customer master key. Enables client-side or server-side encryption. Only the administrator user of the account can use the default customer master key ID to enable client-side encryption. Doesn't apply to streaming ingestion and replication tasks.

Private communication with Amazon S3

You can enable private communication with Amazon S3 by configuring a gateway endpoint or interface endpoint on AWS console and in the Amazon S3 V2 connection.

You can configure Amazon S3 V2 Connector to establish private communication with Amazon S3 without exposing your traffic to the public internet. To access Amazon S3, ensure that the Secure Agent is a part of the subnet in the AWS Virtual Private Cloud (VPC). AWS S3 VPC endpoint enables an S3 request to be routed to the Amazon S3 service, without having to connect a subnet to an internet gateway. You can create an interface endpoint or a gateway endpoint.

For more information, see
Configuring private communication with Amazon S3 using the Amazon S3 V2 Connector.

Server-side encryption with KMS

To use the customer master key managed by AWS Key Management Service (AWS KMS) and enable the encryption with KMS, you need to create a KMS policy.

You can perform the following operations to use the temporary security credentials and enable the encryption with KMS:

- GenerateDataKey
- DescribeKey
- Encrypt
- Decrypt
- ReEncrypt

See the following sample KMS policy for reference:

```
{
"Version":"2012-10-17", "Statement":[{ "Effect":"Allow", "Action":
[ "kms:GenerateDataKey", "kms:DescribeKey", "kms:Encrypt", "kms:Decrypt",
"kms:ReEncrypt*" ], "Resource": ["arn:aws:kms:region:account:key/<KMS_key>"]
    }
  ]
   }
```

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux.

You can use only an unauthenticated proxy server to connect to Informatica Intelligent Cloud Services.

To configure the proxy settings for the Secure Agent, perform the following tasks:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

The proxy settings configured through the Secure Agent Manager are applicable when you create a connection or mapping, and the proxy settings configured through the JVM options are applicable when you run the mapping.

Bypass proxy server

If your organization uses an outgoing proxy server to connect to the internet, the Secure Agent connects to IDMC through the proxy server.

However, if you want to exclude certain IP addresses and host names from the proxy, you can bypass the proxy. Set the *InfaAgent.NonProxyHost* property in the `proxy.ini` file and the *-Dhttp.nonProxyHosts* property in the JVM options of the Secure Agent properties and include the IP addresses and host names that you want to exclude.

The following table shows the proxy setting that you can configure through the proxy.ini file or the JVM options:

Proxy configuration	Proxy Flag Setting

Proxy.ini	InfaAgent.NonProxyHost=localhost|**<your_bucket_name>.s3.**|127.*|[\:\:1]
	For example, to bypass a single S3 Bucket `iam.qa.bucket`, use the following proxy setting:
	`InfaAgent.NonProxyHost=localhost|`**`iam.qa.bucket.s3.`**`|127.*|[\:\:1]`
	To bypass all S3 buckets, use the following proxy setting:
	`InfaAgent.NonProxyHost=localhost|`**`*.s3.*`**`|127.*|[\:\:1]`

JVM option	-Dhttp.nonProxyHosts=localhost|**<your_bucket_name>.s3.**|127.* |[\:\:1]
	For example, to bypass a single S3 Bucket, `iam.qa.bucket`, use the following proxy setting:
	`-Dhttp.nonProxyHosts=localhost|`**`iam.qa.bucket.s3.`**`|127.*|[\:\:1]`
	To bypass all S3 buckets, use the following proxy setting:
	`-Dhttp.nonProxyHosts=localhost|`**`*.s3.*`**`|127.*|[\:\:1]`

Bypass the proxy server in advanced mode

To bypass the proxy server, you must update the `NonProxyHost` value in the `proxy.ini` file. You can set the property in the agent core path to configure the `NonProxyHost` in the advanced cluster configuration.

To bypass the proxy at the Amazon S3 endpoint, perform the following steps:

1. Edit the `proxy.ini` file and set the property in the `NonProxyHost` with the cluster region.
2. Enter the appropriate region name in the property in the following format:

   ```
   InfaAgent.NonProxyHost=localhost|127.*|[\:\:1]|
   169.254.169.254|.<REGION_NAME>.elb.amazonaws.com|*.<REGION_NAME>.elb.amazonaws.com
   ```

   The following example shows how you can update the `NonProxyHost` for the US West region in the `proxy.ini` file:

   ```
   InfaAgent.NonProxyHost=localhost|127.*|[\:\:1]|169.254.169.254|.us-
   west-2.elb.amazonaws.com|*.us-west-2.elb.amazonaws.com|s3.us-west-2.amazonaws.com|
   *.s3.us-west-2.amazonaws.com|s3.amazonaws.com
   ```

3. After you edit the `proxy.ini` file, you must set the property `ccs.enable.storage.proxy.settings` to false in the runtime properties of the advanced cluster.
   Perform the following steps to set the property:
   a. Go to **Administrator**.
   b. In the **Advanced Clusters** page, select the name of the configuration that you want to edit from the list of advanced configurations.

c. Set the property `ccs.enable.storage.proxy.settings` to false and save the cluster configuration in the **Runtime Properties** for the particular cluster.
The following image shows the configured cluster runtime properties:



Rules and guidelines for AssumeRole via IAM user authentication

Consider the following guidelines for Assume Role via IAM user authentication:

- The IAM user or IAM role that requests for the temporary security credentials must not have access to any AWS resources.
- Only authenticated IAM users or IAM roles can request for the temporary security credentials from the AWS Security Token Service (AWS STS).
- Before you run a task, ensure that you have enough time to use the temporary security credentials for running the task. You cannot extend the time duration of the temporary security credentials for an ongoing task.
  For example, when you read from and write to Amazon S3 and if the temporary security credentials expire, you cannot extend the time duration of the temporary security credentials which causes the task to fail.
- After the temporary security credentials expire, AWS does not authorize the IAM users or IAM roles to access the resources using the credentials. You must request for new temporary security credentials before the previous temporary security credentials expire in a mapping.
- Do not use the root user credentials of an AWS account to use the temporary security credentials. You must use the credentials of an IAM user to use the temporary security credentials.
- If both the source and target in a mapping point to the same Amazon S3 bucket, use the same Amazon S3 connection in the Source and Target transformations. If you use two different Amazon S3 connections, configure the same values in the connection properties for both the connections.
- If the source and target in a mapping point to different Amazon S3 buckets, you can use two different Amazon S3 connections.
  You can configure different values in the connection properties for both the connections. However, you must select the **Use EC2 Role to Assume Role** check box in the connection property. You must also specify the same value for the **Temporary Credential Duration** field in the source and target properties.
- In a mapping, if you configure two or more Amazon S3 data sources from the same Amazon S3 bucket with different IAM roles, each IAM role must be able to access the data source of the other IAM role.
- In a mapping with two data sources, if you set up one Amazon S3 data source to use user credentials and another to use an IAM role, consider the following rules:
  - The IAM user for the first data source must also be able to assume the IAM role of the second Amazon S3 data source.

- The IAM role that you configured for the second data source must also have access to the first Amazon S3 data source.

Rules and guidelines for AWS regions

Consider the following rules and guidelines when you configure the region name of the bucket in the connection properties:

- When you change the runtime environment of an existing connection, the region is changed to the default region US East (N. Virginia). Select the region manually to change the default region.
- When you edit an existing connection, you see duplicate entries for regions. Use the regions that contain spaces because these regions are populated from AWS SDK. For example, use US West (Oregon) instead of US West(Oregon).

Rules and guidelines for S3 compatible storage

Consider the following rules and guidelines when you configure S3 compatible storage in an Amazon S3 V2 connection:

- You can only configure basic authentication when you use S3 compatible storage.
- You cannot configure SSE-KMS encryption for the Scality RING S3 compatible storage. You cannot configure SSE and SSE-KMS encryption for MinIO S3 compatible storage.
- You cannot configure SQL ELT optimization to load data from Amazon S3 sources to Amazon Redshift.

AMQP connection properties

When you set up an AMQP connection, you must configure the connection properties.

The following table describes the AMQP connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. Description that you can use to identity the connection. The description cannot exceed 4,000 characters.
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Host Name	Network address of the AMQP broker.
Port	Port number of the AMQP broker to which the underlying TCP connection is made. Default is 5672.

Property	Description
Virtual Host	Virtual host name that identifies the AMQP system. Use the virtual host name for enhanced security.
Username	Username for the AMQP broker.
Password	Password for the AMQP broker.
Use SSL	Enable this option to use SSL for secure transmission. If you enable the SSL authentication, ensure that you provide both keystore and truststore details for using the AMQP connection in a streaming ingestion and replication task.
Keystore File Name	Contains the keys and certificates required for secure communication.
Keystore Password	Password for the keystore filename.
Keystore Type	Type of keystore that you want to use. Keystore type defines the storage and data format of the keystore information and the algorithms used to protect private keys in the keystore. Use one of the following types: - JKS. Stores private keys and certificates. - PKCS12. Stores private keys, secret keys, and certificates.
Truststore File Name	Name of the truststore file.
Truststore Password	Password for the truststore file.
Truststore Type	Type of truststore that you want to use. Use one of the following types: - JKS - PKCS 12
TLS Protocol	Transport protocols that you want to use. Use one of the following types: - SSL - SSLv2Hello - SSLv3 - TLS - TLSv1 - TLSv1.1 - TLSv1.2
Client Authentication	Client authentication policy when connecting to the secured AMQP broker. Use one of the following property values when you define and enable an SSL context. - WANT - REQUIRED - NONE

Business 360 Events connection properties

When you create the Business 360 Events connection, you must configure the connection properties.

The following table describes the Business 360 Events connection properties:

Property	Description
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 100 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.
Runtime Environment	The name of the runtime environment where you want to run the mappings. Specify a Secure Agent, Hosted Agent, or a serverless runtime environment.
Start Timestamp	A system-generated timestamp variable to set the start of a time range for which you want to get events from the Business 360 data store. **Note:** You can't modify this attribute.
End Timestamp	A system-generated timestamp variable to set the end of a time range for which you want to get events from the Business 360 data store. **Note:** You can't modify this attribute.

Cloud Integration Hub connection properties

You can view the Cloud Integration Hub connection only if your organization is provisioned with Cloud Integration Hub. Do not edit, modify, or delete this connection. Do not modify any connection property apart from the **Do not use intermediate staging for subscription flows** and **Use JDBC for Private Publication Repository** properties.

The following table describes the connection properties for a Cloud Integration Hub connection:

Connection Property	Description	Editable
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /	Do not edit.
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.	Yes

Connection Property	Description	Editable
Enable Secret Vault	Stores the publication repository password for the connection in the Secret Manager in the runtime environment that is configured for your organization. This property appears only if secrets manager is set up for your organization. Select this option to use the credentials from the Secret Manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.	Do not edit.
Runtime Environment	Name of the runtime environment where you want to run the tasks.	Do not edit.
Do not use intermediate staging for subscription flows	Disables writing to intermediate staging. Enable this property if you do not want to write to the intermediate staging, the Data Integration task reads the data from Cloud Integration Hub and then writes the data directly to the target location. Disabling writing to intermediate staging might affect system performance.	Yes
Use JDBC for Private Publication Repository	To configure zero downtime for a private publication repository. Enable this property to ensure uninterrupted access to data on the private publication repository. You can enable zero downtime for publications and subscriptions that trigger a Data Integration task. On a hosted publication repository, Cloud Integration Hub applies zero downtime by default for all publication and subscription types.	Yes

Databricks connection properties

Create a Databricks connection to securely read data from or write data to Databricks.

Staging prerequisites

Before you create a connection, you must perform certain prerequisite tasks to configure the staging environment to connect to SQL warehouse, all-purpose cluster, or job cluster.

SQL warehouse

Configure AWS, Azure, or Databricks Volume as the staging environment for the SQL warehouse based on your deployment environment. Additionally, configure the Spark parameters for the SQL warehouse to use the Azure or AWS staging environment.

You can use a SQL warehouse on the Windows and Linux operating systems.

For more information on the types of SQL warehouses that you can connect to, see the Databricks SQL warehouses Knowledge Base article.

Configure AWS staging

Configure IAM AssumeRole authentication to use AWS staging for the SQL warehouse.

Create a minimal Amazon IAM policy

To stage the data in Amazon S3, use the following minimum required permissions: :

- PutObject
- GetObject
- DeleteObject
- ListBucket

You can use the following sample Amazon IAM policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:DeleteObject",
                "s3:ListBucket",

                        ],
            "Resource": [
                "arn:aws:s3:::<bucket_name>/*",
                "arn:aws:s3:::<bucket_name>"
            ]
        }
    ]
}
```

Note: The **Test Connection** does not validate the IAM policy assigned to users. You can specify the Amazon S3 bucket name in the source advanced properties.

Configure Spark parameters for AWS staging

On the Databricks SQL Admin console, navigate to **SQL Warehouse Settings > Data Security**, and then configure the Spark parameters for AWS under **Data access configuration**.

Add the following Spark configuration parameters and restart the SQL warehouse:

- `spark.hadoop.fs.s3a.access.key <S3 Access Key value>`
- `spark.hadoop.fs.s3a.secret.key <S3 Secret Key value>`
- `spark.hadoop.fs.s3a.endpoint <S3 Staging Bucket endpoint value>`

For example, the S3 staging bucket warehouse value is *s3.ap-south-1.amazonaws.com*.

Ensure that the configured access key and secret key have access to the S3 buckets where you store the data for Databricks tables.

Configure Azure staging

Before you use Microsoft Azure Data Lake Storage Gen2 to stage files, perform the following tasks:

- Create a storage account to use with Microsoft Azure Data Lake Storage Gen2 and enable **Hierarchical namespace** in the Azure portal.
  You can use role-based access control to authorize the users to access the resources in the storage account. Assign the Contributor role or Reader role to the users. The contributor role grants you full access to manage all resources in the storage account, but does not allow you to assign roles. The reader role allows you to view all resources in the storage account, but does not allow you to make any changes.

  Note: To add or remove role assignments, you must have write and delete permissions, such as an Owner role.
- Register an application in Azure Active Directory to authenticate users to access the Microsoft Azure Data Lake Storage Gen2 account.
  You can use role-based access control to authorize the application. Assign the Storage Blob Data Contributor or Storage Blob Data Reader role to the application. The Storage Blob Data Contributor role lets you read, write, and delete Azure Storage containers and blobs in the storage account. The Storage Blob Data Reader role lets you only read and list Azure Storage containers and blobs in the storage account.
- Create an Azure Active Directory web application for service-to-service authentication with Microsoft Azure Data Lake Storage Gen2.

  Note: Ensure that you have superuser privileges to access the folders or files created in the application using the connector.
- To read complex files, set the JVM options for type DTM to increase the -Xms and -Xmx values in the system configuration details of the Secure Agent to avoid java heap space error. The recommended -Xms and -Xmx values are 512 MB and 1024 MB respectively.

Configure Spark parameters for Azure staging

On the Databricks SQL Admin console, navigate to **SQL Warehouse Settings > Data Security**, and then configure the Spark parameters for Azure under **Data access configuration**.

Add the following Spark configuration parameters and restart the SQL warehouse:

- `spark.hadoop.fs.azure.account.oauth2.client.id.<storage-account-name>.dfs.core.windows.net <value>`
- `spark.hadoop.fs.azure.account.auth.type.<storage-account-name>.dfs.core.windows.net OAuth`
- `spark.hadoop.fs.azure.account.oauth2.client.secret.<storage-account-name>.dfs.core.windows.net <Value>`
- `spark.hadoop.fs.azure.account.oauth.provider.type.<storage-account-name>.dfs.core.windows.net org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider`
- `spark.hadoop.fs.azure.account.oauth2.client.endpoint.<storage-account-name>.dfs.core.windows.net https://login.microsoftonline.com/<Tenant ID>/oauth2/token`

Ensure that the configured client ID and client secret have access to the file systems where you store the data for Databricks tables.

Stage data in Databricks Volumes

You can stage data in a Volume in Databricks. Volumes are Unity Catalog objects used to manage and secure non-tabular datasets such as files and directories.
You can use a Volume on a Linux or Windows machine and with the Databricks JDBC driver.

To use a Volume, complete the following prerequisites:

- Ensure that your Databricks workspace is enabled for Unity Catalog.
- Get the path where you want to stage the data in a Volume. If you do not specify a volume path in the Databricks connection properties, the Secure Agent creates a managed volume in Databricks, which is automatically deleted once the job is completed. Ensure you have the create and delete permissions for the managed volume if you do not want to specify a path.

All-purpose cluster

Enable the Secure Agent properties for design-time processing on the all-purpose cluster.
You can use all-purpose cluster only on the Linux operating system.

Configure Secure Agent properties

To connect to all-purpose cluster, enable the Secure Agent properties for design time.

1. In **Administrator**, select the Secure Agent listed on the **Runtime Environments** tab.
2. Click **Edit**.
3. In the **System Configuration Details** section, select Data Integration Server as the **Service** and Tomcat JRE as the **Type**.
4. Edit the **JRE_OPTS** field and set the value to *-DUseDatabricksSql=false*.



Job cluster

Configure the Spark parameters for job cluster to use Azure and AWS staging based on where the cluster is deployed.

You also need to enable the Secure Agent properties for runtime processing on the job cluster.

You can use job cluster only on the Linux operating system.

Spark configuration

Before you connect to the job cluster, you must configure the Spark parameters on AWS and Azure.

Configuration on AWS

Add the following Spark configuration parameters for the job cluster and restart the cluster:

- `spark.hadoop.fs.s3a.access.key <value>`
- `spark.hadoop.fs.s3a.secret.key <value>`
- `spark.hadoop.fs.s3a.endpoint <value>`

Ensure that the access and secret key configured has access to the buckets where you store the data for Databricks tables.

Configuration on Azure

Add the following Spark configuration parameters for the job cluster and restart the cluster:

- `fs.azure.account.oauth2.client.id.<storage-account-name>.dfs.core.windows.net <value>`
- `fs.azure.account.auth.type.<storage-account-name>.dfs.core.windows.net <value>`

- `fs.azure.account.oauth2.client.secret.<storage-account-name>.dfs.core.windows.net <Value>`
- `fs.azure.account.oauth.provider.type.<storage-account-name>.dfs.core.windows.net org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider`
- `fs.azure.account.oauth2.client.endpoint.<storage-account-name>.dfs.core.windows.net https://login.microsoftonline.com/<Tenant ID>/oauth2/token`

Ensure that the client ID and client secret configured has access to the file systems where you store the data for Databricks tables.

Configure Secure Agent properties

To connect to job cluster, enable the Secure Agent properties for runtime.

Note: This topic does not pertain to Data Ingestion and Replication.

1. In **Administrator**, select the Secure Agent listed on the **Runtime Environments** tab.
2. Click **Edit**.
3. In the **System Configuration Details** section, select Data Integration Server as the **Service** and DTM as the **Type**.
4. Edit the **JVMOption** field and set the value to *-DUseDatabricksSql=false*.



Connect to Databricks

Let's configure the Databricks connection properties to connect to Databricks.

Before you begin

You can use a Databricks connection to read from and write to Databricks tables.

You can configure the following compute resources to connect to Databricks:

- **SQL warehouse (Recommended)**
  The Secure Agent connects to the SQL warehouse at design time and runtime.
- **All-purpose cluster and job cluster**
  The Secure Agent connects to the all-purpose cluster to import the metadata at design time and to the job cluster to run the mappings.

Note: If you're using an all-purpose or job cluster, Informatica recommends transitioning to the SQL warehouse. The all-purpose and job clusters will no longer receive new feature updates or enhancements, although they will still receive critical security updates to maintain their stability and safety. By switching to the SQL warehouse, you will benefit from the latest features and enhancements.

Before you get started, you'll need to configure the AWS or Azure staging environment to use Databricks connection.

To learn about prerequisites for the Azure or AWS environment, check out "Staging prerequisites" on page 73.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use a Secure Agent or a serverless runtime environment. The tasks must have a source type supported for serverless use. For a file ingestion and replication task or streaming ingestion and replication task, use a Secure Agent. You can't run an application ingestion and replication task, database ingestion and replication task, file ingestion and replication task, or streaming ingestion and replication task on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
SQL Warehouse JDBC URL	Databricks SQL Warehouse JDBC connection URL. This property is required only for Databricks SQL warehouse. Doesn't apply to all-purpose cluster and job cluster. To get the SQL Warehouse JDBC URL, go to the Databricks console and select the JDBC driver version from the JDBC URL menu. Application ingestion and replication and database ingestion and replication tasks can use JDBC URL version **2.6.25 or later** or **2.6.22 or earlier**. The URLs must begin with the prefix `jdbc:databricks://`, as follows: `jdbc:databricks://<Databricks Host>:443/default;transportMode=http;ssl=1;AuthMech=3;httpPath=/sql/1.0/endpoints/<SQL endpoint cluster ID>;` Ensure that you set the required environment variables in the Secure Agent. Also specify the correct **JDBC Driver Class Name** under advanced connection settings. **Note:** Specify the database name in the Database Name connection property. If you specify the database name in the JDBC URL, it is not considered.

Authentication type

You can configure personal access token and OAuth machine-to-machine authentication types to access Databricks.

Note: Data Ingestion and Replication can use only the personal access token authentication type.

Select the required authentication method and then configure the authentication-specific parameters.

Personal access token authentication requires the personal access token. OAuth machine-to-machine authentication requires the client ID and client secret of your Databricks account.

For more information on how to get the personal access token, client ID, and client secret, see the Databricks documentation.

Personal access token authentication

Personal access token authentication requires the personal access token of your Databricks account.
The following table describes the connection properties for personal access token authentication:

Property	Description
Databricks Token	Personal access token to access Databricks. This property is required for SQL warehouse, all-purpose cluster, and job cluster.
Catalog Name	The name of an existing catalog in the metastore when you use Unity Catalog. This property is optional for SQL warehouse. Doesn't apply to all-purpose cluster and job cluster. The catalog name cannot contain special characters. For more information about Unity Catalog, see the *Databricks documentation*.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Database	The name of the schema in Databricks. The name can contain only alphanumeric characters and hyphen (-). This property is optional for SQL warehouse, all-purpose cluster, and job cluster. If you do not specify a value, all databases available in the workspace are listed. The value you specify overrides the schema specified in the **SQL Warehouse JDBC URL** connection property.
JDBC Driver Class Name	The name of the JDBC driver class. This property is optional for SQL warehouse, all-purpose cluster, and job cluster. Default is `com.databricks.client.jdbc.Driver`

Property	Description
Staging Environment	The staging environment where your data is temporarily stored before processing. This property is required for SQL warehouse, all-purpose cluster, and job cluster. Select one of the following options as the staging environment: - AWS. If Databricks is hosted on the AWS platform. - Azure. If Databricks is hosted on the Azure platform. - Personal Staging Location. To stage data in a local personal storage location. If you select Personal Staging Location for a connection that Data Ingestion and Replication uses, the Parquet data files for application ingestion and replication jobs or database ingestion and replication jobs can be staged to a local personal storage location, with a data retention period of 7 days. You must also specify a Database Host value. If you use Unity Catalog, a personal storage location is automatically provisioned. However, you cannot use a personal staging location with Databricks unmanaged tables. - Volume. To stage data in a volume in Databricks. Volumes are Unity Catalog objects used to manage and secure non-tabular datasets such as files and directories. **Note:** When you configure a Databricks connection to connect to Databricks endpoints hosted on Google Cloud Platform, you must use a volume to stage data. Both volume and personal staging location do not apply to all-purpose clusters and job clusters. Default is Volume. **Note:** You cannot switch between clusters after you establish a connection. **Important:** Effective in the October 2024 release, personal staging location is deprecated. Deprecated functionality is supported, but Informatica intends to drop support in a future release. Informatica requests that you use a volume to stage the data.
Volume Path	The absolute path in Volume where you want to stage the data temporarily. Specify the path in the following format: `/Volumes/<catalog_identifier>/<schema_identifier>/<volume_identifier>/` If you do not specify a volume path, the Secure Agent creates a managed volume in Databricks.

AWS staging environment

The following table describes the properties for the AWS staging environment:

Property	Description
S3 Authentication Mode	The authentication mode to connect to Amazon S3. Select one of the following authentication modes: - Permanent IAM credentials. Uses the S3 access key and S3 secret key to connect to Amazon S3. - IAM Assume Role. Uses the AssumeRole for IAM authentication to connect to Amazon S3. This authentication mode applies only to SQL warehouse.
S3 Access Key	The key to access the Amazon S3 bucket.
S3 Secret Key	The secret key to access the Amazon S3 bucket.
S3 Data Bucket	The existing S3 bucket to store the Databricks data.
S3 Staging Bucket	The existing bucket to store the staging files.

Property	Description
S3 VPC Endpoint Type	This property applies only to IAM Assume Role authentication for Amazon S3. The type of Amazon Virtual Private Cloud endpoint for Amazon S3. You can use a VPC endpoint to enable private communication with Amazon S3. You must specify the IAM Role ARN to use a VPC endpoint. Select one of the following options: - None. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon S3 through an interface endpoint which uses a private IP address from the IP address range of your subnet. It serves as an entry point for traffic destined to an AWS service.
Endpoint DNS Name for S3	The DNS name for the Amazon S3 interface endpoint. Replace the asterisk symbol with the bucket keyword in the DNS name. Enter the DNS name in the following format: `bucket.<DNS name of the interface endpoint>` For example, `bucket.vpce-s3.us-west-2.vpce.amazonaws.com`
IAM Role ARN	The Amazon Resource Number (ARN) of the IAM role assumed by the user to use the dynamically generated temporary security credentials. Set the value of this property if you want to use the temporary security credentials to access the Amazon S3 staging bucket. For more information about how to get the ARN of the IAM role, see the *AWS documentation*.
Use EC2 Role to Assume Role	Optional. Select the check box to enable the EC2 role to assume another IAM role specified in the IAM Role ARN option. The EC2 role must have a policy attached with a permission to assume an IAM role from the same or different AWS account.
STS VPC Endpoint Type	This property applies only to IAM Assume Role authentication for Amazon S3. The type of Amazon Virtual Private Cloud endpoint for AWS Security Token Service. You can use a VPC endpoint to enable private communication with Amazon Security Token Service. You must specify the IAM Role ARN to use a VPC endpoint. Select one of the following options: - None. Select if you do not want to use a VPC endpoint. - Interface Endpoint. Select to establish private communication with Amazon Security Token Service through an interface endpoint which uses a private IP address from the IP address range of your subnet.
Endpoint DNS Name for AWS STS	The DNS name for the AWS STS interface endpoint. For example, `vpce-01f22cc14558c241f-s8039x4c.sts.us-west-2.vpce.amazonaws.com`
S3 Service Regional Endpoint	The S3 regional endpoint when the S3 data bucket and the S3 staging bucket need to be accessed through a region-specific S3 regional endpoint. This property is optional for SQL warehouse. Doesn't apply to all-purpose cluster and job cluster. Default is `s3.amazonaws.com`.
S3 Region Name	The AWS cluster region in which the bucket you want to access resides. Select a cluster region if you choose to provide a custom JDBC URL that does not contain a cluster region name in the JDBC URL connection property.

Property	Description
Zone ID	The zone ID for the Databricks job cluster. This property is optional for job cluster. Doesn't apply to SQL warehouse and all-purpose cluster. Specify the Zone ID only if you want to create a Databricks job cluster in a particular zone at runtime. For example, us-west-2a. **Note:** The zone must be in the same region where your Databricks account resides.
EBS Volume Type	The type of EBS volumes launched with the cluster. This property is optional for job cluster. Doesn't apply to SQL warehouse and all-purpose cluster.
EBS Volume Count	The number of EBS volumes launched for each instance. You can choose up to 10 volumes. This property is optional for job cluster. Doesn't apply to SQL warehouse and all-purpose cluster. **Note:** In a Databricks connection, specify at least one EBS volume for node types with no instance store. Otherwise, cluster creation fails.
EBS Volume Size	The size of a single EBS volume in GiB launched for an instance. This property is optional for job cluster. Doesn't apply to SQL warehouse and all-purpose cluster.

Azure staging environment

The following table describes the properties for the Azure staging environment:

Property	Description
ADLS Storage Account Name	The name of the Microsoft Azure Data Lake Storage account.
ADLS Client ID	The ID of your application to complete the OAuth Authentication in the Active Directory.
ADLS Client Secret	The client secret key to complete the OAuth Authentication in the Active Directory.
ADLS Tenant ID	The ID of the Microsoft Azure Data Lake Storage directory that you use to write data.
ADLS Endpoint	The OAuth 2.0 token endpoint from where authentication based on the client ID and client secret is completed.
ADLS Filesystem Name	The name of an existing file system to store the Databricks data. Enter only the filesystem name. Do not include the full path.
ADLS Staging Filesystem Name	The name of an existing file system to store the staging data. Enter only the filesystem name. Do not include the full path.

JDBC URL parameters

You can utilize the additional JDBC URL parameters field in the Databricks connection to customize and set any additional parameters required to connect to Databricks.

You can configure the following properties as additional JDBC URL parameters in the Databricks connection:

- To connect to Databricks using the proxy server, enter the following parameters:
  `jdbc:databricks://<Databricks Host>:443/default;transportMode=http;ssl=1;AuthMech=3;httpPath=/sql/1.0/warehouses/219fe3013963cdce;`**`UseProxy=<Proxy=true>;ProxyHost=<proxy host IPaddress>;ProxyPort=<proxy server port number>;ProxyAuth=<Auth_true>;`**

  Note: Data Ingestion and Replication does not support use of a proxy server to connect to Databricks.
- To connect to SSL-enabled Databricks, specify the value in the JDBC URL in the following format:
  `jdbc:databricks://<Databricks Host>:443/default;transportMode=http;`**`ssl=1;`**`AuthMech=3;httpPath=/sql/1.0/endpoints/<SQL endpoint cluster ID>;`

Rules and guidelines for personal staging location

When you select the personal staging location as a staging environment, the data is first staged in a Java temporary location and then copied to a personal staging location of the unity catalog. Both the staged files will be deleted after the task runs successfully.
However, to stage the data in a different directory, configure the DTM property `-Djava.io.tmpdir=/my/dir/path` in the JVM options in the system configuration settings of the Administrator service.

To enable data staging in a different directory, you should have read and write permissions and enough disk space to stage the data in the directory.

When you specify a personal staging location in the Databricks connection properties for staging, consider the following rules and guidelines:

- You can only specify unity enabled catalog in the SQL warehouse JDBC URL.
- You cannot use personal staging location as the staging environment with OAuth machine-to-machine authentication.
- Mappings run without SQL ELT optimization.
- The data is staged in the folder `stage://tmp/<user_name>` where the `<user_name>` is picked from the Databricks token provided in the connection and requires read and write access to the personal staging location in the root location of AWS and Azure.

Important: Effective in the October 2024 release, personal staging location entered deprecation. While you can use the functionality in the current release, Informatica intends to drop support for the functionality in a future release. Informatica recommends that you use a Volume to stage the data.

Db2 for i Database Ingestion connection properties

To define a Db2 for i Database Ingestion connection, configure the connection properties. You can use this connection type in database ingestion and replication tasks configured in Data Ingestion and Replication.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	The name of the runtime environment where you want to run database ingestion and replication tasks. You define runtime environments in Administrator. You can use either a local Secure Agent installation or serverless runtime environment. You can use a serverless runtime environment for cloud source and target types. You cannot run database ingestion and replication tasks on a Hosted Agent.
User Name	The user name to use for connecting to the Db2 for i instance.
Password	The password to use for connecting to the Db2 for i instance.
Host	The name of the machine that hosts the database server.
Port	The network port number used to connect to the database server.
Location Name	The name of the Db2 for i location that you want to access. Your system administrator can determine the name of the Db2 location by using the WRKRDBDIRE command. In the output, find the name of the database that is listed as *LOCAL and then use that value as the value of this property.
JDBC Driver	The type of JDBC driver to use for connecting to the Db2 for i source. Options are: - **Progress DataDirect Db2 JDBC**. Use the DataDirect JDBC driver for Db2. - **IBM JTOpen JDBC**. Use the IBM JTOpen JDBC driver. Default is Progress DataDirect Db2 JDBC.
Code Page for Bit Data	The code page that Database Ingestion and Replication uses to read character data that is stored as bit data. This value must be a canonical name for the java.io API and java.lang API. For more information, see the supported encodings in the Oracle Java documentation. Specify this property if you have FOR BIT DATA source columns.

Property	Description
Advanced Connection Properties	Advanced properties for the JDBC driver that is used to connect to the Db2 for i source. If you specify more than one *property=value* entry, separate them with a semicolon (;). For information about the Progress DataDirect Db2 JDBC driver connection properties, see Progress DataDirect documentation. For example, you can set the ConnectionRetryCount property to control the number of times the driver retries attempts to connect to the primary database server. For information about the IBM JTOpen JDBC driver connection properties, see IBM Toolbox for Java JDBC properties.

Encryption Method	The data encryption method to use for the IBM JTOpen JDBC driver. Options are:
	- **No Encryption**
	- **SSL**
	Default is No Encryption.
	If you select SSL, you must add the required certificates to the Informatica Cloud Secure Agent JRE cacerts keystore in one of the following locations:
	For Linux:
	`Secure Agent Directory\jdk\jre\lib\security\cacerts`
	For Windows:
	`Secure Agent Directory\apps\jdkLatestVersion\jre`

Db2 for LUW Database Ingestion connection properties

To define a Db2 for LUW Database Ingestion connection, configure the connection properties. You can use this connection type in database ingestion and replication tasks configured in Data Ingestion and Replication.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	The name of the runtime environment where you want to run database ingestion and replication tasks. You define runtime environments in Administrator.
User Name	The user name to use for connecting to the Db2 for LUW instance.
Password	The password to use for connecting to the Db2 for LUW instance.
Host	The name of the machine that hosts the database server.
Port	The network port number used to connect to the database server.

Property	Description
Database Name	The name of the Db2 for LUW database that you want to access.
Advanced Connection Properties	Advanced properties for the Progress DataDirect JDBC DB2 driver, which is used to connect to the Db2 for LUW source. If you specify more than one *property=value* entry, separate them with a semicolon (;). The driver properties that you can enter in this field are described in the Progress DataDirect for JDBC connection properties. For example, you can set the EncryptionMethod property to control whether data is encrypted and decrypted when transmitted over the network between the driver and database server.

Db2 for zOS Database Ingestion connection properties

To define a Db2 for zOS Database Ingestion connection, configure the connection properties. You can use this connection type in database ingestion and replication tasks configured in Data Ingestion and Replication.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	The name of the runtime environment where you want to run database ingestion and replication tasks. You define runtime environments in Administrator. You can use either a local Secure Agent installation or serverless runtime environment. You can use a serverless runtime environment for cloud source and target types. You cannot run database ingestion and replication tasks on a Hosted Agent.
User Name	The user name to use for connecting to the Db2 for z/OS instance.
Password	The password to use for connecting to the Db2 for z/OS instance.
Host	The name of the machine that hosts the database server.
Port	The network port number used to connect to the database server.
Location Name	The name of the Db2 for z/OS location that you want to access. For Db2 for z/OS, your system administrator can determine the name of your Db2 location using the command DISPLAY DDF.
Code Page for Bit Data	The code page that Database Ingestion and Replication uses to read character data that is stored as bit data. This value must be a canonical name for the java.io API and java.lang API. For more information, see the supported encodings in the Oracle Java documentation. Specify this property if you have FOR BIT DATA source columns.

Property	Description
CDC Stored Procedure Schema	For incremental change data capture processing, the name of the schema for the z/OS stored procedure that is required to collect change data from the Db2 log. This value is specified in the #STPINST data set that you customized when setting up the stored procedure on z/OS. No default value is provided.
CDC Stored Procedure Name	For incremental change data capture processing, the name of the z/OS stored procedure that is required to collect change data from the Db2 log. This value is specified in the #STPINST data set that you customized when setting up the stored procedure on z/OS. The default value is INFALOG.
Advanced Connection Properties	Advanced properties for the Progress DataDirect JDBC Db2 driver, which is used to connect to the Db2 for z/OS source. If you specify more than one *property=value* entry, separate them with a semicolon (;). The driver properties that you can enter in this field are described in the Progress DataDirect documentation at https://docs.progress.com/bundle/datadirect-connect-jdbc-51/page/Connection-Properties_10.html. For example, you can set the ConnectionRetryCount property to control the number of times the driver retries attempts to connect to the primary database server.

Flat file connection properties

Defines the properties you need to assign to for a flat file source connection.

The following table describes the flat file connection properties:

Connection Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.

Connection Property	Description
Runtime Environment	Runtime environment that contains the Secure Agent to use for accessing the flat files. **Note:** Do not select a runtime environment with Secure Agents that run on NTT. A flat file connection cannot use a Secure Agent that runs on NTT.

Directory	The directory accessible by all Secure Agents in the selected runtime environment where the flat file is stored.
	Ensure that the Secure Agent is up and running to fetch the remote directory details.
	The directory is the service URL for this connection type.
	Enter the full directory or click **Browse** to locate and select the directory. You can then select a file contained in that directory or in any of its subdirectories when you use the connection.
	Note: If you click **Browse** and receive the error message, "The runtime environment cannot get the requested file system information...," see the Error while clicking Browse on the Flat File connection Knowledge article.
	Maximum length is 100 characters. Directory names can contain alphanumeric characters, spaces, and the following special characters:
	`/ \ : _ ~`
	You don't need to specify the flat file name in the connection. You can specify the file name when you create new files in the directory in a mapping task.
	In an elastic runtime environment, store the flat file in the EFS file system and specify the following directory path:
	`/etc/infa/pod/<elastic runtime environment ID>/`
	For more information about uploading flat files in an elastic runtime enviornment, see *Uploading flat files for flat file connections* in the Administrator help.
	Note: On Windows, the **Browse for Directory** dialog box doesn't display mapped drives. You can browse My Network Places in Windows Explorer to locate the directory and copy the location from the address bar or enter the directory name in the following format:
	`\\<server_name>\<directory_path>`
	If network directories do not display, you can configure a login for the Secure Agent service. This functionality might not be available on newer versions of Windows.

Browse button	Use to locate and select the directory where flat files are stored.
Date Format	Date format for date fields in the flat file. Select one of the following date formats: - MM/dd/yyyy HH:mm:ss - YYYY-MM-DD HH24:MI:SS.US

Connection Property	Description
Code Page	The code page of the system that hosts the flat file. Select one of the following code pages: - MS Windows Latin 1. Select for ISO 8859-1 Western European data - UTF-8. Select for Unicode data - UTF-16 encoding of Unicode (Big Endian) - UTF-16 encoding of Unicode (Lower Endian) - Shift-JIS. Select for double-byte character data. - ISO 8859-15 Latin 9 (Western European) - ISO 8859-2 Eastern European - ISO 8859-3 Southeast European - ISO 8859-5 Cyrillic - ISO 8859-9 Latin 5 (Turkish) - IBM EBCDIC International Latin-1 - Japanese EUC (with \ <-> Yen mapping - IBM EBCDIC Japanese - IBM EBCDIC Japanese CP939 - PC Japanese SJIS-78 syntax (IBM-942) - PC Japanese SJIS-90 (IBM-943) - MS Windows Traditional Chinese, superset of Big 5 - Taiwan Big-5 (w/o euro update) - Chinese EUC - ISO 8859-8 Hebrew - PC Hebrew (old) - PC Hebrew (w/o euro update) - EBCDIC Hebrew (updated with new sheqel, control characters) - IBM EBCDIC US English IBM037 - UTF-32 encoding of Unicode (Lower Endian) - ISO 8859-1 Western European. - IBM EBCDIC French - ISO 8859-10 Latin 6 (Nordic) * - EBCDIC Finland, Sweden - MOS-DOS Thai, superset of TIS 620 - 7-bit ASCII - EBCDIC Finland, Sweden (w/euro update) - MS-DOS Windows Latin 2 (Central Europe) - Japanese EBCDIC-Kana Fujitsu - ISO 8859-7 Greek Characters in the file must be 2 bytes or less. In advanced mappings, flat file objects in cloud storage connections must use UTF-8 encoding. If the file contains supplementary characters with UTF-16 encoding, the task fails. **Note:** When you use a flat file connection with the Shift-JIS code page and a UTF data object, be sure to install fonts that fully support Unicode.
* Data preview uses a similar ISO 8859-4 Scandinavian/Baltic code page, but runtime processing uses ISO 8859-10 Latin 6 (Nordic), so data preview and runtime encoding won't match.	

Google BigQuery V2 connection properties

Create a Google BigQuery V2 connection to securely read data from or write data to Google BigQuery.

Connect to Google BigQuery

Let's configure the Google BigQuery V2 connection properties to connect to Google BigQuery.

Before you begin

Before you configure a connection, ensure that you download the Google service account key file in JSON format. The service account key file is created when you create a Google service account.

You require the client email, private key, and project ID from the service account key JSON file to create a Google BigQuery connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use a Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. You cannot run an application ingestion and replication task, database ingestion and replication task, file ingestion or replication task, or streaming ingestion and replication task on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

Google BigQuery V2 Connector connects to Google BigQuery with a service account key or with Workload Identity Federation (WIF).

Select one of the following authentication types in the connection:

Authentication type	Description
Service Account	Authenticate with the credentials in a Google service account key file.
WIF (Authorization Code)	Authenticate with WIF and an identity provider that uses the OAuth 2.0 authorization code grant. You sign in to the identity provider to generate an access token. You can configure Okta as the identity provider.
WIF (Client Credentials)	Authenticate with WIF and an identity provider that uses the OAuth 2.0 client credentials grant. The connection doesn't require a user to sign in. You can also impersonate a service account and access datasets in a different project. You can configure Microsoft Entra ID as the identity provider.

Select the required authentication type and configure the authentication-specific parameters.

Default is Service Account authentication type.

Service Account authentication

Service Account authentication requires at a minimum your Google BigQuery service account email, service account key, and project ID.

The following table describes the basic connection properties for Service Account authentication:

Property	Description
Service Account Email	The client_email value from the Google service account key JSON file.
Service Account Key	The private_key value from the Google service account key JSON file. To use the private key from the JSON file, enter the entire block of text found within the double quotation marks in the `private_key` field, starting with `-----BEGIN PRIVATE KEY-----\n` and ending at `-----END PRIVATE KEY-----\n`.
Project ID	The project_id value from the Google service account key JSON file. If you have created multiple projects with the same service account, enter the ID of the project that contains the dataset that you want to connect to.

Note: If you want to validate the credentials for the Service Account Email, Service Account Key, and Project ID during a test connection, set the flag `CredentialValidation:true` in the **Provide Optional Properties** field in advanced settings.

Advanced settings

The following table describes the advanced connection properties for Service Account authentication:

Property	Description
Storage Path	Path in Google Cloud Storage where the agent creates a local stage file to store the data temporarily. The agent uses this storage when it reads data in staging mode or writes data in bulk mode. Use one of the following formats: - gs://<bucket_name> - gs://<bucket_name>/<folder_name When you enable cross-region replication in Google BigQuery, enter a Google Cloud Storage path to a dual-region bucket.
Connection Mode	The mode that you want to use to read data from or write data to Google BigQuery. Select one of the following connection modes: - Simple. Flattens each field within the Record data type field as a separate field in the mapping. - Hybrid. Displays all the top-level fields in the Google BigQuery table including Record data type fields. Google BigQuery V2 Connector displays the top-level Record data type field as a single field of the String data type in the mapping. - Complex. Displays all the columns in the Google BigQuery table as a single field of the String data type in the mapping. Default is Simple.
Use Legacy SQL for Custom Query	Select this option to use legacy SQL to define a custom query. If you clear this option, use standard SQL to define a custom query. This property doesn't apply if you configure the Google BigQuery V2 connection in hybrid or complex mode.
Dataset Name for Custom Query	The name of the Google BigQuery dataset that you want to specify in a custom query.
Schema Definition File Path	Directory on the Secure Agent machine where the Secure Agent creates a JSON file with the sample schema of the Google BigQuery table. The JSON file name is the same as the Google BigQuery table name. Alternatively, you can specify a storage path in Google Cloud Storage where the Secure Agent creates a JSON file with the sample schema of the Google BigQuery table. You can download the JSON file from the specified storage path in Google Cloud Storage to a local machine. The schema definition file is required if you configure complex connection mode in the following scenarios: - You add a Hierarchy Builder transformation in a mapping to read data from relational sources and write data to a Google BigQuery target. - You add a Hierarchy Parser transformation in a mapping to read data from a Google BigQuery source and write data to relational targets.
Region ID	The region name where the Google BigQuery dataset that you want to access resides. **Note:** Ensure that you specify a bucket name or the bucket name and folder name in the **Storage Path** property that resides in the specified region. For the list of Google BigQuery regions, see Dataset locations.

Property	Description
Staging Dataset	The Google BigQuery dataset name where you want to create the staging table to stage the data. You can define a Google BigQuery dataset that is different from the source or target dataset.
Provide Optional Properties	Comma-separated key-value pairs of custom properties in the Google BigQuery V2 connection to configure certain source and target functionalities. For more information about the list of custom properties that you can specify, see Optional Properties configuration Knowledge article.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux. You can connect to Google BigQuery through an authenticated or unauthenticated HTTP proxy server.

To configure the proxy settings for the Secure Agent, perform the following tasks:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

The proxy settings configured in the `proxy.ini` configuration file are applicable at design time, and the proxy settings configured through the JVM options are applicable at runtime.

Configure proxy settings for NTLM authentication

You can use a proxy server that uses NTLM authentication to connect to Google BigQuery. To configure the proxy settings for NTLM authentication, perform the following steps:

1. In Administrator, select **Runtime Environments**.
2. Select the Secure Agent for which you want to configure from the list of available Secure Agents.
3. In the upper-right corner, click **Edit**.
4. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Server.
5. Edit the **JVMOption1** and add the following value:
   `-Dhttp.auth.ntlm.domain=<domain name>`
6. Select the **Type** as **Platform** for the Data Integration Server.
7. Edit the **INFA_DEBUG** property and add the following value:
   `-Dhttp.auth.ntlm.domain=<domain name>`
8. Click **Save**.
9. Restart the Secure Agent.

Google Cloud Storage V2 connection properties

Create a Google Cloud Storage V2 connection to securely read data from or write data to Google Cloud Storage files.

Prerequisites for Service Account authentication

Before you configure a Google Cloud Storage V2 connection to use Service Account authentication, complete the following prerequisite tasks.

1. Ensure that you have a Google service account to access Google Cloud Storage.
2. Ensure that you have the client email, project ID, and private key values for the service account. Enter these details when you create a Google Cloud Storage V2 connection.
3. Enable the Google Cloud Storage JSON API for your service account. Google Cloud Storage V2 Connector uses the Google API to integrate with Google Cloud Storage.
4. Verify that you have read and write access to the Google Cloud Storage bucket that contains the source and target files.
5. To run a mapping that reads data from or writes data to a Google Cloud Storage file, ensure that you have the required permissions.
6. To use the Informatica encryption method, complete the following tasks:
   - Enable the Informatica crypto library license in Administrator.
   - Add the following firewall rules to the Virtual Private Cloud (VPC) network on the Google Cloud Platform advanced cluster:
     - List of Secure Agent IP address ranges in the Source Filter section.
     - Google Cloud Platform cluster IP address ranges in the Source Filter section.
     - Ports from 0 through 65535 and udp:443 in the Protocols and Ports section.
   - Ensure that the VPC network tag on the Secure Agent virtual machine matches the source tag and target tag of the firewall rules.

Connect to Google Cloud Storage V2

Let's configure the Google Cloud Storage V2 connection properties to connect to Google Cloud Storage.

Connection details

The following table describes the Google Cloud Storage V2 connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use the Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. For file ingestion and replication tasks and streaming ingestion and replication tasks, you can use only the Secure Agent. You cannot run tasks on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can connect to Google Cloud Storage with a service account key.

The following table describes the authentication types that you can select in the connection:

Authentication Type	Description
Service Account	Authenticate with the credentials in a Google service account key file.
WIF (Client Credentials)	**Note:** Data Ingestion and Replication can't use the WIF (Client Credentials) authentication type.

The default is Service Account.

After you select an authentication type, configure the properties for that type.

Service Account authentication

Configure the basic and advanced connection properties for Service Account authentication to Google Cloud Storage.

Service Account authentication requires a service account email, a service account key, and a project ID.

The following table describes the basic connection properties for Service Account authentication:

Property	Description
Service Account Email	The client_email value from the Google service account key JSON file.
Service Account Key	The private_key value from the Google service account key JSON file. Enter the entire block of text within the double quotation marks in the private_key field. Start with `-----BEGIN PRIVATE KEY-----\n` and end with `-----END PRIVATE KEY-----\n`.
Project ID	The project_id value from the Google service account key JSON file. If you created multiple projects with the same service account, enter the ID of the project that contains the bucket.
Optimize Object Metadata Import	Imports metadata for the selected object without parsing other objects, folders, or subfolders in the bucket. Importing only the selected object reduces the time that the connector spends parsing the bucket. Default is not selected. This property does not apply to a file ingestion and replication task.

Advanced settings

The following table describes the advanced connection properties for Service Account authentication:

Property	Description
Service Account Private Key ID	The `private_key_id` value from the Google service account key JSON file. This property applies only to a database ingestion and replication or streaming ingestion and replication task.
Service Account Client ID	The `client_id` value from the Google service account key JSON file. This property applies only to a database ingestion and replication or streaming ingestion and replication task.
Is Encrypted File	Specifies whether a file is encrypted. Select this option when you import an encrypted file from Google Cloud Storage. Default is not selected.
Bucket Name	The name of the Google Cloud Storage bucket that you want to connect to. When you select a source object , the Package Explorer lists files and folders available in the specified Google Cloud Storage bucket. If you don't specify a bucket name, you can select a bucket from the Package Explorer to select a source.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux. You can connect to Google Cloud Storage through an authenticated or unauthenticated HTTP proxy server.

To configure the proxy settings for the Secure Agent, perform the following tasks:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

Configure proxy settings for NTLM authentication

You can use a proxy server that uses NTLM authentication to connect to Google Cloud Storage.

To configure the proxy settings for NTLM authentication, perform the following steps:

1. In Administrator, select **Runtime Environments**.
2. Select the Secure Agent for which you want to configure from the list of available Secure Agents.
3. In the upper-right corner, click **Edit**.
4. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Server.
5. Edit the **JVMOption1** and add the following value:
   `-Dhttp.auth.ntlm.domain=<domain name>`
6. Select the **Type** as **Platform** for the Data Integration Server.
7. Edit the **INFA_DEBUG** property and add the following value:
   `-Dhttp.auth.ntlm.domain=<domain name>`
8. Click **Save**.
9. Restart the Secure Agent.

Google PubSub - Streaming Ingestion and Replication connection properties

When you define a Google PubSub Streaming Ingestion and Replication connection, you must configure connection properties. You can use this connection type in streaming ingestion and replication tasks, which you configure in the Data Ingestion and Replication service.

The following table describes the Google PubSub connection properties:

Property	Description
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. Description that you use to identity the connection. The description must not exceed 4,000 characters.
Runtime Environment	Name of the runtime environment where you want to run the tasks.

Property	Description
Client Email	The `client_email` value available in the JSON file that you download after you create a service account.
Client ID	The `client_id` value available in the JSON file that you download after you create a service account.
Private Key ID	The `private_key_id` value available in the JSON file that you download after you create a service account.
Private Key	The `private_key` value available in the JSON file that you download after you create a service account.
Project ID	The `project_id` value available in the JSON file that you download after you create a service account.

Note: The test connection for the Google PubSub connector does not fail even if you enter incorrect values for **Client ID** and **Private Key ID**.

Hadoop Files V2 connection properties

When you set up a Hadoop Files V2 connection, you must configure the connection properties.

The following table describes the Hadoop Files V2 connection properties:

Connection property	Description
Connection Name	Name of the Hadoop Files V2 connection.
Description	Description of the connection. The description cannot exceed 765 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run the tasks. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Connection property	Description
User Name	Required to read data from HDFS. Enter a user name that has access to the single-node HDFS location to read data from or write data to.

NameNode URI	The URI to access HDFS.
	Use the following format to specify the name node URI in Cloudera, Amazon EMR, and Hortonworks distributions:
	`hdfs://<namenode>:<port>/`
	where,
	- `<namenode>` is the host name or IP address of the name node.
	- `<port>` is the port that the name node listens for remote procedure calls (RPC).
	To connect to the Hadoop cluster, specify the name node port `fs.defaultFS`.
	If the Hadoop cluster is configured for high availability, you must copy the `fs.defaultFS` value in the `core-site.xml` file and append `/` to specify the name node URI.
	For example, the following snippet shows the `fs.defaultFS` value in a sample `core-site.xml` file:
	`<property>`
	`<name>fs.defaultFS</name>`
	`<value>hdfs://nameservice1</value>`
	`<source>core-site.xml</source>`
	`</property>`
	In the above snippet, the `fs.defaultFS` value is
	`hdfs://nameservice1`
	and the corresponding name node URI is
	`hdfs://nameservice1/`
	Note: Specify either the name node URI or the local path. Do not specify the name node URI if you want to read data from or write data to a local file system path.

Local Path	A local file system path to read and write data. Read the following conditions to specify the local path: - You must enter **NA** in local path if you specify the name node URI. If the local path does not contain **NA**, the name node URI does not work. - If you specify the name node URI and local path, the local path takes the preference. The connection uses the local path to run all tasks. - If you leave the local path blank, the agent configures the root directory (/) in the connection. The connection uses the local path to run all tasks. - If the file or directory is in the local system, enter the fully qualified path of the file or directory. For example, `/user/testdir` specifies the location of a directory in the local system. Default value for Local Path is NA.
Configuration Files Path	The directory that contains the Hadoop configuration files. **Note:** Copy the core-site.xml, hdfs-site.xml, and hive-site.xmlfrom the Hadoop cluster and add them to a folder on the Linux Box.
Keytab File	The file that contains encrypted keys and Kerberos principals to authenticate the machine.

Connection property	Description
Principal Name	Users assigned to the superuser privilege can perform all the tasks that a user with the administrator privilege can perform.
Impersonation Username	You can enable different users to run mappings in a Hadoop cluster that uses Kerberos authentication or connect to sources and targets that use Kerberos authentication. To enable different users to run mappings or connect to big data sources and targets, you must configure user impersonation.

Note: When you read from or write to remote files, the **NameNode URI** and **Configuration Files Path** fields are mandatory. When you read from or write to local files, you require only the **Local Path** field.

JDBC V2 connection properties

Create a JDBC V2 connection to access data from Aurora PostgreSQL or any database that supports the Type 4 JDBC driver.

Prerequisites

Before you create a JDBC V2 connection to read from or write to databases that support the JDBC Type 4 driver, complete the prerequisites.

Install the Type 4 JDBC driver

To read from or write to JDBC V2 objects, you need to install the Type 4 JDBC driver on the Secure Agent machine.

1. Download the latest Type 4 JDBC driver version that your database supports from the third-party vendor site.
   If you want to use JDBC V2 Connector to connect to Aurora PostgreSQL, download the Aurora PostgreSQL driver. Informatica has certified Aurora PostgreSQL driver 42.2.6 for JDBC V2 Connector.
2. Install the Type 4 JDBC driver for the database on the Secure Agent machine and perform the following tasks:
   a. Navigate to the following directory on the Secure Agent machine:
      `<Secure Agent installation directory>/ext/connectors/thirdparty/`
3. Restart the Secure Agent.
   If you update the driver on the Secure Agent machine while the mapping in advanced mode runs, you need to restart the Secure Agent.

Connect to JDBC V2

Let's configure the JDBC V2 connection properties to connect to JDBC-compliant databases.

Before you begin

Before you get started, you'll need to install the Type 4 JDBC driver on the Secure Agent machine to establish a JDBC V2 connection.

Check out "Prerequisites" on page 100 to learn more about the configuration prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. In Data Ingestion and Replication, you cannot run a streaming ingestion and replication task on a Hosted Agent, serverless runtime environment, or elastic runtime environment.
JDBC Driver Class Name	Name of the JDBC driver class. For example, to connect to Aurora PostgreSQL, specify the following driver class name: org.postgresql.Driver For more information about which driver class to use with specific databases, see the corresponding third-party vendor documentation.
Connection String	Connection string to connect to the database. Use the following format to specify the connection string: `jdbc:<subprotocol>:<subname>` For example, the connection string for the Aurora PostgreSQL database type is `jdbc:postgresql://<host>:<port>[/dbname]`. For more information about the connection string to use with specific drivers, see the corresponding third-party vendor documentation.
User Name	The user name to connect to the database.
Password	The password to connect to the database.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Database Type	The database type to which you want to connect. Select one of the following database types: - PostgreSQL. Connect to the Aurora PostgreSQL database hosted in the Amazon Web Services or the Microsoft Azure environment. - Azure SQL Database. Connect to Azure SQL Database hosted in the Microsoft Azure environment. - Others. Connect to any database that supports the Type 4 JDBC driver, such as Salesforce Data Cloud. Default is Others.
Schema Name	The schema name used for the JDBC object. If you don't specify the schema name, all the schemas available in the database are listed. To read from or write to Oracle public synonyms, enter PUBLIC.
Connection Environment SQL	The SQL statement to set up the database environment when you connect to a PostgreSQL database. The database environment applies for the entire session that uses this connection. For example, you can enter this statement to set the time zone: `SET timezone to 'America/New_York';`
Additional Security Properties	Masks sensitive and confidential data of the connection string that you don't want to display in the session log. Specify the part of the connection string that you want to mask. When you create a connection, the string you enter in this field appends to the string that you specified in the **Connection String** field.
Enable Auto Commit	Specifies whether the driver supports connections to automatically commit data to the database when you run an SQL statement. When disabled, the driver does not support connections to automatically commit data even if the auto-commit mode is enabled in the JDBC driver. Default is disabled.
Support Mixed-Case Identifiers	Indicates whether the database supports case-sensitive identifiers. When enabled, the Secure Agent encloses all identifiers within the character selected for the SQL Identifier Character property. Default is disabled.
SQL Identifier Character	Type of character that the database uses to enclose delimited identifiers in SQL queries. The available characters depend on the database type. Select **None** if the database uses regular identifiers. When the Secure Agent generates SQL queries, it does not place delimited characters around any identifiers. Select a character from the list based on what delimiter the database uses for identifiers. When the Secure Agent generates SQL queries, it encloses delimited identifiers within this character.

JMS connection properties

Create a JMS connection to securely read data from or write data to JMS.

Prerequisites

Before you configure the connection properties, make sure to copy the JMS provider-specific third-party client libraries to the following location and restart the Secure Agent after copying the client libraries:

```
<Informatica Secure Agent installation directory>/ext/connectors/thirdparty/infa.jms
```

Create the folder structure if it does not already exist.

For more information about the required third-party client libraries, see the JMS provider documentation.

Connect to JMS

Let's configure the JMS connection properties to connect to JMS.

Before you begin

Before you get started, you need to keep the details of the JNDI naming provider, JNDI context factory, and credentials for JMS connection handy.

Check out "Prerequisites" on page 103 to learn about the requirements before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select Secure Agent for Streaming Ingestion and Replication. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Connection URL	The URL of the JNDI naming provider.

Property	Description
JNDI Username	The user name to connect to the JNDI context factory.
JNDI Password	The password to connect to the JNDI context factory.
JNDI Context Factory	A fully qualified class name that allows you to connect to the JNDI service. For example, the class name of the Initial Context Factory for Apache Active MQ is `org.apache.activemq.jndi.ActiveMQInitialContextFactory` For example, the class name of the Initial Context Factory for RabbitMQ is `com.sun.jndi.fscontext.RefFSContextFactory` For more information, see the documentation of the JMS provider.
JNDI Package Prefixes	A colon-separated list of package name prefixes provided to the JNDI context factory to identify and load the appropriate context factory classes. For example, `java.naming.factory.url.pkgs=org.apache.activemq.jndi` For more information about the values, see the documentation of the JMS provider.
JMS Connection Factory	The name of the JNDI object used by a JMS client to create a connection. For example, `jms/QCF` or `jmsSalesSystem`.
Connection Username	The user name to authenticate the JMS provider to connect to the JMS connection factory.
Connection Password	The password to authenticate the JMS provider to connect to the JMS connection factory.

Kafka connection properties

When you set up a Kafka connection, configure the connection properties.

The following table describes the Kafka connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive. It must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
Description	Optional. Description that you use to identity the connection. The description cannot exceed 4,000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	Name of the runtime environment where you want to run tasks. In Data Ingestion and Replication, you can use a Secure Agent in application ingestion and replication, file ingestion and replication, and streaming ingestion and replication tasks. For database ingestion and replication tasks, you can use a Secure Agent or serverless runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Kafka Broker List	Comma-separated list of the Kafka brokers. To list a Kafka broker, use the following format: `<HostName>:<PortNumber>` **Note:** When you connect to a Kafka broker over SSL, you must specify the fully qualified domain name for the host name. Otherwise, the test connection fails with SSL handshake error.
Kafka Broker Version	Kafka message broker version. The only valid value is Apache 0.10.1.1 and above. Optional for a streaming ingestion and replication task.
Additional Connection Properties	Optional. Comma-separated list of additional configuration properties of the Kafka producer or consumer. For a streaming ingestion and replication task, ensure that you set the `<kerberos name>` property if you configure `<Security Protocol>` as `SASL_PLAINTEXT` or `SASL_SSL`. For a database ingestion and replication task, if you want to specify a security protocol and properties, specify them here instead of in the **Additional Security Properties** property. For example: `security.protocol=SSL,ssl.truststore.location=/opt/kafka/config/kafka.truststore.jks,ssl.truststore.password=<trustore_password>`.

Confluent connection properties

Property	Description

Confluent Schema Registry URL	Location and port of the Confluent schema registry service to access Avro sources and targets in Kafka.
	To list a schema registry URL, use the following format:
	`<https>://<HostName or IP>:<PortNumber>`
	or
	`<http>://<HostName or IP>:<PortNumber>`
	Example for the schema registry URL:
	`https://kafkarnd.informatica.com:8082`
	or
	`http://10.65.146.181:8084`
	Applies only when you import a Kafka topic in Avro format that uses the Confluent schema registry to store the metadata.
	This property is not used by Database Ingestion and Replication. You can specify an equivalent Kafka property in **Additional Connection Properties**.

Security configuration properties

Property	Description
SSL Mode	Required. Determines the encryption type to use for the connection. You can choose a mode from the following SSL modes: - Disabled. Establishes an unencrypted connection to the Kafka broker. - One-way. Establishes an encrypted connection to the Kafka broker using truststore file and truststore password. - Two-way. Establishes an encrypted connection to the Kafka broker using truststore file, truststore password, keystore file, and keystore password. This property is not used by Database Ingestion and Replication. You can specify an equivalent Kafka property in **Additional Connection Properties**.
SSL TrustStore File Path	Required when you use the one-way or two-way SSL mode. Absolute path and file name of the SSL truststore file that contains the SSL certificate to connect to the Kafka broker.
SSL TrustStore Password	Required when you use the one-way or two-way SSL mode. Password for the SSL truststore.
SSL KeyStore File Path	Required when you use the two-way SSL mode. Absolute path and file name of the SSL keystore file that contains private keys and certificates to connect to the Kafka broker.

Property	Description
SSL KeyStore Password	Required when you use the two-way SSL mode. Password for the SSL keystore.
Additional Security Properties	Comma-separated list of additional configuration properties to connect to the Kafka broker in a secure way. To configure one-way or two-way SSL mode, enter the following value: `security.protocol=SSL` If you specify two different values for the same property in **Additional Connection Properties** and **Additional Security Properties Schema Registry**, the value in **Additional Security Properties Schema Registry** overrides the value in **Additional Connection Properties**.

Schema registry security configuration properties

When you configure the **Schema Registry URL** connection property, you can configure the schema registry security configuration properties. You can configure one-way SSL, two-way SSL, and basic authentication to connect to the Confluent schema registry in a secure way.

The following table describes the security properties for the Kafka connection when you use the Confluent schema registry:

Property	Description
SSL Mode Schema Registry[1]	Required. Determines the encryption type to use for the connection. You can choose a mode from the following SSL modes: - Disabled. Establishes an unencrypted connection to the Confluent schema registry. - One-way. Establishes an encrypted connection to the Confluent schema registry using truststore file and truststore password. - Two-way. Establishes an encrypted connection to the Confluent schema registry using truststore file, truststore password, keystore file, and keystore password. This property is not used by Database Ingestion and Replication. You can specify an equivalent Kafka property in **Additional Connection Properties**.
SSL TrustStore File Path Schema Registry[1]	Required when you use the one-way or two-way SSL mode. Absolute path and file name of the SSL truststore file that contains the SSL certificate to connect to the Confluent schema registry.
SSL TrustStore Password Schema Registry[1]	Required when you use the one-way or two-way SSL mode. Password for the SSL truststore.
SSL KeyStore File Path Schema Registry[1]	Required when you use the two-way SSL mode. Absolute path and file name of the SSL keystore file that contains private keys and certificates to connect to the Confluent schema registry.

Property	Description
SSL KeyStore Password Schema Registry[1]	Required when you use the two-way SSL mode. Password for the SSL keystore.
Additional Security Properties Schema Registry[2]	Optional. Comma-separated list of additional security properties to connect to the Confluent schema registry in a secure way. For example, when you configure basic authentication to establish a secure communication with Confluent schema registry, specify the following value: `basic.auth.credentials.source=USER_INFO,basic.auth.user.info=<username>:<password>` If you specify two different values for the same property in **Additional Connection Properties** and **Additional Security Properties Schema Registry**, the value in **Additional Security Properties Schema Registry** overrides the value in **Additional Connection Properties**. This property is not used by Database Ingestion and Replication.
[1] Applies only to mappings in advanced mode. [2] Applies to both mappings and mappings in advanced mode.	

Configuring the krb5.conf file to read data from or write to a Kerberised Kafka cluster

To read from or write to a Kerberised Kafka cluster, configure the default realm, KDC, and Kafka advanced source or target properties.

You can configure Kerberos authentication for a Kafka client by placing the required Kerberos configuration files on the Secure Agent machine and specifying the required JAAS configuration in the Kafka connection. The JAAS configuration defines the keytab and principal details that the Kafka broker must use to authenticate the Kafka client.

Before you read from or write to a Kerberised Kafka cluster, perform the following tasks:

1. Ensure that you have the `krb5.conf` file for the Kerberised Kafka cluster.
2. Configure the default realm and KDC. If the default `/etc/krb5.conf` file is not configured or you want to change the configuration, add the following lines to the `/etc/krb5.conf` file:

```
[libdefaults]
default_realm = <REALM NAME>
dns_lookup_realm = false
dns_lookup_kdc = false
ticket_lifetime = 24h
renew_lifetime = 7d
forwardable = true

[realms]
<REALM NAME> = {
kdc = <Location where KDC is installed>
admin_server = <Location where KDC is installed>
                                }
 [domain_realm]
.<domain name or hostname> = <KERBEROS DOMAIN NAME>
<domain name or hostname> = <KERBEROS DOMAIN NAME>
```

3. To pass a static JAAS configuration file into the JVM using the `java.security.auth.login.config` property at runtime, perform the following tasks:

   a. Ensure that you have JAAS configuration file.

      For information about creating JAAS configuration and configuring keytab for Kafka clients, see the Apache Kafka documentation at https://kafka.apache.org/0101/documentation/#security

      For example, the JAAS configuration file can contain the following lines of configuration:

      ```
      //Kafka Client Authentication. Used for client to kafka broker connection
      KafkaClient {
      com.sun.security.auth.module.Krb5LoginModule required
      doNotPrompt=true
      useKeyTab=true
      storeKey=true
      keyTab="<path to Kafka keytab  file>/<Kafka keytab file name>"
      principal="<principal name>"
      client=true
      };
      ```

   b. Place the JAAS config file and keytab file in the same location on all the secure agents.

      Informatica recommends that you place the files in a location that is accessible by all the secure agents in the runtime environment. For example, `/etc` or `/temp`.

   c. Configure the following properties:

      Kafka connection

      Configure the **Additional Connection Properties** in a Kafka connection and specify the value in the following format:

      ```
      security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechan
      ism=GSSAPI
      ```

      Sources

      Configure the **Consumer Configuration Properties** in the advanced source properties to override the value specified in **Additional Connection Properties** in a Kafka connection. Specify the value in the following format:

      ```
      security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechan
      ism=GSSAPI
      ```

      Targets

      Configure the **Producer Configuration Properties** in the advanced target properties to override the value specified in **Additional Connection Properties** in a Kafka connection. Specify the value in the following format:

      ```
      security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechan
      ism=GSSAPI
      ```

4. To embed the JAAS configuration in the `sasl.jaas.config` configuration property, configure the following properties:

   Kafka connection

   Configure the **Additional Connection Properties** in a Kafka connection and specify the value in the following format:

   ```
   security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=
   GSSAPI,
   sasl.jaas.config=com.sun.security.auth.module.Krb5LoginModule required
   useKeyTab=true
   storeKey=true doNotPrompt=true serviceName="<service_name>" keyTab="<location of
   keytab file>"
   client=true principal="<principal_name>";
   ```

Sources

Configure the **Consumer Configuration Properties** in the advanced source properties to override the value specified in **Kerberos Configuration Properties** in a Kafka connection. Specify the value in the following format:

```
security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=
GSSAPI,
sasl.jaas.config=com.sun.security.auth.module.Krb5LoginModule required
useKeyTab=true
storeKey=true doNotPrompt=true serviceName="<service_name>" keyTab="<location of
keytab file>"
client=true principal="<principal_name>";
```

Targets

Configure the **Producer Configuration Properties** in the advanced target properties to override the value specified in **Kerberos Configuration Properties** in a Kafka connection. Specify the value in the following format:

```
security.protocol=SASL_PLAINTEXT,sasl.kerberos.service.name=kafka,sasl.mechanism=
GSSAPI,
sasl.jaas.config=com.sun.security.auth.module.Krb5LoginModule required
useKeyTab=true
storeKey=true doNotPrompt=true serviceName="<service_name>" keyTab="<location of
keytab file>"
client=true principal="<principal_name>";
```

Configuring SASL PLAIN authentication for a Kafka cluster

In the Kafka connection, you can configure PLAIN security for the Kafka broker to connect to a Kafka broker. To read data from or write data to a Kafka broker with SASL PLAIN authentication, configure the Kafka connection properties. To override the properties defined in the Kafka connection, you can configure the advanced source or target properties.

You can configure SASL PLAIN authentication so that the Kafka broker can authenticate the Kafka producer and the Kafka consumer. Kafka uses the Java Authentication and Authorization Service (JAAS) for SASL PLAIN authentication. To enable SASL PLAIN authentication, you must specify the SASL mechanism as PLAIN. You must also provide the formatted JAAS configuration that the Kafka broker must use for authentication. The JAAS configuration defines the username, password, that the Kafka broker must use to authenticate the Kafka client.

Configure the following properties:

Kafka connection

Configure the **Additional Connection Properties** property in the Kafka connection and specify the value in the following format:

```
security.protocol=SASL_SSL,sasl.mechanism=PLAIN,sasl.jaas.config=org.apache.kafka.com
mon.security.plain.PlainLoginModule required username="<username>"
password="<password>";
```

In the **Security Configuration Section**, select **One-Way** as the **SSL Mode** and specify the SSL TrustStore File Path and SSL TrustStore Password.

Sources

Configure the **Consumer Configuration Properties** property in the advanced source properties to override the value that you specified in the **Additional Connection Properties** property in the Kafka connection. Specify the value in the following format:

```
security.protocol=SASL_SSL,sasl.mechanism=PLAIN,sasl.jaas.config=org.apache.kafka.com
mon.security.plain.PlainLoginModule required username="<username>"
password="<password>";
```

Targets

Configure the **Producer Configuration Properties** property in the advanced target properties to override the value that you specified in the **Additional Connection Properties** property in the Kafka connection. Specify the value in the following format:

```
security.protocol=SASL_SSL,sasl.mechanism=PLAIN,sasl.jaas.config=org.apache.kafka.com
mon.security.plain.PlainLoginModule required username="<username>"
password="<password>";
```

Configuring SASL PLAIN authentication for an Azure Event Hub Kafka broker

In the Kafka connection, you can configure PLAIN security for the Kafka broker to connect to an Azure Event Hub Kafka broker. When you connect to an Azure Event Hub Kakfa broker, the password defines the endpoint URL that contains the fully qualified domain name (FQDN) of the Event Hub namespace, shared access key name, and shared access key required to connect to an Azure Event Hub Kafka broker. Configure the SSL Mode as One-Way and provide the path to a trusted root certificate on your file system for SSL TrustStore File Path.

To connect to an Azure Event Hub Kafka broker, configure any of the above properties and specify the value in the following format:

```
security.protocol=SASL_SSL,sasl.mechanism=PLAIN,sasl.kerberos.service.name=Kafka,sasl.jaa
s.config=org.apache.kafka.common.security.plain.PlainLoginModule required
username="$ConnectionString" password="Endpoint=sb://<FQDN>/;SharedAccessKeyName=<key
name>;SharedAccessKey=<shared access key>=";
```

Configuring SASL_SSL authentication for a Cloud Confluent Kafka cluster

In the Kafka connection, you can configure SSL security for encryption and authentication while connecting to a Kafka broker. To read data from or write data to a Confluent Kafka broker with `SASL_SSL` authentication, configure the Kafka connection properties. To override the properties defined in the Kafka connection, you can configure the advanced source or target properties.

Configure the following properties:

Property	Values
Additional Connection Properties	`security.protocol=SASL_SSL,sasl.kerberos.service.name=kafka, ssl.endpoint.identification.algorithm=https,sasl.mechanism=PLAIN, sasl.jaas.config=org.apache.kafka.common.security.plain.PlainLoginModule required username=<> password=<>`
SSL Mode	One-way
SSL TrustStore File Path	Path to a trusted root cert available in either of the following locations within your Secure Agent installation directory: - `<AGENT_HOME>/jdk/jre/lib/security/cacerts` - `<AGENT_HOME>/jdk/lib/security/cacerts` - `<AGENT_HOME>/jdk8/jre/lib/security/cacerts`
SSL TrustStore Password	Password for the SSL truststore.

Connecting to Amazon Managed Streaming for Apache Kafka

In the Kafka connection, you can configure PLAINTEXT or TLS encryption to connect to an Amazon Managed Streaming for Apache Kafka broker. To read data from or write data to an Amazon Managed Streaming for Apache Kafka broker, configure the Kafka connection properties.

Configure the **Kafka Broker List** property in the Kafka connection and specify the comma-separated list of Kafka brokers that you want to connect to in the following format:

```
<HostName>:<PortNumber>
```

Configure TLS encryption to securely connect the Kafka broker to the Kafka producer and the Kafka consumer. To configure TLS encryption for an Amazon Managed Streaming for Apache Kafka broker, configure the following properties:

Property	Values
Additional Connection Properties	`security.protocol=SSL`
SSL Mode	One-way or Two-way.
SSL TrustStore File Path	Required when you use the one-way or two-way SSL mode. Absolute path and file name of the SSL truststore file.
SSL TrustStore Password	Required when you use the one-way or two-way SSL mode. Password for the SSL truststore.
SSL KeyStore File Path	Required when you use the two-way SSL mode. Absolute path and file name of the SSL keystore file that contains private keys and certificates that the Kafka broker validates against the Kafka cluster certificate.
SSL KeyStore Password	Required when you use the two-way SSL mode. Password for the SSL keystore.

When you run a mapping that runs on an advanced cluster and connect to an Amazon Managed Streaming for Apache Kafka broker, configure the Kafka broker using SASL_SSL authentication with Salted Challenge Response Authentication Mechanism (SCRAM). To read data from or write data to an Amazon Managed Streaming for Apache Kafka broker with SASL_SSL authentication, configure the following properties:

Property	Values
Additional Connection Properties	`security.protocol=SASL_SSL,sasl.mechanism=SCRAM-SHA-512,sasl.jaas.config=org.apache.kafka.common.security.scram.ScramLoginModule required username="<username>" password="<password>";`
SSL Mode	One-way or Two-way.
SSL TrustStore File Path	Required when you use the one-way or two-way SSL mode. Absolute path and file name of the SSL truststore file.
SSL TrustStore Password	Required when you use the one-way or two-way SSL mode. Password for the SSL truststore.

Property	Values
SSL KeyStore File Path	Required when you use the two-way SSL mode. Absolute path and file name of the SSL keystore file that contains private keys and certificates that the Kafka broker validates against the Kafka cluster certificate.
SSL KeyStore Password	Required when you use the two-way SSL mode. Password for the SSL keystore.

Marketo V3 connection properties

Create a Marketo V3 connection to securely read data from or write data to Marketo.

You can use Marketo V3 Connector in multiple services. For service-specific information, see the following documentation:

- Application Integration
- Data Integration
- Data Ingestion and Replication

Connect to Marketo

Let's configure the Marketo V3 connection properties to connect to Marketo.

Before you begin

Before you configure a connection, create an API user and associate it with an API role to grant access permissions for the Marketo REST APIs. You also need to get the client ID, client secret, and REST API URL from your Marketo account to authenticate the Marketo custom service.
For more information on how to create an API user and generate the client ID and client secret, see Custom Services in the Marketo documentation.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. **Note:** For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Client ID	The client ID of the Marketo service. You need the client ID to generate an access token to authenticate access to the Marketo service.
Client Secret	The client secret of the Marketo service.
REST API URL	Enter the URL in the following format: https://<Host name of the Marketo Rest API Server>

Advanced settings

The following table describes the advanced connection properties:

Property	Description
grant_type	Enter client_credentials as the grant type. If you don't specify the grant type, an error occurs and the connection fails.
Bypass Proxy	**Note:** This property doesn't apply to connections configured for application ingestion and replication tasks.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux.

You can use the unauthenticated or authenticated proxy server. You can configure proxy for connections used both in mappings and in mappings in advanced mode.

To configure the proxy settings for the Secure Agent, use one of the following methods:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

To bypass the proxy server settings defined for the Secure Agent, select Bypass Proxy in the advanced settings for the connection.

Microsoft Azure Blob Storage V3 connection properties

Create a Microsoft Azure Blob Storage V3 connection to securely read data from or write data to Microsoft Azure Blob Storage.

Prepare for authentication

You can use shared key authentication or shared access signature authentication in the Microsoft Azure Blob Storage V3 connection to connect to Microsoft Azure Blob Storage.

Before you configure authentication, create a storage account to use with Microsoft Azure Blob Storage and create a blob container in the storage account. For more information on how to create a storage account and a blob container, see the Prerequisites to create a Microsoft Azure Blob Storage V3 connection Informatica How-To Library article.

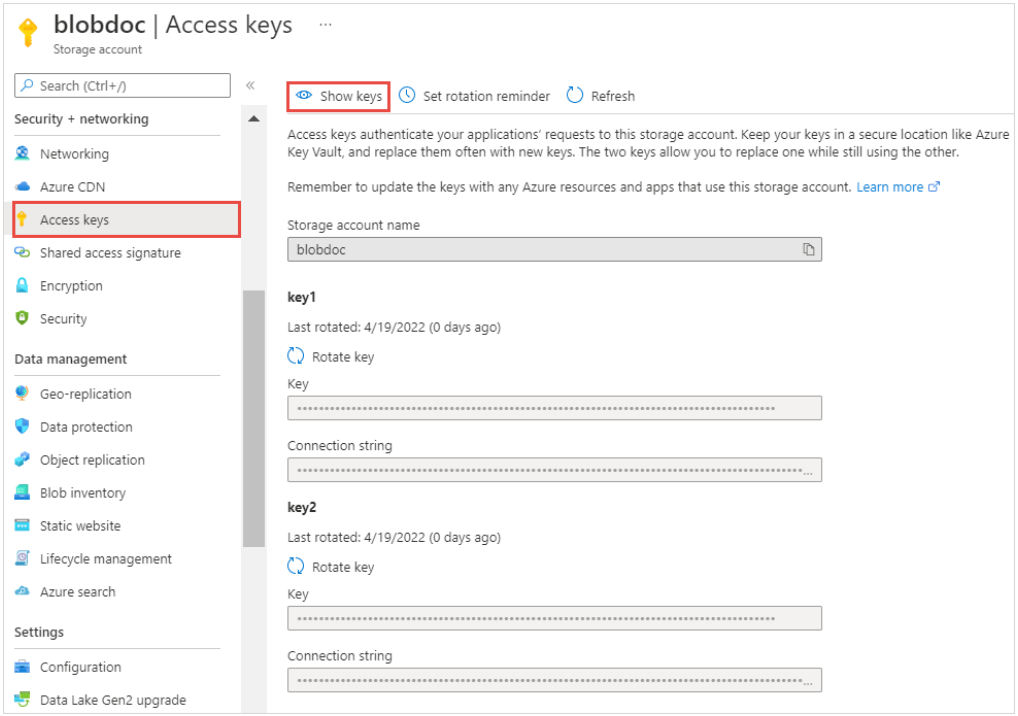
Before you configure the connection properties, you also need to keep the authentication details handy based on the authentication type that you want to use.

Shared key authentication

To connect to Microsoft Azure Blob Storage using shared key authentication, you need the storage account name and account key.

1. Open the storage account.
2. Under **Security + Networking**, click **Access keys**.

3. Click **Show keys**.



4. Make a note of the storage account name and account key. You can use key1 or key2.

Shared access signature

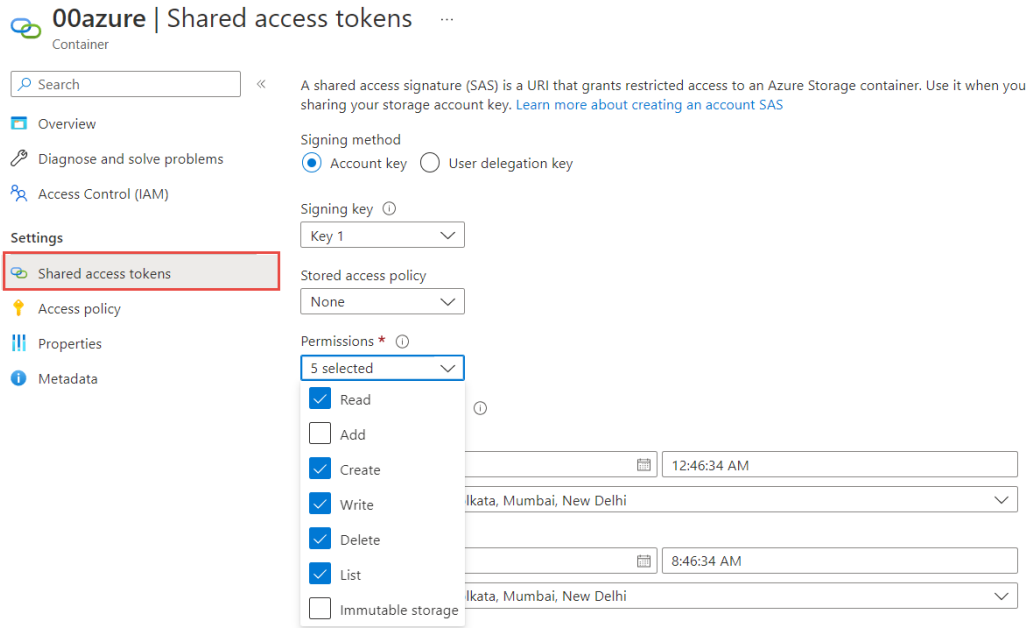
To connect to Microsoft Azure Blob Storage using shared access signature, you need to configure the minimum permissions for shared access signature authentication and generate the SAS token in the Azure portal.

You can generate the SAS token for the storage account or for the container.

- To generate the SAS token for the storage account, on the Azure portal, go to **Security + Networking**, and click **Shared access signature**.
  Select the minimum permissions required for shared access signature authentication, as shown in the following image:



- To generate the SAS token for the Blob container, go to **Settings** of the container, and click **Shared access tokens**. You can use either the Account key or User delegation key signing method. If you use the User delegation key signing method, ensure that you have the **Storage Blob Data Owner** role for the container or the storage account.
  Select the minimum permissions required for shared access signature authentication, as shown in the following image:



Connect to Microsoft Azure Blob Storage V3

Let's configure the Microsoft Azure Blob Storage V3 connection properties to connect to Microsoft Azure Blob Storage.

Before you begin

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Account Name	Microsoft Azure Blob Storage account name.

Authentication types

You can configure shared key authentication and shared access signature authentication types to access Microsoft Azure Blob Storage.

Select the required authentication method and then configure the authentication-specific parameters.

Shared key authentication

Shared key authentication uses the storage account name and account key to connect to Microsoft Azure Blob Storage.

The following table describes the connection properties for shared key authentication:

Property	Description
Account Key	The account key for the Microsoft Azure Blob Storage account.
Container Name	The name of the blob container in the Microsoft Azure Blob Storage account.
Endpoint Suffix	Types of Microsoft Azure endpoints. Select one of the following options: - core.windows.net. Connects to Azure endpoints. - core.usgovcloudapi.net. Connects to Azure Government endpoints. - core.chinacloudapi.cn. Not applicable. Default is core.windows.net.

Shared access signature authentication

Shared access signature authentication uses the SAS token to connect to Microsoft Azure Blob Storage. Use the SAS token to grant access to the resources in the storage account or container for a specific time range without sharing the account key.

Note: The file ingestion task fails if this option is on a container level and if you use a different container.

The following table describes the connection properties for shared access signature authentication:

Property	Description
SAS Token	The shared access signature token generated in the Azure portal to authenticate successfully and gain access to the Microsoft Azure Blob Storage resources.
Container Name	The name of the blob container in the Microsoft Azure Blob Storage account.
Endpoint Suffix	Types of Microsoft Azure endpoints. Select one of the following options: - core.windows.net. Connects to Azure endpoints. - core.usgovcloudapi.net. Connects to Azure Government endpoints. - core.chinacloudapi.cn. Not applicable. Default is core.windows.net.

Proxy Server Settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

To configure proxy settings for the Secure Agent, use one of the following methods:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure proxy server through the JVM options. To do this, perform the following steps:
  1. Log in to IDMC.
  2. Open Administrator and select **Runtime Environments**.
  3. Select the Secure Agent for which you want to configure the proxy server.
  4. On the upper-right corner of the page, click **Edit**.
  5. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Service.
  6. Add the following parameters in any **JVMOption** field and specify appropriate values for each parameter:

Parameter	Description
-DproxyEnabled=	Required. Set the value to true to enable proxy server.
-Dhttp.proxyHost=	Required. Host name of the outgoing HTTP proxy server.
-Dhttp.proxyPort=	Required. Port number of the outgoing HTTP proxy server.

Example for HTTP:

```
JVMOption1=-DproxyEnabled=true
JVMOption2=-Dhttp.proxyHost=<proxy_server_hostname>
JVMOption3=-Dhttp.proxyPort=8081
```

7. Click **Save**.
   The Secure Agent restarts to apply the settings.

Microsoft Azure Data Lake Storage Gen2 connection properties

Create a Microsoft Azure Data Lake Storage Gen2 connection to securely read data from or write to Microsoft Azure Data Lake Storage Gen2.

Prepare for authentication

You can configure Shared Key, Managed Identity, and Service Principal authentication types to access Microsoft Azure Data Lake Storage Gen2. Before you configure the authentication, you need to set up your environment and keep the authentication details handy.

Create storage account and configure access

To access Microsoft Azure Data Lake Storage Gen2, follow these steps to set up your environment:

1. Set up a storage account to use with Microsoft Azure Data Lake Storage Gen2 and create a blob container in the storage account. You can use role-based access control or access control lists to authorize users to access the resources in the storage account.
2. Register the application in Azure Active Directory to authenticate users to access the Microsoft Azure Data Lake Storage Gen2 account. You can use role-based access control or access control lists to authorize the application.
3. Create an Azure Active Directory web application for service-to-service authentication with Microsoft Azure Data Lake Storage Gen2. Ensure that you have superuser privileges to access the folders or files created in the application.
   For more information about these prerequisite tasks, see the Informatica How-To Library article, Prerequisites to create a Microsoft Azure Data Lake Storage Gen2 connection.

Get authentication details

Ensure you get all the required authentication details based on the authentication method you want to use in the connection:

Service principal authentication

You need the client ID, client secret, and tenant ID for your application registered in the Azure Active Directory.

Shared key authentication

You need the account key for the Microsoft Azure Data Lake Storage Gen2 account.

Managed identity authentication

You need the client ID or application ID for your application registered in the Azure Active Directory. Before you get the client ID or application ID, be sure to complete certain prerequisites.

Managed identity authentication

Managed Identity authentication uses managed identities in Azure Active Directory to authenticate and authorize access to Azure resources securely.

Before you use managed identity authentication to connect to Microsoft Azure Data Lake Storage Gen2, be sure to complete certain prerequisites.

1. Create an Azure virtual machine.
   To configure managed identity authentication in a Microsoft Azure Data Lake Storage Gen2 connection, select the Azure virtual machine on which you have installed the Secure Agent.
2. Install the Secure Agent on the Azure virtual machine.
3. Enable system assigned identity or user assigned identity for the Azure virtual machine.
   If you enable system assigned identity, assign the required role or permissions to the Azure virtual machine to run mappings and tasks. If you enable user assigned identity, assign the required role or permissions to the user assigned identity. For example, if you use role-based access control, assign the Storage Blob Data Contributor role and if you use access control lists, assign the read, write, and execute permissions. If you enable both and do not specify the client ID, the system assigned identity is used for authentication.
4. After you add or remove a managed identity, restart the Azure virtual machine.

Connect to Microsoft Azure Data Lake Storage Gen2

Let's configure the Microsoft Azure Data Lake Storage Gen2 connection properties to connect to Microsoft Azure Data Lake Storage Gen2.

Before you begin

Before you get started, you'll need to get information from your Microsoft Azure Data Lake Storage Gen2 account based on the authentication type that you want to configure.

Check out "Prepare for authentication" on page 120 to learn more about the authentication prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use the Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. For file ingestion and replication tasks and streaming ingestion and replication tasks, you can use only the Secure Agent. You cannot run tasks on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Account Name	Microsoft Azure Data Lake Storage Gen2 account name or the service name.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.
Directory Path	The path of a directory without the file system name. You can select from the following directory structures: - `/` for root directory - `/dir1` - `dir1/dir2` Default is `/`.

Authentication types

You can select service principal authentication, shared key authentication, and managed identity authentication to access the Microsoft Azure Data Lake Storage Gen2 account.

Note: Data Ingestion and Replication supports managed identity authentication. However, Streaming Ingestion and Replication does not support shared key authentication or managed identity authentication.

Select your preferred authentication type and then configure the authentication-specific parameters.

Service principal authentication

Service principal authentication uses the client ID, client secret, and tenant ID to connect to Microsoft Azure Data Lake Storage Gen2.

The following table describes the basic connection properties for service principal authentication:

Property	Description
Client ID	The client ID of your application. Specify the client ID for your application registered in the Azure Active Directory.
Client Secret	The client secret key generated for the client ID. Specify the client secret key to complete the OAuth authentication in the Azure Active Directory.
Tenant ID	The directory ID of the Azure Active Directory.
Endpoint Suffix	The type of Microsoft Azure endpoints. Select one of the following endpoints: - core.windows.net. Connects to Azure endpoints. - core.usgovcloudapi.net. Connects to US government Microsoft Azure Data Lake storage Gen2 endpoints. - core.chinacloudapi.cn. Connects to Microsoft Azure Data Lake storage Gen2 endpoints in the China region. Default is core.windows.net.

Shared key authentication

Shared key authentication uses the account key to connect to Microsoft Azure Data Lake Storage Gen2.

The following table describes the basic connection properties for shared key authentication:

Property	Description
Account Key	The account key for the Microsoft Azure Data Lake Storage Gen2 account.
Endpoint Suffix	The type of Microsoft Azure endpoints. Select one of the following endpoints: - core.windows.net. Connects to Azure endpoints. - core.usgovcloudapi.net. Connects to US government Microsoft Azure Data Lake storage Gen2 endpoints. - core.chinacloudapi.cn. Connects to Microsoft Azure Data Lake storage Gen2 endpoints in the China region. Default is core.windows.net.

Managed identity authentication

Managed identity authentication authenticates using identities that are assigned to applications in Azure to access Azure resources in Microsoft Azure Data Lake Storage Gen2.

The following table describes the basic connection properties for managed identity authentication:

Property	Description
Client ID	The client ID of your application. To use managed identity authentication, specify the client ID for the user-assigned managed identity. Leave the field blank in the following scenarios: - If the permission is provided by system-assigned managed identity. - If there is no system-assigned identity but only a single user-assigned managed identity.
Endpoint Suffix	The type of Microsoft Azure endpoints. Select one of the following endpoints: - core.windows.net. Connects to Azure endpoints. - core.usgovcloudapi.net. Connects to US government Microsoft Azure Data Lake storage Gen2 endpoints. - core.chinacloudapi.cn. Connects to Microsoft Azure Data Lake storage Gen2 endpoints in the China region. Default is core.windows.net.

Proxy Server Settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

Note: You cannot use a proxy server with managed identity authentication.

You can use one of the following types of proxy servers:

- Unauthenticated proxy - Requires only the host and port address for configuration.
- Authenticated proxy - Requires the host address, port address, user name, and password for configuration.

To configure proxy settings for the Secure Agent, use one of the following methods:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

Bypass the proxy server

You can bypass the proxy server settings configured for the Secure Agent.

Perform the following steps to bypass the proxy server:

1. Navigate to the following directory:
   `<Secure Agent installation directory>/apps/agentcore`
2. Enter the following command in the `proxy.ini` file:
   `InfaAgent.NonProxyHost=localhost|{*}core.windows.net|127.|[\:\:1]*`
   - To bypass proxy server for service principal authentication, append `login.microsoftonline.com` to the command.

- To bypass proxy server for managed identity authentication, append `169.254.169.254` to the command.

  For example, `InfaAgent.NonProxyHost=localhost|127.*|[\:\:1]|<accountname>.blob.core.windows.net|<accountname>.dfs.core.windows.net|<accountname>.blob.core.windows.net|login.microsoftonline.com|169.254.169.254`

3. Restart the Secure Agent.

Microsoft Azure Event Hub connection properties

When you set up an Azure Event Hub connection, you must configure the connection properties.

The following table describes the Azure Event Hub connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive. It must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. Description that you use to identity the connection. The description cannot exceed 4,000 characters.
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Tenant ID	The ID of the tenant that the data belongs to. This ID is the Directory ID of the Azure Active Directory.
Subscription ID	The ID of the Azure subscription.
Resource Group Name	The name of the Azure resource group associated with the Event Hub namespace.
Client Application ID	The ID of the application created under the Azure Active Directory.
Client Secret Key	The secret key generated for the application.
Event Hub Namespace	The name of the Event Hub namespace that is associated with the resource group name.
Shared Access Policy Name	Optional. The name of the Event Hub Namespace Shared Access Policy. The policy must apply to all data objects that are associated with this connection. To read from Event Hubs, you must have Listen permission. To write to an Event Hub, the policy must have Send permission.
Shared Access Policy Primary Key	Optional. The primary key of the Event Hub Namespace Shared Access Policy.

Microsoft Azure Synapse Analytics Database Ingestion connection properties

When you define a Microsoft Azure Synapse Analytics Database Ingestion connection, you must configure connection properties. You can use this connection type in application ingestion and replication tasks and database ingestion and replication tasks, which you configure in the Data Ingestion and Replication service.

Note: Some properties are for Microsoft Azure Data Lake Storage Gen2. Application Ingestion and Replication and Database Ingestion and Replication use Microsoft Azure Data Lake Storage Gen2 to stage data in files before sending the data to the Microsoft Azure Synapse Analytics target tables.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	The name of the runtime environment where you want to run the application ingestion and replication tasks or database ingestion and replication tasks. You define runtime environments in Administrator. For application ingestion and replication tasks and database ingestion and replication tasks that have a supported source type, you can use the serverless runtime environment. You cannot run application ingestion and replication and database ingestion and replication tasks on a Hosted Agent.

Azure Synapse Analytics JDBC URL	The Microsoft Azure Synapse Analytics (formerly SQL Data Warehouse) JDBC connection string.
	Example connection string for Microsoft SQL Server authentication:
	`jdbc:sqlserver://server.database.windows.net:1433;database=database`
	Example connection string for Azure Active Directory (AAD) authentication:
	`jdbc:sqlserver://server.database.windows.net:1433;database=database;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;Authentication=ActiveDirectoryPassword;`
	Note: The default authentication type is Microsoft SQL Server authentication.

Azure Synapse Analytics JDBC Username	The user name to use for connecting to the Microsoft Azure Synapse Analytics account. Provide the AAD user name for AAD authentication.
Azure Synapse Analytics JDBC Password	The password to use for connecting to the Microsoft Azure Synapse Analytics account.
Azure Synapse Analytics Schema Name	The name of the schema in the Microsoft Azure Synapse Analytics target.

Property	Description
ADLS Gen2 Account Name	The name of the Microsoft Azure Data Lake Storage Gen2 account.
Client Id	The ID of your client application for completing the OAuth Authentication in the Active Directory.
Client Secret	The client secret key for completing the OAuth Authentication in the Active Directory.
Directory	The Microsoft Azure Data Lake Storage Gen2 directory that Application Ingestion and Replication and Database Ingestion and Replication use to stage data in files. The default is the root directory.
Filesystem Name	The name of an existing file system in the Microsoft Azure Data Lake Storage Gen2 account.
Tenant ID	The Directory ID of the Azure Active Directory.

Microsoft Azure Synapse SQL connection properties

Create a Microsoft Azure Synapse SQL connection to securely read data from or write data to Microsoft Azure Synapse SQL.

Prerequisites

You can configure Microsoft SQL Server, Azure Active Directory, Managed Identity, and Service Principal authentication types to access Microsoft Azure Synapse SQL.

Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

Azure Active Directory authentication

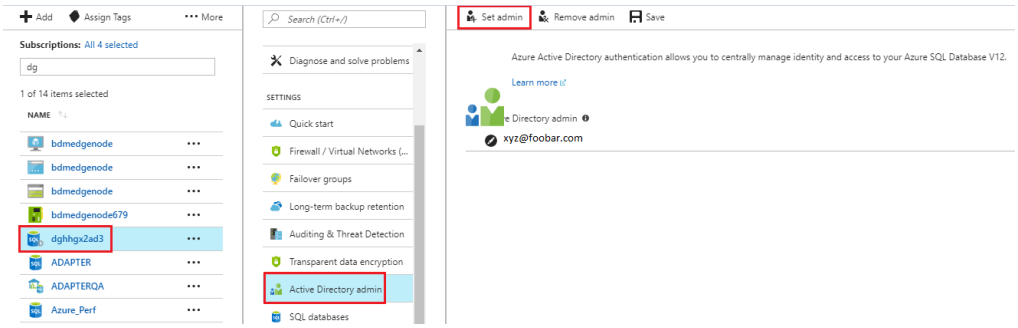
To connect to Microsoft Azure Synapse SQL using Azure Active Directory (AAD) authentication, you need to create an Azure Active Directory administrator and an Azure Active Directory user.

Create an Azure Active Directory administrator

To add new users to your Azure Active Directory, you must have an administrator role.

To set up an Azure Active Directory administrator for AAD and Microsoft SQL Server that hosts your Microsoft Azure Synapse SQL, perform the following steps:

1. Log on to the Microsoft Azure portal using your credentials.
   The Dashboard page appears.
2. From the All Resources page, select the Microsoft SQL Server that hosts Microsoft Azure Synapse SQL.
3. Under Settings displayed for Microsoft SQL Server, select the **Active Directory admin** option.
   The image shows the Active Directory admin settings:



4. Click **Set admin**.
   The Add admin page appears.
5. Enter the email ID that you want to use as admin, and then click **Select**.
6. Click **Save**.

Create an Azure Active Directory user

Create an AAD user and use the AAD user credentials when you configure a Microsoft Azure Synapse SQL connection with AAD authentication.
Perform the following steps to create an AAD user:

1. Connect to Microsoft Azure Synapse SQL using the Azure Active Directory administrator created in the previous steps.
   You can use Microsoft SQL Server Management Studio to connect to the Microsoft Azure Synapse SQL.
2. In a new query window in Microsoft SQL Server Management Studio, run the following command to create an AAD user:
   ```
   create user [user@foobar.com] from external provider;
   ```
3. Assign the following privileges to the user:
   ```
   CREATE USER [username] FROM EXTERNAL PROVIDER;

   ALTER ROLE db_datareader ADD MEMBER [username]

   ALTER ROLE db_datawriter ADD MEMBER [username]

   GRANT EXECUTE TO [username]

   grant ALTER ANY EXTERNAL DATA SOURCE to [username];

   grant create table to [username];

   grant create schema to [username];

   grant select to [username];

   grant update to [username];

   grant insert to [username];

   grant delete to [username];

   grant create view to [username];

   grant select on schema :: sys to [username];

   grant control to [username];

   EXEC sp_addrolemember 'db_owner','[username]';

   ALTER ROLE db_owner ADD MEMBER [username]
   ```

Service principal authentication

Service Principal authentication involves the use of a service principal identity to authenticate and authorize access to Azure resources. Before you use service principal authentication to connect to Microsoft Azure Synapse SQL, be sure to complete certain prerequisites.

1. Register a service principal application.
2. Configure a service principal user for a dedicated SQL pool.

For more information, see Prerequisites to use service principal authentication Informatica How-To Library article.

Managed Identity authentication

Managed Identity authentication uses managed identities in Azure Active Directory to authenticate and authorize access to Azure resources securely.

When you use managed identity authentication to connect to Microsoft Azure Synapse SQL, the user for the system assigned identity is the virtual machine for which you enable the identity. The user for the user assigned identity is the user identity that you create in the Azure portal.

Before you use managed identity authentication to connect to Microsoft Azure Synapse SQL or Microsoft Azure Data Lake Storage Gen2, be sure to complete certain prerequisites.

1. Create an Azure virtual machine.
2. Install the Secure Agent on the Azure virtual machine.
3. Enable system assigned identity or user assigned identity for the Azure virtual machine.
   If you enable both and do not specify the client ID, the system assigned identity is used for authentication.
4. After you add or remove a managed identity, restart the Azure virtual machine.

Permissions

Permissions define the level of access for the operations that you can perform in Microsoft Azure Synapse SQL.

You must verify the following permissions:

- Ensure that a default schema is present at the account level or user or group level in Microsoft Azure SQL Data Warehouse.
- Verify that either the `db_owner` privilege or the following more granular privileges are granted to the user to connect to Microsoft Azure SQL Data Warehouse and perform operations successfully:

  - `EXEC sp_addrolemember 'db_datareader', '<user>';` // Alternately assign permission to the individual table.
  - `EXEC sp_addrolemember 'db_datawriter', '<user>';` // Alternately assign permission to the individual table.
  - `GRANT ALTER ANY EXTERNAL DATA SOURCE TO <user>;`
  - `GRANT ALTER ANY EXTERNAL FILE FORMAT TO <user>;`
  - `GRANT CONTROL TO <user>;` // To grant all permissions on the database.
    or
    `GRANT ALTER ANY SCHEMA TO <user>;` // To grant permissions only on the schema.
  - `GRANT CREATE TABLE TO <user>;`

- Assign required privileges for tasks performed through Pre-SQL and Post-SQL commands.

- If you configure the staging schema name in the connection properties, ensure the following additional privileges are granted to the user:

  - `ALTER ROLE db_datareader ADD MEMBER <user>;`
  - `GRANT ALTER ANY EXTERNAL DATA SOURCE TO <user>;`
  - `GRANT ALTER ANY EXTERNAL FILE FORMAT TO <user>;`
  - `GRANT CREATE TABLE TO <user>;`
  - `GRANT ALTER ON SCHEMA::<staging_schema_name> TO <user>;`// Ensures this user does not have access to create or drop an external table in another schema.
  - `GRANT REFERENCES ON DATABASE SCOPED CREDENTIAL::<db_credential_name> TO <user>;`// Ensures this user does not have access to drop the credentials.
    For example, `GRANT REFERENCES ON DATABASE SCOPED CREDENTIAL::db_creds1 TO srvls;`

- If you have the ALTER ANY SCHEMA permissions, you must create the Master Key, Database Scoped Credential, and External Data Source in Microsoft Azure Synapse SQL that require the CONTROL permission on the database and specify the external data source when you create a connection.
  Also, Microsoft Azure Synapse SQL Connector does not delete the Database Scoped Credential and External Data Source. You must manually delete the Database Scoped Credential and External Data Source.

- When you use managed identity authentication to connect to Microsoft Azure Synapse SQL, grant permissions to the virtual machine and user identity. For example, `GRANT CONTROL TO <virtual machine name>;` and `GRANT CONTROL TO <user identity name>;`

Connect to Microsoft Azure Synapse SQL

Let's configure the Microsoft Azure Synapse SQL connection properties to connect to Microsoft Azure Synapse SQL.

Before you begin

Before you get started, you'll need to get information from your Microsoft Azure Synapse SQL account based on the authentication type that you want to configure.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For file ingestion and replication tasks, you can use the Hosted Agent but not serverless environments. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Property	Description

Azure DW JDBC URL	The Microsoft Azure Synapse SQL JDBC connection string.
	Use the following string to connect to Microsoft Azure Synapse SQL:
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>`
	You can include an authentication parameter in the connection string to specify the authentication type. You can configure the following authentication types to connect to Microsoft Azure Synapse SQL:
	- Microsoft SQL Server
	- Azure Active Directory
	- Managed Identity
	- Service Principal
	If you don't include an authentication parameter in the connection string, the Secure Agent uses Microsoft SQL Server authentication as the authentication type.
	Connection string format for Microsoft SQL Server authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>`
	Connection string format for Azure Active Directory (AAD) authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;Authentication=ActiveDirectoryPassword;`
	Connection string format for Service Principal authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30; Authentication=ActiveDirectoryServicePrincipal;`
	Connection string format for Managed Identity authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;Authentication=ActiveDirectoryMsi;`

Property	Description
Azure DW JDBC Username	User name to connect to the Microsoft Azure Synapse SQL account. - For AAD authentication, provide your AAD user name. - For Microsoft SQL server authentication, provide your SQL auth user name. - For service principal authentication, provide the application ID or client ID for your application registered in Azure Active Directory. This property doesn't apply to Managed Identity authentication.
Azure DW JDBC Password	Password to connect to the Microsoft Azure Synapse SQL account. - For AAD authentication, provide the password of the AAD user. - For Microsoft SQL server authentication, provide the password of SQL auth user. - For service principal authentication, provide the client secret for your application registered in the Azure Active Directory. This property doesn't apply to Managed Identity authentication.
Azure DW Client ID	Required if you want to use the user-assigned managed identity for Managed Identity Authentication to connect to Microsoft Azure Synapse SQL. The client ID of the user-assigned managed identity. If you use system-assigned managed identity, leave the field empty.
Azure DW Schema Name	Name of the schema in Microsoft Azure Synapse SQL.

Azure storage types

You can select Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 as the Azure storage type to stage the data files. Default is Azure Blob.

Select your preferred storage type and then configure the storage-specific parameters.

To get credentials for shared key authentication when you connect to Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 to stage files, see Get credentials for shared key authentication How-To Library article.

To get credentials for service principal authentication when you connect to Microsoft Azure Data Lake Storage Gen2 to stage files, see Get credentials for service principal authentication How-To Library article.

Azure Blob storage

When you select Microsoft Azure Blob as the storage type, you can configure Shared Key Authentication as the authentication type to stage the files.

The following table describes the authentication type that you can configure for Microsoft Azure Blob storage:

Property	Description
Authentication Type	Authentication type to connect to Microsoft Azure Blob storage to stage the files. You can configure Shared Key Authentication as the authentication type to stage the files.

Shared Key Authentication

Uses the storage account name and account key to connect to Microsoft Azure Blob storage.

The following table describes the basic connection properties for shared key authentication:

Property	Description
Azure Blob Account Name	Name of the Microsoft Azure Blob Storage account to stage the files.
Azure Blob Account Key	The Microsoft Azure Blob Storage access key to stage the files.
Container Name	The name of the container in the Azure Blob Storage account.

ADLS Gen2 storage

When you select Microsoft Azure Data Lake Storage Gen2 as the storage type, you can configure various authentication types to stage the files.

The following table describes authentication types that you can configure for Microsoft Azure Data Lake Storage Gen2 storage:

Property	Description
Authentication Type	Authentication type to connect to Azure storage to stage the files. Select one of the following options: - Shared Key Authentication - Service Principal Authentication - Managed Identity Authentication For more information on how to configure the authentication types, see Setting up authentication to connect to Microsoft Azure Synapse SQL.

Shared Key Authentication

Uses the storage account name and account key to connect to Microsoft Azure Data Lake Storage Gen2.

The following table describes the basic connection properties for shared key authentication:

Property	Description
ADLS Gen2 Storage Account Name	Name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
ADLS Gen2 Account Key	The Microsoft Azure Data Lake Storage Gen2 access key to stage the files.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.

Service Principal Authentication

Uses the account name, client ID, client secret, and tenant ID to connect to Microsoft Azure Data Lake Storage Gen2.
The following table describes the basic connection properties for service principal authentication:

Property	Description
ADLS Gen2 Storage Account Name	Name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
Client ID	The client ID of your application. Enter the application ID or client ID for your application registered in the Azure Active Directory.
Client Secret	The client secret for your application.
Tenant ID	The directory ID or tenant ID for your application.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.

Managed Identity Authentication

Select this authentication type to authenticate using system-assigned or user-assigned identities that are assigned to applications in Azure to access Azure resources in Microsoft Azure Data Lake Storage Gen2.

The following table describes the basic connection properties for managed identity authentication:

Property	Description
ADLS Gen2 Storage Account Name	Name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
Client ID	The client ID of your application. Enter the client ID for the user-assigned managed identity. If the managed identity is system-assigned, leave the field empty.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.

In a file ingestion and replication task, if you select Microsoft Azure Synapse SQL with Managed Identity authentication type as the target, then you must select Microsoft Azure Data Lake Storage Gen2 as the source.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
External Data Source	The data source to create the external table. Ensure that the external data source exists in Microsoft Azure Synapse SQL and you have the permission to access the external data source. When you use the copy command method to load data from the staging location to Microsoft Azure Synapse SQL, you don't need to specify an external data source.
Staging Schema Name	The name of the schema that the Secure Agent uses to create external tables for staging data files. If you do not specify the staging schema name, the Secure Agent considers the configured Azure DW Schema Name.
Blob End-point	Type of Microsoft Azure endpoint. Select one of the following endpoints: - core.windows.net. Connects to Azure endpoints. - core.usgovcloudapi.net. Connects to US Government Microsoft Azure Synapse SQL endpoints. - core.chinacloudapi.cn. Connects to Microsoft Azure Synapse SQL endpoints in the China region. Default is core.windows.net.
VNet Rule	Enable to connect to a Microsoft Azure Synapse SQL endpoint residing in a virtual network (VNet). This property doesn't apply to a serverless runtime environment.

Microsoft Dynamics 365 Mass Ingestion connection properties

When you set up a Microsoft Dynamics 365 Mass Ingestion connection, you must configure the connection properties.

The Microsoft Dynamics 365 Mass Ingestion connection requires a native application that is registered in Azure Active Directory (Azure AD) to access the Microsoft Dynamics 365 data. Before you configure the connection, you must register an application in Azure AD to allow the connection to access the Microsoft Dynamics 365 data. For more information about registering an application in Azure AD, see the Microsoft documentation.

The properties of a Microsoft Dynamics 365 Mass Ingestion connection vary based on the authentication method that you specify for the connection. When you create a connection, you can select one of the following authentication methods:

- **OAuth 2.0 Username-Password Flow**: Authenticates the connection by using the Microsoft Dynamics 365 account login credentials and the client ID of the application registered in Azure AD.
- **OAuth 2.0 Client Secret Flow**: Authenticates the connection by using the client ID and client secret of the application registered in Azure AD.
- **OAuth 2.0 JWT Bearer Flow**: Authenticates the connection by using a X509 Public Key Infrastructure (PKI) certificate and a JSON Web Token (JWT). Use this authentication method to gain secured access to Microsoft Dynamics 365 without sharing sensitive information, such as client secret and Microsoft Dynamics 365 account login credentials.

Connection properties for OAuth 2.0 Username-Password Flow authentication

The following table describes the connection properties for a Microsoft Dynamics 365 Mass Ingestion connection configured with OAuth 2.0 Username-Password Flow authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
User Name	User name of the Microsoft Dynamics 365 account.
Password	Password for the Microsoft Dynamics 365 account.
Client ID	Client ID of the application registered in Azure AD.
Resource URL	URL of the Microsoft Dynamics 365 organization. You must enter the resource URL in the following format: `https://<Microsoft_Dynamics_365_org_name>.api.crm8.dynamics.com`
OAuth Token URL	OAuth 2.0 token endpoint of the Microsoft Dynamics 365 organization. The application that is registered in Azure AD sends the access token requests to this endpoint. You must enter the following value in this field: `https://login.windows.net/common/oauth2/token`

Note: For more information about the OAuth 2.0 Username-Password Flow authentication method, see the Microsoft Dynamics 365 documentation.

Connection properties for OAuth 2.0 Client Secret Flow authentication

The following table describes the connection properties for a Microsoft Dynamics 365 Mass Ingestion connection configured with OAuth 2.0 Client Secret Flow authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.

Connection property	Description
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
Client ID	Client ID of the application registered in Azure AD.
Client Secret	Client secret of the application registered in Azure AD.
Resource URL	URL of the Microsoft Dynamics 365 organization. You must enter the resource URL in the following format: `https://<Microsoft_Dynamics_365_org_name>.api.crm8.dynamics.com`
OAuth Token URL	OAuth 2.0 token endpoint of the Microsoft Dynamics 365 organization. The application that is registered in Azure AD sends the access token requests to this endpoint. You must enter the following value in this field: `https://login.microsoftonline.com/<tentant_id>/oauth2/token`

Note: For more information about the OAuth 2.0 Client Secret Flow authentication method, see the Microsoft Dynamics 365 documentation.

Connection properties for OAuth 2.0 JWT Bearer Flow authentication

The following table describes the connection properties for a Microsoft Dynamics 365 Mass Ingestion connection configured with OAuth 2.0 JWT Bearer Flow authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
Client ID	Client ID of the application registered in Azure AD.
Certificate Signature	Base64URL string that encodes the hexadecimal value which represents the SHA-1 thumbprint of the X509 certificate.
Keystore Path	Absolute path to the keystore file that contains the X509 certificate required to validate a JSON Web Token (JWT) and establish a secure connection with Microsoft Dynamics 365. The keystore file must be in the Java KeyStore (JKS) format.
Keystore Password	Password for the keystore file.

Connection property	Description
Private Key Alias	Alias name of the private key used to sign the JWT.
Private Key Password	Password for the private key.
Audience for JWT	URL of the Microsoft Dynamics 365 resource server to which the application that is registered in Azure AD sends the JWT for validation. You must enter the address in the following format: `https://login.microsoftonline.com/<tentant_id>/oauth2/token`
Resource URL	URL of the Microsoft Dynamics 365 organization. You must enter the resource URL in the following format: `https://<Microsoft_Dynamics_365_org_name>.api.crm8.dynamics.com`
OAuth Token URL	OAuth 2.0 token endpoint of the Microsoft Dynamics 365 organization. The application that is registered in Azure AD sends the access token requests to this endpoint. You must enter the following value in this field: `https://login.microsoftonline.com/<tentant_id>/oauth2/token`

Note: For more information about the OAuth 2.0 Client Secret Flow authentication method, see the Microsoft Dynamics 365 documentation.

Microsoft Fabric Data Warehouse connection properties

Create a Microsoft Fabric Data Warehouse connection to securely load data to Microsoft Fabric Data Warehouse.

Prerequisites

Before you configure the connection properties, make sure to complete the following prerequisites:

1. Power BI prerequisites:
   a. Create a workspace, if it does not already exist.
   b. Create a lakehouse in the workspace. Make sure you have the required admin privileges to create a lakehouse.

   For more information about creating a workspace and lakehouse, see Creating a workspace and lakehouse.
2. Microsoft Azure portal prerequisites:
   a. Register your application in Microsoft Entra ID and generate client ID, tenant ID, and client secret. Microsoft Fabric Data Warehouse Connector does not require any API permissions assigned to your application. Additionally, you can remove any API permissions that are assigned to the application.

b. Create a security group with assigned owners or members in Microsoft Entra ID.
For more information about creating a security group, see Creating a security group.

3. To allow the Microsoft Entra app to use the Power BI APIs, enable the Power BI service admin settings, and then add the Microsoft Entra app to the workspace in Power BI.

Note: The Manage access pane in the Microsoft Entra app allows you to add roles based on the required capabilities. The pane displays admin, member, contributor, and viewer roles. The maximum number of capabilities are associated with the admin role. However, to access Microsoft Fabric OneLake and perform read and write operations, you need at least the Contributor role.

For more information about enabling access in the Power BI admin portal and adding the Microsoft Entra app to the workspace, see Enabling access in the Power BI portal.

Connect to Microsoft Fabric Data Warehouse

Let's configure the Microsoft Fabric Data Warehouse connection properties to connect to Microsoft Fabric Data Warehouse.

Before you begin

Before you get started, you'll need to keep the client ID, client secret, workspace name, database name, and tenant ID details handy.

Check out "Prerequisites" on page 139 to make sure that the setup is complete and all the prerequisites are in place before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.

Property	Description
Runtime Environment	The name of the runtime environment where you want to run tasks. You cannot run an ingestion and replication task on a Hosted Agent or in a serverless runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
SQL Connection String	The SQL connection string to connect to Microsoft Fabric Data Warehouse. Specify the connection string in the following format: `<Server>.datawarehouse.pbidedicated.windows.net`
Client ID	The application ID or client ID of your application registered in Azure Active Directory for service principal authentication.
Client Secret	The client secret for your application registered in Azure Active Directory.
Tenant ID	The tenant ID of your application registered in Azure Active Directory.
Workspace	The name of the workspace in Microsoft Fabric Data Warehouse that you want to connect.
Database	The name of the database in Microsoft Fabric Data Warehouse that you want to connect.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Schema Name	The name of the schema in Microsoft Fabric Data Warehouse where the tables are stored.

Additional Properties	Additional connection properties that you configure in key-value pairs.
	You can configure the following properties:
	- connectRetryCount. The maximum number of attempts to establish or re-establish a connection.
	You can configure an integer between 0 and 255. A value of 0 indicates that a retry won't be attempted.
	Default is 3.
	- connectRetryInterval. The time, in seconds, between connection retry attempts.
	You can configure an integer between 1 and 60. This property is ignored if connectRetryCount is set to 0.
	Default is 10 seconds.
	- loginTimeout. The number of seconds to wait for a connection to be established.
	Connection attempts continue up to the limit defined by connectRetryCount. However, if the total retry time exceeds the loginTimeout, any attempt to establish the connection stops when the loginTimeout duration is reached.
	Maximum is 65535 seconds. Default is 30 seconds.
	- WorkspaceId. The ID of the workspace that you want to connect to in Microsoft Fabric Data Warehouse.
	- LakehouseId. The ID of the lakehouse used to stage data.
	Configure this property when you enable the Copy command to stage data as Parquet files and load staging data to Microsoft Fabric Warehouse.
	Enter the property in the following format:
	`LakehouseId=<ID>`
	For example, in the following URL, the lakehouse ID is the value that follows `/lakehouse/`:
	`https://app.fabric.microsoft.com/workloads/lakehouse/<lakehouse-ID>`
	Note: Ensure that the lakehouse ID that you specify in the connection is valid.

Private links to access Microsoft Fabric Data Warehouse

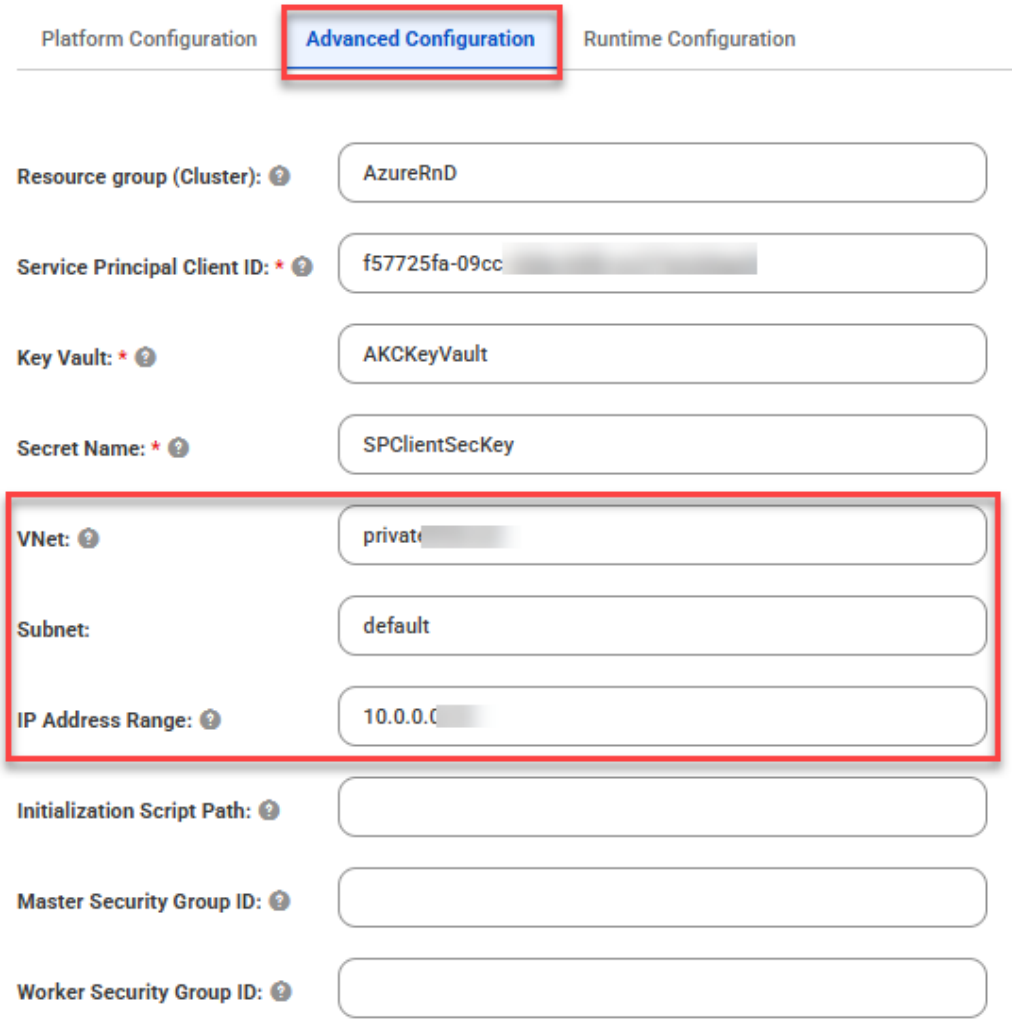
You can access Microsoft Fabric Data Warehouse using Azure Private Link endpoints.

To connect to the Microsoft Fabric Data Warehouse account over the private Azure network, see Secure connectivity to Microsoft Fabric.

To run mappings in advanced mode over the private Azure network, additionally you must perform the following steps after you complete the required prerequisites:

1. Log in to Intelligent Data Management Cloud (IDMC), and then click **Administrator**.
2. In **Administrator**, navigate to **Advanced Clusters** to create or modify an Advanced Configuration.

3. In the **Advanced Configuration** tab, add the Vnet, Subnet, and IP Address Range properties that is linked with the private DNS zone. This helps to spawn the cluster resources under this subnet and provide access to the private DNS zone.



Microsoft Fabric Lakehouse connection properties

Create a Microsoft Fabric Lakehouse connection to securely read from or write to Microsoft Fabric Lakehouse.

Prerequisites

Before you configure the connection properties, make sure to complete the following prerequisites:

1. Power BI prerequisites:
   a. Create a workspace, if it does not already exist.

b. Create a lakehouse in the workspace. Make sure you have the required admin privileges to create a lakehouse.

For more information about creating a workspace and lakehouse, see Creating a workspace and lakehouse.

2. Microsoft Azure portal prerequisites:
   a. Register your application in Microsoft Entra ID and generate client ID, tenant ID, and client secret. The connector does not require any API permissions assigned to your application. Additionally, you can remove any API permissions that are assigned to the application.
   b. Create a security group with assigned owners or members in Microsoft Entra ID.
   For more information about creating a security group, see Creating a security group.
3. To allow Microsoft Entra app to use the Power BI APIs, enable the Power BI service admin settings, and then add the Microsoft Entra app to the workspace in Power BI.

   Note: The Manage access pane in the Microsoft Entra app allows you to add roles based on the required capabilities. The pane displays admin, member, contributor, and viewer roles. The maximum number of capabilities are associated with the admin role. However, to access Microsoft Fabric OneLake and perform read and write operations, you need at least the Contributor role.

   For more information about enabling access in the Power BI admin portal and adding the Microsoft Entra app to the workspace, see Enabling access in the Power BI portal.

Connect to Microsoft Fabric Lakehouse

Let's configure the Microsoft Fabric Lakehouse connection properties to connect to Microsoft Fabric Lakehouse.

Before you begin

Before you get started, you'll need to keep the client ID, client secret, tenant ID, workspace, and database details handy.

Check out "Prerequisites" on page 143 to make sure that the setup is complete and all the prerequisites are in place before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
SQL Connection String	The SQL connection string to connect to Microsoft Fabric Lakehouse. Specify the connection string in the following format: `<Server>.lakehouse.pbidedicated.windows.net`
Client ID	The application ID or client ID of your application registered in Azure Active Directory for service principal authentication.
Client Secret	The client secret for your application registered in Azure Active Directory.
Tenant ID	The tenant ID of your application registered in Azure Active Directory.
Workspace	The name of the workspace in Microsoft Fabric Lakehouse that you want to connect.
Database	The name of the database in Microsoft Fabric Lakehouse that you want to connect.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Lakehouse schemas	Indicates if the schemas are enabled for Microsoft Fabric Lakehouse.
Schema Name	The name of the schema in Microsoft Fabric Data Lakehouse where the tables are stored.

Microsoft Fabric OneLake connection properties

Create a Microsoft Fabric OneLake connection to securely read from or write to Microsoft Fabric OneLake.

Prerequisites

Before you configure the connection properties, make sure to complete the following prerequisites:

1. Power BI prerequisites:
   a. Create a workspace, if it does not already exist.
   b. Create a lakehouse in the workspace. Make sure you have the required admin privileges to create a lakehouse.

   For more information about creating a workspace and lakehouse, see Creating a workspace and lakehouse.
2. Microsoft Azure portal prerequisites:
   a. Register your application in Microsoft Entra ID and generate client ID, tenant ID, and client secret. The connector does not require any API permissions assigned to your application. Additionally, you can remove any API permissions that are assigned to the application.
   b. Create a security group with assigned owners or members in Microsoft Entra ID.
   For more information about creating a security group, see Creating a security group.
3. To allow Microsoft Entra app to use the Power BI APIs, enable the Power BI service admin settings, and then add the Microsoft Entra app to the workspace in Power BI.

   Note: The Manage access pane in the Microsoft Entra app allows you to add roles based on the required capabilities. The pane displays admin, member, contributor, and viewer roles. The maximum number of capabilities are associated with the admin role. However, to access Microsoft Fabric OneLake and perform read and write operations, you need at least the Contributor role.

   For more information about enabling access in the Power BI admin portal and adding the Microsoft Entra app to the workspace, see Enabling access in the Power BI portal.

Connect to Microsoft Fabric OneLake

Let's configure the Microsoft Fabric OneLake connection properties to connect to Microsoft Fabric OneLake.

Before you begin

Before you get started, you'll need to keep the workspace name, lakehouse path, client ID, client secret, and tenant ID details handy.

Check out "Prerequisites" on page 146 to make sure that the setup is complete and all the prerequisites are in place before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run the tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use the Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. For file ingestion and replication tasks, you can use only the Secure Agent. You cannot run tasks on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Workspace Name	Name of the workspace in Microsoft Fabric OneLake. Workspace name cannot contain special characters or spaces.
Lakehouse Path	Path or name of the lakehouse present in the workspace. You can specify the path in one of the following ways: - *root directory (/)* to access the files in the workspace. - *lakehouse name/Files* to access the files present in the lakehouse.
Authentication Type	Authentication type to access Microsoft Fabric OneLake. Service Principal Authentication uses the client ID, client secret, and tenant ID to connect to Microsoft Fabric OneLake.
Client ID	The application ID or client ID of your application registered in the Azure Active Directory.
Client Secret	The client secret of your application registered in the Azure Active Directory.
Tenant ID	The ID of the Azure Active Directory instance in which you created the application.
Microsoft Fabric OneLake Endpoint	The type of Microsoft Fabric OneLake endpoint that you want to connect to. Default is **fabric.microsoft.com**.

Note: You can access Microsoft Fabric OneLake using Azure Private Link endpoints for connections used in file ingestion and replication tasks.

Proxy Server Settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux. You can use the unauthenticated or authenticated proxy server.

You can use one of the following types of proxy servers:

- Unauthenticated proxy - Requires only the host and port address for configuration.
- Authenticated proxy - Requires the host address, port address, user name, and password for configuration.

To configure proxy settings for the Secure Agent, use one of the following methods:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.
- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.

Bypass the proxy server

You can bypass the proxy server settings configured for the Secure Agent.

Perform the following steps to bypass the proxy server:

1. Navigate to the following directory:
   `<Secure Agent installation directory>/apps/agentcore`
2. Specify the following command in the `proxy.ini` file:
   `InfaAgent.NonProxyHost=localhost|{*}onelake.blob.fabric.microsoft.com|onelake.dfs.fabric.microsoft.com |127.|[\:\:1]*`
3. Restart the Secure Agent.

Microsoft SQL Server connection properties

Create a Microsoft SQL Server connection to read from or write data to Microsoft SQL Server.

Prepare for authentication

You can configure Database or Kerberos authentication method to connect to Microsoft SQL Server. Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use. For Kerberos authentication, you need to configure certain prerequisites.

Note: Data Ingestion and Replication does not support Kerberos authentication.

Connect to Microsoft SQL Server

Let's configure the Microsoft SQL Server connection properties to connect to Microsoft SQL Server databases.

Before you begin

Before you get started, you'll need to get information from your SQL Server DB account based on the authentication method and the type of SQL server DB to which you want to connect.

Check out "Prepare for authentication" on page 148 to learn more about the authentication prerequisites.

Connection details

The following table describes the Microsoft SQL Server connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run the tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use a Secure Agent or serverless runtime environment. You can't run and application ingestion and replication tasks or database ingestion and replication tasks on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
SQL Server Version	This property is no longer used. If you select a version, it is ignored.

Authentication modes

You can configure one of the following authentication modes to connect to Microsoft SQL Server databases:

- SQL Server authentication
- Windows authentication (Deprecated)
- Active Directory Password authentication
- Windows Authentication V2
- Kerberos authentication
- Service Principal authentication

Select the required authentication mode and then configure the authentication-specific parameters.

For SQL Server sources in database ingestion and replication tasks, you must select SQL Server Authentication, Windows Authentication v2, or Active Directory Password. Don't use the Kerberos or Service Principal Authentication mode.

For SQL Server targets in application ingestion and replication tasks and database ingestion and replication tasks, you must select either SQL Server Authentication or Windows Authentication v2. Don't use any other authentication type.

SQL Server authentication

SQL Server authentication uses your Microsoft SQL Server user name and password to access Microsoft SQL Server.

The following table describes the basic connection properties for SQL Server authentication:

Property	Description
User Name	User name for the database login. The user name can't contain a semicolon. To connect to Microsoft Azure SQL Database, specify the user name in the following format: `username@host`
Password	Password for the database login. You can use special characters in the password. For more information about the special characters that you can use, see "Using special characters in the password" on page 155.
Host	Name of the machine hosting the database server. To connect to Microsoft Azure SQL Database, specify the fully qualified host name. For example, `vmjcmwxsfboheng.westus.cloudapp.azure.com`.
Port	Network port number used to connect to the database server. Default is 1433.
Instance Name	Instance name of the Microsoft SQL Server database.
Database Name	Database name for the Microsoft SQL Server target connection.Database name is case-sensitive if the database is case-sensitive. Maximum length is 100 characters. Database names can include alphanumeric and underscore characters.
Schema	The schema name to select tables during object selection in a mapping. When you select objects in a mapping, all schemas you have access to are displayed, and you can select tables from the available schemas.
Code Page	The code page of the database server.
Domain	This property is not applicable for SQL Server authentication.

Windows authentication

Windows Authentication (Deprecated) uses the Microsoft Windows authentication to access Microsoft SQL Server. This option is available when you access Data Integration by using Microsoft Windows.

When you choose this option, you don't need to enter credentials to access Microsoft SQL Server and ensure that the user account that starts the Secure Agent service is available in the Microsoft SQL Server database.

Note: Windows authentication is not certified for Microsoft SQL Server 2017 version hosted on Linux.

For Data Ingestion and Replication, use Windows Authentication v2 authentication.

The following table describes the basic connection properties for Windows authentication:

Property	Description
Host	Name of the machine hosting the database server. To connect to Microsoft Azure SQL Database, specify the fully qualified host name. For example, `vmjcmwxsfboheng.westus.cloudapp.azure.com`.
Port	Network port number used to connect to the database server. Default is 1433.
Instance Name	Instance name of the Microsoft SQL Server database.
Database Name	Database name for the Microsoft SQL Server target connection. Database name is case-sensitive if the database is case-sensitive. Maximum length is 100 characters. Database names can include alphanumeric and underscore characters.
Schema	The schema name to select tables during object selection in a mapping. When you select objects in a mapping, all schemas you have access to are displayed, and you can select tables from the available schemas.
Code Page	The code page of the database server.
Domain	This property is not applicable for Windows authentication.

Active Directory Password authentication

Active Directory Password authentication uses the Azure Active Directory user name and password to authenticate and access the Microsoft Azure SQL Database.

Note: Database Ingestion and Replication supports this authentication mode for sources in initial load jobs and for sources in incremental load and combined initial and incremental load jobs that use the **CDC Tables** or **Log-based** capture method. If you use this option for these jobs, verify that the **Validate Server Certificate** value is False.

The following table describes the basic connection properties for Active Directory Password authentication:

Property	Description
User Name	User name for the database login. The user name can't contain a semicolon. To connect to Microsoft Azure SQL Database, specify the user name in the following format: `username@host`
Password	Password for the database login. The password can't contain a semicolon.
Host	Name of the machine hosting the database server. To connect to Microsoft Azure SQL Database, specify the fully qualified host name. For example, `vmjcmwxsfboheng.westus.cloudapp.azure.com`.
Port	Network port number used to connect to the database server. Default is 1433.
Instance Name	Instance name of the Microsoft SQL Server database.

Property	Description
Database Name	Database name for the Microsoft SQL Server target connection. Database name is case-sensitive if the database is case-sensitive. Maximum length is 100 characters. Database names can include alphanumeric and underscore characters.
Schema	The schema name to select tables during object selection in a mapping. When you select objects in a mapping, all schemas you have access to are displayed, and you can select tables from the available schemas.
Code Page	The code page of the database server.
Domain	This property is not applicable for Active Directory Password authentication.

Windows Authentication V2

Windows Authentication V2 uses this authentication method to access Microsoft SQL Server from Data Integration or Data Ingestion and Replication using an agent hosted on a Linux or Windows machine.

When you choose this option on Linux, enter your domain name and Microsoft Windows credentials to access Microsoft SQL Server.

When you choose this option on Windows, the agent uses the user credentials specified in the connection only to test the connection. During runtime, the agent uses the credentials of the user who started the Secure Agent service. Ensure that the user account that starts the Secure Agent service is available in the Microsoft SQL Server database.

The following table describes the basic connection properties for Windows Authentication V2:

Property	Description
User Name	User name for the database login. The user name can't contain a semicolon. To connect to Microsoft Azure SQL Database, specify the user name in the following format: `username@host` If you use Windows Authentication v2 on Windows, the user name is used as follows: - During design time, the agent uses the user name specified here to test the connection. - During runtime, the Microsoft SQL server driver ignores the user name specified in this field and uses the credentials of the user who started the Secure Agent service. If you use Windows Authentication v2 on Linux, the user name specified here is used both during design time and runtime.
Password	Password for the database login. The password can't contain a semicolon. If you use Windows Authentication v2 on Windows, the password is used as follows: - During design time, the agent uses the password specified here to test the connection. - During runtime, the Microsoft SQL server driver ignores the password specified in this field and uses the credentials of the user who started the Secure Agent service. If you use Windows Authentication v2 on Linux, the password specified here is used both during design time and runtime.
Host	Name of the machine hosting the database server. To connect to Microsoft Azure SQL Database, specify the fully qualified host name. For example, `vmjcmwxsfboheng.westus.cloudapp.azure.com`.

Property	Description
Port	Network port number used to connect to the database server. Default is 1433.
Instance Name	Instance name of the Microsoft SQL Server database.
Database Name	Database name for the Microsoft SQL Server target connection. Database name is case-sensitive if the database is case-sensitive. Maximum length is 100 characters. Database names can include alphanumeric and underscore characters.
Schema	The schema name to select tables during object selection in a mapping. When you select objects in a mapping, all schemas you have access to are displayed, and you can select tables from the available schemas.
Code Page	The code page of the database server.
Domain	The domain name of the Windows user.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Encryption Method	The method that the Secure Agent uses to encrypt the data sent between the driver and the database server. You can use the encryption method to connect to Microsoft Azure SQL Database. Default is None.
Crypto Protocol Version	Cryptographic protocols to use when you enable SSL encryption.
Validate Server Certificate	When set to True, Secure Agent validates the certificate that is sent by the database server. If you specify the HostNameInCertificate parameter, Secure Agent also validates the host name in the certificate. When set to false, the Secure Agent doesn't validate the certificate that is sent by the database server.
Trust Store	The location and name of the truststore file. The truststore file contains a list of Certificate Authorities (CAs) that the driver uses for SSL server authentication. For the serverless runtime environment, specify the following certificate path in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<TrustStore_filename>`
Trust Store Password	The password to access the contents of the truststore file.
Host Name in Certificate	Host name of the machine that hosts the secure database. If you specify a host name, the Secure Agent validates the host name included in the connection with the host name in the SSL certificate.

Property	Description
Metadata Advanced Connection Properties	Additional properties for the JDBC driver to fetch the metadata. Enter properties in the following format: <parameter name>=<parameter value> If you enter more than one property, separate each key-value pair with a semicolon. For example, enter the following property to configure the connection timeout when you test a connection: LoginTimeout=<value_in_seconds> **Note:** The default connection timeout is 270 seconds.
Runtime Advanced Connection Properties	Additional properties for the ODBC driver required at run time. If you specify more than one property, separate each key-value pair with a semicolon.

Configure SSL with the serverless runtime environment

You can use the serverless runtime environment with Microsoft SQL Server Connector to connect to an SSL-enabled Microsoft SQL Server database.

Before you configure a secure Microsoft SQL Server connection using the serverless runtime environment, complete the following prerequisite tasks to add the SSL certificates to the serverless runtime location:

1. Create the following structure for the serverless agent configuration on AWS or Azure: <Supplementary file location>/serverless_agent_config

   Note: Data Ingestion and Replication does not support serverless runtime environments hosted on AWS.
2. Add the truststore certificate in the Amazon S3 bucket or Azure container in the following location in your AWS or Azure account: <Supplementary file location>/serverless_agent_config/SSL
3. Copy the following code snippet to a text editor:

   ```
   version: 1
   agent:
     agentAutoApply:
       general:
         sslStore:
           - fileCopy:
               sourcePath: SSL/<TrustStore_filename>
   ```

   where the source path is the directory path of the certificate file on AWS or Azure.
4. Ensure that the syntax and indentations are valid, and then save the file as serverlessUserAgentConfig.yml in the following AWS or Azure location: <Supplementary file location>/serverless_agent_config
   When the .yml file runs, the SSL certificate is copied from the AWS or Azure location to the serverless agent directory.
5. In the Microsoft SQL Server connection properties, specify the following certificate path in the serverless agent directory in the **Trust Store** field: /home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<cert_filename>

Using special characters in the password

When you use SQL Server authentication to connect to Microsoft SQL Server, you can use the following special characters in the password for the database user name:

- Special characters: `!@#\$%^'&*()_-;+=:,.<>?/~{][)(`
- Multibyte characters: `ÖÅÑÜČßΣ`
- Chinese characters: `Ψ龙爱福桜神空`

To use these special characters in the password, you need to set the `EnableMSSQLPasswordSpecialChars` custom property in the Secure Agent properties.

To configure the custom property in the Secure Agent, perform the following steps:

1. Log in to Informatica® Intelligent Data Management Cloud™ (IDMC).
2. Select **Administrator** > **Runtime Environments**.
3. On the **Runtime Environments** page, select the Secure Agent.
4. Click **Edit**.
5. In the **Custom Configuration Details** section, select **Data Integration Server** as the service and **DTM** as the type.
6. Enter `EnableMSSQLPasswordSpecialChars` in the **Name** field and set the value to `true`.
7. Click **Save**.

MongoDB Mass Ingestion connection properties

To define a MongoDB Mass Ingestion connection, configure the connection properties. You can use this connection for sources in database ingestion and replication tasks configured in Data Ingestion and Replication.

The following table describes the connection properties:

Connection property	Description
Connection Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 255 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
Description	Optional. A description of the connection. The description cannot exceed 4,000 characters.
Runtime Environment	The name of the runtime environment where you want to run database ingestion and replication tasks. You define runtime environments in Administrator. You can use either a local Secure Agent installation or serverless runtime environment. You cannot run a database ingestion and replication task on a Hosted Agent or elastic runtime environment.
Host and Port	An SRV record or a comma-separated list of *host_name*:*port* pairs. **Note:** If you are using the MongoDB replica set mode, you can enter multiple host names for resilience. If one host is not available, another specified host will be used.

Connection property	Description
SRV	Select this check box if you specified an SRV record in **Host and Port** property.
Authentication	The authentication method for establishing a secure connection. Select one of the following authenication modes: - **Username and Password**. Uses the user name and password credentials to connect to the MongoDB server. - **X.509**. Uses an X.509 certificate to connect to the MongoDB server. Default is **Username and Password**.
User Name	If you selected **Username and Password** authentication, enter the user name to use for logging in to the database.
Password	If you selected **Username and Password** authentication, enter the password for the specified database user.
SSL KeyStore File Path	If you selected **X.509** authentication, the absolute path of the keystore file on the Secure Agent machine that contains the keys and certificates required for secure communication. The keystore file must be in JKS format. Ensure that you download the certificates and place them in the Secure Agent machine before you specify this property. To successfully test the connection, the Secure Agent must have access to the specified keystore and truststore files using the passwords provided. If you use a Secure Agent group, all agents must have access to these files. Place a copy of the files on every machine where a Secure Agent runs.
SSL KeyStore Password	If you selected **X.509** authentication, the password for the keystore file that is required for secure communication.
SSL TrustStore File Path	If you selected **X.509** authentication, the absolute path of the truststore file on the Secure Agent machine.
SSL TrustStore Password	If you selected **X.509** authentication, the password for the truststore file.
Authentication Database	The name of the authentication database associated with the specified user.
Replica Set Name	The name of the replica set that is composed of the MongoDB servers with replicas of the source data. This field is relevant if you are using the MongoDB replica set mode.
Additional Connection Properties	One or more additional MongoDB connection string options that you want to use. Specify the properties as key-value pairs. If you specify more than one property, separate them with the ampersand symbol (&). The connection properties are case sensitive. Example: `authSource=admin&replicaSet=rsprimary` For more information about the MongoDB connection string options, refer to: https://www.mongodb.com/docs/v5.2/reference/connection-string/#connection-string-options

MQTT connection properties

When you set up an MQ Telemetry Transport (MQTT) connection, you must configure the connection properties.

Important: MQTT Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release.

The following table describes the MQTT connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
Description	Optional. Description that you can use to identity the connection. The description cannot exceed 4,000 characters.
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Broker URI	The connection URL of the MQTT broker. If specified, this value overrides the URL specified in the main portion of the URL. Sample URL: `tcp://<IP Address>:<port>`
Client Id	Client identifier of your MQTT client. If this value is left blank, the MQTT server assigns a unique value. This property value must be unique for each MQTT client connecting to a specific MQTT server. Sharing projects without changing the Client ID can lead to connection issues, including disconnections and missing updates.
Username	Username to use when connecting to the broker.
Password	Password to use when connecting to the broker.
Connection Timeout	Maximum time interval the client will wait for the connection to the MQTT server to be established. Default timeout is 30 seconds. A value of 0 disables timeout processing. That is, the client waits until the network connection is made successfully or fails.
Use SSL	Enable this option to use SSL for secure transmission. If you enable the SSL authentication, ensure to provide both keystore and truststore details for using the MQTT connection in a streaming ingestion and replication task.
Keystore Filename	Contains the keys and certificates required for secure communication.
Keystore Password	Password for the keystore filename.

Property	Description
Keystore Type	Type of keystore to use. Keystore type defines the storage and data format of the keystore information and the algorithms used to protect private keys in the keystore. Use one of the following types: - JKS. Stores private keys and certificates. - PKCS12. Stores private keys, secret keys. and certificates.
Truststore Filename	File name of the truststore file.
Truststore Password	Password for the truststore file name.
Truststore Type	Type of truststore to use. Use one of the following types: - JKS - PKCS 12
TLS Protocol	Transport protocols to use. Use one of the following types: - SSL - SSLv3 - TLS - TLSv1 - TLSv1.1 - TLSv1.2

MySQL connection properties

When you set up a MySQL connection, configure the connection properties.

The following table describes the MySQL connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run the tasks. For Data Ingestion and Replication, you can use either a local Secure Agent installation or serverless runtime environment. You can use a serverless runtime environment for cloud source types. You cannot run database ingestion and replication tasks on a Hosted Agent.
User Name	User name for the database login. The user name can't contain a semicolon.
Password	Password for the database login. The password can't contain a semicolon.
Host	Name of the machine that hosts the database server.
Port	Network port number used to connect to the database server. Default is 3306.
Database Name	Name of the MySQL database that you want to connect to. **Note:** The database name is case-sensitive. Maximum length is 64 characters. Database name can contain alphanumeric and underscore characters.
Code Page	The code page of the database server.

Metadata Advanced Connection Properties	Additional properties for the JDBC driver to fetch the metadata. Enter properties in the following format:
	`<parameter name>=<parameter value>`
	If you enter more than one property, separate each key-value pair with an ampersand (&).
	For example, enter the following property to configure the connection timeout when you test a connection:
	`connectTimeout=<milliseconds>`
	Note: The default connection timeout is 270000 miliseconds.

Runtime Advanced Connection Properties	Additional properties for the ODBC driver to run ingestion and replication jobs. If you specify more than one property, separate each key-value pair with a semicolon.

SSL properties

You can configure a MySQL connection to use SSL to securely communicate with the MySQL database.

Note: You can enable SSL for a MySQL connection only when you use the 8.x MySQL JDBC and ODBC drivers. Ensure that both the MySQL JDBC and ODBC drivers are of 8.x version.

To configure SSL, you must first download and install the MySQL ODBC and JDBC drivers, version 8.x. For information about installing the MySQL ODBC and JDBC drivers, version 8.x, see the Knowledge Base article: 561573

After you install the drivers, in the MySQL connection properties, enable SSL and specify the TLS protocols that you want to use for the secure communication.

When you enable SSL for the MySQL connection, you must configure the SSL properties for both the MySQL JDBC and ODBC drivers. Configure the required SSL properties for the JDBC driver, so that the Secure Agent can access metadata securely from MySQL. Also, configure the required SSL properties for the ODBC driver, so that the Secure Agent runs mappings to securely read from or write data to MySQL.

Note: SSL is not applicable when you use the Hosted Agent. You can configure SSL when you use the Secure Agent or the serverless runtime environment.

The following table describes the MySQL connection SSL properties:

Connection property	Description
Use SSL	Determines whether the Secure Agent establishes a secure connection to the MySQL database. When you select this option and the database server supports SSL, the Secure Agent establishes an encrypted connection. If the MySQL database server cannot configure SSL, the connection either fails or the Secure Agent establishes an unencrypted connection depending on whether you enable or disable the **Require SSL** checkbox. If you do not select the **Use SSL** checkbox, the Secure Agent attempts to establish an unencrypted connection.
Verify Server Certificate	If you select **Use SSL** and select this option, the client validates the server certificate that is sent by the database server.
Require SSL	Applicable only if you select **Use SSL**. If you select the **Require SSL** checkbox, and the MySQL database supports SSL, the Secure Agent establishes an SSL connection. If you select the **Require SSL** checkbox, and the MySQL database cannot configure SSL, the Secure Agent attempts to establish an SSL connection but fails. If you clear the **Require SSL** checkbox, and the MySQL database cannot configure SSL, the Secure Agent establishes an unencrypted connection.
TLS Protocols	The TLS protocols used for the secure communication when you select **Use SSL**. You can select from the following protocols: - TLSv1 - TLSv1.1 - TLSv1.2 Default is TLSv1.2. The TLSv1 and TLSv1.1 protocols are not applicable.

The following table describes the MySQL connection properties for the JDBC driver version 8.x when you enable **Use SSL**:

Connection property	Description
Trust Certificate Key Store	The path and file name of the truststore file. You must prefix the file path with file colon (`file:`). For example, `file:C:\SSL\mysql_new\truststore` For the serverless runtime environment, specify the following certificate path in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<TrustStore_filename>`
Trust Certificate Key Store Password	The password for the truststore file.
Client Certificate Key Store	The path and file name of the keystore file. You must prefix the file path with file colon (`file:`). For example, `file:C:\SSL\mysql_new\keystore` For the serverless runtime environment, specify the following certificate path in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<KeyStore_filename>`
Client Certificate Key Store Password	The password to access the keystore file.
JDBC Cipher Suites	Colon-separated cipher suite values in RFC format. For example: `TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256:TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256`

The following table describes the MySQL connection properties for the ODBC driver version 8.x when you enable **Use SSL**:

Connection property	Description
SSL Certificate Authority	The path and name of the CA certificate. For example, `C:\SSL\mysql_new\ca.pem`
SSL Certificate	The path and name of the client certificate. For example, `C:\SSL\mysql_new\client-cert.pem`
SSL Key	The path and the name of the private key of the client. For example, `C:\SSL\mysql_new\client-key.pem`

Connection property	Description

SSL Cipher	Colon-separated cipher-suite values in OpenSSL format.
	For example:
	`ECDHE-ECDSA-AES128-GCM-SHA256:`
	`ECDHE-ECDSA-AES256-GCM-SHA384:`
	`ECDHE-RSA-AES128-GCM-SHA256:`

Verify Server's Identity	Verifies the host name in the certificate while verifying the server CA certificate. This property is applicable only when you enable **Verify Server Certificate** in the SSL properties.

Use the serverless runtime environment

In Data Ingestion and Replication, you can use a serverless runtime environment hosted on Azure to connect to the MySQL database.

Before you configure a MySQL connection using the serverless runtime environment, perform the following tasks:

- Add the My SQL JDBC driver JAR files in the Azure container in your Azure account.
- Configure the .yml serverless configuration file.

Add the JDBC driver as a JAR file in the Azure container in your Azure account

Perform the following steps:

1. Create the following structure for the serverless agent configuration in Azure: `<Supplementary file location>/serverless_agent_config/`
2. Add the JDBC driver files in the Azure container in the following location in your Azure account: `<Supplementary file location>/serverless_agent_config/mysql`

Configure the .yml serverless configuration file

Perform the following steps to configure the .yml serverless configuration file in the serverless runtime environment:

1. Copy the following code snippet to a text editor and specify the driver file name:

```
version: 1
agent:
  databaseIngestion:
    autoDeploy:
      mysql:
        jdbc:
          drivers:
            - fileCopy:
                sourcePath: mysql/jdbc/<file_name>.jar
```

   where the sourcePath is the directory path of the driver files in Azure.

2. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the JDBC driver files are copied from the Azure location to the following serverless agent directory: `<AGENT_DIR>/ext/connectors/thirdparty/com.mysql`

Netezza connection properties

Create a Netezza connection to securely read data from or write data to Netezza.

Prerequisites

Before you can use Netezza Connector, install the Netezza client on Windows or Linux.

Ensure that you have access to the Secure Agent directory that contains the success and error files. This directory path must be the same on each Secure Agent machine in the runtime environment selected for your connection.

Before you create a Netezza connection, be sure to configure the Netezza ODBC and JDBC drivers.

Download the Netezza JDBC Driver

1. Download the Netezza JDBC driver version from the IBM website:
   To download the Netezza JDBC driver on Windows, follow the instructions in the How to download the Netezza JDBC driver Knowledge article.

   If you want to use the Netezza JDBC driver on Linux, you can use the Netezza JDBC driver downloaded for Windows on the Linux machine.
2. After you download the Netezza JDBC driver, navigate to the following location:
   `<Secure Agent installation directory>/apps/Data_Integration_Server/ext/`
3. Manually create the following directory structure:
   `deploy_to_main/bin/rdtm-extra/Netezza`
4. Copy the Netezza JDBC driver jar file, `nzjdbc.jar`, to the following directory you created on the Secure Agent machine:
   `<Secure Agent installation directory>/apps/Data_Integration_Server/ext/deploy_to_main/bin/rdtm-extra/Netezza`
5. Restart the Secure Agent.

Download the Netezza ODBC Driver

1. Download the Netezza ODBC driver version from the IBM website:
   To download the Netezza ODBC driver, follow the instructions in the How to download the Netezza ODBC driver Knowledge article.
2. Perform the following tasks to use the Netezza ODBC driver based on the operating system:
   - On Windows, verify if the NetezzaSQL driver appears in the ODBC Data Source Administrator driver list.
   - On Linux, add the driver entries in the `odbcinst.ini` file in the Secure Agent installation directory. The following code shows a sample entry:

```
[NetezzaSQL]
Driver          = /data/home/adputf_9/cloud_td/Netezza/installer/linux64/lib64/
libnzodbc.so
Setup           = /data/home/adputf_9/cloud_td/Netezza/installer/linux64/lib64/
libnzodbc.so
APILevel        = 1
ConnectFunctions = YYN
Description     = Netezza ODBC driver
DriverODBCVer   = 03.51
DebugLogging    = true
```

```
LogPath          = /tmp
UnicodeTranslationOption = utf8
CharacterTranslationOption = all
PreFetch         = 256
Socket           = 16384
```

Connect to Netezza

Let's configure the Netezza connection properties to connect to Netezza.

Before you begin

Before you get started, you need to get Netezza database and authentication details from your Netetzza account. You also need to install Netezza client and drivers.

Check out "Prerequisites" on page 163 to learn more about the tasks you must perform.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Specify a Secure Agent.
Database	The name of the Netezza database.
Schemaname	The schema used for the Netezza source or target. Schema name is case sensitive.
Servername	The Netezza database host name.
Port	Network port number used to connect to the database server. Default is 1521.
Username	Database user name with the appropriate read and write database permissions to access the database.
Password	Password for the database user name.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Driver	The Netezza ODBC driver name that is used to connect to the Netezza database. The Netezza ODBC driver name is NetezzaSQL.
Runtime Additional Connection Configuration	Additional runtime attributes required to fetch data. For example, `securityLevel=preferredUnSecured;caCertFile =`
Metadata Additional Connection Configuration	The values to set the optional properties of the JDBC driver to fetch the metadata.

Database privileges

Database privileges define the level of access for the operations that you can perform in the Netezza database.

Verify that you have the following privileges on the Netezza database:

- CREATE TABLE
- CREATE EXTERNAL TABLE
- DELETE
- DROP
- INSERT
- LIST
- SELECT
- TRUNCATE
- UPDATE

NetSuite Mass Ingestion connection properties

When you set up a NetSuite Mass Ingestion connection, you must configure the connection properties.

Note: Before you configure the connection properties, install the SuiteAnalytics Connect JDBC driver and copy the NQjc.jar file to the following directory: `<Secure_Agent>\ext\connectors\thirdparty\informatica.netsuiteami`

For more information about installing the SuiteAnalytics Connect JDBC driver, see the SuiteAnalytics Connect documentation.

The following table describes the connection properties for a NetSuite Mass Ingestion connection:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
Email ID	User name of the NetSuite account. The user name is an email address.
Password	Password for the NetSuite account.
Service Host	Name of the SuiteAnalytics Connect Service host. The value in this field must match the value specified in the **Service Host** field under the **Your Configuration** section of the **SuiteAnalytics Connect Driver Download** page in NetSuite. To access the **SuiteAnalytics Connect Driver Download** page, log in to NetSuite and click the Set Up SuiteAnalytics Connect link in the Settings portlet.
Service Port	TCP/IP port on which the SuiteAnalytics Connect server is listening. Default is 1708.
Service Datasource	Data source that you want to use to access NetSuite data. You can select one of the following data sources: - NetSuite.com - NetSuite2.com Default is NetSuite2.com. **Note:** - In connections configured before the August 2022 release, the default value for this field is NetSuite.com. - To use a NetSuite2.com data source, the NetSuite user account must be configured with some specific roles and permissions. For more information about the roles and permissions required to access NetSuite2.com data sources, see the NetSuite documentation.
Account ID	NetSuite account ID. To find your account ID, log in to NetSuite and click **Setup** > **Integration** > **Web Services Preferences**. If you cannot access the **Setup** menu, navigate to **Support** > **Go to Suite Answers** > **Contact support by phone**. The page displays your account ID.

Connection property	Description
Role ID	Role ID associated with the NetSuite account.
Additional Connection Properties	Additional properties for the SuiteAnalytics Connect Driver that is used to connect to the NetSuite service data source. Specify the properties in `<property>=<value>` format. If you want to specify multiple properties, separate each property-value pair with a semicolon (;). You can specify the following connection properties in this field: - **ValidateServerCertificate**: Determines whether the driver validates the certificate sent by the SuiteAnalytics Connect server. During SSL server authentication, the SuiteAnalytics Connect server sends a certificate issued by a trusted Certificate Authority (CA). The required CAs are usually included in the Java truststore but you can also specify them using the TrustStore property. Valid values for the ValidateServerCertificate property are *true* and *false*. - **TrustStore**: Contains the path to a valid truststore containing the security certificates to be used for server authentication. The TrustStore property is ignored if the ValidateServerCertificate property is set to *false*. **Note:** For more information about the additional connection properties, see the NetSuite documentation.

Use the serverless runtime environment

In Data Ingestion and Replication, you can use a serverless runtime environment hosted on Azure to connect to the NetSuite database.

Before you configure a NetSuite Mass Ingestion connection using the serverless runtime environment, perform the following tasks:

- Add the JDBC driver JAR files in the Azure container in your Azure account.
- Configure the .yml serverless configuration file.

Add the JDBC driver as a JAR file in the Azure container in your Azure account

Perform the following steps:

1. Create the following structure for the serverless agent configuration in Azure: `<Supplementary file location>/serverless_agent_config/`
2. Add the JDBC driver files in the Azure container in the following location in your Azure account: `<Supplementary file location>/serverless_agent_config/netsuiteami`

Configure the .yml serverless configuration file

Perform the following steps to configure the .yml serverless configuration file in the serverless runtime environment:

1. Copy the following code snippet to a text editor and specify the driver file name:

```
version: 1
agent:
  databaseIngestion:
    autoDeploy:
      netsuiteami:
        jdbc:
          drivers:
            - fileCopy:
                sourcePath: netsuite/<file_name>.jar
```

where the sourcePath is the directory path of the driver files in Azure.

2. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the JDBC driver files are copied from the Azure location to the following serverless agent directory: `<AGENT_DIR>/ext/connectors/thirdparty/informatica.netsuiteami`

OPC UA connection properties

When you set up an OPC UA connection, you must configure the connection properties.

Important: OPC UA Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release.

The following table describes the OPC UA connection properties:

Property	Description
Connection Name	Name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:`~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
Description	Optional. Description of the connection. The description cannot exceed 4,000 characters.
Runtime Environment	Name of the runtime environment where you want to run the tasks.
Endpoint URL	The unique URL to connect to the OPC UA server. The endpoint URL identifies the specific instance of a server and a security policy type. A valid endpoint URL consists of the endpoint type (opc.tcp), the endpoint host name (IP address, URL, or DSN), and the endpoint port number. For example, `opc.tcp://opcuaserver.com:48010`
Security Policy	The security policy used to connect to the OPC UA server. The security policy parameters specify the security algorithms that the OPC UA server supports. You can choose one of the following security policies: - None. No security provided. - Basic128Rsa15 - Basic256 - Basic256Sha256 - Aes128_Sha256_RsaOaep - Aes256_Sha256_RsaPss **Note:** The OPC Foundation deprecated the security policies, Basic128Rsa15 and Basic256 as of OPC UA specification version 1.04. The encryption provided by these policies is less secure. Use these security policies only to provide backward compatibility.

Property	Description
Security Mode	The security mode used to connect to the OPC UA server. The security mode is valid only when security policy is not set to None. You can choose one of the following security policies: - Sign. Transfer unencrypted data, but with digital signatures that allow verification of data integrity. - SignAndEncrypt. Transfer signed and encrypted data.
Application URI	Optional. A unique identifier that the OPC UA application can use to connect to the OPC UA server. Enter a unique ID in the following format: `urn:aaa:bbb` For example, `urn:nifi:opcua` The unique identifier must match the URI of the Subject Alternative Name of your OPC UA client certificate.
Client Keystore Location	Optional. Absolute path and file name of the keystore file that contains private keys and certificates for the OPC UA server. Enter the path in the following format: `/root/opcua/client.jks` The keystore must contain only one keypair entry of private key and certificate. If multiple keypair entries exist, the first entry is used.
Client Keystore Password	Optional. Password for the client keystore.
Require server authentication	Optional. Enable if you require server authentication of client certificates, client authentication of server certificates, or both.
Trust store Location	Optional. The absolute path of the truststore file that contains the trusted certificate. Enter the path in the following format: `/root/opcua/trust.jks`
Trust store Password	Password for the truststore file.
Authentication Policy	Authentication settings required to establish the connections. You can choose one of the following authentication policies: - Anon. Anonymous authentication. Anonymous tokens are associated with servers that do not require user authentication. - UserName. User name and password tokens are associated with servers with any password based system, such as Windows.
User Name	User name to access the OPC UA server if you choose authentication policy as **UserName**.
Password	Password to access the OPC UA server if you choose authentication policy as **UserName**.

Open Table connection properties

Create an Open Table connection to securely load data to Open Table formats available in a catalog.

Prerequisites

Before you create an Open Table connection, complete the prerequisites.

If you use an AWS Glue Catalog and Amazon S3 Storage to interact with Apache Iceberg, you need to have access to the following AWS services that manage the tables on AWS:

- AWS Glue Catalog: AWS Glue Catalog manages the metadata associated with the Apache Iceberg tables.
- Amazon S3 Storage: Amazon S3 stores the Apache Iceberg tables containing actual records in columnar format, organized in partitioned directories.
- Amazon Athena: Amazon Athena uses the AWS Glue Data Catalog to store metadata such as table and column names for your data stored in Amazon S3. Open Table Connector uses the Amazon Athena JDBC driver to connect to the AWS Glue Catalog to access Apache Iceberg table metadata.

You need to create separate policies to access these services.

Create minimal IAM policies

You need to create IAM policies with the minimum required permissions to interact with Apache Iceberg tables managed by AWS Glue Catalog. For more information on configuring these policies, refer to the AWS documentation.

Minimum policy for Amazon Athena

The following sample policy shows the minimal Amazon IAM policy to access Amazon Athena:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "athena:CreatePreparedStatement",
                "athena:GetPreparedStatement",
                "athena:GetWorkGroup",
                "athena:GetTableMetadata",
                "athena:StartQueryExecution",
                "athena:GetQueryResultsStream",
                "athena:ListDatabases",
                "athena:GetQueryExecution",
                "athena:GetQueryResults",
                "athena:GetDatabase",
                "athena:ListTableMetadata",
                "athena:GetDataCatalog",
                "athena:DeletePreparedStatement"
            ],
            "Resource": [
                "arn:aws:athena:*:*:workgroup/*",
                "arn:aws:athena:*:*:datacatalog/*"
            ]
        },
        {
            "Sid": "VisualEditor1",
            "Effect": "Allow",
            "Action": [
                "athena:ListDataCatalogs",
                "athena:GetQueryExecution",
                "athena:ListWorkGroups",
                "athena:GetPreparedStatement"
            ],
            "Resource": "*"
        }
    ]
}
```

Minimum policy for AWS Glue

The following sample policy shows the minimal Amazon IAM policy to access AWS Glue Catalog:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "glue:*"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Minimum policy for AWS S3

The following sample policy shows the minimal Amazon IAM policy to read from or write data to an Amazon S3 bucket:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket",
                "s3:DeleteObject"
            ],
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "s3:ListBucket",
                "s3:GetBucketLocation",
                "s3:ListAllMyBuckets",
                "s3:GetBucketAcl"
            ],
            "Resource": [
                "*"
            ]
        }
    ]
}
```

Configure EC2 role to assume role

You can configure an EC2 role to assume an IAM role and generate temporary security credentials to connect to Amazon S3 from the same or different AWS accounts.

When you configure EC2 role to assume role, ensure that you have the **sts:AssumeRole** permission and a trust relationship established within the AWS accounts to use the temporary security credentials. The trust relationship is defined in the trust policy of the IAM role when you create the role. The IAM role adds the EC2 role as a trusted entity allowing the EC2 role to use the temporary security credentials and access the AWS accounts.

When the trusted EC2 role requests for the temporary security credentials, the AWS Security Token Service (AWS STS) dynamically generates the temporary security credentials that are valid for a specified period and provides the credentials to the trusted EC2 role.

Before you use the EC2 Role to Assume Role authentication, consider the following prerequisites:

- Install the Secure Agent on the AWS EC2 instance.
- The EC2 role attached to the AWS EC2 instance must have permissions to assume another IAM role. The following is a sample permission policy of EC2 role that is attached to the AWS EC2 instance:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "sts:AssumeRole",
            "Resource": "arn:aws:iam::001234567890:role/open_table_rolearn"
        }
    ]
}
```

  Resource value must include the ARN of IAM role that the EC2 role needs to assume.

- The IAM role that the EC2 role needs to assume must have a permission policy and a trust policy attached to access AWS Glue Catalog, Amazon Athena, and Amazon S3.
  You can also specify the external ID of your AWS account for a more secure access. The external ID must be a string.

  The following sample shows the assumed IAM role's trust policy with the external ID:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::001234567890:root" //anyone in this account
001234567890 can assume this role, this can also be limited to one role.
            },
            "Action": "sts:AssumeRole",
            "Condition": {
                "StringEquals": {
                    "sts:ExternalId": "aws_externalid"
                }
            }
        }
    ]
}
```

For more information about the minimum permission policies, see “Create minimal IAM policies” on page 171.

Connect to Open Table

Let's configure the Open Table connection properties to connect to AWS Glue Catalog or Nessie catalog.

Before you begin

Before you get started, you will need to create the minimal IAM policies to interact with Apache Iceberg tables managed by AWS Glue Catalog. You also need to configure the authentication-specific prerequisites to connect to Amazon S3.

Permanent IAM Credentials authentication for Amazon S3 requires the access key and secret key values of the IAM user. EC2 Role to Assume Role authentication for Amazon S3 requires the ARN of the IAM role that the EC2 role assumes to generate temporary security credentials.

Check out “Prerequisites” on page 171 to learn more about how to configure policies and role to access Apache Iceberg tables.

Connection details

The following table describes the Open Table connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. You cannot run a database ingestion task on a Hosted Agent or in a serverless runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Open Table Format	The Open Table format that you want to use to read from or write data to a catalog. Select Apache Iceberg from the list.

Catalog types

You can select AWS Glue Catalog or Nessie catalog as the catalog type to manage the metadata of the Open Table format that you selected.

Select the catalog type that your Open Table format uses and then configure the catalog specific parameters.

AWS Glue Catalog

If the Apache Iceberg Open Table format uses AWS Glue Catalog as the catalog type, configure the properties specific to AWS Glue Catalog.

The following table describes the property to configure AWS Glue Catalog:

Property	Description
Athena JDBC URL	Enter the JDBC URL in the following format: `jdbc:athena://Region=<AWS_Region>;OutputLocation=<S3_Location>` For example, `jdbc:athena://Region=us-west1;OutputLocation=s3://working/dir`.
Catalog Authentication Type	The authentication method to connect to the catalog. Select one of the following options: - None. Connects to AWS Glue Catalog without any authentication credentials. - OAuth 2.0 Client Credentials. Connects to a REST catalog or Nessie catalog using a Client ID and Client Secret to obtain an access token from the authorization server.

Storage types

You can choose Amazon S3 or MinIO as the storage type to store the Open Table format tables.

Select the storage type and configure the storage specific authentication parameters.

Amazon S3

If you use AWS Glue Catalog as the catalog type, configure the properties specific to Amazon S3 storage.

AWS Permanent IAM Credentials authentication

You can use AWS Permanent IAM Credentials authentication for Amazon S3 storage when you connect to an AWS Glue Catalog.

The following table describes the properties to configure AWS Permanent IAM Credentials authentication:

Property	Description
Access Key	The key to access the AWS Glue Catalog.
Secret Key	The secret key to access the AWS Glue Catalog. The secret key is associated with the access key and uniquely identifies the account.

AWS EC2 Role to Assume Role authentication

You can use AWS EC2 Role to Assume Role authentication for Amazon S3 storage only when you read Apache Iceberg tables from AWS Glue Catalog.

The following table describes the properties to configure AWS EC2 Role to Assume Role authentication:

Property	Description
IAM Role ARN	The ARN of the IAM role assumed by the EC2 role to generate the temporary session credentials.
External ID	A unique, user-defined string value that the IAM role requires the EC2 role to provide when calling the sts:AssumeRole API.

Oracle Autonomous Database connection properties

Create an Oracle Autonomous Database connection to load data to an Oracle Autonomous Database target.

Prerequisites

Before you create an Oracle Autonomous Database connection, be sure to complete the prerequisites.

Prepare for object storage authentication

You can configure the following object storage authentication methods for Oracle Autonomous Database Connector:

ConfigFile authentication

The ConfigFile authentication uses identity credentials of Oracle Cloud Infrastructure (OCI) account provided through a configuration file for authentication. This authentication method is based on the profile selected in the configuration file.

You can create a configuration file in the following format:

```
[<profile name>]
user=<user ocid>
fingerprint=<fingerprint>
tenancy=<tenancy ocid>
region=<region>
key_file=<private key file location>
```

You require the user OCID, fingerprint, and tenancy OCID information from the OCI account for the configuration file.

For more information about the steps to extract the identity credentials from the Oracle Cloud Infrastructure Console, see the Oracle Cloud Infrastructure documentation.

By default, the OCI configuration file is located at `~/.oci/config` on the Secure Agent machine. The `~/.oci/config` file can contain several profiles. The default profile name is `DEFAULT`. You can change the default profile name to any new profile names based on the profiles that you add to the `~/.oci/config` file. The `~/.oci/config` file cannot contain two profiles with the same name.

Simple authentication

The simple authentication uses API keys for authentication. You can provide the authentication details in the Oracle Autonomous Database connection. You need to place the private key file in the Secure Agent machine.

You require the user OCID, fingerprint, and tenancy OCID information from the Oracle Cloud Infrastructure account to create an Oracle Autonomous Database connection.

For more information about the steps to extract the identity credentials from the Oracle Cloud Infrastructure Console, see the Oracle Cloud Infrastructure documentation.

Connect to Oracle Autonomous Database

Let's configure the Oracle Autonomous Database connection properties to connect to Oracle Autonomous Database hosted on Oracle Cloud Infrastructure.

Before you begin

Before you get started, get the required information from your Oracle Cloud Infrastructure account based on the object storage authentication type that you want to configure.

Connection details

When you create an Oracle Autonomous Database connection, configure the connection properties.

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For database ingestion and replication tasks, you can use only the Secure Agent. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
User Name	The user name to access the Oracle Autonomous Database. It is used to authenticate the connection.
Password	The password associated with the user name.

Authentication types

You can configure only mutual TLS authentication to connect to Oracle Autonomous Database.
Default is mutual TLS authentication.

Mutual TLS authentication

The following table describes the basic connection properties for mutual TLS authentication:

Property	Description
Service Name	The specific database service accessed for the secured connection.
Wallet Path	The location of the wallet file for the secured connection.

Object storage authentication types

You can configure ConfigFile or simple authentication for staging the connection when you connect to Oracle Autonomous Database. Select the required authentication type and then configure the authentication-specific parameters.

Default is ConfigFile authentication.

ConfigFile authentication

The following table describes the basic connection property for ConfigFile authentication:

Property	Description
Region	The Oracle Cloud Infrastructure region where the object storage bucket resides. Select the Oracle Cloud Object Storage region from the list.

Advanced settings

The following table describes the advanced connection properties for ConfigFile authentication:

Property	Description
Configuration File Location	The absolute path of the configuration file on the Secure Agent machine. If you do not enter the value, the Secure Agent uses the following configuration file path: `~/.oci/config`
Profile Name	The name of the profile in the configuration file that you want to use. Default is `DEFAULT`.

Simple authentication

The following table describes the basic connection properties for simple authentication:

Property	Description
User OCID	The unique identifier of the user in Oracle Cloud Infrastructure. For example, `ocid1.user.oc1..aaaaaaaaherdgpjknqzrwbdc7n5ksokkot7c5jngtx3pgolr7oqbw7xzksza`
Fingerprint	The fingerprint of the public key.
Tenancy OCID	The unique identifier of the tenancy in Oracle Cloud Infrastructure. The tenancy is the globally unique name of the Oracle Cloud Infrastructure account. For example, `ocid1.tenancy.oc1..aaaaaaaaba3pv6wkcr4jqae5f44n2b2m2yt2j6rx32uzr4h25vqstifsfdsq`
Private Key File Location	The location of the private key file in .PEM format on the Secure Agent machine.
Region	The Oracle Cloud Infrastructure region where the object storage bucket resides. Select the Oracle Cloud Object Storage region from the list.

Oracle Cloud Object Storage connections

Create a Oracle Cloud Object Storage connection to read data from or write data to Oracle Cloud Object Storage files.

Prerequisites

Before you create an Oracle Cloud Object Storage connection to read from or write to Oracle Cloud Object Storage, be sure to complete the prerequisites.

Configure Oracle Cloud Infrastructure policies

As a user, you can use Oracle Cloud Object Storage Connector after the organization administrator creates a minimal Oracle Cloud Infrastructure (OCI) Identity and Access Management (IAM) policy for Oracle Cloud Object Storage Connector.

The Oracle Cloud Infrastructure policy defines the resources that users and groups can access in an OCI account and how to access them. You can use policies to manage certain types of resources in a specific compartment in certain ways.

You need to perform the following tasks:

1. Define users, groups, and one or more compartments to hold the cloud resources for your organization.
2. Create the policies.
3. Place users into the appropriate groups depending on the compartments and resources they need to work with.
4. Provide the users with the one-time passwords that they need to access the console and work with the compartments.

For more information about adding users, groups, and policies, see Oracle Cloud Infrastructure documentation.

You can create a policy in the following format:

```
Allow group <group_name> to <verb> <resource-type> in compartment <compartment_name>
```

For example,

```
Allow group ObjectReaders to read buckets in compartment ABC
```

```
Allow group ObjectWriters to manage objects in compartment ABC where any
{request.permission='OBJECT_CREATE', request.permission='OBJECT_INSPECT'}
```

You need to add the following policies to configure the Oracle Cloud Object Storage connection, access objects, and run mappings:

- Policies for Oracle Cloud Object Storage test connection

  ```
  Allow group <group_name> to inspect object-family in compartment <compartment_name>
  ```

  ```
  Allow group <group_name> to inspect buckets in compartment <compartment_name>
  ```

- Policies for Oracle Cloud Object Storage sources

  ```
  Allow group <group_name> to inspect buckets in compartment <compartment_name>
  ```

  ```
  Allow group <group_name> to read object-family in compartment <compartment_name>
  ```

- Policies for Oracle Cloud Object Storage targets

  ```
  Allow group <group_name> to manage inspect buckets in compartment <compartment_name>
  ```

  ```
  Allow group <group_name> to manage object-family in compartment <compartment_name>
  ```

Prepare for authentication

You can configure the following authentication methods for Oracle Cloud Object Storage Connector:

ConfigFile authentication

The ConfigFile authentication uses identity credentials of Oracle Cloud Infrastructure (OCI) account provided through a configuration file for authentication. This authentication method is based on the profile selected in the configuration file.

You can create a configuration file in the following format:

```
[<profile name>]
user=<user ocid>
fingerprint=<fingerprint>
tenancy=<tenancy ocid>
region=<region>
key_file=<private key file location>
```

You require the user OCID, fingerprint, and tenancy OCID information from the OCI account for the configuration file.

For more information about the steps to extract the identity credentials from the Oracle Cloud Infrastructure Console, see the Oracle Cloud Infrastructure documentation.

By default, the OCI configuration file is located at `~/.oci/config` on the Secure Agent machine. The `~/.oci/config` file can contain several profiles. The default profile name is `DEFAULT`. You can change the default profile name to any new profile names based on the profiles that you add to the `~/.oci/config` file. The `~/.oci/config` file cannot contain two profiles with the same name.

Simple authentication

The simple authentication uses API keys for authentication. You can provide the authentication details in the Oracle Cloud Object Storage connection. You need to place the private key file in the Secure Agent machine.

You require the user OCID, fingerprint, and tenancy OCID information from the Oracle Cloud Infrastructure account to create an Oracle Cloud Object Storage connection.

For more information about the steps to extract the identity credentials from the Oracle Cloud Infrastructure Console, see the Oracle Cloud Infrastructure documentation.

Connect to Oracle Cloud Object Storage

Let's configure the Oracle Cloud Object Storage connection properties to connect to Oracle Cloud Object Storage.

Before you begin

Before you get started, configure the Oracle Cloud Infrastructure policies and get the required information from your Oracle Cloud Infrastructure account based on the authentication type that you want to configure.

Check out "Prerequisites" on page 179 to learn more about these tasks.

Connection details

When you create an Oracle Cloud Object Storage connection, configure the connection properties.

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment-	The name of the runtime environment where you want to run tasks. For more information about runtime environments, see *Runtime Environments* in the Administrator help. For application ingestion and replication tasks and database ingestion and replication tasks, you can use the Secure Agent or serverless runtime environment. The tasks must have a source type supported for serverless use. For file ingestion and replication tasks, you can use only the Secure Agent. You can't run tasks on a Hosted Agent or elastic runtime environment.

Authentication types

You can configure ConfigFile or simple authentication to connect to Oracle Cloud Object Storage. Select the required authentication type and then configure the authentication-specific parameters.
Default is ConfigFile Authentication.

ConfigFile authentication

The following table describes the basic connection properties for ConfigFile authentication:

Property	Description
Region	The Oracle Cloud Infrastructure region where the object storage bucket resides. Before you select a region, ensure that the Secure Agent is up and running. If you still don't see your region in the list, restart the Data Integration Server. Select the Oracle Cloud Object Storage region from the list.
Bucket Name	The Oracle Cloud Object Storage bucket name that contains the objects.

Advanced settings

The following table describes the advanced connection properties for ConfigFile authentication:

Property	Description
Configuration File Location	The absolute path of the configuration file on the Secure Agent machine. If you don't enter the value, the Secure Agent uses the following configuration file path: `~/.oci/config`
Profile Name	The name of the profile in the configuration file that you want to use. Default is `DEFAULT`.
Folder Path	The folder under the specified Oracle Cloud Object Storage bucket. For example, `bucket/Dir_1/Dir_2/FileName.txt`. Here, `Dir_1/Dir_2` is the folder path.

Simple authentication

The following table describes the basic connection properties for simple authentication:

Property	Description
User OCID	The unique identifier of the user in Oracle Cloud Infrastructure. For example, `ocid1.user.oc1..aaaaaaaaherdgpjknqzrwbdc7n5ksokkot7c5jngtx3pgolr7oqbw7xzksza`
Fingerprint	The fingerprint of the public key.
Tenancy OCID	The unique identifier of the tenancy in Oracle Cloud Infrastructure. The tenancy is the globally unique name of the Oracle Cloud Infrastructure account. For example, `ocid1.tenancy.oc1..aaaaaaaaba3pv6wkcr4jqae5f44n2b2m2yt2j6rx32uzr4h25vqstifsfdsq`
Private Key File Location	The location of the private key file in .PEM format on the Secure Agent machine.
Region	The Oracle Cloud Infrastructure region where the object storage bucket resides. Before you select a region, ensure that the Secure Agent is up and running. If you still don't see your region in the list, restart the Data Integration Server. Select the Oracle Cloud Object Storage region from the list.
Bucket Name	The Oracle Cloud Object Storage bucket name that contains the objects.

Advanced settings

The following table describes the advanced connection property for simple authentication:

Property	Description
Folder Path	The folder under the specified Oracle Cloud Object Storage bucket. For example, `bucket/Dir_1/Dir_2/FileName.txt`. Here, `Dir_1/Dir_2` is the folder path.

Oracle Database Ingestion connection properties

To define an Oracle Database Ingestion connection, configure the connection properties. You can use the connection for sources or targets in database ingestion and replication tasks configured in Data Ingestion and Replication.You can also use the connection for targets in application ingestion and replication tasks.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	The name of the runtime environment where you want to run tasks. You define runtime environments in Administrator. For application ingestion and replication tasks and database ingestion and replication tasks that have a supported source type, you can use the serverless runtime environment. You cannot run and application ingestion and replication or a database ingestion and replication tasks on a Hosted Agent.
Authentication Mode	The authentication mode that the connector must use to log in to Oracle. Options are: - **Oracle Database Authentication**. Uses your Oracle user name and password to connect to Oracle. - **Kerberos**. Uses Kerberos authentication to connect to Oracle. **Note:** If you select Kerberos, the **Schema** list that's displayed when you define an Oracle source or target in an application or database ingestion and replication task will include the schemas of other Kerberos users as well as your schemas. **Note:** You cannot use a serverless runtime environment with Kerberos authentication. Default is Oracle Database Authentication.

Property	Description
Driver Type	The type of JDBC driver to use for connecting to an Oracle source or target. Options are: - **Progress DataDirect Oracle JDBC**. Use the Progress DataDirect for JDBC for Oracle Driver. - **Native Oracle JDBC**. Use the Oracle JDBC driver. This driver has demonstrated better performance when reading data from the source. Default is Progress DataDirect Oracle JDBC. **Note:** If you select the Native Oracle JDBC driver, use the connection in new database ingestion and replication jobs only. For existing jobs that have run with the DataDirect driver, switching to the Native Oracle JDBC driver is not recommended.
User Name	If you use Oracle Database Authentication, the user name for the Oracle database login. The user name cannot contain a semicolon. **Note:** This property is not displayed if you use Kerberos authentication.
Password	If you use Oracle Database Authentication, the password for the Oracle database login. The password cannot contain a semicolon. **Note:** This property is not displayed if you use Kerberos authentication.
Host	Host name of the database server.
Port	Network port number used to connect to the database server. Default is 1521.
Service Name	Service name or System ID (SID) that uniquely identifies the Oracle database. Specify the SID in the following format, including the leading semicolon (;), to connect to Oracle databases: `;SID=<ORACLE_SID>`
Code Page	The code page of the database server. Database ingestion and replication tasks use the UTF-8 code page. Default is UTF-8.
Encryption Method	For initial load jobs, determines whether the data exchanged between the Secure Agent and the Oracle database server is encrypted: Options are: - **SSL**. Establishes a secure connection using SSL for data encryption. If the Oracle database server cannot configure SSL, the connection fails. - **No Encryption**. Establishes a connection without using SSL. Data is not encrypted. Default is No Encryption. **Note:** To enable SSL, import the Oracle SSL certificate to the cacerts file located in the appropriate JDK or JRE folder specific for your Secure Agent installation.

Property	Description
Crypto Protocol Version	If you selected SSL as the encryption method, you must specify a cryptographic protocol or a list of cryptographic protocols supported by your server to use with an encrypted connection. Options are: - SSLv2 - SSLv3 - TLSv1.2 Default is TLSv1.2.
Validate Server Certificate	If you selected SSL as the encryption method, controls whether the Secure Agent validates the server certificate that is sent by the Oracle database server. - **True**. Validate the server certificate. - **False**. Do not validate the server certificate. Default is False. If you also specify the **Host Name in Certificate** property, the Secure Agent also validates the host name in the certificate.
Trust Store	If you selected SSL as the encryption method and enabled validation of the server certificate, specify the path and name of the truststore file, which contains the list of the Certificate Authorities (CAs) that the client trusts for SSL authentication.
Trust Store Password	If you selected SSL as the encryption method and enabled validation of the server certificate, specify a password for accessing the contents of the truststore file.
Host Name in Certificate	If you selected SSL as the encryption method and enabled validation of the server certificate, specify the host name of the machine that hosts the Oracle database to provide for additional security. The Secure Agent validates the host name included the connection with the host name in the SSL certificate.
Key Store	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the path and name of the keystore file. The keystore file contains the certificates that the client sends to the Oracle server in response to the server's certificate request. For more information, see "Creating a Key Store Certificate" on page 192.
Key Store Password	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the password for the keystore file.
Key Password	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the password for the keys in the keystore file. Use this property when the keys have a different password than the keystore file.
Database Connect String	A TNS name, an Oracle Net keyword-value pair, or a SQL connect string URL that OCI uses to connect to Oracle.

Property	Description
TDE Wallet Directory	The path to the directory that contains the Oracle wallet file used for Oracle Transparent Data Encryption (TDE). Specify this property value only if you capture change data from TDE-encrypted tablespaces and one of the following conditions are true: - The Oracle wallet is not available to the database. - The Oracle database is running on a server that is remote from Oracle redo logs. - The wallet directory is not in the default location on the database host or the wallet name is not the default name of ewallet.p12. - The wallet directory is not available to the Secure Agent host.
TDE Wallet Password	A clear text password that is required to access the Oracle TDE wallet and get the master key. This property value is required if you need to read and decrypt data from TDE-encrypted tablespaces in the Oracle source database.
Directory Substitution	A local path prefix to substitute for the server path prefix of the redo logs on the Oracle server. This substitute local path is required when the log reader runs on a system other than the Oracle server and uses a different mapping to access the redo log files. Use this property in the following situations: - The redo logs reside on shared disk. - The redo logs have been copied to a system other than the Oracle system. - The archived redo logs are accessed by using a different NFS mount. Do *not* use this statement if you use Oracle Automatic Storage Management (ASM) to manage the redo logs. You can define one or more substitutions in the following format: `server_path_prefix,local_path_prefix;server_path_prefix,local_path_prefix;...` **Note:** This property does not apply to Oracle targets.
Reader Active Log Mask	A mask that the log reader uses for selecting active redo logs when the Oracle database uses multiplexing of redo logs. The log reader compares the mask against the member names in an active redo log group to determine which log to read. In the mask, you can use the asterisk (*) wildcard to represent zero or more characters. The mask can be up to 128 characters in length. It is case-sensitive on Linux or UNIX systems but not on Windows systems. **Note:** This property does not apply to Oracle targets.

Property	Description
Reader Archive Destination 1	The primary log destination from which the log reader reads archived logs, when Oracle is configured to write more than one copy of each archived redo log. Enter a number that corresponds to a *n* value in an Oracle LOG_ARCHIVE_DEST_*n* initialization parameter, where *n* is a value from 1 to 10. If you set only one of the Reader Archive Destination 1 and Destination 2 properties, the log reader uses that property setting. If you specify neither property, the archive log queries are not filtered by the log destination. **Note:** This property does not apply to Oracle targets.
Reader Archive Destination 2	The secondary log destination from which the log reader reads archived logs when the primary destination becomes unavailable or when the logs at the primary destination cannot be read. For example, logs might have been corrupted or deleted. Enter a number that corresponds to the *n* value in an Oracle LOG_ARCHIVE_DEST_*n* initialization parameter, where *n* is a value from 1 to 10. Usually, this value is a number greater than 1. **Note:** This property does not apply to Oracle targets.
Reader ASM Connect String	In an Oracle ASM environment, the Oracle connection string, defined in TNS, that the log reader uses to connect to the ASM instance that manages storage of active and archived redo logs for the source database. **Note:** This property does not apply to Oracle targets.
Reader ASM User Name	In an Oracle ASM environment, an Oracle user ID that the log reader uses to connect to the ASM instance that manages storage of active and archived redo logs for the source database. This user ID must have SYSDBA or SYSASM authority. To use SYSASM authority, set the **Reader ASM Connect As SYSASM** property to Y. **Note:** This property does not apply to Oracle targets.
Reader ASM Password	In an Oracle ASM environment, a clear text password for the user that is specified in the **Reader ASM User Name** property. The log reader uses this password and the ASM user name to connect to the ASM instance that manages storage of active and archived redo logs for the source database. **Note:** This property does not apply to Oracle targets.
Reader ASM Connect As SYSASM	If you use Oracle 11*g* ASM or later and want the log reader to use a user ID that has SYSASM authority to connect to the ASM instance, select this check box. Also specify a user ID that has SYSASM authority in the **Reader ASM User Name** property. To use a user ID that has SYSDBA authority, clear this check box. By default, this check box is cleared. **Note:** This property does not apply to Oracle targets.

Property	Description
Reader Mode	Indicates the source of and types of Oracle redo logs that the log reader reads. Valid options are: - **ACTIVE**. Read active and archived redo logs from the Oracle online system. Optionally, you can use the **Reader Active Log Mask** property to filter the active redo logs and use the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties to limit the archived log destinations from which to read archived logs. - **ARCHIVEONLY**. Read only archived redo logs. Optionally, you can use the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties to limit the archived log destinations from which to read archived logs. - **ARCHIVECOPY**. Read archived redo logs that have been copied to an alternate file system. For combined initial and incremental load jobs, you must also set the source custom property pwx.cdcreader.oracle.reader.additional with the dir and file parameters, at the direction of Informatica Global Customer Support. You can use this option in the following situations: - You do not have the authority to access the Oracle archived redo logs directly. - The archived redo logs are written to ASM, but you do not have access to ASM. - The archived log retention policy for the database server causes the archived logs to not be retained long enough. With this option, the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties are ignored. Default is **ACTIVE**. **Note:** This property does not apply to Oracle targets.
Reader Standby Log Mask	A mask that the log reader uses for selecting redo logs for an Oracle physical standby database when the database uses multiplexing of redo logs. The log reader compares the mask against the member names in an redo log group to determine which log to read. In the mask, you can use the asterisk (*) wildcard to represent zero or more characters. The mask can be up to 128 characters in length. It is case-sensitive on Linux or UNIX systems but not on Windows systems. **Note:** This property does not apply to Oracle targets.
Standby Connect String	An Oracle connection string, defined in TNS, that the log reader uses to connect to the Oracle physical standby database for change capture when the database is not open with read only access. **Note:** This property does not apply to Oracle targets.
Standby User Name	A user ID that the log reader uses to connect to the Oracle physical standby database for change capture. This user ID must have SYSDBA authority. **Note:** This property does not apply to Oracle targets.

Property	Description
Standby Password	A password that the log reader uses to connect to the Oracle physical standby database for change capture. **Note:** This property does not apply to Oracle targets.
RAC Members	The maximum number of active redo log threads, or *members*, in an Oracle Real Application Cluster (RAC) that can be tracked. For a Data Guard physical standby database that supports a primary database in a RAC environment, this value is the number of active threads for the primary database. Valid values are 1 to 100. Default is 0, which causes an appropriate number of log threads to be determined automatically. If this value is not appropriate for your environment, set this property to a value greater than 0. **Note:** This property does not apply to Oracle targets.
BFILE Access	Select this check box in the following circumstances: - You use BFILE access to redo logs in physical directories on the local Oracle server file system. BFILE access uses Oracle directory objects to remotely access the redo logs in the file system. This method is an alternative to other log access methods such as ASM or NFS mounts. - You have an Amazon Relational Database Service (RDS) for Oracle source. In this case, this option enables access to the redo logs of a cloud-based database instance deployed in RDS. By default, this check box is cleared. **Note:** This property does not apply to Oracle targets.

Use the serverless runtime environment

You can use the serverless runtime environment with the Oracle Database Ingestion connector to connect to the Oracle database.

Before you configure an Oracle Database Ingestion connection using the serverless runtime environment, perform the following tasks:

- Prepare the Oracle OCI driver as a zip file.
- Add the Oracle OCI driver as a zip file in the Azure container in your Azure account.
- Configure the .yml serverless configuration file.

Prepare the Oracle OCI driver zip file

Perform the following steps:

1. Download the required version of the Oracle OCI drivers for Linux x86_64 (64 bit).
2. Extract the downloaded zip file on a Linux machine.
3. Add the `tnsnames.ora` file to the the `network/admin` folder within the internal folder in the extracted zip file.
4. Reassemble the zip file with the following command:

```
zip -yr <file_name>.zip <file_name>
```

Add the Oracle OCI driver as a zip file in the Azure container in your Azure account

Perform the following steps:

1. Create the following structure for the serverless agent configuration in Azure: `<Supplementary file location>/serverless_agent_config/oci`
2. Add the Oracle OCI driver as a zip file to an internal subfolder.

Configure the .yml serverless configuration file

Perform the following steps to configure the .yml serverless configuration file in the serverless runtime environment:

1. Copy the following code snippet to a text editor and specify the driver file name:

```
version: 1
agent:
  databaseIngestion:
    autoDeploy:
      oracle:
        oci:
          drivers:
          - fileCopy:
              sourcePath: oci/<file_name>.zip
```

   where the sourcePath specifies the directory path of the Oracle OCI drivers in Azure.

2. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the Oracle OCI drivers are copied from the Azure location to the serverless agent directory and installed in the following location:

   ```
   <AGENT_DIR/serverless/configurations/odbc/oracle-oci
   ```

   Note: The Database Ingestion agent service ociPath property value in the serverless runtime environment is defaulted with the following path:

   ```
   /home/cldagnt/SystemAgent/serverless/configurations/odbc/oracle-oci/
   ```

Prerequisites for Kerberos authentication

To use Kerberos authentication to connect to Oracle source or target databases, you must place some required configuration files on the Secure Agent machine and set some environment variables.

When you configure Kerberos authentication to connect to Oracle, consider the following guidelines:

- The Secure Agent and database server that you use must be registered in the KDC server.
- You can't add more than one KDC to a krb5.conf file. Multiple KDCs are not supported.
- You can't generate a credential cache file for more than one Kerberos principal user.
- Kerberos cross-realm authentication is not supported.
- The environment variables you define for Kerberos authentication must be consistent with the entries in the Oracle sqlnet.ora and tnsnames.ora files.

Configuring Kerberos authentication

Before you use Kerberos authentication to connect to an Oracle database on Linux or Windows, your organization administrator needs to create a few configuration files and set some environment variables.

1. Configure the Java Authentication and Authorization Service configuration file (JAAS) that the JDBC driver will use for Java client authentication.
   a. Create a JAAS configuration file on the Secure Agent machine.
   b. Add an entry to the JAAS configuration file that specifies the authentication technology to use for a particular driver. For example:

      ```
      JDBC_DRIVER_01 {
        com.sun.security.auth.module.Krb5LoginModule required
        useTicketCache=true
        principal="user@EXAMPLE.COM";
      };
      ```

      The Krb5LoginModule authenticates users by using Kerberos protocols. You can add LoginModule options such as useTicketCache and principal as needed. For more information, see the Oracle Java documentation at https://docs.oracle.com/javase/8/docs/technotes/guides/security/jgss/tutorials/LoginConfigFile.html.
2. Configure the Kerberos configuration file, krb5.conf, which defines Kerberos settings and realm details.
   a. Create a krb5.conf file on the Secure Agent machine.
   b. Add the details for the Key Distribution Center (KDC) and admin server to the krb5.conf file in the following format:

      ```
      [libdefaults]
       default_realm = <realm_name>

      [realms]
      <realm_name> = {
        kdc = <location where KDC is installed>
        admin_server = <location where KDC is installed>
       }
      ```

      Where [libdefaults] sets the default realm, and [realms] specifies the KDC and admin server for the realm.

      For example:

      ```
      [libdefaults]
       default_realm = EXAMPLE.COM

      [realms]
      EXAMPLE.COM = {
        kdc = rnd.EXAMPLE.COM
        admin_server = rnd.EXAMPLE.COM
       }
      ```

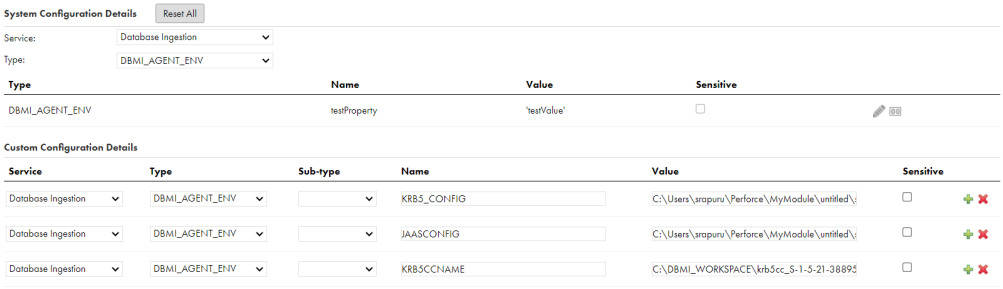
      For more information, see the Oracle documentation at https://docs.oracle.com/cd/E86824_01/html/E54775/krb5.conf-4.html.
3. Set the following environment variables on the machine where Data Ingestion and Replication and Secure Agent run:

   ```
   setenv JAASCONFIG <Absolute path of the JAAS config file>\<File name>.conf>
   setenv KRB5_CONFIG <Absolute path of the Kerberos configuration file>\krb5.conf>
   setenv KRB5CCNAME <Absolute path and file name of the credentials cache file>
   ```

   These variables are required to test Oracle Database Ingestion connections, deploy tasks, and run jobs when Kerberos authentication is in use.

Alternatively, you can specify these environment variables in Administrator for the Secure Agent. If you set environment variables in Administrator and on the Secure Agent machine, the variables that you specify in Administrator take precedence.

To define environment variables for the Seucre Agent in Administrator, go to **Runtime Environments**. Then open a Secure Agent and click **Edit**. Under **System Configuration Details** > **Custom Configuration Details**, enter the variables for the Database Ingestion service and DBMI_AGEN_ENV type. For example:



4. Restart the Secure Agent.
5. Generate the credential cache file by using the kinit or okinit tool.
   - To use the kinit tool on Windows, first install the MIT Kerberos Client and make sure that the KRB5CCNAME system environment variable or the default_ccache_name variable in the [libdefaults] section of krb5.conf is set. Then issue the following command to run the kinit tool. Enter the password when prompted.

     ```
     kinit user@<realm_name>
     ```
   - To use the kinit tool on Linux, first install the kinit tool and make sure that the KRB5CCNAME system environment variable is set. Then issue the following command to run the kinit tool. Enter the password when prompted.

     ```
     kinit user@<realm_name>
     ```
   - To use the okinit tool, first install the Oracle Instant Client. The credential cache file will be created based on the SQLNET.KERBEROS5_CC_NAME property in the sqlnet.ora file. Then issue the following command to run the okinit tool. Enter the password when prompted.

     ```
     okinit user@<realm_name>
     ```

Creating a Key Store Certificate

Create a keystore certificate when client authentication is enabled in the Oracle server. You must create a keystore certificate that contains all the client certificates to establish an Oracle connection.

Perform the following steps to create a keystore certificate:

1. Download and install the Oracle client from the Oracle website.
2. To create an Oracle wallet, run the following command:

   ```
   orapki wallet create -wallet <Path where wallet is to be created> -auto_login -pwd
   <wallet password>
   ```

   Example:

   ```
   C:\app\client\ksuwalka\product\12.1.0\client_1\BIN>orapki wallet create -wallet C:\app
   \client\ksuwalka\product\12.1.0\client_1\owm\wallet -auto_login -pwd oracle4u
   ```
3. To create a self-signed client certificate to the Oracle wallet, run the following command:

   ```
   orapki wallet add -wallet <Path where wallet is to be created> -dn "CN=<common
   name>, OU=<organization unit>, O=<organization>, L=<locality>, ST=<state>,
   C=<country>" -keysize <key size in bits> -self_signed-validity <number of days> -pwd
   <wallet password>
   ```

The command runs and creates the pkcs12 certificate at the specified location.
You must specify the values from the server certificate for CN, OU, O, L, ST, C, keysize, self signed - validity, and pwd.

Example:

```
C:\app\client\ksuwalka\product\12.1.0\client_1\BIN>orapki wallet add -wallet C:\app
\client\ksuwalka\product\12.1.0\client_1\owm\wallet -dn "CN=inw1pc07_kriti, OU=DEV,
O=infa,L=blr, ST=ka, C=IN" -keysize 2048 -self_signed -validity 3650 -pwd oracle4u
```

The `ewallet.p12` certificate is created in the following location: `C:\app\client\ksuwalka\product\12.1.0\client_1\owm\wallet`

4. To export the self-signed client certificate, run the following orapki command:

   ```
   orapki wallet export -wallet <wallet path> -dn "CN=<common name>, OU=<organization
   unit>, O=<organization>, L=<locality>, ST=<state>, C=<country>" -cert <Name of the
   exported certificate with path>
   ```

   The **-dn** command identifies the client certificate uniquely since the server wallet contains multiple installed client certificates.
   Example:

   ```
   C:\app\client\ksuwalka\product\12.1.0\client_1\BIN>orapki wallet export -wallet C:\app
   \client\ksuwalka\product\12.1.0\client_1\owm\wallet -dn "CN=inw1pc07_kriti, OU=DEV,
   O=infa,L=blr, ST=ka, C=IN" -cert C:\Users\ksuwalka\Desktop\client_inw1pc07.cert
   ```

5. Install the self-signed client certificate in the server's Oracle wallet.

   Note: The client authentication fails if you do not add the self-signed client certificate to the server database Oracle wallet.

6. To add the server certificate as a trusted certificate to the Oracle wallet, run the following command:

   ```
   orapki wallet add -wallet <wallet path> -trusted_cert -cert <Name of the server
   certificate with path> -pwd <wallet password>
   ```

   Note: You must use the same wallet password for all orapki commands.

   Example:

   ```
   C:\app\client\ksuwalka\product\12.1.0\client_1\BIN>orapki wallet add -wallet C:\app
   \client\ksuwalka\product\12.1.0\client_1\owm\wallet -trusted_cert -cert C:\SSL\oracle
   \oratls_server.cert -pwd oracle4u
   ```

   This enables you to use the keystore `C:\app\client\ksuwalka\product\12.1.0\client_1\owm\wallet\ewallet.p12` with keystore password `oracle4u`.

7. To extract the keys from the ewallet.p12 file generated in Step 6, run the following command:

   ```
   openssl pkcs12 -in ewallet.p12 -nocerts -nodes -out keys.pem -passin pass:<password>
   ```

8. To extract the certs from the ewallet.p12 file, run the following command:

   ```
   openssl pkcs12 -in ewallet.p12 -nokeys -out certs.pem -passin pass:<password>
   ```

9. To combine the keys and certs files into single file named keystore.p12, run the following command:

   ```
   openssl pkcs12 -export -in certs.pem -inkey keys.pem -out keystore.p12 -name
   "<aliasName>" -passout pass:<password>
   ```

   Note: You can use this keystore.p12 file with either the Progress DataDirect Oracle JDBC or Native Oracle JDBC driver.

Oracle Fusion Cloud Mass Ingestion connection properties

When you set up an Oracle Fusion Cloud Mass Ingestion connection, you must configure the connection properties.

Note: Oracle Fusion Cloud Mass Ingestion connections can access the data of only Enterprise Resource Planning (ERP), Human Capital Management (HCM), and Oracle Supply Chain and Manufacturing (SCM) modules of Oracle Fusion Cloud Applications Suite.

The following table describes the connection properties for an Oracle Fusion Cloud Mass Ingestion connection:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
Authentication	Authentication method of the connection. By default, the connection uses the Basic authentication method.
User Name	User name of the Oracle Cloud account.
Password	Password for the Oracle Cloud account.
Server URL	URL of the Oracle Cloud service that you want to access.
API Version	Version of the Oracle Cloud REST API that you want to use for the connection. Optional for the BICC replication approach.

PostgreSQL connection properties

Create a PostgreSQL connection to securely read data from or write data to PostgreSQL.

Prepare for authentication

You can configure database or Kerberos authentication method to connect to a PostgreSQL database.
Note: Data Ingestion and Replication does not support Kerberos authentication.

Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

To configure database authentication, you need the user name, password, host name, port, and database name from your PostgreSQL account.

To configure Kerberos authentication, you need the service principal name, host name, port, and database name from your PostgreSQL account. You'll also need to perform certain prerequisite tasks.

Connect to PostgreSQL

Let's configure the PostgreSQL connection properties to connect to PostgreSQL.

Before you begin

Before you get started, get the required information from your PostgreSQL account based on the authentication method that you want to use.

Check out "Prepare for authentication" on page 194 to learn more about the authentication prerequisites.

Connection details

The following table describes the PostgreSQL connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use a Secure Agent or serverless runtime environment. You can't run application ingestion and replication or database ingestion and replication tasks on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can configure database or Kerberos authentication methods to connect to PostgreSQL databases.

Note: For Data Ingestion and Replication, you must use database authentication.

Select the required authentication type and then configure the authentication-specific parameters.

Database authentication

To configure database authentication, you need the user name, password, host name, port, and database name from your PostgreSQL account.

The following table describes the basic connection properties for database authentication:

Property	Description
User Name	User name to access the PostgreSQL database.
Password	Password for the PostgreSQL database user name.
Host Name	Host name of the PostgreSQL server to which you want to connect.
Port	Port number for the PostgreSQL server to which you want to connect. Default is 5432.
Database Name	The PostgreSQL database name.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Schema Name	The schema name. If you don't specify the schema name, all the schemas available in the database are listed when you import the source object in Data Integration.
Connection Environment SQL	The SQL statement to set up the database environment when you connect to the database. The database environment applies for the entire session that uses this connection. For example, you can enter this statement to set the time zone: `SET timezone to 'America/New_York';`
Additional Connection Properties	Additional connection parameters that you want to use. Provide the connection parameters as semicolon-separated key-value pairs.

Encryption types

The encryption method determines if the Secure Agent and the PostgreSQL database server exchange encrypted data. If you do not want to establish a connection using SSL, select noEncryption. PostgreSQL establishes a connection without using SSL. Data is not encrypted. Default is noEncryption.

To use SSL, select the required encryption method and then configure the encryption-specific parameters.

SSL

When you use the SSL encryption method, data is encrypted using SSL. If the PostgreSQL database server can't configure SSL, the connection fails.

The following table describes the advanced connection properties for SSL encryption:

Note: You need to select the **Validate Server Certificate** check box for some SSL properties, while others need client authentication enabled on the PostgreSQL server.

Property	Description
Validate Server Certificate	Determines if the Secure Agent validates the server certificate sent by the PostgreSQL database server. If you specify the Host Name In Certificate property, the Secure Agent also validates the host name in the certificate. Select this option to validate the server certificate.
Truststore	This property applies if you select the Validate Server Certificate option. The path and name of the truststore file, which contains the list of the Certificate Authorities (CAs) that the PostgreSQL client trusts. For the serverless runtime environment, specify the following certificate path in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<TrustStore_filename>`
Truststore Password	This property applies if you select the Validate Server Certificate option. The password to access the truststore file that contains the SSL certificate.
Host Name In Certificate	Optional when you select the Validate Server Certificate option. A host name for providing additional security. The Secure Agent validates the host name included in the connection with the host name in the SSL certificate.
Keystore	This property applies when client authentication is enabled on the PostgreSQL database server. The path and the file name of the key store. The keystore file contains the certificates that the PostgreSQL client sends to the PostgreSQL server in response to the server's certificate request. For the serverless runtime environment, specify the following certificate path in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<KeyStore_filename>`
Keystore Password	This property applies when client authentication is enabled on the PostgreSQL database server. The password for the keystore file required for secure communication.
Key Password	This property applies when client authentication is enabled on the PostgreSQL database server. Required when individual keys in the keystore file have a different password than the keystore file.
Use SSLv3	Uses SSLv3 as the cryptographic protocol for an encrypted connection.
Use TLSv1.2	Uses TLSv1.2 as the cryptographic protocol for an encrypted connection.

Request SSL

When you use the requestSSL encryption method, PostgreSQL attempts to establish a connection using SSL. If the PostgreSQL database server can't configure SSL, the Secure Agent establishes an unencrypted connection.

The following table describes the advanced connection properties for Request SSL encryption:

Property	Description
Validate Server Certificate	Determines if the Secure Agent validates the server certificate sent by the PostgreSQL database server. If you specify the Host Name In Certificate property, the Secure Agent also validates the host name in the certificate. Select this option to validate the server certificate.
Truststore	This property applies if you select the Validate Server Certificate option. The path and name of the truststore file, which contains the list of the Certificate Authorities (CAs) that the PostgreSQL client trusts. For the serverless runtime environment, specify the following certificate path in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<TrustStore_filename>`
Truststore Password	This property applies if you select the Validate Server Certificate option. The password to access the truststore file that contains the SSL certificate.
Host Name In Certificate	Optional when you select the Validate Server Certificate option. A host name for providing additional security. The Secure Agent validates the host name included in the connection with the host name in the SSL certificate.
Use SSLv3	Uses SSLv3 as the cryptographic protocol for an encrypted connection.
Use TLSv1.2	Uses TLSv1.2 as the cryptographic protocol for an encrypted connection.

Configure SSL with serverless runtime environment

You can use the serverless runtime environment with PostgreSQL Connector to connect to an SSL-enabled PostgreSQL database.

Before you configure a secure PostgreSQL connection using the serverless runtime environment, complete the following prerequisite tasks to add the SSL certificates to the serverless runtime location:

1. Create the following structure for the serverless agent configuration on AWS or Azure: `<Supplementary file location>/serverless_agent_config`

   Note: Data Ingestion and Replication does not support serverless runtime environments hosted on AWS.
2. Add the truststore and keystore certificates in the Amazon S3 bucket or Azure container in the following location in your AWS or Azure account: `<Supplementary file location>/serverless_agent_config/SSL`
3. Copy the following code snippet to a text editor:

```
version: 1
agent:
  agentAutoApply:
    general:
      sslStore:
        - fileCopy:
            sourcePath: SSL/<TrustStore_filename>
        - fileCopy:
            sourcePath: SSL/<KeyStore_filename>
```

   where the sourcePath is the directory path of the certificate files on AWS or Azure.

4. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following AWS or Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the SSL certificates are copied from the AWS or Azure location to the serverless agent directory.
5. In the PostgreSQL connection properties, specify the following certificate path in the serverless agent directory in the **Trust Store** and **Key Store** fields: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<cert_filename>`

REST V2 connection properties

Create a REST V2 connection to interact with web service applications built on REST architecture.

Prerequisites

Before you configure a REST V2 connection, be sure to complete the prerequisites.

- Install the Secure Agent on a 64-bit machine.
- Ensure that the machine hosting the Secure Agent has a minimum memory size of 2048 MB.

Connect to REST V2

Let's configure the REST V2 connection properties to interact with web service applications built on REST architecture.

Before you begin

Before you get started, be sure to complete the prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	Name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, serverless, or elastic runtime environment. You cannot run a streaming ingestion and replication task on a Hosted Agent, serverless runtime environment, or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can configure standard, OAuth 2.0 client credentials, OAuth 2.0 authorization code, JWT bearer token, and API key authentication types to connect to a REST endpoint.

Select the required authentication method and then configure the authentication-specific parameters.

Standard authentication

Standard authentication requires an authentication user ID and password to connect to a REST endpoint. When you configure a standard authentication type, you can further configure basic and OAuth authentication types.

Note: Digest authentication is not applicable.

The following table describes the basic connection properties for standard authentication:

Property	Description
Authentication Type	The authentication type that you can use when you select the Standard authentication. You can select one of the following authentication types: - BASIC - OAUTH - NONE Default is NONE.
Auth User ID	The user name to log in to the web service application when you select the standard authentication. Required for Basic authentication type.

Property	Description
Auth Password	The password associated with the user name when you select the standard authentication. Required for Basic authentication type.
OAuth Consumer Key	The client key associated with the web service application. Required only for OAuth authentication type.
OAuth Consumer Secret	The client password to connect to the web service application. Required only for OAuth authentication type.
OAuth Token	The access token to connect to the web service application. Required only for OAuth authentication type.
OAuth Token Secret	The password associated with the OAuth token. Required only for OAuth authentication type.
Swagger File Path	The path of the Swagger file or OpenAPI file. You can specify one of the following file paths: - Path and file name of the Swagger or OpenAPI file on the Secure Agent machine. - The URL on which the Swagger or OpenAPI file is hosted. The hosted URL must return the content of the file without prompting for further authentication and redirection. For example, the path of the Swagger file can be: `C:\Swagger\sampleSwagger.json` The user must have the read permission for the folder and the file.

OAuth 2.0 Client Credentials authentication

OAuth 2.0 client credentials authentication requires at a minimum the client ID, access token URL, client secret, scope, and the access token.

The following table describes the basic connection properties for OAuth 2.0 client credentials authentication:

Property	Description
Access Token URL	Access token URL configured in your application.
Client ID	Client ID of your application.
Client Secret	Client secret of your application.
Scope	Specifies access control if the API endpoint has defined custom scopes. Enter scope attributes, each separated by a space. For example: `root_readonly root_readwrite manage_app_users`

Property	Description
Access Token Parameters	Additional parameters to use with the access token URL. Define the parameters in the JSON format. For example, `[{"Name":"resource","Value":"https://<serverName>"}]`
Client Authentication	Select an option to send the client ID and client secret for authorization either in the request body or in the request header. Default is **Send Client Credentials in Body**.
Generate Access Token	Generates an access token based on the information provided in the above fields.
Access Token	The access token value. Enter the access token value or click **Generate Access Token** to populate the access token value. To pass the generate access token call through a proxy server, you must configure an unauthenticated proxy server in the Secure Agent properties. The proxy configured in the connection configuration does not apply to the generate access token call.
Swagger File Path	The path of the Swagger file or OpenAPI file. You can specify one of the following file paths: - Path and file name of the Swagger or OpenAPI file on the Secure Agent machine. - The URL on which the Swagger or OpenAPI file is hosted. The hosted URL must return the content of the file without prompting for further authentication and redirection. For example, the path of the swagger file can be: `C:\swagger\sampleSwagger.json` The user must have the read permission for the folder and the file. **Note:** In a streaming ingestion and replication task, use only a hosted URL of the swagger specification file as the swagger file path.

OAuth 2.0 Authorization Code authentication

To use authorization code authentication, register the following Informatica redirect URL in your application:

`https://<Informatica cloud hosting facility for your organization>/ma/proxy/oauthcallback`

If the access token expires and the error codes `400`, `401`, and `403` are returned in the response, Informatica redirect URL, which is outside the customer firewall, tries to connect to the endpoint and retrieves a new access token.

The following table describes the basic connection properties for OAuth 2.0 authorization code authentication:

Property	Description
Authorization Token URL	Authorization server URL configured in your application.
Access Token URL	Access token URL configured in your application.
Client ID	Client ID of your application.
Client Secret	Client secret of your application.
Scope	Specifies access control if the API endpoint has defined custom scopes. Enter scope attributes, each separated by a space. For example, `root_readonly root_readwrite manage_app_users`
Access Token Parameters	Additional parameters to use with the access token URL. Define the parameters in the JSON format. For example, `[{"Name":"resource","Value":"https://<serverName>"}]`
Authorization Code Parameters	Additional parameters to use with the authorization token URL. Define the parameters in the JSON format. For example, `[{"Name":"max_age","Value":60},{"Name":"state","Value":"test"}]`
Client Authentication	Select an option to send the client ID and client secret for authorization either in the request body or in the request header. Default is **Send Client Credentials in Body**.
Generate Access Token	Generates an access token and refreshes the token based on the information provided in the above fields.
Access Token	The access token value. Enter the access token value or click **Generate Access Token** to populate the access token value. To pass the generate access token call through a proxy server, you must configure an unauthenticated proxy server in the Secure Agent properties. The proxy configured in the connection configuration does not apply to the generate access token call.

Property	Description
Refresh Token	The refresh token value. Enter the refresh token value or click **Generate Access Token** to populate the refresh token value. If the access token is not valid or expires, the Secure Agent fetches a new access token with the help of refresh token. If the refresh token expires, you must either provide a valid refresh token or regenerate a new refresh token by clicking **Generate Access Token**.
Swagger File Path	The path of the Swagger file or OpenAPI file. You can specify one of the following file paths: - Path and file name of the Swagger or OpenAPI file on the Secure Agent machine. - The URL on which the Swagger or OpenAPI file is hosted. The hosted URL must return the content of the file without prompting for further authentication and redirection. For example, the path of the swagger file can be: `C:\swagger\sampleSwagger.json` The user must have the read permission for the folder and the file. **Note:** In a streaming ingestion and replication task, use only a hosted URL of the swagger specification file as the swagger file path.

JWT bearer token authentication

JWT bearer token authentication requires at a minimum the JWT header, JWT payload, and authorization server URL.

The following table describes the basic connection properties for JWT bearer token authentication:

Property	Description

JWT Header	JWT header in JSON format.
	Sample:
	{
	"alg":"RS256",
	"kid":"xxyyzz"
	}
	You can configure HS256 and RS256 algorithms.

JWT Payload	JWT payload in JSON format.
	Sample:
	{
	"iss":"abc",
	"sub":"678",
	"aud":"https://api.box.com/oauth2/token",
	"box_sub_type":"enterprise",
	"exp":"120",
	"jti":"3ee9364e"
	}
	The expiry time represented as **exp** is the relative time in seconds. The expiry time is calculated in the UTC format from the token issuer time (iat).
	When iat is defined in the payload and the expiry time is reached, mappings and Generate Access Token fails. To generate a new access token, you must provide a valid iat in the payload.
	If iat is not defined in the payload, the expiry time is calculated from the current timestamp.
	To pass the expiry time as a string value, enclose the value with double quotes. For example:
	"exp":"120"
	To pass the expiry time as an integer value, do not enclose the value with double quotes.
	For example,
	"exp":"120"

Authorization Server	Access token URL configured in your application.
KeyStore File Path	The absolute path of the keystore file that contains the keys and certificates required to establish a two-way secure communication with the REST API. Specify a directory path that is available on each Secure Agent machine. You can also configure the keystore file name and path as a JVM option or import the certificate to any directory. For the serverless runtime environment, specify the keystore file path in the serverless agent directory. For example, /home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<cert_name>.jks

Property	Description
KeyStore Password	The password for the keystore file required for a secure communication. You can also configure the keystore password as a JVM option.
Private Key Alias	Alias name of the private key used to sign the JWT payload.
Private Key Password	The password for the keystore file required for a secure communication. The private key password must be same as the keystore password.
Access Token	The access token value. Enter the access token value or click **Generate Access Token** to populate the access token value. To pass the generate access token call through a proxy server, you must configure an unauthenticated proxy server in the Secure Agent properties. The proxy configured in the connection configuration does not apply to the generate access token call.
Swagger File Path	The path of the Swagger file or OpenAPI file. You can specify one of the following file paths: - Path and file name of the Swagger or OpenAPI file on the Secure Agent machine. - The URL on which the Swagger or OpenAPI file is hosted. The hosted URL must return the content of the file without prompting for further authentication and redirection. For example, the path of the swagger file can be: `C:\swagger\sampleSwagger.json` The user must have the read permission for the folder and the file. **Note:** In a streaming ingestion and replication task, use only a hosted URL of the swagger specification file as the swagger file path.

Advanced settings

The following table describes the advanced connection properties for JWT bearer token authentication:

Property	Description
Authorization Advanced Properties	Additional parameters to use with the access token URL. Parameters must be defined in the JSON format. For example, `[\{"Name":"client_id","Value":"abc"},\ {"Name":"client_secret","Value":"abc"}]`
TrustStore File Path	The absolute path of the truststore file that contains the TLS certificate to establish a one-way or two-way secure connection with the REST API. Specify a directory path that is available on each Secure Agent machine. You can also configure the truststore file name and password as a JVM option or import the certificate to the following directory: `<Secure Agent installation directory\jre\lib \security\cacerts`. For the serverless runtime environment, specify the truststore file path in the serverless agent directory. For example, `/home/cldagnt/SystemAgent/ serverless/configurations/ssl_store/ <cert_name>.jks`
TrustStore Password	The password for the truststore file that contains the SSL certificate. You can also configure the truststore password as a JVM option.
Proxy Type	Type of proxy. Select one of the following options: - No Proxy. Bypasses the proxy server configured in the agent or the connection properties. - Platform Proxy. Considers the proxy configured in the agent. - Custom Proxy. Considers the proxy configured in the connection properties.

Property	Description
Proxy Configuration	The format required to configure proxy. You can configure proxy using the following format: `<host>:<port>` You cannot configure an authenticated proxy server.
Advanced Fields	Enter the arguments that the agent uses when connecting to a REST endpoint. When you specify multiple arguments, separate each argument by a semicolon. For example, `connectiondelaytime:10000;retryattempts:5` You can specify the following arguments: - **ConnectionTimeout**. The wait time in milliseconds to get a response from a REST endpoint. The connection ends after the connection timeout is over. Default is the timeout defined in the endpoint API. **Note:** If you define both the REST V2 connection timeout and the endpoint API timeout, the connection ends at the shortest defined timeout. - **connectiondelaytime**. The delay time in milliseconds to send a request to a REST endpoint. Default is 10000. - **retryattempts**. Number of times the connection is attempted when 400 and 500 series error codes are returned in the response. Default is 3. Specify 0 to disable the retry attempts. - **qualifiedSchema**. Determines if the schema selected is qualified or unqualified. Default is false. **Note:** In a streaming ingestion and replication task, only `ConnectionTimeout` and `retryattempts` are applicable.

API key authentication

API key authentication allows you to provide a unique key and a corresponding value to authenticate API calls made to the REST endpoint.

The following table describes the basic connection properties for API key authentication:

Property	Description
Key	The unique API key that REST V2 Connector uses to authenticate the API calls made to the REST endpoint.
Value	The value corresponding to the API key that is required to make the API calls.

Property	Description
Add API Key to	Determines if the API key and its corresponding value must be sent as a request header or a query parameter to make API calls to the REST endpoint. Select one of the following options: - Request Header - Query Parameter
Swagger File Path	The path of the Swagger file or OpenAPI file. You can specify one of the following file paths: - Path and file name of the Swagger or OpenAPI file on the Secure Agent machine. - The URL on which the Swagger or OpenAPI file is hosted. The hosted URL must return the content of the file without prompting for further authentication and redirection. For example, the path of the swagger file can be: `C:\swagger\sampleSwagger.json` The user must have the read permission for the folder and the file. **Note:** In a streaming ingestion and replication task, use only a hosted URL of the swagger specification file as the swagger file path.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
TrustStore File Path	The absolute path of the truststore file that contains the TLS certificate to establish a one-way or two-way secure connection with the REST API. Specify a directory path that is available on each Secure Agent machine. You can also configure the truststore file name and password as a JVM option or import the certificate to the following directory: `<Secure Agent installation directory\jre\lib\security\cacerts.` For the serverless runtime environment, specify the truststore file path in the serverless agent directory. For example, `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<cert_name>.jks`
TrustStore Password	The password for the truststore file that contains the SSL certificate. You can also configure the truststore password as a JVM option.
KeyStore File Path	The absolute path of the keystore file that contains the keys and certificates required to establish a two-way secure communication with the REST API. Specify a directory path that is available on each Secure Agent machine. You can also configure the keystore file name and path as a JVM option or import the certificate to any directory. For the serverless runtime environment, specify the keystore file path in the serverless agent directory. For example, `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<cert_name>.jks`

Property	Description
KeyStore Password	The password for the keystore file required for secure communication. You can also configure the keystore password as a JVM option.
Proxy Type	Type of proxy. Select one of the following options: - No Proxy: Bypasses the proxy server configured in the agent or the connection properties. - Platform Proxy: Considers the proxy configured in the agent. - Custom Proxy: Considers the proxy configured in the connection properties.
Proxy Configuration	The format required to configure proxy. Configure proxy using the following format: `<host>:<port>` You cannot configure an authenticated proxy server.
Advanced Fields	Enter the arguments that the agent uses when connecting to a REST endpoint. When you specify multiple arguments, separate each argument by a semicolon. For example, `connectiondelaytime:10000;retryattempts:5` You can specify the following arguments: - ConnectionTimeout. The connection wait time in milliseconds to get a response from a REST endpoint. The connection ends after the connection timeout is over. Default is the timeout defined in the endpoint API. **Note:** If you define both the REST V2 connection timeout and the endpoint API timeout, the connection ends at the shortest defined timeout. REST V2 Connector retrieves the Swagger file content by calling the hosted URL defined in the connection. If the endpoint is temporarily unavailable, you can configure the connection wait time during which REST V2 Connector continuously attempts to connect to the endpoint, until the configured time elapses. - connectiondelaytime. The wait time in milliseconds to send a request to a REST endpoint. Default is 10000. - retryattempts. Number of times the connection is attempted when 400 and 500 series error codes are returned in the response. Default is 3. Specify 0 to disable the retry attempts. - qualifiedSchema. Determines if the schema selected is qualified or unqualified. Default is false. - encodeQueryParams. Determines if query parameters in the URL must be encoded when REST V2 Connector makes API calls to a REST endpoint. Set this argument to false to skip encoding query parameters. Default is true. - OAuth2RefreshURLTimeout. The connection wait time in milliseconds to get a new access token. When an access token expires, REST V2 Connector calls a REST API endpoint to receive a new access token. If the endpoint is temporarily unavailable, you can configure the connection wait time during which REST V2 Connector continuously attempts to connect to the endpoint, until the configured time elapses. **Note:** In a streaming ingestion and replication task, only `ConnectionTimeout` and `retryattempts` are applicable.

Secure communication with TLS authentication

Configure TLS authentication to establish one-way or two-way secure communication between the Secure Agent and the REST API over TLS.

To establish one-way secure communcation, perform the following steps:

1. Generate the truststore. For more information on the steps, see *Generate a Truststore*.
2. Configure the REST V2 connection for one-way SSL. You can specify the truststore file and truststore password in the connection, or set them in the JVM options of the Secure Agent.

To establish two-way secure communcation, you must first configure one-way secure communication, and then perform the following steps:

1. Generate the keystore. For more information on the steps, see *Generate a Keystore*.
2. Configure the REST V2 connection for two-way SSL. You can specify the keystore file and keystore password in the connection, or set them in the JVM options of the Secure Agent.

If you specify keystore and truststore properties in the connection and in the JVM options, the Secure Agent processes the certificates based on the properties configured in the connection.

Generate a truststore

To generate a truststore, you need a server certificate. Get the server certificate and perform the following steps to generate the truststore:

1. Import the server certificate to either of the following directories available within your Secure Agent installation:
   - `<Secure Agent installation directory>\jdk\jre\lib\security\cacerts`
   - `<Secure Agent installation directory>\jdk8\jre\lib\security\cacerts`
2. To generate the truststore, run the following command from the command line:
   `keytool -importcert -alias <Specify alias name here> -file <Specify server certificate here> -keystore <Specify the name of custom truststore to be generated> -storepass <Specify password for the custom truststore>`

   For example, `keytool -importcert -alias RESTV2CACert -file ca.pem -keystore sampletruststore -storepass JKSTrustStorePassword`

   In the example, a truststore file is generated by the name *sampletruststore* and password *JKSTrustStorePassword*.

Generate a keystore

To generate a keystore, you need a client certificate and a client private key. Get the client certificate and client private key, and then perform the following steps to generate the keystore:

1. Import the server certificate to either of the following directories available within your Secure Agent installation:
   - `<Secure Agent installation directory>\jdk\jre\lib\security\cacerts`
   - `<Secure Agent installation directory>\jdk8\jre\lib\security\cacerts`

2. To generate the keystore, run the following command from the command line:
   `openssl pkcs12 -export -in <Specify client certificate here> -inkey <Specify client private key here> -name "<Specify any name here>" -passout pass:<Specify password for the keystore to be generated> -out <Specify name for the keystore with p12 extension>`

   For example, `openssl pkcs12 -export -in /home/samplefolder/certs/client-cert.pem -inkey /home/samplefolder/certs/client-key.pem -name "restclient" -passout pass:PKCSKeyStorePassword -out samplekeystore.p12`

   In the example, a keystore file by the name *samplekeystore.p12* is generated in the PKCS12 format.

   To convert the keystore file from .p12 format to .jks format, run the following command from the command line:

   `keytool -importkeystore -srckeystore <Specify name of the p12 keystore file> -srcstoretype pkcs12 -srcstorepass <Specify password for generated p12 keystore file> -destkeystore <Specify name for the JKS keystore file> -deststoretype JKS -deststorepass <Specify password for the JKS keystore file>`

   Note: Ensure that the password specified in `-srcstorepass` must be the same as the `-deststorepass`.

   For example, `keytool -importkeystore -srckeystore samplekeystore.p12 -srcstoretype pkcs12 -srcstorepass PKCSKeyStorePassword -destkeystore keystore -deststoretype JKS -deststorepass PKCSKeyStorePassword`

   In the example, a keystore file is generated by the name *samplekeystore* and password *PKCSKeyStorePassword*.

Configuring one-way or two-way secure communication

You can configure a connection for one-way or two-way SSL.

Configuring the connection for one-way SSL

You can either specify the name of the truststore file and truststore password in the TrustStore File Name and TrustStore Password fields in the connection properties. Alterntaively, you can set the truststore file name and truststore password in the JVM options in the Secure Agent properties.

1. Click **Administrator** > **Runtime Environments**, and select an agent.
2. Select **Type** as DTM under **System Configuration Details**.
3. Add the following JVM options:
   - `JVMOption1=-Djavax.net.ssl.trustStore=<absolute path of the .jks truststore file>`
   - `JVMOption2=-Djavax.net.ssl.trustStorePassword=<truststore password>`

Configuring the connection for two-way SSL

You can either specify the name of the keystore file and keystore password in the KeyStore File Name and KeyStore Password connection properties. Alternatively, you can set the keystore file and keystore password in the JVM options in the Secure Agent properties.

To use two-way SSL, you must first configure one-way SSL, and then perform the following steps to configure two-way SSL:

1. Click **Administrator** > **Runtime Environments**, and select an agent.
2. Select **Type** as DTM under **System Configuration Details**.
3. Add the following JVM options:
   - `JVMOption3=-Djavax.net.ssl.keyStore=<absolute path of the .jks keystore file>`

- JVMOption4=-Djavax.net.ssl.keyStorePassword=*<keystore password>*

Secure communication in a serverless runtime environment

When you use the serverless runtime environment, you can configure TLS authentication and establish one-way or two-way secure communication with the REST API.

Ensure that the certificates are in the `.jks` format.

To configure a secure REST V2 connection using the serverless runtime environment, complete the following prerequisite tasks to add the TLS certificates to the serverless runtime location:

1. Create the following structure for the serverless agent configuration on AWS:
   `<Supplementary file location>/serverless_agent_config`
2. For one-way secure communication, add the truststore certificates and for the two-way secure communication, add the truststore and keystore certificates in the Amazon S3 bucket in the following location in your AWS account:
   `<Supplementary file location>/serverless_agent_config/SSL`
3. Copy the following code snippet to a text editor:

```
version: 1
agent:
  agentAutoApply:
    general:
      sslStore:
        - fileCopy:
            sourcePath: SSL/<RESTV2_trustStore_cert_name>.jks
        - fileCopy:
            sourcePath: SSL/<RESTV2_keyStore_cert_name>.jks
```

   where the source path is the directory of the certificate files in AWS.
4. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following AWS location:
   `<Supplementary file location>/serverless_agent_config`

   When the .yml file runs, the SSL certificates are copied from the AWS location to the serverless agent directory.
5. In the REST V2 connection properties, specify the following certificate path in the serverless agent directory in the **TrustStore File Path** and **KeyStore File Path** fields:
   `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<cert_name>.jks`

Swagger specification file in a serverless runtime environment

To configure a swagger file in a serverless runtime environment, be sure to complete the prerequisites.

In the serverless runtime environment, you can configure a swagger file in one of the following ways:

- Provide the swagger file public hosted URL in the **Swagger File Path** connection property. Ensure that the URL must return the content of the file without prompting for further authentication and redirection.
- Place the swagger file in the serverless agent directory.

To configure a swagger file in a serverless runtime environment, complete the following prerequisite tasks to add the swagger file to the serverless runtime location:

1. Create the following structure for the serverless agent configuration in AWS or Azure:
   `<Supplementary file location>/serverless_agent_config`

2. Add the swagger specification file in the Amazon S3 bucket or Azure container in the following location in your AWS or Azure account:
   `<Supplementary file location>/serverless_agent_config/restv2`
   a. Copy the following code snippet to a text editor:

```
version: 1
agent:
  dataIntegrationServer:
    autoApply:
      restv2:
        swaggers:
          - fileCopy:
              sourcePath: restv2/<swagger_file_name1>.json
          - fileCopy:
              sourcePath: restv2/<swagger_file_name2>.json
```

   where the source path is the directory path of the swagger files in AWS or Azure.
3. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following AWS or Azure location:
   `<Supplementary file location>/serverless_agent_config`

   When the .yml file runs, the SSL certificates are copied from the AWS or Azure location to the serverless agent directory.
4. In the REST V2 connection properties, specify the following swagger path in the serverless agent directory in the **Swagger File Path** field:
   `/home/cldagnt/SystemAgent/serverless/configurations/restv2/<swagger_file_name>.json`

For more information about the serverless runtime environment, see *Serverless Runtime Environments* in the Administrator help.

Rules and guidelines for runtime environment

Consider the following guidelines when you run tasks in different runtime environments:

- You cannot use a proxy server to connect to IDMC when you use the Hosted Agent or the serverless runtime environment.
- You cannot connect to the REST API endpoints that require custom server certificate signed by CA and are not a part of Informatica cacerts truststore, when you use the Hosted Agent.
- You cannot configure JWT bearer token authentication when you use the Hosted Agent.
- Ensure that the swagger specification file URL is a public URL and returns the content of the file without prompting for further authentication and redirection when you use the Hosted Agent.

Rules and guidelines for a REST V2 connection

Consider the following rules and guidelines for a Rest V2 connection:

- When you test the connection, the Secure Agent validates the following parameters:
  - Path of the local Swagger file or the URL of the hosted Swagger file.
  - JSON format of the Swagger file.

  However, the Secure Agent does not validate endpoint credentials when you test the connection.

- You can configure proxy at the agent level or connection level. See the following table to understand the proxy settings that take precedence when you define the System proxy and proxy at the connection level:

System proxy	REST V2 connection attribute			Result
	No proxy	Platform proxy	Custom proxy	
No	Yes	No	No	Does not consider proxy.
No	No	Yes	No	Does not consider proxy.
No	No	No	Yes	Considers custom proxy.
Yes	Yes	No	No	Does not consider proxy.
Yes	No	Yes	No	Considers platform proxy.
Yes	No	No	Yes	Considers custom proxy.

Salesforce Data 360 connection properties

Create a Salesforce Data 360 connection to load data to Salesforce Data 360.

Connect to Salesforce Data 360

Let's configure the Salesforce Data 360 connection properties to connect to Salesforce Data 360.

Before you begin

Before you get started, make sure that you complete the following prerequisites:

- Create an external client app in Salesforce Data 360, and assign at least the following scopes to the app:
  - Manage user data via APIs (api)
  - Manage Data 360 Ingestion API data (cdp_ingest_api)
  - Perform ANSI SQL queries on Data Cloud data (cdp_query_api)

  For more information about configuring your external client app, see Salesforce Data 360 Quick Start.
- Get the consumer key and consumer secret from the external client app to generate the access token.
- Create an Ingestion API data stream to load data to an existing target.
  For more information about Ingestion API data stream, see Ingestion API.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	This property appears only if secrets manager is set up for your organization. However, this property doesn't apply to a Salesforce Data 360 connection.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, use a Secure Agent. You can't run a task on a Hosted Agent or on a serverless runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Authentication Type	The authentication method to connect to Salesforce Data 360. Default is OAuth 2.0 client credentials.
Access Token URL	The endpoint where OAuth 2.0 requests are sent to obtain an access token to connect to the Salesforce Data 360 instance. Specify the URL in the following format: `https://<Salesforce Data 360 organization ID>.salesforce.com/services/oauth2/token`
Client ID	Client ID of your application to connect to Salesforce Data 360.
Client Secret	The client secret associated with the client ID.
Access Token	The access token value. Click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Retry Count	The total number of retries to get the response from the Salesforce Data 360 endpoint based on the HTTP status codes. Enter a positive integer. Default is 3.
Retry Interval	The wait time in milliseconds before the Salesforce Data 360 connection makes another attempt to receive a response. Enter a positive integer. Default is 5000 milliseconds.

Additional Properties	Additional properties that you can configure in the Salesforce Data 360 connection, such as specifying a custom data space, defining properties for connections and data streams.
	You can specify multiple properties, separated by an ampersand and colon (&:), in the following format:
	`<key1>=<value>&:<key2>=<value>&:<key3>=<value>`
	For example, you can specify a custom data space when you choose query as your source object:
	`customQueryDataSpace=My_Dataspace`
	For the list of properties that you can configure, see "Configure additional properties" on page 217.

Configure additional properties

You can use the additional properties field in the advanced settings to customize and set any additional properties in a Salesforce Data 360 connection.

Specify a custom data space.

- **customQueryDataSpace.** A custom data space that you can use when you choose query as your source object.
  For example, `customQueryDataSpace=My_Dataspace`

  If you do not specify a data space, the default data space of your organization is considered.

Configure status codes and timeout properties.

- **okhttp.retryStatusCodes.** A list of HTTP status codes that indicate transient errors for which a request should be retried.
  Default codes are 429,500,502,503,504.
- **okhttp.connectTimeout.** The maximum wait time in seconds that the Salesforce Data 360 connection waits for a response from the server.
  Default is 30 seconds.
- **okhttp.readTimeout.** The maximum wait time in seconds when an attempt is made to read a file from the server.
  Default is 30 seconds.
- **okhttp.writeTimeout.** The maximum wait time in seconds when an attempt is made to write a file to the server.
  Default is 30 seconds.

Configure data stream ingestion properties.

- **dataStreamIngestReadyInitialPollIntervalSeconds.** The initial poll interval duration in seconds that the Salesforce Data 360 Connector waits before making the first attempt to fetch the refresh date and time for the data stream.
  Default is 120 seconds.
- **dataStreamIngestReadyMaxWaitTimeSeconds.** The maximum wait time in seconds that the Salesforce Data 360 Connector will wait for the data stream to become ready to accept incoming data during the initial data load into the data lake object.
  Default is 1200 seconds.

Configure data stream active time properties.

- **dataStreamActiveInitialPollIntervalSeconds.**The initial poll interval duration in seconds that the Connector waits before making the first attempt to fetch the data stream status.
  Default is 1 second.
- **dataStreamActiveMaxWaitTimeSeconds.**The maximum wait time in seconds for the data stream status to transition to Active after successful creation.
  Default is 60 seconds.

For example,

```
okhttp.retryStatusCodes=429,500,502,503,504&:okhttp.connectTimeout=30&:dataStreamActiveMaxWaitTimeSeconds=60&:dataStreamActiveInitialPollIntervalSeconds=1
```

Salesforce Marketing Cloud connection properties

When you set up a Salesforce Marketing Cloud connection, configure the connection properties.

The following table describes the Salesforce Marketing Cloud connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.

Property	Description
Runtime Environment	The name of the runtime environment where you want to run the tasks. For application ingestion and replication tasks, you can select a Secure Agent, or for initial load jobs only, you can select a serverless runtime environment. You can't use a Hosted Agent or an elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Salesforce Marketing Cloud Url	The URL that the agent uses to connect to the Salesforce Marketing Cloud WSDL.
Username	Applies to basic authentication. The user name of the Salesforce Marketing Cloud account. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.
Password	Applies to basic authentication. The password for the Salesforce Marketing Cloud account. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.
Client ID	The client ID of Salesforce Marketing Cloud required to generate a valid access token.
Client Secret	The client secret of Salesforce Marketing Cloud required to generate a valid access token.
Use Proxy Server	Connects to Salesforce Marketing Cloud though proxy. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.
Enable Logging	Enables logging for the task. When you enable logging, you can view the session log for the log details. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.
UTC offset	Uses the UTC offset connection property to read data from and write data to Salesforce Marketing Cloud in the UTC offset time zone. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.
Batch Size	Number of rows that the agent writes in a batch to the target. When you insert or update data and specify the contact key, the data associated with the specified contact ID is inserted or updated in a batch to Salesforce Marketing Cloud. When you upsert data to Salesforce Marketing Cloud, do not specify the contact key. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.
Enable Multiple BU	Uses the Salesforce Marketing Cloud connection to access data across all business units. Select this option if there are multiple business units in your Salesforce Marketing Cloud account. **Note:** This property is not applicable to connections configured for application ingestion and replication tasks.

Salesforce Mass Ingestion connection properties

When you set up a Salesforce Mass Ingestion connection, configure the connection properties.

Before you configure the connection, perform the following tasks based on the authentication that you want to use:

- Create an app in Salesforce to access data. For OAuth 2.0 Username-Password Flow, create a connected app. For OAuth 2.0 JWT Bearer Flow, you can create either a connected app or an External Client App (ECA). For more information, see Create a Connected App or Create an External Client App.

  Important: You can no longer create new connected apps in Salesforce. Existing connected apps continue to work, but you cannot create new ones.

- To use OAuth 2.0 JWT Bearer Flow with the ECA, perform the following tasks:
  - Configure the JWT Bearer OAuth flow for the ECA. For more information, see Configure OAuth 2.0 JWT Bearer Flow for External Client Apps.
  - Add the required OAuth scopes to the ECA to avoid connection test failures. At a minimum, include the *Manage user data via APIs (api)* and *Perform requests at any time (refresh_token, offline_access)* scopes.

The properties of a Salesforce Mass Ingestion connection vary based on the authentication method that you specify for the connection. When you create a connection, you can select one of the following authentication methods:

- **OAuth 2.0 Username-Password Flow**: Authenticates the connection by using the Salesforce account login credentials and the consumer key and consumer secret that Salesforce generates for the connected app.

  Note: Effective in the October 2026 release, OAuth 2.0 Username-Password Flow authentication in Salesforce Mass Ingestion Connector is deprecated because Salesforce is retiring the SOAP API login() call in API versions 31.0 through 64.0 to align with enhanced cybersecurity standards. For more information, see Platform SOAP API login() Retirement. Informatica intends to drop support for OAuth 2.0 Username-Password Flow on May 1, 2027. In the interim, transition to OAuth 2.0 JWT Bearer Flow authentication.

  Deprecation Notice: Deprecated functionality is supported, but Informatica intends to drop support in a future release. Informatica requests that you transition to different functionality before the functionality is dropped.

- **OAuth 2.0 JWT Bearer Flow**: Authenticates the connection by using the Salesforce account user name, private key alias, private key password, and the consumer key that Salesforce generates for the connected app or ECA. Informatica recommends that you use this authentication method because this method provides secured access to Salesforce without sharing sensitive information, such as consumer secret and Salesforce account password.

Connection properties for OAuth 2.0 Username-Password Flow authentication

The following table describes the connection properties for a Salesforce Mass Ingestion connection configured with OAuth 2.0 Username-Password Flow authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
User Name	User name of the Salesforce account.
Password	Password for the Salesforce account.
Security Token	Security token associated with the Salesforce account. You can configure the connection without specifying the security token if there are no IP restrictions specified for the connected app. However, you must specify the security token if IP restrictions are enforced for the app and if the Secure Agent is not running on the trusted IP range specified for your Salesforce organization. **Note:** If you do not have the security token, reset the security token in Salesforce. For more information about resetting the security token, see the Salesforce documentation.
Consumer Key	Consumer key that Salesforce generates when you enable OAuth 2.0 authentication for the connected app.
Consumer Secret	Consumer secret that Salesforce generates when you enable OAuth 2.0 authentication for the connected app.
API Version	Version of the Salesforce API that you want to use to access the source data. Default is 51.0. **Note:** You cannot use a version older than 51.0.

OAuth token URL	OAuth 2.0 token endpoint of the Salesforce organization. The connected app sends access token requests to this endpoint.
	Default value is:
	`https://login.salesforce.com/services/oauth2/token`
	This default URL is used for all Salesforce instances.
	Alternatively, you can enter an instance-specific URL:
	`https://<instance domain URL>/services/oauth2/token`
	An instance-specific URL can establish a more direct and faster connection to the Salesforce host server. If the load on the common default endpoint is heavy and ingestion jobs fail with an authentication error when using it, use this alternative URL instead.

Note: For more information about the OAuth 2.0 Username-Password Flow authentication method, see the Salesforce documentation.

Connection properties for OAuth 2.0 JWT Bearer Flow authentication

The following table describes the connection properties for a Salesforce Mass Ingestion connection configured with OAuth 2.0 JWT Bearer Flow authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
User Name	User name of the Salesforce account.
Consumer Key	Consumer key that Salesforce generates when you enable OAuth 2.0 authentication for the ECA or connected app.
Keystore Path	Absolute path to the keystore file that contains the X509 certificate required to validate a JSON Web Token (JWT) and establish a secure connection with Salesforce. The keystore file must be in the Java KeyStore (JKS) format.
Keystore Password	Password for the keystore file.
Private Key Alias	Alias name of the private key used to sign the JWT.
Private Key Password	Password for the private key.
API Version	Version of the Salesforce API that you want to use to access the source data. Default is 51.0. **Note:** You cannot use a version older than 51.0.

OAuth token URL	OAuth 2.0 token endpoint of the Salesforce organization. The ECA or connected app sends access token requests to this endpoint.
	Default value is:
	`https://login.salesforce.com/services/oauth2/token`
	This default URL is used for all Salesforce instances.
	Alternatively, you can enter an instance-specific URL:
	`https://<instance domain URL>/services/oauth2/token`
	An instance-specific URL can establish a more direct and faster connection to the Salesforce host server. If the load on the common default endpoint is heavy and ingestion jobs fail with an authentication error when using it, use this alternative URL instead.
	If you plan to use Bulk API 1.0 in your task to ingest data from a Salesforce source, you need to specify the instance-specific URL.

Note: For more information about the OAuth 2.0 JWT Bearer Flow authentication method, see the Salesforce documentation.

SAP HANA Database Ingestion connection properties

To define an SAP HANA Database Ingestion connection, configure the connection properties. You can use this connection for sources in database ingestion and replication tasks configured in Data Ingestion and Replication.

The following table describes the SAP HANA connection properties:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	The name of the runtime environment where you want to run database ingestion and replication tasks. You define runtime environments in Administrator. You can use either a local Secure Agent installation or serverless runtime environment. You can use the serverless runtime environment for cloud source types for initial loads only. You cannot run a database ingestion and replication task on a Hosted Agent or elastic runtime environment.
User Name	The user name for connecting to the SAP HANA instance. Enter the user name in the same case as in the database user name specified in SAP HANA.
Password	The password for connecting to the SAP HANA instance.
Host	The name of the machine that hosts the SAP HANA database server.
Port	The port number for the SAP HANA server to which you want to connect. Default is 30015.
Database Name	The SAP HANA source database name.
Advanced Connection Properties	Optional advanced properties for the SAP HANA JDBC driver, which is used to connect to the SAP HANA source. If you specify more than one *property=value* entry, separate them with an ampersand (&). The JDBC connection properties that you can enter in this field are described in the SAP JDBC Connection Properties documentation. For example: encrypt=true.
Capture Type	Select one of the following options to indicate the capture method that database ingestion incremental load jobs use to capture change data from SAP HANA databases: - **Trigger Based**. Capture change data from SAP HANA source tables in the schema by using AFTER DELETE, AFTER INSERT, and AFTER UPDATE triggers. The triggers get before images and after images of DML changes for each source table and write entries for the changes to the PKLOG and shadow _CDC tables. - **Log Based**. Capture change data from the SAP HANA database logs. **Note:** To use the Log Based capture type, a flag must be set for your organization. If you can't select this option, contact Informatica Global Customer Support. Default is Trigger Based.

Connection property	Description
Log Clear	Required for incremental loads. Depending on the capture type, set this property as follows: - For **Trigger Based** change capture, enter the time interval, in days, after which the PKLOG table entries and shadow _CDC table entries are purged. The purging occurs only while an incremental load job is running. Valid values for a database ingestion job are 0 to 366. Any positive value in this range cause automatic housekeeping to run while the incremental job is running. Default is 14. A value of 0 means that the table entries are not purged. For manual housekeeping, enter 0 and use your in-house process. Any value outside the range of 0 to 366, including a negative number or non-numeric value, causes database ingestion jobs that use the connection to fail with the following error: `LogClear contains a non numeric number. Caused by: LogClear contains a non numeric number.` - For **Log Based** change capture, enter the number of days for which data records are stored in the HANA or Oracle database cache before they're compacted. When this time interval is met, the records are compacted but not deleted. Default is 14. A value of 0 means compaction does not occur. No maximum value is in effect.
Trigger Prefix	For the **Trigger Based** capture type, you can add a prefix to the names of the AFTER DELETE, AFTER INSERT, and AFTER UPDATE triggers that the CDC script generates for each source table to get before images and after images of the DML changes. Enter any prefix value up to 16 characters in length. An underscore (_) follows the prefix in the trigger name, for example, **TX_**SAP_DEMO_TABLE_DBMI_USER_t_d. You can use the prefix to comply with your site's trigger naming conventions.
Cache Type	For the **Log Based** capture type, select **Hana** or **Oracle** as the cache type. The Secure Agent stores change data in the cache until the change records are ready to write to the target.
Cache Host	For the **Log Based** capture type, enter the host name of the machine that hosts the cache database.
Cache Port	For the **Log Based** capture type, enter the port number for the cache database server.
Cache User Name	For the **Log Based** capture type, enter the user name to use for connecting to the cache database.
Cache Password	For the **Log Based** capture type, enter the password to use for connecting to the cache database.
Cache Database/ Service Name	For the **Log Based** capture type, enter either the HANA cache database name or the Oracle cache service name, depending on the cache type you selected.
Cache Additional Connection Properties	For the **Log Based** capture type, you can enter a list of optional cache connection properties. If you use a HANA cache, use the ampersand (&) separator. If you use an Oracle cache, use the semicolon (;) separator. Examples: HANA: `latency=0&communicationtimeout=0` Oracle: `EncryptionMethod=SSL;CryptoProtocolVersion=TLSv1.1`

Connection property	Description
Cache Security Connection Properties	For the **Log Based** capture type, you can enter a list of optional security properties for the cache connection. If you use HANA cache, use the ampersand (&) separator. If you use Oracle cache, use the semicolon (;) separator. Examples: HANA: `encrypt=true&validateCertificate=false` Oracle: `KeyStorePassword=xyz;TrustStorePassword=xy`
Server Log Path	For the **Log Based** capture type, enter the log path for the SAP HANA database server.
Client Log Path	For the **Log Based** capture type, enter the mapping of the Secure Agent machine mount path to the source database log location.
Client Archive Log Path	For the **Log Based** capture type, enter the mapping of the Secure Agent machine mount path to the source database archive log location.
Key Backup Password	If you use the **Log Based** capture type with encrypted SAP HANA log files, enter that password that you created or generated in SAP HANA for accessing the encryption root keys file that contains the keys needed to decrypt the online redo and archive logs. **Note:** To use the Log Based capture type,with or without encrypted logs, a flag must be set for your organization. If you can't select Log Based capture, contact Informatica Global Customer Support. You must also specify the **Key Backup Data Path** property to capture change data from encrypted log files. If only one of these fields is set, job deployment will fail. If both fields are not set, Database Ingestion and Replication assumes the logs are not encrypted.
Key Backup Data Path	If you use the **Log Based** capture type with encrypted SAP HANA log files, enter the path to the encryption root key file that contains keys that are required to decrypt the online redo logs and archive logs for CDC processing. This file must reside on the Secure Agent machine. In a Secure Agent group, the file must reside on each Secure Agent machine in the group under the same file name. **Note:** To use the Log Based capture type,with or without encrypted logs, a flag must be set for your organization. If you can't select Log Based capture, contact Informatica Global Customer Support. You must also specify the **Key Backup Password** property to capture change data from encrypted log files. If only one of these fields is set, job deployment will fail. If both fields are not set, Database Ingestion and Replication assumes the logs are not encrypted.

Note: If you test the connection and the test fails, check that the SAP HANA JDBC driver file, ngdbc.jar, has been installed at *`Secure Agent installation directory`*`>/ext/connectors/thirdparty/informatica.hanami`.

Use the serverless runtime environment

In Data Ingestion and Replication, you can use a serverless runtime environment hosted on Azure to connect to the SAP HANA database.

Before you configure a SAP HANA Database Ingestion connection using the serverless runtime environment, perform the following tasks:

- Add the JDBC driver JAR files in the Azure container in your Azure account.
- Configure the .yml serverless configuration file.

Add the JDBC driver as a JAR file in the Azure container in your Azure account

Perform the following steps:

1. Create the following structure for the serverless agent configuration in Azure: `<Supplementary file location>/serverless_agent_config/`
2. Add the JDBC driver files in the Azure container in the following location in your Azure account: `<Supplementary file location>/serverless_agent_config/hanami`

Configure the .yml serverless configuration file

Perform the following steps to configure the .yml serverless configuration file in the serverless runtime environment:

1. Copy the following code snippet to a text editor and specify the driver file name:

```
version: 1
agent:
  databaseIngestion:
    autoDeploy:
      hanami:
        drivers:
        - fileCopy:
            sourcePath: hanami/<file_name>.jar
```

   where the sourcePath is the directory path of the driver files in Azure.

2. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the JDBC driver files are copied from the Azure location to the following serverless agent directory: `<AGENT_DIR>/ext/connectors/thirdparty/informatica.hanami`

SAP Mass Ingestion connection properties

When you set up a SAP Mass Ingestion connection, you must configure the connection properties.

The following table describes the connection properties for a SAP Mass Ingestion connection:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent. **Note:** You can use the serverless runtime environment only for tasks that have an SAP ECC source on Oracle database.
User Name	User name of the SAP instance.

Connection property	Description
Password	Password for the SAP instance.
Language Code	Language code that corresponds to the SAP language.
System Number	System number of the SAP server.
Client Number	Client number of the SAP server.
Port Range	HTTP port range to run the Netty server.
Connection Type	Type of connection to access the ABAP application server. Options are: - **Direct Connection**: Accesses a single ABAP application server using the server host. - **Load Balancing Connection**: Accesses a group of ABAP application servers through the message server.
Application Server	Name of the SAP application server host. **Note:** This field appears only for the **Direct Connection** type.
Use HTTPS	Determines whether the connection uses HTTPS to securely connect to SAP. Select this option to enable HTTPS for the connection, By default, HTTPS is disabled. For more information about configuring HTTPS, see "Configure HTTPS to connect to SAP" on page 233. **Note:** This field appears only for the **Direct Connection** type.
Keystore Location	Absolute path and file name of the keystore file to connect to SAP. Specify both the path and file name in the following format: `<directory>/<keystore file name>.jks` **Note:** This field appears only for the **Direct Connection** type.
Keystore Password	The destination password to access the keystore file. **Note:** This field appears only for the **Direct Connection** type.
Private Key Password	The password to export the private key from the `.p12` keystore file. **Note:** This field appears only for the **Direct Connection** type.
Message Server	IP address or name of the SAP message server. **Note:** This field appears only for the **Load Balancing Connection** type.
SAP Logon Group	Name of the group of servers that belong to the SAP system you want to access. **Note:** This field appears only for the **Load Balancing Connection** type.
SAP System ID	ID of the SAP system that you want to access. **Note:** This field appears only for the **Load Balancing Connection** type.
Message Server Port	Port number on which the SAP message server is listening. **Note:** This field appears only for the **Load Balancing Connection** type.
Database	The name of the underlying database. Select one of the following options: - Oracle - SAP HANA (S/4 trigger based)
For Oracle database	

Connection property	Description
User Name	User name of the database instance.
Password	Password for the database instance.
Host	Host name of the database server.
Port	Network port number used to connect to the database server. Default is 1521.
Service Name	Service name or System ID (SID) that uniquely identifies the Oracle database. Specify the SID in the following format to connect to Oracle databases: `SID:<ORACLE_SID>`
Code Page	The code page of the database server. Application ingestion and replication tasks use the UTF-8 code page. Default is UTF-8.
Encryption Method	For initial load jobs, determines whether the data exchanged between the Secure Agent and the Oracle database server is encrypted: Select one of the following options: - **SSL**. Establishes a secure connection using SSL for data encryption. If the Oracle database server cannot configure SSL, the connection fails. - **No Encryption**. Establishes a connection without using SSL. Data is not encrypted. Default is **No Encryption**.
Crypto Protocol Version	If you selected SSL as the encryption method, you must specify a cryptographic protocol or a list of cryptographic protocols supported by your server to use with an encrypted connection. Select one of the following options: - **SSLv2** - **SSLv3** - **TLSv1.2** Default is **TLSv1.2**.
Validate Server Certificate	If you selected SSL as the encryption method, this option controls whether the Secure Agent validates the server certificate that is sent by the Oracle database server. Select one of the following options: - **True**. Validate the server certificate. - **False**. Do not validate the server certificate. Default is **False**. If you also specify the **Host Name in Certificate** property, the Secure Agent also validates the host name in the certificate.
Trust Store	If you selected SSL as the encryption method and enabled validation of the server certificate, specify the path and name of the truststore file, which contains the list of the Certificate Authorities (CAs) that the client trusts for SSL authentication.
Trust Store Password	If you selected SSL as the encryption method and enabled validation of the server certificate, specify a password for accessing the contents of the truststore file.

Connection property	Description
Host Name in Certificate	If you selected SSL as the encryption method and enabled validation of the server certificate, specify the host name of the machine that hosts the Oracle database to provide for additional security. The Secure Agent validates the host name included in the connection with the host name in the SSL certificate.
Key Store	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the path and name of the keystore file. The keystore file contains the certificates that the client sends to the Oracle server in response to the server's certificate request.
Key Store Password	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the password for the keystore file.
Key Password	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the password for the keys in the keystore file. Use this property when the keys have a different password than the keystore file.
Database Connect String	An Oracle connection string, defined in TNS, that application ingestion and replication tasks use to connect to the Oracle database.
TDE Wallet Directory	The path to the directory that contains the Oracle wallet file used for Oracle Transparent Data Encryption (TDE). Specify this property value only if you capture change data from TDE-encrypted table spaces and one of the following conditions are true: - The Oracle wallet is not available to the database. - The Oracle database is running on a server that is remote from Oracle redo logs. - The wallet directory is not in the default location on the database host or the wallet name is not the default name of ewallet.p12. - The wallet directory is not available to the Secure Agent host.
TDE Wallet Password	A clear text password that is required to access the Oracle TDE wallet and get the master key. This property value is required if you need to read and decrypt data from TDE-encrypted tablespaces in the Oracle source database.
Directory Substitution	A local path prefix to substitute for the server path prefix of the redo logs on the Oracle server. This substitute local path is required when the log reader runs on a system other than the Oracle server and uses a different mapping to access the redo log files. Use this property in the following situations: - The redo logs reside on shared disk. - The redo logs have been copied to a system other than the Oracle system. - The archived redo logs are accessed by using a different NFS mount. **Note:** Do not use this property if you use Oracle Automatic Storage Management (ASM) to manage the redo logs. You can define one or more substitutions. Use the following format: `server_path_prefix,local_path_prefix;server_path_prefix,local_path_prefix;...`

Connection property	Description
Active Log Mask	A mask that the log reader uses for selecting active redo logs when the Oracle database uses multiplexing of redo logs. The log reader compares the mask against the member names in an active redo log group to determine which log to read. In the mask, you can use the asterisk (*) wildcard to represent zero or more characters. The mask can be up to 128 characters in length. It is case-sensitive on Linux or UNIX systems but not on Windows systems.
Archive Destination 1	The primary log destination from which the log reader reads archived logs, when Oracle is configured to write more than one copy of each archived redo log. Enter a number that corresponds to an *n* value in an Oracle LOG_ARCHIVE_DEST_*n* initialization parameter, where *n* is a value from 1 to 10. If you set only one of the Reader Archive Destination 1 and Destination 2 properties, the log reader uses that property setting. If you specify neither property, the archive log queries are not filtered by the log destination.
Archive Destination 2	The secondary log destination from which the log reader reads archived logs when the primary destination becomes unavailable or when the logs at the primary destination cannot be read. For example, logs might have been corrupted or deleted. Enter a number that corresponds to the *n* value in an Oracle LOG_ARCHIVE_DEST_*n* initialization parameter, where *n* is a value from 1 to 10. Usually, this value is a number greater than 1.
ASM Connect String	In an Oracle ASM environment, the Oracle connection string, defined in TNS, that the log reader uses to connect to the ASM instance that manages storage of active and archived redo logs for the source database.
ASM User Name	In an Oracle ASM environment, an Oracle user ID that the log reader uses to connect to the ASM instance that manages storage of active and archived redo logs for the source database. This user ID must have SYSDBA or SYSASM authority. To use SYSASM authority, set the **Reader ASM Connect As SYSASM** property to Y.
ASM Password	In an Oracle ASM environment, a clear text password for the user that is specified in the **Reader ASM User Name** property. The log reader uses this password and the ASM user name to connect to the ASM instance that manages storage of active and archived redo logs for the source database.
ASM Connect As SYSASM	If you use Oracle 11*g* ASM or later and want the log reader to use a user ID that has SYSASM authority to connect to the ASM instance, select this check box. Also specify a user ID that has SYSASM authority in the **Reader ASM User Name** property. To use a user ID that has SYSDBA authority, clear this check box. By default, this check box is cleared.

Connection property	Description
Mode	Indicates the source of and types of Oracle redo logs that the log reader reads. Select one of the following options: - **ACTIVE**. Read active and archived redo logs from the Oracle online system. Optionally, you can use the **Reader Active Log Mask** property to filter the active redo logs and use the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties to limit the archived log destinations from which to read archived logs. - **ARCHIVEONLY**. Read only archived redo logs. Optionally, you can use the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties to limit the archived log destinations from which to read archived logs. - **ARCHIVECOPY**. Read archived redo logs that have been copied to an alternate file system. Use this option in the following situations: - You do not have the authority to access the Oracle archived redo logs directly. - The archived redo logs are written to ASM, but you do not have access to ASM. - The archived log retention policy for the database server causes the archived logs to not be retained long enough. With this option, the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties are ignored. Default is **ACTIVE**.
Standby Log Mask	A mask that the log reader uses for selecting redo logs for an Oracle physical standby database when the database uses multiplexing of redo logs. The log reader compares the mask against the member names in an redo log group to determine which log to read. In the mask, you can use the asterisk (*) wildcard to represent zero or more characters. The mask can be up to 128 characters in length. It is case-sensitive on Linux or UNIX systems but not on Windows systems.
Standby Connect String	An Oracle connection string, defined in TNS, that the log reader uses to connect to the Oracle physical standby database for change capture when the database is not open with read only access.
Standby User Name	A user ID that the log reader uses to connect to the Oracle physical standby database for change capture. This user ID must have SYSDBA authority.
Standby Password	A password that the log reader uses to connect to the Oracle physical standby database for change capture.
RAC Members	The maximum number of active redo log threads, or *members*, in an Oracle Real Application Cluster (RAC) that can be tracked. For a Data Guard physical standby database that supports a primary database in a RAC environment, this value is the number of active threads for the primary database. Valid values are 1 to 100. Default is 0, which causes an appropriate number of log threads to be determined automatically. If this value is not appropriate for your environment, set this property to a value greater than 0.

Connection property	Description
BFILE Access	Select this check box in the following circumstances: - You use BFILE access to redo logs in physical directories on the local Oracle server file system. BFILE access uses Oracle directory objects to remotely access the redo logs in the file system. This method is an alternative to other log access methods such as ASM or NFS mounts. - You have an Amazon Relational Database Service (RDS) for Oracle source. In this case, this option enables access to the redo logs of a cloud-based database instance deployed in RDS. By default, this check box is cleared.
For SAP HANA (S/4 trigger based) database	
User Name	The user name to connect to the SAP HANA instance.
Password	The password to connect to the SAP HANA instance.
Host	The name of the machine that hosts the SAP HANA database server.
Port	The port number of the SAP HANA server that you want to connect to. Default is 30015.
Database Name	The SAP HANA source database name.
Advanced Connection Properties	Advanced properties for the SAP HANA JDBC driver, which is used to connect to the SAP HANA source. If you specify more than one *property=value* entry, separate them with an ampersand (&). The JDBC connection properties that you can enter in this field are described in the SAP JDBC Connection Properties documentation. For example: encrypt=true.
Log Clear	Required for incremental loads. The time interval, in days, after which the PKLOG table entries and shadow _CDC table entries are purged. The purging occurs only while an incremental load job is running. Valid values for a database ingestion job are 0 to 366. Any positive value in this range causes automatic housekeeping to run while the incremental job is running. Default is 14. A value of 0 means that the table entries are not purged. For manual housekeeping, enter 0 and use your in-house process. Any value outside the range of 0 to 366, including a negative number or non-numeric value, causes database ingestion jobs that use the connection to fail with the following error: `LogClear contains a non numeric number. Caused by: LogClear contains a non numeric number.`
Trigger Prefix	If you use the trigger-based capture type, you can add a prefix to the names of the AFTER DELETE, AFTER INSERT, and AFTER UPDATE triggers that the CDC script generates for each source table to get before images and after images of the DML changes. Enter any prefix value up to 16 characters in length. An underscore (_) follows the prefix in the trigger name, for example, **TX_**SAP_DEMO_TABLE_DBMI_USER_t_d. You can use the prefix to comply with your site's trigger naming conventions.

Configure HTTPS to connect to SAP

To connect to SAP through HTTPS, ensure that an OpenSSL certificate is available on both the Secure Agent machine and the SAP system.

Create an OpenSSL certificate in the Secure Agent machine. Then, import the created certificate in the PSE format to the SAP system truststore.

Additionally, to enable HTTPS in an SAP Mass Ingestion connection, you need to specify the generated keystore password and private key password of the keystore file both in the SAP Mass Ingestion connection properties and in the SAP system.

Create an OpenSSL certificate

Before you create an OpenSSL certificate, you need to perform the prerequisite tasks.

- Download and install OpenSSL on the Secure Agent machine.
- Based on the operating system of the machine that hosts the Secure Agent and the SAP system, download the latest available patch of the SAPGENPSE Cryptography tool from the SAP Service Marketplace.
  By default, the SAPGENPSE files are extracted to the `nt-x86_64` directory.
- Configure the following SAP parameters: `icm/server_port`, `ssl/ssl_lib`, `sec/libsapsecu`, `ssf/ssfapi_lib`, `ssf/name`, `icm/HTTPS/verify_client`, `ssl/client_pse`, and `wdisp/ssl_encrypt`.
  For more information, see the SAP documentation.

To create a self-signed certificate using OpenSSL, perform the following tasks:

1. From the command line, set the `OPENSSL_CONF` variable to the absolute path to the `openssl.cfg` file.
   For example, run the following command: `set OPENSSL_CONF= C:\OpenSSL-Win64\bin\openssl.cfg`
2. Navigate to the `<openSSL installation directory>\bin` directory.
3. To generate a 2048-bit RSA private key, run the following command:
   `openssl.exe req -new -newkey rsa:2048 -sha1 -keyout <RSAkey File_Name>.key -out <RSAkey File_Name>.csr`
4. When prompted, enter the following values:
   - Private key password (PEM pass phrase). Enter a phrase that you want to use to encrypt the secret key. Re-enter the password for verification.

     Important: Make a note of this PEM password. You need to keep this password handy while creating a self-signed key and PKCS#12 certificate.
   - Two-letter code for country name.
   - State or province name.
   - Locality name.
   - Organization name
   - Organization unit name.
   - Common name (CN). Mandatory.

     Important: Enter the fully qualified host name of the machine that hosts the Secure Agent.
   - Email address.
5. Optionally, enter the following attributes that you want to pass along with the certificate request:
   - Challenge password.
   - Optional company name.

A RSA private key of 2048-bit size is created. The `<RSAkey File_Name>.key` and `<RSAkey File_Name>.csr` files are generated in the specified directory.

6. To generate a self-signed key using the RSA private key, run the following command:
   `openssl x509 -req -days 11499 -in <RSAkey File_Name>.csr -signkey <RSAkey File_Name>.key -out <Certificate File_Name>.crt`
7. When prompted, enter the PEM pass phrase for the RSA private key.
   The `<Certificate File_Name>.crt` file is generated in the specified directory.
8. To concatenate the contents of the `<Certificate File_Name>.crt` file and the `<RSAkey File_Name>.key` file to a `.pem` file, perform the following tasks:
   a. Open the `<Certificate File_Name>.crt` file and the `<RSAkey File_Name>.key` files in a Text editor.
   b. Create a file and save it as `<PEM File_Name>.pem`.
   c. Copy the contents of the `<Certificate File_Name>.crt` file and paste it in the `.pem` file.
   d. Copy the contents of the `<RSAKey_Name>.key` file and append it to the existing contents of the `.pem` file.
   e. Save the `<PEM file name>.pem` file.
9. To create a PKCS#12 certificate, run the following command from the command line:
   `openssl pkcs12 -export -in <PEM File_Name>.pem -out <P12 File_Name>.p12 -name "domain name"`
10. When prompted, enter the following details:
    - The PEM pass phrase for the `.pem` file.
    - An export password for the P12 file. Re-enter the password for verification.

      Important: Make a note of this export password for the P12 file. You need to keep this password handy while creating a Java keystore file to connect to SAP through HTTPS.

    The `<P12 File_Name>.p12` file is generated in the specified directory.
11. To create a Java keystore file, enter the following command:
    `keytool -v -importkeystore -srckeystore <P12 File_Name>.p12 -srcstoretype PKCS12 -destkeystore <JKS File_Name>.jks -deststoretype JKS -srcalias "source alias" -destalias "destination alias"`
12. When prompted, enter the following details:
    - Password for the destination keystore, the JKS file.

      Important: Make a note of this password. You need to keep this password handy while creating an SAP Table connection.
    - Password for the source keystore, the P12 file. Enter the Export password for the P12 file.

    The `<JKS File_Name>.jks` file is generated in the specified directory.

    While enabling HTTPS in a SAP Mass Ingestion connection, specify the path and file name of this keystore file. You also need to specify the destination keystore password as the Keystore Password and the source keystore password as the Private Key Password both in the SAP Mass Ingestion connection properties and in the SAP system.

Convert an OpenSSL certificate to PSE format

After you create an OpenSSL certificate, you need to convert the OpenSSL certificate to PSE format using the SAPGENPSE tool.

1. From the command line, navigate to the `<SAPGENPSE Extraction Directory>` directory.

2. To generate a PSE file, run the following command:
   `sapgenpse import_p12 -p <PSE_Directory>\<PSE File_Name>.pse <P12 Certificate_Directory>\<P12 File_Name>.p12`
3. When prompted, enter the following details:
   - Password for the P12 file. Enter the Export password for the P12 file.
   - Personal identification number (PIN) to protect the PSE file. Re-enter the PIN for verification.

   The `<PSE File_Name>.pse` file is generated in the specified directory.
4. To generate the certificate based on the PSE format, run the following command:
   `sapgenpse export_own_cert -p <PSE File_Directory>\<PSE File_Name>.pse -o <Certificate_Name>.crt`
5. When prompted, enter the PSE PIN number.
   The `<Certificate_Name>.crt` file is generated in the specified directory. Import this certificate file to the SAP system trust store.

Enable the HTTPS service on the SAP system

To configure HTTPS to connect to an SAP system, you need to enable the HTTPS service from the transaction code SAP ICM Monitor (SMICM) in the SAP system

For more information about how to enable the HTTPS service on the SAP system, see the SAP documentation.

Import a certificate to the SAP system trust store

You need to import the certificate in PSE format to the SAP system trust store to connect to SAP through HTTPS.

1. Log in to SAP and go to the STRUST transaction.
2. Select SSL Client (Standard) and specify the password.
   In the **Import Certificate** dialog, you need to select Base64 format as the certificate file format.
3. Click the **Import** icon, and select the `<Certificate_Name>.crt` file in PSE format.

   Note: If a user is on a different SAP network, you might need to add a DNS entry of the agent host on the SAP app server.
4. Click **Add to Certificate List**.
5. Restart the Internet Communication Manager.

SAP OData V2 connection properties

Create an SAP OData V2 connection to securely read from or write to OData V2-compliant applications in SAP deployed in the cloud or on-premises.

Prepare for authentication

You can configure basic, API key, authorization code, and client credentials authentication types to access OData V2-compliant applications in SAP. Only basic authentication is applicable for Data Ingestion and Replication.

Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

Basic

To connect to SAP OData V2 services using basic authentication, you need the SAP account user name and password.

Get the required details from the SAP application to which you want to connect.

For more information about basic authentication in SAP, see Basic authentication in the SAP documentation.

Connect to SAP OData V2

Let's configure the SAP OData V2 connection properties to connect to SAP OData V2 services, and read from or write to OData V2-compliant applications in SAP.

Before you begin

Before you get started, you'll need to configure the SAP system and get the relevant configuration details from your SAP account based on the authentication type that you want to configure.

Check out "Prerequisites" on page 236 to learn more about the configuration prerequisites.

Prerequisites

To connect to SAP OData V2 services, the SAP administrator needs to configure the SAP user account in the SAP system.

Additionally, when you configure the connection properties, you need to complete the authentication prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication jobs, you can select either a Secure Agent or serverless runtime environment. You can't use a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Service Type	The service type of the OData V2 application endpoint to which you want to connect. Select one of the following service types from the list: - Default. Connects to a specific SAP service. - SAP Gateway Catalog. Connects to the SAP catalog service that provides a list of all available services on SAP Gateway. Default is Default.

Service URL	The service URL for the selected service type.
	If you select the **Default** service type, enter the root URL of the service in the following format:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata/sap/<Service name>/`
	For example, if you want to connect to the ZALL_DATATYPE_SRV service in SAP when the host name is http://invs15con01.informatica.com and port number is 8081, enter the following service URL:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata/sap/ZALL_DATATYPE_SRV/`
	Note: If you use the connection for application ingestion and replication tasks, don't add the trailing '/' at the end of the URL for the default service type, as it prevents the task from retrieving schemas from the source.
	If you select the **SAP Gateway Catalog** service type, enter the service URL in the following format:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata/IWFND/<Catalog service name>`
	For example, if you want to connect to the CATALOGSERVICE;v=2 catalog service in SAP when the host name is abcd01con02.example.com and port number is 8081, enter the following service URL:
	`http://abcd01con02.example.com:8081/sap/opu/odata/IWFND/CATALOGSERVICE;v=2`

Authentication types

You can configure basic, API key, authorization code, and client credentials authentication types to access OData V2-compliant applications in SAP. Only basic authentication is applicable for Data Ingestion and Replication.

Select the required authentication method and then configure the authentication-specific parameters.

Basic authentication

Basic authentication is the default type which requires at a minimum your SAP account name and password.

The following table describes the basic connection properties for basic authentication:

Property	Description
Username	The user name to connect to the SAP OData V2 service.
Password	The password to connect to the SAP OData V2 service.

Advanced settings

The following table describes the advanced connection properties for basic authentication:

Property	Description

SAP Custom Query Options	Additional custom queries that you can use when you connect to the SAP OData V2 service.
	You can enter multiple SAP custom queries, separated by an ampersand (&), in the following format:
	`<Custom query1>=<value>&<Custom query2>=<value>&<Custom query3>=<value>....`
	For example, if you want to pass the SAP client number and language code when you connect to the SAP OData V2 service, enter the following custom queries:
	`sap-client=400&sap-language=DE`
	When you add queries, ensure that there is no space before and after the equal sign (=).
	For more information about the list of SAP custom queries that you can configure, see SAP URL parameters in the SAP documentation.
	This property is not applicable for Data Ingestion and Replication.

Use the serverless runtime environment

In Data Ingestion and Replication, you can use a serverless runtime environment hosted on Azure to connect to the SAP S4/HANA database with SAP OData V2 connector.

Before you configure a SAP OData V2 connection using the serverless runtime environment, perform the following tasks:

- Add the JDBC driver JAR files in the Azure container in your Azure account.
- Configure the .yml serverless configuration file.

Add the JDBC driver as a JAR file in the Azure container in your Azure account

Perform the following steps:

1. Create the following structure for the serverless agent configuration in Azure: `<Supplementary file location>/serverless_agent_config/`
2. Add the JDBC driver files in the Azure container in the following location in your Azure account: `<Supplementary file location>/serverless_agent_config/odp`

Configure the .yml serverless configuration file

Perform the following steps to configure the .yml serverless configuration file in the serverless runtime environment:

1. Copy the following code snippet to a text editor and specify the driver file name:

```
version: 1
agent:
  databaseIngestion:
    autoDeploy:
      odp:
        jdbc:
          drivers:
            - fileCopy:
                sourcePath: odp/<file_name>.jar
```

   where the sourcePath is the directory path of the driver files in Azure.

2. Ensure that the syntax and indentations are valid, and then save the file as `serverlessUserAgentConfig.yml` in the following Azure location: `<Supplementary file location>/serverless_agent_config`
   When the .yml file runs, the JDBC driver files are copied from the Azure location to the following serverless agent directory: `<AGENT_DIR>/ext/connectors/thirdparty/infa.odp`

SAP ODP Extractor connection properties

Create an SAP ODP Extractor connection to securely read data from SAP ODP objects.

You can use an SAP ODP Extractor connection to read data from the following applications:

- SAP S/4HANA
- ECC
- Applications enabled with Operational Data Provisioning (ODP)
- Applications enabled with Operational Delta Queue (ODQ)

Prerequisites

Before you use an SAP ODP Extractor connection, the SAP administrator needs to perform certain prerequisite tasks to configure the Secure Agent machine and SAP system.

To process SAP ODP data, you also need to verify if the required licenses are enabled for the SAP system.

Verify the required SAP Notes in the SAP server

To read data from the SAP ODP objects, you need to verify that the required SAP Notes are available in the SAP server.

- 1931427 - ODP Data Replication API 2.0
- 2232584 - Release of SAP extractors for ODP replication (ODP SAPI)

SAP ODP Extractor Connector uses the ODP Replication APIs version 2.0 when you read data from SAP ODP objects.

Download and configure the SAP libraries

To read data from SAP ODP objects, you need to download and configure the SAP JCo libraries on the Secure Agent machine. If you encounter any issues while you download libraries, contact SAP Customer Support.

1. Go to the SAP Support Portal, and then click **Software Downloads**.

   Note: You need to have SAP credentials to access **Software Downloads** from the SAP Support Portal.

2. Download the latest version of the 64-bit SAP JCo libraries based on the operating system on which the Secure Agent runs.

Operating System	SAP JCo Libraries
Windows	- sapjco3.jar - sapjco3.dll
Linux	- sapjco3.jar - libsapjco3.so

3. Copy the JCo libraries to the following directory:
   `<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\ext\deploy_to_main\bin\rdtm-extra\tpl\sap`

   Create the `deploy_to_main\bin\rdtm-extra\tpl\sap` directory if it does not already exist.

4. Log in to Informatica Intelligent Cloud Services and configure the JAVA_LIBS property for the Secure Agent.
   a. Select **Administrator** > **Runtime Environments**.
   b. Click **Runtime Environments** to access the **Runtime Environments** page.
   c. To the left of the agent name, click **Edit Secure Agent**.
   d. From the **Service** list, select **Data Integration Server**.
   e. From the **Type** list, select **Tomcat JRE**.
   f. Enter the JAVA_LIBS value based on the operating system on which the Secure Agent runs.

Operating System	Value
Windows	..\bin\rdtm-extra\tpl\sap\sapjco3.jar;..\bin\rdtm\javalib\sap\sap-adapter-common.jar
Linux	../bin/rdtm-extra/tpl/sap/sapjco3.jar:../bin/rdtm/javalib/sap/sap-adapter-common.jar

   Warning: If you copy the value directly from the table, the hyphens (-) in the value might be incorrectly copied. Copy the value to a text editor and make sure that the value you copied is not corrupted.



   g. Click **Save**.

5. After you save the JAVA_LIBS value, configure the JVMClassPath property for the Secure Agent.
   a. From the **Service** list, select **Data Integration Server**.
   b. From the **Type** list, select **DTM**.
   c. Enter the JVMClassPath value based on the operating system on which the Secure Agent runs.

Operating System	Value
Windows	pmserversdk.jar;..\..\bin\rdtm-extra\tpl\sap\sapjco3.jar;..\..\bin\rdtm\javalib\sap\sap-adapter-common.jar
Linux	pmserversdk.jar:../../bin/rdtm-extra/tpl/sap/sapjco3.jar:../../bin/rdtm/javalib/sap/sap-adapter-common.jar

Warning: If you copy the value directly from the table, the hyphens (-) in the value might be incorrectly copied. Copy the value to a text editor and make sure that the value you copied is not corrupted.



   d. Click **Save**.
   e. Repeat steps 2 through 5 on every machine where you installed the Secure Agent.
6. Restart the Secure Agent.

Configure SAP user authorization

Configure the SAP user account in the SAP system to process SAP ODP data.

For more information about how to configure SAP user authorization in the SAP system, see SAP user authorizations.

The following table describes the required authorization to read from SAP ODP objects:

Read Object Name	Authorization Values	Value	Activity	Design Time/Run Time
S_RFC	RFC_TYPE - Function Group(FUGR)	SYST	16	Both
	RFC_TYPE - Function Module(FUGR)	RFC1	16	Both
	RFC_TYPE - Function Module(FUNC)	RFCPING	16	Both
	RFC_TYPE - Function Group(FUGR)	RFC_METADATA	16	Both

Read Object Name	Authorization Values	Value	Activity	Design Time/Run Time
	RFC_TYPE - Function Module(FUNC)	RFC_METADATA_GET	16	Both
	RFC_TYPE - Function Module(FUNC)	RFC_GET_FUNCTION_INTERFACE	16	Both
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_CONTEXT_GET_LIST	16	Both
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_ODP_GET_DETAIL	16	Both
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_ODP_GET_LIST	16	Both
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_ODP_OPEN	16	Both
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_ODP_CLOSE	16	Both
	RFC_TYPE - Function Module(FUNC)	/INFADI/ODP_FETCH_XML	16	Run Time
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_ODP_FETCH	16	Run Time
	RFC_TYPE - Function Module(FUNC)	RODPS_REPL_ODP_FETCH_XML	16	Run Time
	RFC_TYPE - Function Module(FUNC)	DDIF_FIELDINFO_GET	16	Both
S_BTCH_ADM	FIELD NAME - BTCADMIN	Y	N/A	Both
S_BTCH_JOB	FIELD NAME - JOBACTION	RELE	RELE(Release Jobs)	Both
	FIELD NAME - JOBGROUP	' '	N/A	Both
S_RS_ODP_H	FIELD NAME - RSODPHNAME	*	3	Both
	FIELD NAME - RSODPHPKG	*	3	Both
S_RO_OSOA	FIELD NAME - OLTPSOURCE	*	3	Both
	FIELD NAME - OSOAAPCO	*	3	Both
	FIELD NAME - OSOAPART	Data, Definition	3	Both

Read Object Name	Authorization Values	Value	Activity	Design Time/Run Time
S_RS_HYBR	FIELD NAME - RSHYBRPROV	'*'	3	Both
	FIELD NAME - RSHYBRPROJ	Definition	3	Both
S_RS_ICUBE	FIELD NAME - OLTPSOURCE	*	3	Both
	FIELD NAME - OSOAAPCO	*	3	Both
	FIELD NAME - OSOAPART	Data, Definition	3	Both
S_RS_IOMAD	FIELD NAME - RSINFOAREA	*	3	Both
	FIELD NAME - RSAPPLNM	*	3	Both
	FIELD NAME - RSIOBJNM	*	3	Both
S_RS_MPRO	FIELD NAME - RSINFOAREA	*	3	Both
	FIELD NAME - RSMPRO	*	3	Both
	FIELD NAME - RSMPROOBJ	Data	3	Both
S_RS_ODSO	FIELD NAME - RSINFOAREA	*	3	Both
	FIELD NAME - RSODSOBJ	*	3	Both
	FIELD NAME - RSODSPART	Data	3	Both
S_ADMI_FCD	FIELDNAME - S_ADMI_FCD	PADM	N/A	Both

Configure the Secure Network Communication protocol

To use the SAP ODP Extractor connection with the Secure Network Communication (SNC) protocol, you need to configure the SNC protocol on both the SAP server and Secure Agent machine.
To create a connection with the SNC protocol, use the application server SNC connection and load balancing server SNC connection.

For more information about the prerequisites and steps to configure an SAP SNC connection, see Configure the SAP Secure Network Communication protocol Informatica How-To Library article.

Connect to SAP ODP

Let's configure the SAP ODP Extractor connection properties to connect to SAP ODP objects.

Before you begin

Before you get started, you'll need to configure the Secure Agent machine and SAP system to establish an SAP ODP Extractor connection.

Check out "Prerequisites" on page 239 to learn more about these tasks.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks to access SAP S/4HANA or SAP ECC. For application ingestion and replication tasks, you can't use a Hosted Agent, serverless runtime environment, or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

SAP server connection types

You can configure application server, application server SNC, load balancing server, and load balancing server SNC connection types to access SAP ODP.
Select the required connection type and then configure the connection-specific parameters.

Application server connection

Application server connection is the default type which requires your SAP client details.

The following table describes the basic connection properties for an application server connection:

Connection property	Description
SAP Client Number	The client number of the SAP application server. Get the required client number from the SAP system to which you want to connect.
SAP Language	Language code that corresponds to the SAP language. Get the required language code from the SAP system to which you want to connect.
SAP Application Server	The host name of the SAP application Server.
SAP System Number	The system number of the SAP application server to connect. Get the required system number from the SAP system to which you want to connect.
SAP Username	The user name with the appropriate user authorization to connect to the SAP account.
SAP Password	The password to connect to the SAP account.
Subscriber Name	A name that defines the Secure Agent as a unique subscriber in the SAP system. SAP uses this name to define unique operational delta queue (ODQ) in case of delta read from ODP.

Application server SNC connection

Application server SNC connection requires the Secure Agent PSE certificate name, SAP server PSE certificate name, and path to the X509 certificate file along with your SAP client details.

The following table describes the basic connection properties for an application server SNC connection:

Connection property	Description
SAP Client Number	The client number of the SAP application server. Get the required client number from the SAP system to which you want to connect.
SAP Language	Language code that corresponds to the SAP language. Get the required language code from the SAP system to which you want to connect.
SAP Application Server	The host name of the SAP application Server.
SAP System Number	The system number of the SAP application server to connect. Get the required system number from the SAP system to which you want to connect.
SNC My Name	The agent Personal Security Environment (PSE) or certificate name generated for the Secure Agent. Default length is 256.
SNC Partner Name	The server PSE or certificate name generated on the SAP server. Default length is 256.

Connection property	Description
SNC Quality of Protection (QoP)	The level of protection applied to a communication path when you create an SAP SNC connection. Select one of the following options from the list: - 1 - Apply authentication only - 2 - Apply authentication and integrity protection - 3 - Apply authentication, integrity, and privacy protection (encryption) - 8 - Apply global default protection (usually 3) - 9 - Apply the maximum protection Default is 3 - Apply authentication, integrity, and privacy protection (encryption).
Use X509 Certificate	Method of logging in. Select **Use X509 Certificate** to log in with SNC encryption that uses the X.509 certificate. If you don't select this option, you need to enter your SAP user name in the **X509 Certificate Path or SAP Username** property. Default is disabled.
X509 Certificate Path or SAP Username	The path and file name of the X509 certificate file. If the X509 certificate file name is abc.crt and the path is `\root\<folder name>`, enter both the path and file name in the following format: `\root\<folder name>\abc.crt` If you select to use the X509 certificate, you don't need to enter the SAP user name. If you don't want to use the X509 certificate, enter the SAP user name for which SNC is configured in the SAP server.
Subscriber Name	A name that defines the Secure Agent as a unique subscriber in the SAP system. SAP uses this name to define unique operational delta queue (ODQ) when the Secure Agent reads delta data from ODP.

The following table describes the advanced connection properties for application server SNC connection:

Property	Description
SAP Cryptographic Library Path	The path and file name of the SAP cryptographic library. Enter both the path and file name in the following format based on the operating system on which the Secure Agent runs: - On Windows: `\root\<folder name>\sapcrypto.dll` - On Linux: `/root/<folder name>/libsapcrypto.so`

Load balancing server connection

Create a load balancing server connection when you want to connect to the SAP system with the least load at run time.

Load balancing server connection requires your SAP client details and message server group name.

The following table describes the basic connection properties for a load balancing server connection:

Connection property	Description
SAP Client Number	The client number of the SAP message server. Get the required client number from the SAP system to which you want to connect.
SAP Language	Language code that corresponds to the SAP language. Get the required language code from the SAP system to which you want to connect.
SAP Message Server	The host name of the SAP message server.
SAP System ID	The system ID of the SAP message server. Get the required system ID from the SAP system to which you want to connect.
SAP Group	The name of the SAP logon group through which you want to connect. For example, PUBLIC.
SAP Username	The user name with the appropriate user authorization to connect to the SAP account.
SAP Password	The password to connect to the SAP account.
Subscriber Name	A name that defines the Secure Agent as a unique subscriber in the SAP system. SAP uses this name to define unique operational delta queue (ODQ) when you read delta data from ODP.

Load balancing server SNC connection

Create a load balancing server SNC connection when you want to connect to the SAP system using the (SNC) protocol with the least load at run time.

Load balancing server SNC connection requires the Secure Agent PSE certificate name, SAP server PSE certificate name, and path to the X509 certificate file along with your SAP client details and message server group name.

The following table describes the basic connection properties for a load balancing server SNC connection:

Connection property	Description
SAP Client Number	The client number of the SAP message server. Get the required client number from the SAP system to which you want to connect.
SAP Language	Language code that corresponds to the SAP language. Get the required language code from the SAP system to which you want to connect.
SAP Message Server	The host name of the SAP message server.
SAP System ID	The system ID of the SAP message server. Get the required system ID from the SAP system to which you want to connect.
SAP Group	The name of the SAP logon group through which you want to connect. For example, PUBLIC.

Connection property	Description
SNC My Name	The agent PSE or certificate name generated for the Secure Agent. Default length is 256.
SNC Partner Name	The server PSE or certificate name generated on the SAP server. Default length is 256.
SNC Quality of Protection (QoP)	The level of protection applied to a communication path when you create an SAP SNC connection. Select one of the following options from the list: - 1 - Apply authentication only - 2 - Apply authentication and integrity protection - 3 - Apply authentication, integrity, and privacy protection (encryption) - 8 - Apply global default protection (usually 3) - 9 - Apply the maximum protection Default is 3 - Apply authentication, integrity, and privacy protection (encryption).
Use X509 Certificate	Method of logging in. Select **Use X509 Certificate** to log in with SNC encryption that uses the X.509 certificate. If you don't select this option, you need to enter your SAP user name in the **X509 Certificate Path or SAP Username** property. Default is disabled.
X509 Certificate Path or SAP Username	The path and file name of the X509 certificate file. If the X509 certificate file name is abc.crt and the path is `\root\<folder name>`, enter both the path and file name in the following format: `\root\<folder name>\abc.crt` If you select to use the X509 certificate, you don't need to enter the SAP user name. If you don't want to use the X509 certificate, enter the SAP user name for which SNC is configured in the SAP server.
Subscriber Name	A name that defines the Secure Agent as a unique subscriber in the SAP system. SAP uses this name to define unique operational delta queue (ODQ) when you read delta data from ODP.

The following table describes the advanced connection properties for load balancing server SNC connection:

Property	Description
SAP Cryptographic Library Path	The path and file name of the SAP cryptographic library. Enter both the path and file name in the following format based on the operating system on which the Secure Agent runs: - On Windows: `\root\<folder name>\sapcrypto.dll` - On Linux: `/root/<folder name>/libsapcrypto.so`

Advanced settings

The following table describes the advanced connection properties:

Property	Description

Additional Parameters	Additional SAP parameters that you can use when you connect to the SAP system.
	You can enter multiple additional parameters, separated by semicolon, in the following format:
	`<parameter name1>=<value1>;<parameter name2>=<value2>;<parameter name3>=<value3>....`
	For example, to generate SAP JCo and SAP CPIC trace files, enter the following additional parameters:
	`jco.client.trace="1";jco.client.cpic_trace="3";`
	During the run time, the SAP JCo and SAP CPIC trace files are generated in the following location:
	`<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\<DIS version>\ICS\main\bin\rdtm`
	During the design time, the SAP CPIC traces are generated in the `tomcat.out` files at the following location:
	`<Informatica Secure Agent installation directory>\apps\Data_Integration_Server\<DIS version>tomcat.out`

Display Delta Fields	Specifies whether the mapping displays the operation modes that caused the changed data on ODP sources. When enabled, the mapping generates the ODQ_CHANGEMODE and ODQ_ENTITYCNTR fields on the **Fields** tab for ODP sources that are enabled with Operational Delta Queue (ODQ). Default is disabled.

Hierarchical data extraction from SAP ODP objects

Before you use an SAP ODP Extractor connection to extract hierarchical data from SAP ODP objects in a Unicode SAP system, you need to install the SAP ODP Extractor transport files that you get from the Secure Agent directory to the SAP system.

Prerequisites to install the transport files

Before you install the SAP ODP Extractor transports, make sure to perform the following prerequisite tasks:

- Ensure that the transport files you install on the SAP machines are the latest. Get the latest transport files from the following directory:
  `<Informatica Secure Agent installation directory>\downloads\package-SAPODP.<Latest version>\package\sapodp\sap-transport`
- Verify that the transport files are applicable for SAP version ERP 6.0 EHP7 system or later.
- Before you install the transports on your production system, install and test the transports in a development system.

The following table lists the transports that you need to install to read data from the SAP ODP objects:

Data and Cofile Names	Transport Request	Functionality
- K900861.N75 - R900861.N75	N75K900861	Install the transports only when you want to read from an SAP ODP that supports hierarchy. If the SAP ODP objects that do not contain hierarchical data, you can use SAP ODP Extractor Connector without installing the SAP ODP Extractor transport files.

Installing transport files

To install the SAP ODP Extractor transport files, perform the following steps:

1. Find the transport files in the following directory on the Secure Agent machine:
   `<Informatica Secure Agent installation directory>\downloads\package-SAPODP.<Latest version>\package\sapodp\sap-transport`
2. Copy the cofile transport file to the `Cofile` directory in the SAP transport management directory on each SAP machine that you want to access.
   The cofile transport file uses the following naming convention: `<number>.<sap system>`.
3. Copy the data transport file to the `Data` directory in the SAP transport management directory on each SAP machine that you want to access.
   The data transport file uses the following naming convention: `<number>.<sap-system>`.
4. To import the transports to SAP, in the STMS, click **Extras** > **Other Requests** > **Add** and add the transport request to the system queue.
5. In the **Add Transport Request to Import Queue** dialog box, enter the request number for the cofile transport.
   The request number inverts the order of the renamed cofile as follows: `<sap-system><number>`.
6. In the Request area of the import queue, select the transport request number that you added, and click **Import**.
7. If you want to upgrade from a previous version of the Informatica Transports, select the **Overwrite Originals** option.

ServiceNow Mass Ingestion connection properties

When you set up a ServiceNow Mass Ingestion connection, you must configure the connection properties.

The properties of a ServiceNow Mass Ingestion connection vary based on the authentication method that you specify for the connection. When you create a connection, you can select one of the following authentication methods:

- **OAuth 2.0**: Authenticates the connection by using the details of the OAuth API endpoint that is created for the connection in ServiceNow. To use this method, you must create OAuth API endpoint in ServiceNow and then specify the client ID and client secret of the API endpoint in the connection properties. For more information about creating an OAuth API endpoint in ServiceNow, see the ServiceNow documentation.
- **Basic**: Authenticates the connection by validating the login credentials of the ServiceNow account.

Connection properties for OAuth 2.0 authentication

The following table describes the connection properties for a ServiceNow Mass Ingestion connection configured with OAuth 2.0 authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
User Name	User name of the ServiceNow account.
Password	Password for the ServiceNow account.
Client Secret	Client secret of the API endpoint created for the connection in ServiceNow.
Client ID	Client ID of the API endpoint created for the connection in ServiceNow.
Base URI	URL of the ServiceNow instance. You must enter the base URI in the following format: `https://{your_servicenow_instance}.service-now.com/`
OAuth Token URL	OAuth token endpoint of the ServiceNow instance. The API client associated with the connection sends the access token requests to this endpoint.

Connection properties for Basic authentication

The following table describes the connection properties for a ServiceNow Mass Ingestion connection configured with Basic authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
User Name	User name of the ServiceNow account.

Connection property	Description
Password	Password for the ServiceNow account.
Base URI	URL of the ServiceNow instance. You must enter the base URI in the following format: `https://{your_servicenow_instance}.service-now.com/`

Snowflake Data Cloud connection properties

Create a Snowflake Data Cloud connection to securely read data from or write data to Snowflake.

Prepare for authentication

You can configure standard, authorization code, key pair, and programmatic access token authentication types to access Snowflake. Consider using authorization code, key pair, or programmatic access token authentication to connect more securely to Snowflake.

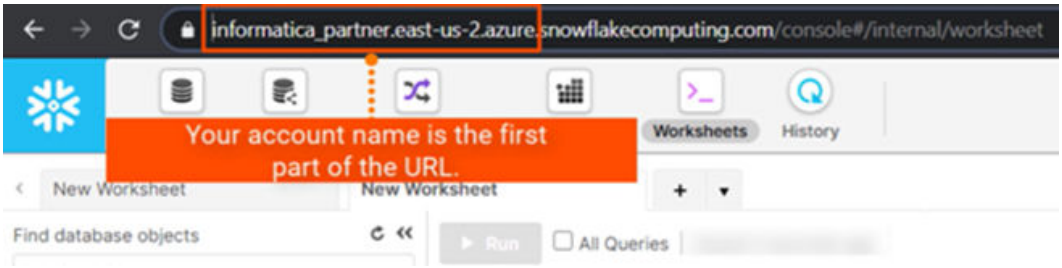
Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

Standard

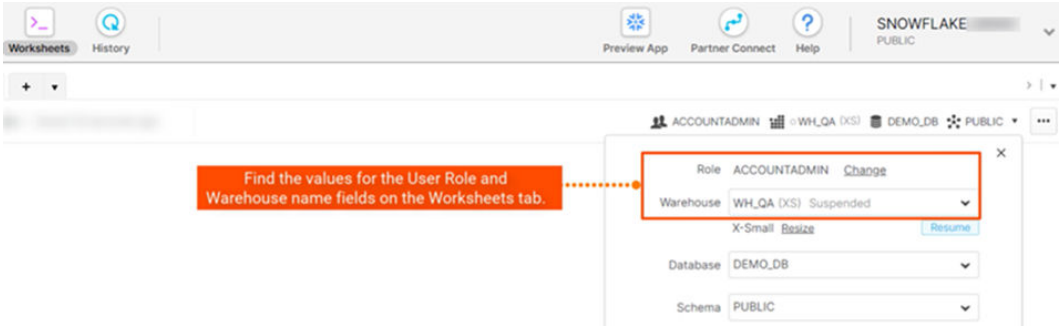
To connect to Snowflake using standard authentication, you need the Snowflake user name, password, account name, and warehouse name.

Let's get the required details such as the Snowflake account name, warehouse, and role details from the Snowflake account.

The following image shows you where you can find the name of your Snowflake account:



The following image shows you where you can find the name of the warehouse and role details of your Snowflake account:



Authorization code

To connect to Snowflake using the OAuth 2.0 authorization code, you need the Snowflake account name, warehouse name, authorization URL, access token URL, client ID, client secret, and access token.

To get the authorization details, create a security integration in Snowflake that enables OAuth access, which acts as a secure gateway for your application to connect to Snowflake and register the following Informatica redirect URL in Security Integration:

`https://<Informatica cloud hosting facility for your organization>/ma/proxy/oauthcallback`

If the access token expires, Informatica redirect URL, which is outside the customer firewall, tries to connect to the endpoint and retrieves a new access token.

When you use the OAuth 2.0 authorization code to connect to Snowflake, you can use either the Snowflake OAuth provider or an external OAuth authorization server, such as Okta or Microsoft Entra ID, that uses the OAuth protocol for accessing Snowflake.

For more information about how to create a security integration and get the authorization details, see Create security integration in the Snowflake documentation.

Key pair

To connect to Snowflake using key pair authentication, you need the private key file and private key file password, along with your Snowflake account user name.

Generate the public and private key pair using OpenSSL. The key pair authentication method requires a 2048-bit RSA key pair. Specify the path to the private key file and password in the connection properties to access Snowflake.

Generate the public and private key

Before you generate the public and private key for key pair authentication, you need to have the security admin role or higher in Snowflake.

1. From the OpenSSL command line, generate a private key:
   - To generate a decrypted private key, run the following command, and provide a passphrase when prompted:

     ```
     $ openssl genrsa 2048 | openssl pkcs8 -topk8 -<cryptographic algorithm> -inform PEM -out rsa_key.p8 -nocrypt
     ```

- To generate an encrypted private key, run the following command, and provide a passphrase when prompted:

```
$ openssl genrsa 2048 | openssl pkcs8 -topk8 -<cryptographic algorithm> -inform
PEM -out rsa_key.p8
```

Note: Check out the Snowflake documentation to use the recommended cryptographic algorithm in the command to generate a private key.

The passphrase is used to encrypt the private key file while connecting to Snowflake.

2. Generate the public key. Run the following command and specify the encrypted private key located in the file, for example, `rsa_key.p8`:

```
openssl rsa -in rsa_key.p8 -pubout -out rsa_key.pub
```

3. Copy the public and private key files in a directory that the Secure Agent can access.
For example, `C:\Program Files\Informatica Cloud Secure Agent\apps\Data_Integration_Server\data\snowflake\rsa_key.p8`

You require the path details when you configure the Snowflake connection.

4. In Snowflake, assign the public key to the Snowflake user using the ALTER USER command:

```
alter user <user> set rsa_public_key='<content of the public key
after removing the header and footer lines>';
```

For example, `alter user jsmith set rsa_public_key='MIIXBIjABCdef...';`

For more information about configuring a key pair authentication for Snowflake, see the Snowflake documentation.

Configure the private key on an advanced cluster

After you generate the public and private key pair using OpenSSL, you need to additionally perform certain tasks for the connection to work in a mapping in advanced mode.

Before you run mappings with the configured connection on an advanced cluster, set the properties for the cluster application in the mapping task.

The following list describes the properties that you need to set in the advanced session properties in a mapping task:

Spark.NeedUserCredentialFileForAdapter=true

Copies the contents of the private key from the location you specify in `Spark.UserCredentialDirOnDIS` from the Secure Agent machine to the Spark driver and executers. The folder that contains the credential file does not have the 1 MB limit. You need to ensure that the credential file of the secret key content that you copy to the cluster application does not exceed 1 MB. You need to set the value to true. Default is false.

If you do not set this flag or you set this flag to false, the private key file is not copied to the cluster application and the mapping fails.

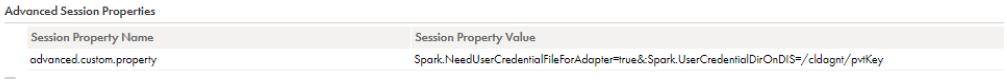
Spark.UserCredentialDirOnDIS=<private key file directory>

Overrides the default Secure Agent directory that contains the private key with the directory that you specify for copying the private key contents to the cluster application. The default directory is `/infa/user/credentials`. Ensure that the directory does not include the private key file name.

If you do not set this flag, the default location is used. To use the default location, create the `/infa/user/credentials` directory on the Secure Agent machine and the copy the private key file here.

If you set the flag to override the location specified in the advanced session properties of the mapping task, make sure that the override location that you specify in *Spark.UserCredentialDirOnDIS* contains the private key file. Ensure that the override location and the private key file have the write permissions.

The following image shows the configured advanced custom property in the mapping task:



Programmatic access token

To connect to Snowflake using programmatic access token authentication, you need the Snowflake user name, programmatic access token, account name, and warehouse name.

If you use the Secure Agent deployed in your environment, serverless runtime environment, or elastic runtime environment, you need to allow the range of IP addresses to connect to Snowflake using a PAT.

To allow the range of IP addresses in Snowflake, perform the following tasks:

1. Create a network rule for the allowed IP addresses.
   For more information about creating a network rule, see Working with network rules in the Snowflake documentation.
2. Create a network policy for the network rule that you created.
   For more information about creating a network policy, see Working with network policies in the Snowflake documentation.

For the detailed steps to generate a PAT, see Generating a programmatic access token in the Snowflake documentation.

Connect to Snowflake

Let's configure the Snowflake Data Cloud connection properties to connect to Snowflake.

Before you begin

Before you get started, you'll need to get information from your Snowflake account based on the authentication type that you want to configure.

Check out "Prepare for authentication" on page 252 to learn more about the authentication prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	This property applies to standard, key pair, and programmatic access token authentication types. Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help. If you use the key pair authentication type to connect to Snowflake, enter the password for the private key file in the **Private Key File Password** advanced connection property.
Runtime Environment	The name of the runtime environment where you want to run tasks. For application ingestion and replication tasks and database ingestion and replication tasks, you can use the Secure Agent or serverless runtime environment. You can't run an application ingestion and replication, a database ingestion and replication, or a streaming ingestion and replication task on a Hosted Agent or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Authentication types

You can configure standard, authorization code, key pair, and programmatic access token authentication types to access Snowflake.

Select the required authentication method and then configure the authentication-specific parameters.

Standard authentication

Standard authentication is the default type and the minimum requirement for a standard authentication is the Snowflake user name, password, account name, and warehouse name.

Note:

The following table describes the basic connection properties for standard authentication:

Property	Description
Username	The user name to connect to the Snowflake account.
Password	The password to connect to the Snowflake account.

Property	Description
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Advanced settings

The following table describes the advanced connection properties for standard authentication:

Property	Description
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).
	For the list of additional JDBC parameters that you can configure, see "JDBC URL parameters" on page 262.

Authorization code authentication

OAuth 2.0 authentication requires the OAuth 2.0 protocol with Authorization Code grant type to connect to Snowflake. Authorization Code allows authorized access to Snowflake without the need to share or store your login credentials.

The following table describes the basic connection properties for OAuth 2.0 authorization code authentication:

Property	Description
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.
Authorization URL	The Snowflake server endpoint that is used to authorize the user request. The authorization URL is https://<account name>.snowflakecomputing.com/oauth/authorize, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/authorize **Note:** If the account name contains underscores, use the alias name. You can also use the Authorization Code grant type that supports the authorization server in a Virtual Private Cloud network.
Access Token URL	The Snowflake access token endpoint that is used to exchange the authorization code to get an access token. The access token URL is https://<account name>.snowflakecomputing.com/oauth/token-request, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/token-request **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Client ID	Client ID of your application generated when you create a security integration of type OAuth in Snowflake.

Property	Description
Client Secret	Client secret generated for the client ID.
Access Token	The access token value. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties for OAuth 2.0 authorization code authentication:

Property	Description

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).
	For the list of additional JDBC parameters that you can configure, see "JDBC URL parameters" on page 262.

Scope	Determines the access control when the API endpoint has defined custom scopes. For example, specify `session:role:CQA_GCP` as the scope to override the value of the default user role. The value needs to be one of the roles assigned in Security Integration. To enter multiple scope attributes, separate each scope attribute with a space.

Access Token Parameters	Additional parameters to use with the access token URL.
	Define the access token parameters in the following JSON format:
	`[{"Name":"<Parameter name>","Value":"<Parameter value>"}]`
	For example, you can use the following code_verifier parameter when you connect to Snowflake:
	`[{"Name":"code_verifier","Value":"5PMddu6Zcg6Tc4sbg"}]`
	For more information about access token parameters that you can define, see Introduction to OAuth in the Snowflake documentation.

Property	Description

Authorization Code Parameters	Additional parameters to use with the authorization token URL.
	Define multiple parameters, separated by comma, in the following JSON format:
	`[{"Name":"<Parameter name>","Value":"<Parameter value>"}, {"Name":"<Parameter name>","Value":"<Parameter value>"}]`
	For example, you can use the following code_challenge and code_challenge_method parameters when you connect to Snowflake:
	`[{"Name":"code_challenge","Value":"Ikr-vv52th0UeVRi4"}, {"Name":"code_challenge_method","Value":"S256"}]`

Refresh Token	The refresh token value. Enter the populated refresh token value that you get from the OAuth endpoint, or click **Generate AccessToken** to populate the refresh token value. If the access token is not valid or expires, the Secure Agent fetches a new access token with the help of the refresh token. **Note:** If the refresh token expires, provide a valid refresh token or regenerate a new refresh token by clicking **Generate Access Token**.

Key pair authentication

Key pair authentication requires the private key file and private key file password, along with your Snowflake account user name to connect to Snowflake.

When you configure key pair authentication to connect to Snowflake, don't use a Hosted Agent as the runtime environment.

The following table describes the basic connection properties for key pair authentication:

Property	Description
Username	The user name to connect to the Snowflake account.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.

Property	Description
Warehouse	The Snowflake warehouse name.
Private Key File	Path to the private key file, including the private key file name, that the Secure Agent uses to access Snowflake. For example, specify the following path and key file name in the Secure Agent machine: - On Windows: `C:\Users\path_to_key_file\rsa_key.p8` - On Linux: `/export/home/user/path_to_key_file/rsa_key.p8` To use the serverless runtime environment, specify the following path and key file name in the serverless agent directory: `/home/cldagnt/SystemAgent/serverless/configurations/ssl_store/<Private key file name>` **Note:** Verify that the keystore is FIPS-certified.

Advanced settings

The following table describes the advanced connection properties for key pair authentication:

Property	Description

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).
	For the list of additional JDBC parameters that you can configure, see "JDBC URL parameters" on page 262.

Private Key File Password	This property is required when you enable the secret vault in the connection. Password for the private key file.

Programmatic access token authentication

The minimum requirement for an programmatic access token authentication is the Snowflake user name, programmatic access token, account name, and warehouse name.

The following table describes the basic connection properties for programmatic access token authentication:

Property	Description
Username	The user name to connect to the Snowflake account.
Programmatic Access Token	The programmatic access token to connect to the Snowflake account. Enter the programmatic access token granted by Snowflake.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. If the account name contains underscores, you need to use the alias name. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Advanced settings

The following table describes the advanced connection properties for programmatic access token authentication:

Property	Description
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	You can specify multiple JDBC connection parameters, separated by ampersand (&), in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, you can pass the following database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	When you add parameters, ensure that there is no space before and after the equal sign (=).
	For the list of additional JDBC parameters that you can configure, see "JDBC URL parameters" on page 262.

JDBC URL parameters

You can use the additional JDBC URL parameters field in the Snowflake Data Cloud connection to customize and set any additional parameters when you connect to Snowflake.

The Snowflake warehouse, database, schema, table, and storage integration names are case-sensitive. Be sure to take this into account when you define properties in the additional JDBC parameters field.

You can configure the following properties as additional JDBC URL parameters in the Snowflake Data Cloud connection:

- To override the database and schema name used to create temporary tables in Snowflake, enter the database and schema name in the following format:

  ```
  ProcessConnDB=<DB name>&ProcessConnSchema=<Schema name>
  ```

- To view only the specified database and schema while importing a Snowflake table, enter the database and schema name in the following format:

  ```
  db=<Database name>&schema=<Schema name>
  ```

- To read UDF string and numeric data from Snowflake, enter the database and schema where the UDF is created in Snowflake in the following format:

  ```
  db=<Database name>&schema=<Schema name>
  ```

- To access Snowflake through Okta SSO authentication, enter the web-based IdP implementing SAML 2.0 protocol in the following format:

  ```
  authenticator=https://<Your Okta account name>.okta.com
  ```

  Note: Microsoft ADFS is not applicable.

  For more information about configuring Okta authentication, see Configuring Snowflake to use federated authentication.

- To load data from Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2 to Snowflake for SQL ELT optimization, enter the Cloud Storage Integration name created for the Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2 account in Snowflake in the following format:

  ```
  storage_integration=<Storage Integration name>
  ```

  The storage integration name is case-sensitive. For example, if the storage integration name you created for Amazon S3, Google Cloud Storage, or Microsoft Azure Data Lake Storage Gen2 in Snowflake is *STORAGE_INT*, you need to specify the same integration name:

  ```
  storage_integration=STORAGE_INT
  ```

  Note: You can also load data from Amazon S3 to Snowflake for SQL ELT optimization without using storage integration.

- To connect to Snowflake using the proxy server, enter the following parameters:

  ```
  useProxy=true&
  proxyHost=<Proxy host IP address>&
  proxyPort=<Proxy server port number>&
  proxyUser=<Proxy server user name>&
  proxyPassword=<Proxy server password>
  ```

- To ignore double quotes in the table and treat all tables as case-insensitive, enter the following parameter:

  ```
  QUOTED_IDENTIFIERS_IGNORE_CASE=true
  ```

  When you set this property in the connection to true, Snowflake ignores the double quotes in the table and treats all tables as case-insensitive.
  If you have set this property to true, you cannot access case-sensitive tables with the same connection. You need to create a new connection to fetch any existing case-sensitive tables.

- To filter queries that are executed in a Snowflake job on the Snowflake web interface, enter the tag name in the following format:

  ```
  query_tag=<Tag name>
  ```

  You have an option to override the query_tag parameter that is defined in the Snowflake connection when you run a mapping task.

To override the query_tag parameter, click the **Runtime Options** tab of the mapping task. In the **Advanced Session Properties** section, select **Custom Properties** from the **Session Property Name** list, and then enter the following value:

```
snowflake_query_tag=<Tag name>
```

Note: In advanced mode, you can't override the query_tag parameter.

- To write data of the TIMESTAMPNTZ data type with microsecond or nanosecond precision to a Snowflake target when the Secure Agent is configured with the *INFA_DTM_STAGING_ENABLED_CONNECTORS* staging property for the write operation, enter the following parameter with its corresponding values:

  - For microsecond precision, enter the following value:

    ```
    DTM_STAGING_TIMESTAMPNTZ_FORMAT=YYYY-MM-DD HH24:MI:SS.US
    ```

  - For nanosecond precision, enter the following value:

    ```
    DTM_STAGING_TIMESTAMPNTZ_FORMAT=YYYY-MM-DD HH24:MI:SS.NS
    ```

  This parameter doesn't apply to mappings in advanced mode.

- To read or write data of the VARCHAR data type with a maximum precision value of 104,857,600, enter the following parameter:

  ```
  VARCHAR_MAX_PRECISION=true
  ```

  When you increase the maximum precision value to 104,857,600, you might need to increase the Java heap space memory.
  For more information about how to increase the Java heap space memory, see the Increase Java heap size Knowledge article.

In addition to the parameters listed, this field provides you the flexibility to configure other Snowflake parameters based on your requirements.

Microsoft Azure Active Directory for external OAuth authorization

You can use Microsoft Azure Active Directory as an external OAuth authorization server to authenticate Snowflake.

To use Microsoft Azure Active Directory as an external OAuth authorization server, select **Authorization Code** as the authentication type in the connection properties. Provide the account name, warehouse, authorization URL, access token URL, client ID, client secret, access token, and scope details from the Microsoft Azure Active Directory OAuth authorization server.

To configure the Microsoft Azure Active Directory OAuth authorization server, see Configure Microsoft Entra ID for external OAuth in the Snowflake documentation.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use the proxy server on Windows and Linux. You can use the unauthenticated or authenticated proxy server. You can configure proxy for connections used both in mappings and in mappings in advanced mode.

To configure proxy settings for the Secure Agent, use one of the following methods:

- Configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

  For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.

- Configure the JVM options for the DTM in the Secure Agent properties. For instructions, see the Proxy server settings Knowledge article.
- Configure the proxy server properties in the additional JDBC URL parameters in the Snowflake connection. For more information, see "JDBC URL parameters" on page 262.

Private links to access Snowflake

You can enable private communication between the Secure Agent and Snowflake endpoint hosted on the Azure, AWS, or Google Cloud platform.

When you create a Snowflake Data Cloud connection, specify the Snowflake private link account name in the **Account** field in the connection properties.

The private communication setup ensures that the connection to Snowflake uses the internal network of AWS, Azure, or Google Cloud and doesn't take place over the public Internet.

To connect to the Snowflake account over the private AWS network, see AWS Private Link and Snowflake.

To connect to the Snowflake account over the private Azure network, see Azure Private Link and Snowflake.

To connect to the Snowflake account over the private Google Cloud network, see Google Cloud Private Service Connect and Snowflake.

Teradata connection properties

Create a Teradata connection to securely read data from or write data to Teradata.

Prerequisites

Before you can use Teradata Connector, ensure that you meet the prerequisite tasks.

Perform the following prerequisite tasks:

1. Install the Teradata Parallel Transporter utilities and set the environment variables.
2. Ensure that you have access to the Secure Agent directory that contains the success and error files. This directory path must be the same on each Secure Agent machine in the runtime environment. The Teradata JDBC driver is packaged with the Secure Agent. When you install the Secure Agent, the JDBC driver is installed and the JDBC jars are copied to the Secure Agent machine.
3. Configure the authentication prerequisites to connect to a Teradata database. You can configure Native, LDAP, or KRB5 authentication. Keep the authentication details handy based on the authentication type that you want to use.
   For Data Ingestion and Replication, you can use Native authentication only.

Teradata Parallel Transporter Utilities

Before you use Teradata Connector, install the Teradata Parallel Transporter utilities on the Secure Agent machine.

You must install the following Teradata Parallel Transporter utilities:

- Teradata Parallel Transporter Base
- Teradata Parallel Transporter Stream Operator

- Teradata CLIv2
- Teradata Generic Security Services
- Shared ICU Libraries for Teradata

Setting Environment Variables

You must configure Java and Teradata environment variables before you can use Teradata Connector.

The following table describes the environment variables you must set on UNIX:

Environment Variable	Value
THREADONOFF	On UNIX and Linux, set the `THREADONOFF` environment variable to 1 to enable multithreading support for Teradata processes.
NLSPATH	Set the NLSPATH variable to the location of the `opermsgs.cat` file. For example, `/opt/teradata/client/15.10/msg/%N`

Also, set the shared library environment variable based on the operating system.

The following table describes the shared library variables for each operating system:

Operating System	Value
Windows	`PATH`
Linux	`LD_LIBRARY_PATH`

For example, use the following syntax for Linux:

- Using a Bourne shell:

```
export LD_LIBRARY_PATH="/opt/teradata/client/15.10/lib64:${LD_LIBRARY_PATH}"
```

- Using a C shell:

```
LD_LIBRARY_PATH="/opt/teradata/client/15.10/lib64:${LD_LIBRARY_PATH}
```

After you set the environment variables, restart the Secure Agent.

Connect to Teradata

Let's configure the Teradata connection properties to connect to Teradata.

Before you begin

Before you get started, be sure to complete the prerequisites.

Check out "Prerequisites" on page 265 to learn more about the authentication prerequisites and tasks that you must perform.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. This property is not supported by Data Ingestion and Replication and the Data Access Management services. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Specify a Secure Agent. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
TDPID	The name or IP address of the Teradata database machine. If you configure KRB5 authentication to connect to the Teradata database, specify the fully qualified hostname of the Teradata database machine.
Database Name	The Teradata database name. If you do not enter a database name, Teradata PT API uses the default login database name.
Code Page	Code page associated with the Teradata database. Select one of the following code pages: - MS Windows Latin 1. Select for ISO 8859-1 Western European data. - UTF-8. Select for Unicode and non-Unicode data. When you run a task that extracts data from a Teradata source, the code page of the Teradata PT API connection must be the same as the code page of the Teradata source.

Authentication types

You can configure the Native, LDAP, or KRB5 authentication type to connect to the Teradata database.

Note: For Data Ingestion and Replication, you must use Native authentication. Do not use the KRB5 or LDAP authentication type.

Select the required authentication type and then configure the authentication-specific parameters.

Native authentication

To configure native authentication, you need the user name and password from your Teradata account.

The following table describes the basic connection properties for Native authentication:

Property	Description
Username	User name with the appropriate read and write database permissions to access the Teradata database.
Password	Password for the database user name.

Advanced settings

The following table describes the advanced connection properties:

Property	Description
Tenacity	Amount of time, in hours, that the Teradata PT API continues attempting to log in again when the maximum number of operations run on the Teradata database. Specify a positive integer. Default is 4.
Max Sessions	Maximum number of sessions allowed for the Teradata PT API to establish connection with the Teradata database. Specify a positive, non-zero integer. Default is 4.
Min Sessions	Minimum number of Teradata PT API sessions required for the Teradata PT API job to continue. Specify a positive integer between 1 and the value specified in the **Max Sessions** field. Default is 1.
Sleep	Amount of time, in minutes, that the Teradata PT API waits before it attempts to log in again when the maximum number of operations run on the Teradata database. Specify a positive, non-zero integer. Default is 6.
Data Encryption	Enables full security encryption of SQL requests, responses, and data. Default is disabled.
Block Size	The size of each data block, in bytes, that the Teradata PT API reads from the Teradata source through the Export operator. Maximum is 16775168 bytes for Teradata database version 16.20 and later. If the Teradata database version is earlier than 16.20, Teradata scales down the block size from 16775168 bytes to the maximum allowed value. The block size 16775168 is not allowed in the Spool mode. For more information about the block size, see the Teradata logs and verify the Teradata documentation of the same version.

Property	Description

Metadata Advanced Connection Properties	The optional properties for the JDBC driver to fetch the metadata from Teradata.
	If you add multiple properties, make sure to separate each property with a comma.
	Example: `tmode=ANSI,JAASCONFIG=<path to JAASCONFIG>/TeraJDBC.config`
	The following snippet is an example of JAASCONFIG with mandatory attributes:
	`JDBC_DRIVER_01 {`
	`com.sun.security.auth.module.Krb5LoginModule required`
	`useTicketCache=true;`
	`};`

Enable Metadata Quotification	Determines if a Teradata connection reads reserved words used in table or column names from the Teradata database. Select the **Enable Metadata Quotification** checkbox for the Secure Agent to read reserved words from Teradata.

Database privileges

Before you use Teradata Connector, ensure you have select privileges on specific Data Dictionary tables.

Verify that you have the following database privileges:

- Select privileges on DBC.Tables, DBC.Columns, DBC.UDTInfo, and DBC.Databases.
  For information about select privileges, see the Teradata JDBC Driver documentation.

Workday Mass Ingestion connection properties

When you set up a Workday Mass Ingestion connection, you must configure the connection properties.

The properties of a Workday Mass Ingestion connection vary based on the authentication method that you specify for the connection. When you create a connection, you can select one of the following authentication methods:

- **Basic**: Authenticates the connection by validating the login credentials of the Workday account.
- **OAuth 2.0 Refresh Token Flow**: Authenticates the connection by using an application that is registered in Workday. To use this method, you must register an application in Workday and then specify the client ID and client secret of the application in the connection properties. For more information about registering an application in Workday, see the Workday documentation.

Connection properties for Basic authentication

The following table describes the connection properties for a Workday Mass Ingestion connection configured with Basic authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
Domain Name	Name of the Workday domain that contains the resources that you want to access.
Tenant Name	Identifier of the Workday tenant that you want to access.
Version	Optional. Web Service Description Language (WSDL) version for the endpoints that the connection must use to retrieve Workday data. The list of operations supported for a web service depends on the WSDL version that you specify in this field. **Note:** Informatica recommends that you use WSDL v37.0 because Workday Mass Ingestion connections might not read data from the services that are not part of WSDL v37.0. For more information on the WSDL versions, see the Workday Web Services (WWS) documentation.
User Name	User name of the Workday account.
Password	Password for the Workday account.

Note: If you configure a connection with the Basic authentication method and then test the connection, the test is always successful even if the connection property values that you specified are incorrect. Therefore, ensure that you specify correct values for the connection properties before you save the connection.

Connection properties for OAuth 2.0 Refresh Token Flow authentication

The following table describes the connection properties for a Workday Mass Ingestion connection configured with OAuth 2.0 Refresh Token Flow authentication:

Connection property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.

Connection property	Description
Runtime Environment	Name of the runtime environment where you want to run the ingestion tasks. For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent.
Domain Name	Name of the Workday domain that contains the resources that you want to access.
Tenant Name	Identifier of the Workday tenant that you want to access.
Version	Optional. Web Service Description Language (WSDL) version for the endpoints that the connection must use to retrieve Workday data. The list of operations supported for a web service depends on the WSDL version that you specify in this field. **Note:** Informatica recommends that you use WSDL v37.0 because Workday Mass Ingestion connections might not read data from the services that are not part of WSDL v37.0. For more information on the WSDL versions, see the Workday Web Services (WWS) documentation.
User Name	Optional. User name of the Workday account.
Client ID	Client ID of the application registered in Workday.
Client Secret	Private key of the application registered in Workday.
Refresh Token	Refresh token string that Workday generates for the registered application.
Token Endpoint	OAuth token endpoint of the Workday instance. The registered application sends the access token requests to this endpoint.

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